LilyPond Contemporary Notation Cookbook: Snippets and Their Grammars

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Contents

Foreword

0.1 Preamble

This document houses all the codes I built on LilyPond since September 2024. Because I deal with contemporary notations in my compositional practice, I found myself creating codes and turning them into variables in order to repeatedly use them in my projects. I created a dedicated `.ly` file to store these codes for use, which quickly became very lengthy. I thought it would be useful to organize them into a document where I could easily consult and remind myself what they are and how to use them. This is that document.

Because I use LilyPond actively in my daily compositional and musical typesetting activities, this document is a work in progress.

0.2 README

This document and the codes contained herein are under the MIT License. So long as you include the copyright as well as the MIT License permission notices, please feel free to use my codes in your LilyPond files or modify them according to your specific need. Furthermore, crediting in the following manner is greatly appreciated:

```
% Original Code written by Yoshiaki Onishi
% https://github.com/yoshiakionishi/lilypond-snippets
```

I make this document public because I wish to return something useful to the LilyPond community, but also to seek and implement any improvements other users may find in my codes. Please feel free to reach out to the email address shown on the title page of this document.

In the interest of making the codes found in this document available to as many people as possible, I have avoided using copyrighted musical examples. However, wherever appropriate, I have provided bibliographical sources. Furthermore, I acknowledge that, just as academic work in humanities goes, my ideas are built on those that are formulated by others; as such, whenever there is a direct source of inspiration for formulating a code, I provide sources.

In creating this document, I make no claim that my notational choices represent an absolute standard that everyone should adhere to. Once the basic principles of notation and typesetting are established (e.g., avoiding collisions, etc.), notation becomes a personal decision for each composer, shaped by careful study of preexisting scores and an evaluation of their musical contexts.

For example, in his book *The Bass Clarinet – A Personal History*, Harry Sparnaay lists nine variants of noteheads for the slap tongue technique.[1] In my work, I created two subcategories of the slap tongue technique: one followed by a pitch and another followed by an air sound (which produces the slap tongue effect that sounds "empty"). To distinguish between the two, I decided to use encircled noteheads—both filled and hollow—and attribute them to each subcategory. Again, this is a method that I have found works for my music, but I would be reluctant to suggest that others should follow the same.[2]

Readers are encouraged to modify my codes in order to suit their desired techniques. This document serves as a record of how I arrived at certain notational choices, because learning LilyPond meant that I would also need to become familiar with Scheme, which proved to be somewhat challenging—even though I have used Common Lisp before owing to programming in OpenMusic—because I had to make many guessworks as I navigated various Scheme codes in other snippets available online. I have also gained familiarity in PostScript language as I continued to familiarize myself with LilyPond.[3]

0.3 Background

After MakeMusic announced that they would cease development of the music notation software program Finale, which I had used for the past twenty-four years, I decided to explore a few other music notation programs to determine the best alternative. At the time of writing this document in late November 2024, a little under three months have passed since I started using LilyPond for my daily typesetting needs. I now open LilyPond more often than Finale and am committed to using it for the foreseeable future. LilyPond appears to me as the way forward both as a composer and a musical typesetter, as other proprietary notation programs, such as Dorico (which MakeMusic has claimed to be the leading program in the industry) and Sibelius, fall short of what I wish to accomplish.

While LilyPond is "just" a music notation software program that I happened to choose, it is, in a way, more than a toolkit for a composer. It appears that way to me, at least, because choosing an open-source platform with strong community support and engagement, rather than a proprietary program where desired functionality is subject to the priorities of a small group of salaried developers, reflects a critique of the capitalist/commercialist mindset that often pervades a composer's life.

For example, before transitioning to LilyPond, I briefly explored Dorico. However, as of late September 2024, its functionality for displaying straight flags was very limited; the angle of the straight flags provided by the software was too steep. I consulted the online forum and discovered that another user had posted a question similar to mine. The chief developer of Dorico responded to that post, noting that implementing improvements to this feature was possible but "not currently a high priority."[4] In this tiered structure typical of capitalism, composers may find themselves with increasingly limited creative "freedom."

MakeMusic has heavily advertised on social media platforms that Finale users should migrate to Dorico because it is the "next industry standard." However, this advertising seems to discourage

1. Harry Sparnaay, *The Bass Clarinet: A Personal History* (Periferia Sheet Music, 2012), 66.
2. This particular notation becomes quickly problematic in terms of rhythmic notation when a bar is longer than a half note (e.g. 1/2, 2/4, 4/8...) For this reason, I tend to favor time signatures that avoids the use of a half note, such as 3/8 or 5/16.
3. See Appendix A for some resources I referred to for Scheme- and PostScript-related matters.
4. See: https://forums.steinberg.net/t/straight-flags-angle/766503.

thoughtful consideration of alternatives, leaving little room for reflection or exploration. I became increasingly disillusioned as I witnessed the coercion to invest in a program—however exciting it may appear—with no definite promise of its long-term security and stability.

Of course, it is not my intent to claim that all composers should abandon their proprietary programs of choice, particularly those they have invested money in and/or have been using for many years. It is, however, important to note that:

1. All proprietary programs are at the mercy of the executives who run the companies behind them. "Oh, [insert the name of a proprietary program] is operated by [insert the name of its company], and I just don't see them closing the program down," someone might say. Yet, it happened to Finale.

2. All notation programs, owing to the ways they operate, exert some degree of influence on the way composers compose. As early as the 1980s, Finale's *Mass Mover*, *Note Mover*, and MIDI playback features were already influential in shaping the way composers worked on their music.[5] On the one hand, these features may have helped composers save time. On the other hand, their ready availability may have invited overuse.

3. The lack or underdevelopment of certain functionalities may also push composers to work in certain ways rather than others. Finale benefitted from having the flexibility to implement graphical notation, but even then, many of my composer friends found it practical to use external graphical editing programs to further refine their scores. Even from my personal experience using Finale, I encountered situations where I had to devise creative alternatives to meet my notational goals.

These points implicitly highlight the benefits of learning an additional notation program, ideally an open-source one, alongside the program one primarily uses. LilyPond resonated with me most because of its text-based interface, which I have become increasingly familiar with through my involvement in computer programming. As other users have remarked, I have also found it to be very flexible and extensible. All the snippets I list in this document can be reused with relative ease, allowing me to save time in the long run when using specialized notations in my music. This was not necessarily the case when working on the music notation of extended techniques in Finale.

0.4 How This Document Is Structured

Each chapter of this document addresses a specific element of music notation, such as noteheads, stems, beams, and so on. Some chapters, however, cover topics specific to LilyPond coding, such as Markups and Spanners. Snippets that use more than one snippet covered in earlier chapters, thus simulating practical applications of these snippets, are covered in the chapter *Combinations*. Snippets that do not appear to belong to earlier chapters find their home in the chapter *Miscellanies*.

Each snippet entry includes a musical example, a description, the relevant grammar, the code required for the snippet to function, and, whenever necessary, a "Discussion" section.

0.5 LilyPond Version Used

The version of LilyPond used to create these snippets is `2.26.0`.

5. For example, watch from 15:20 of https://youtu.be/T1IRlg87Qks.

0.6 AI/LLM Statement

In coming up with the codes found in this document, I did not use commercially available AI Chatbot or Assistant. I underwent many hours of trials and errors (mostly errors) on each of the codes to create them. When I am stuck, which happens all the time, I refer to the LilyPond Manuals, as well as manuals for programming languages Scheme and PostScript. I have consulted other manuals that are relevant to the creation of these codes as appropriate. Sometimes I ask skilled programmers in the mailing list `lilypond-user` for help and hint. I primarily consult examples found on, among others, `lilypond-user` mailing list and `LilyPond Wiki` Website. When I quote/reuse significant portion of the code or the idea that may be considered non-trivial, I credit them in my codes.

0.7 Acknowledgements

I thank the supportive community of LilyPond users, whose exchanges on `lilypond-user` mailing list have inspired me greatly.

Even though I have not met him, I am grateful to Ben Lemon for his generosity in creating and sharing his LilyPond tutorial videos on YouTube. These videos were immensely helpful during the initial stages of learning LilyPond.

I also want to thank my friends who inspired me to start using LilyPond. It was Cole Ingraham who first introduced me to the program in 2016. My initial attempt at using it was not successful, but more recently, Santiago Beis composed and typeset his orchestral piece *Spletna* entirely in LilyPond, which compelled me to give it another try.

I extend my gratitude to my former composition students at the University of Delaware School of Music, with whom I embarked on this journey of learning LilyPond. Even though they were not directly affected by Finale's discontinuation, they remained curious and enthusiastic about exploring this program. I hope that if the programs of their choice ever face a fate similar to Finale (though I sincerely hope they do not), they will be better equipped to adapt without the annoyance and arduous work often associated with transitioning to a new tool.

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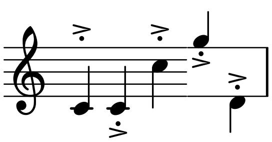
Chapter 1

Articulations

1.1 Accent-Staccato



1.1.1 Description

While *accent-staccato* is not specific to contemporary music, in LilyPond, specifying accent and staccato via `->-.` could cause the two articulation marks to be separated from each other. This happens because of the default setting for accents is to have them placed *outside* the staff line. For example, if you wrote `\stemUp g''->-.`, the following results:

This code implements a combination of the music glyphs `scripts.sforzato` and `scripts.staccato` as one entity.

1.1.2 Grammar

```
NOTE \accentStaccato
NOTE \accentStaccatoUp
NOTE \accentStaccatoDown
```

1.1.3 Code

```
\version "2.26.0"

#(append! default-script-alist
          (list
           `(accentStaccatoUp
             . (
```

```
                (stencil . ,ly:text-interface::print)
                (text . ,#{ \markup
                            \center-align
                            \combine \musicglyph "scripts.sforzato"
                            \translate #'(0 . -0.75)
                            \musicglyph "scripts.staccato"
                            #})
                ; any other properties
                (toward-stem-shift-in-column . 1.0)
                (outside-staff-priority . #t)
                (padding . 0.5)
                (avoid-slur . around)
                (side-axis . ,RIGHT)
                ; new in 2.26.0, side-axis value needs to be specified
                (direction . ,UP))))

          (list
           `(accentStaccatoDown
             . (
                (stencil . ,ly:text-interface::print)
                (text . ,#{ \markup \center-align
                            \combine \musicglyph "scripts.staccato"
                            \translate #'(0 . -0.75)
                            \musicglyph "scripts.sforzato"
                            #})
                ; any other properties
                (toward-stem-shift-in-column . 1.0)
                (outside-staff-priority . #t)
                (padding . 0.5)
                (avoid-slur . around)
                (side-axis . ,RIGHT)
                ; new in 2.26.0, side-axis value needs to be specified
                (direction . ,DOWN)))))

accentStaccato =  #(make-articulation 'accentStaccatoUp)
accentStaccatoUp = #(make-articulation 'accentStaccatoUp)
accentStaccatoDown = #(make-articulation 'accentStaccatoDown)


{
 \override Staff.TimeSignature.stencil = ##f
 \time 5/4

 c'4\accentStaccato c'4 \accentStaccatoDown c''\accentStaccatoUp
 \stopStaff
 \override Staff.StaffSymbol.line-positions = #'(4 -4)
```

```
  \startStaff

  \stemUp g'' \tweak Y-offset #1.5 \accentStaccatoDown
  \stemDown d' \tweak Y-offset #-0.5 \accentStaccatoUp
}
```

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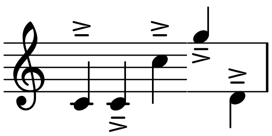
1.2 Accent-Tenuto



1.2.1 Description

For the same rationale as explained in the Accent-Staccato entry, this code implements a combination of the music glyphs `scripts.sforzato` and `scripts.tenuto` as one entity.

1.2.2 Grammar

```
NOTE \accentTenuto
NOTE \accentTenutoUp
NOTE \accentTenutoDown
```

1.2.3 Code

```
\version "2.26.0"

#(append! default-script-alist
          (list
           `(accentTenutoUp
             . (
                (stencil . ,ly:text-interface::print)
                (text . ,#{ \markup


                            \center-align
                            \combine \musicglyph "scripts.sforzato"
                            \translate #'(0 . -0.75)
                            \musicglyph "scripts.tenuto"
                            #})
                ; any other properties
                (toward-stem-shift-in-column . 1.0)
                (outside-staff-priority . #t)
                (padding . 0.5)
                (avoid-slur . around)
                (side-axis . ,RIGHT)
                ; new in 2.26.0, side-axis value needs to be specified
                (direction . ,UP))))

          (list
           `(accentTenutoDown
             . (
                (stencil . ,ly:text-interface::print)
                (text . ,#{ \markup \center-align
```

```
                              \combine \musicglyph "scripts.tenuto"
                              \translate #'(0 . -0.75)
                              \musicglyph "scripts.sforzato"
                              #})
                ; any other properties
                (toward-stem-shift-in-column . 1.0)
                (outside-staff-priority . #t)
                (padding . 0.5)
                (avoid-slur . around)
                (side-axis . ,RIGHT)
                ; new in 2.26.0, side-axis value needs to be specified
                (direction . ,DOWN)))))

accentTenuto =  #(make-articulation 'accentTenutoUp)
accentTenutoUp = #(make-articulation 'accentTenutoUp)
accentTenutoDown = #(make-articulation 'accentTenutoDown)


{
 \override Staff.TimeSignature.stencil = ##f
 \time 5/4

 c'4\accentTenuto c'4 \accentTenutoDown c''\accentTenutoUp
 \stopStaff
 \override Staff.StaffSymbol.line-positions = #'(4 -4)
 \startStaff

 \stemUp g'' \tweak Y-offset #1.5 \accentTenutoDown
 \stemDown d' \tweak Y-offset #-0.5 \accentTenutoUp
}
```

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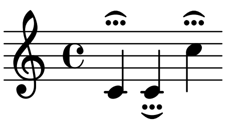
1.3 Jeté (Ricochet)



1.3.1 Description

I use this notation to designate jeté/ricochet for string instruments, adding that the number of bounces are undetermined.[1]

I apply this indication *above* the note regardless of how high or low the note is; however, in case of need, I have supplied the version to be used *under* the note.

1.3.2 Grammar

```
NOTE \jete
NOTE \jeteUp
NOTE \jeteDown
```

1.3.3 Code

```
\version "2.26.0"

jeteDesign =
\markup
\center-align
\combine \combine \combine
\override #'(filled . #t)
\path  #0.1
#'((moveto    -0.25 0.5)
   (curveto   0.35 1.1 0.85 1.1 1.45 0.5)
   (curveto   0.85 0.8 0.35 0.8 -0.25 0.5)
   (closepath))
\draw-circle #0.2 #0 ##t
\translate #'(0.6 . 0) \draw-circle #0.2 #0 ##t
\translate #'(1.2 . 0)\draw-circle #0.2 #0 ##t
#(append! default-script-alist
          (list
           `(jetelistUp
             . (
                (stencil . ,ly:text-interface::print)
                (text . ,#{ \markup \jeteDesign #})
                ; any other properties
                (toward-stem-shift-in-column . 1.0)
                (outside-staff-priority . #t)
```

1. Concerning the technique of adding articulation designs to an internal alist, I was inspired by the following thread on `lilypond-user` mailing list: https://lists.gnu.org/archive/html/lilypond-user/2015-04/msg00105.html

```
                (padding . 0.5)
                (avoid-slur . around)
                (side-axis . ,RIGHT)
                ; new in 2.26.0, side-axis value needs to be specified
                (direction . ,UP))))

          (list
           `(jetelistDown
             . (
                (stencil . ,ly:text-interface::print)
                (text . ,#{ \markup \rotate #180 \jeteDesign #})
                ; any other properties
                (toward-stem-shift-in-column . 1.0)
                (outside-staff-priority . #t)
                (padding . 0.5)
                (avoid-slur . around)
                (side-axis . ,RIGHT)
                ; new in 2.26.0, side-axis value needs to be specified
                (direction . ,DOWN)))))

jete =  #(make-articulation 'jetelistUp)
jeteUp = #(make-articulation 'jetelistUp)
jeteDown = #(make-articulation 'jetelistDown)


{c'4\jete c'4 \jeteDown c''\jeteUp }

```

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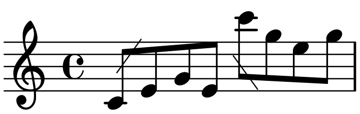
Chapter 2

Beams

2.1 Slashed Beam



2.1.1 Description

This function adds a slash to an ordinary beam in order to simulate a group of grace notes. See **Repeat Note Patterns As Fast As possible** in Chapter 12 for an example of using this function.

This function adds the slash with the correct angle according to the stem direction.

2.1.2 Grammar

```
\slashBeam NOTES
```

2.1.3 Code

```
\version "2.26.0"

\pointAndClickOff

slashBeam = {
 \once \override Stem.stencil =
 #(grob-transformer
   'stencil
   (lambda (grob original)
    (let* ((added-markup
            #{
             #(case (ly:grob-property grob 'direction)
               ( (1) #{ \markup \general-align #X #-0.5
```

```
                        {
                         \path #0.1
                         #'((rmoveto 0 0)
                            (rlineto 2 2.75))
                     } #}
                     )
               ( (-1) #{ \markup \general-align #X #-0.5
                         {
                          \path #0.1
                          #'((rmoveto 0 0)
                             (rlineto 2 -2.75))
                      } #}
                      )
               )
            #}
            )
           (added-stencil (grob-interpret-markup grob added-markup)))

     (if (ly:stencil? original)
         (case  (ly:grob-property grob 'direction)
          ((1) (ly:stencil-combine-at-edge
                original 1 1 added-stencil -1.75))
          ((-1) (ly:stencil-combine-at-edge
                 original 1 -1 added-stencil -1.75))
          )
         added-stencil))))
}

{
 \slashBeam c'8 e' g' e'
 \slashBeam c''' g'' e'' g''
}

```

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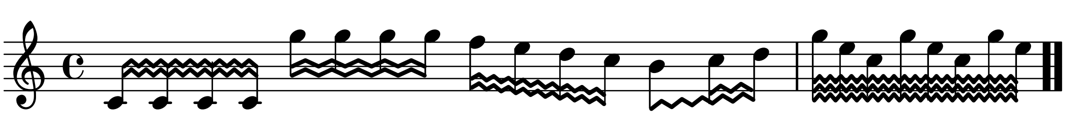
2.2 Wiggle Beam (zig-zag shaped beam)



2.2.1 Description

Ordinary beams are replaced with zig-zag beams. A set of forward then backward beams are printed in the amount specified in the argument. I use this notation in such pieces as *jeux enjeux* (2022) for brass quintet, in order to designate somewhat uneven rhythmic figures, which are nonetheless to be played within the time frame indicated.

`\wiggleBeamOne` replaces an 8th-note beam.

`\wiggleBeamTwo` replaces a 16th-note beam.

`\wiggleBeamThree` replaces a 32nd-note beam.

`\wiggleBeam_markup` adds a zig-zag beam at will. This allows beaming of mixed note durations, such as: 

`\wiggleBeamStemAdjust` allows the adjustment of a stem length, in the event the wiggle beam and the stem do not touch each other.

2.2.2 Grammar

`\wiggleBeamOne #vOffset #howMany #width #rotation`
`\wiggleBeamTwo #vOffset #howMany #width #rotation`
`\wiggleBeamThree #vOffset #howMany #width #rotation`

NB

- `hOffset` = (`\wiggleBeam_markup` only) the horizontal offset value originating from where the ordinary beam is placed.
- `vOffset` = the vertical offset value originating from where the ordinary beam is placed.
- `howMany` = how many ”wiggles” to print. It only accepts integers.
- `width` = how wide each ”wiggle” should appear. When in doubt, start with `#1`.
- `rotation` = a positive value would rotate the beam upward, and the negative value would rotate the beam downward.

`NOTE \wiggleBeam_markup #hOffset #vOffset #howMany #width #rotation`

NB

- `hOffset` = the horizontal offset value originating from where the ordinary beam is placed.
- `vOffset` = the vertical offset value originating from where an above-staff markup is placed. Thus, `#0` would place a wiggle beam above the staff line.
- `howMany` = how many ”wiggles” to print. It only accepts integers.
- `width` = how wide each ”wiggle” should appear. When in doubt, start with `#1`.
- `rotation` = a positive value would rotate the beam upward, and the negative value would rotate the beam downward.
- More than one `\wiggleBeam_markup` may be added in sequence, provided that for each instance all the arguments are defined.

`\wiggleBeamStemAdjust #fromMiddleLine #howFar NOTE`

NB

- `fromMiddleLine` = (`\wiggleBeamStemAdjust` only) = determines one end of the stem, `#0` being the middle line of an ordinary 5-line staff.
- `howFar` = (`\wiggleBeamStemAdjust` only) = computes how long the stem should be extended. A positive value would draw the stem upward, and a negative value would draw the stem downward. An integer corresponds to the distance between two staff lines of an ordinary 5-line staff.

2.2.3 Code

```
\version "2.26.0"
\language "english"

wiggleBeamOne =
#(define-music-function (vOffset howMany howWide howTilted)
  (number? number? number? number?) #{
   \once \override Voice.Beam.stencil = #ly:text-interface::print
   \once \override Voice.Beam.text = \markup {
    \translate #(cons 0 vOffset)
    \postscript #(string-append
                  "newpath
              1 setlinejoin
              1 setlinecap
              0.35 setlinewidth
              0.13 0 moveto "
                  (number->string howMany)
                  " {" (number->string (* 0.6 howWide))  " "
                  (number->string (+ 0.5 howTilted)) " rlineto "
                  (number->string (* 0.6 howWide))
                  " -0.5 rlineto} repeat
              stroke"
                  )

   }
  #})


wiggleBeamTwo =
#(define-music-function (vOffset howMany howWide howTilted )
  (number? number? number? number?) #{
   \once \override Voice.Beam.stencil = #ly:text-interface::print
   \once \override Voice.Beam.text = \markup {
    \translate #(cons 0 vOffset)
    \postscript #(string-append
                  "newpath
              1 setlinejoin
              1 setlinecap
              0.35 setlinewidth
```

```
              0.13 0 moveto "
                  (number->string howMany)
                  " {" (number->string (* 0.6 howWide)) " "
                  (number->string (+ 0.5 howTilted)) " rlineto "
                  (number->string (* 0.6 howWide))
                  " -0.5 rlineto} repeat
              stroke newpath
              0.35 setlinewidth
              1 setlinejoin
              0.13 -0.75 moveto "
                  (number->string howMany)
                  " {" (number->string (* 0.6 howWide))  " "
                  (number->string (+ 0.5 howTilted)) " rlineto "
                  (number->string (* 0.6 howWide))
                  " -0.5 rlineto} repeat
              stroke"
                  )
   }
  #})


wiggleBeamThree =
#(define-music-function (vOffset howMany howWide howTilted )
  (number? number? number? number?)
  #{
   \once \override Voice.Beam.stencil = #ly:text-interface::print
   \once \override Voice.Beam.text = \markup         {
    \translate #(cons 0 vOffset)
    \postscript #(string-append
                  "newpath
              1 setlinejoin
              1 setlinecap
              0.35 setlinewidth
              0.13 0 moveto "
                  (number->string howMany) " {"
                  (number->string (* 0.6 howWide))  " "
                  (number->string (+ 0.5 howTilted)) " rlineto "
                  (number->string (* 0.6 howWide))
                  " -0.5 rlineto} repeat
              stroke
              newpath
              0.35 setlinewidth
              1 setlinejoin
              0.13 -0.75 moveto "
                  (number->string howMany) " {"
                  (number->string (* 0.6 howWide))  " "
                  (number->string (+ 0.5 howTilted)) " rlineto "
                  (number->string (* 0.6 howWide))
```

```
                  " -0.5 rlineto} repeat
              stroke
              newpath
              0.35 setlinewidth
              1 setlinejoin
              0.13 -1.5 moveto "
                  (number->string howMany) " {"
                  (number->string (* 0.6 howWide)) " "
                  (number->string (+ 0.5 howTilted)) " rlineto "
                  (number->string (* 0.6 howWide))
                  " -0.5 rlineto} repeat
              stroke"
                  )
   }
  #})


wiggleBeam_markup =
#(define-music-function (hOffset vOffset howMany howWide howTilted )
  (number? number? number? number? number?)
  #{
   ^\markup         {
    \translate #(cons hOffset vOffset)
    \postscript #(string-append
                  "newpath
              1 setlinejoin
              1 setlinecap
              0.35 setlinewidth
              0.17 0 moveto "
                  (number->string howMany) " {"
                  (number->string (* 0.6 howWide))  " "
                  (number->string (+ 0.5 howTilted)) " rlineto "
                  (number->string (* 0.6 howWide))
                  " -0.5 rlineto} repeat
              stroke"
                  )

   }
  #})

wiggleBeamStemAdjust =
#(define-music-function (fromMiddleLine howFar)
  (number? number?)
  #{
   \once \override Stem.stencil = #ly:text-interface::print
   \once \override Stem.text = \markup {
    \postscript #(string-append
                  "newpath
```

```
              0.12 setlinewidth
              0 " (number->string fromMiddleLine) " moveto
              0 " (number->string howFar) " rlineto
              stroke"
                  )
   }
  #})


{
 \wiggleBeamTwo #0 #9 #1.01 #0 c'16 c'
 \wiggleBeamStemAdjust #-3 #3.4 c' c'
 \wiggleBeamTwo #0 #5 #1.82 #0 g''
 \wiggleBeamStemAdjust #2.5 #-3 g''
 \wiggleBeamStemAdjust #2.5 #-3 g'' g''
 \wiggleBeamTwo #-1 #9 #1.01 #-0.15 f''
 \wiggleBeamStemAdjust #1.5  #-3.5 e''
 \wiggleBeamStemAdjust #1 #-3.5 d''
 \wiggleBeamStemAdjust #0.5 #-3.5 c''
 \wiggleBeamOne #-3.5 #5 #1.4 #0.15 b'8
 c''16 \wiggleBeam_markup #0 #-4.8 #2 #1.4 #0.15 d''
 \wiggleBeamThree #-1.3 #19 #0.73 #0  g''32
 \wiggleBeamStemAdjust #1.5  #-4 e''
 \wiggleBeamStemAdjust #0.5  #-3 c'' g'' e''
 \wiggleBeamStemAdjust #0.5  #-3  c''
 \wiggleBeamStemAdjust #2.5  #-5 g'' e''
 \bar ".."
}
```

2.2.4 Discussion

1. Admittedly, while the current setup allows great flexibility in making the wiggle beams appear, it is entirely possible that some of the parameters be automated.
2. When using many wiggle beams, it may be easier to make the score proportionally notated, in order to avoid the micromanagement of the parameters.

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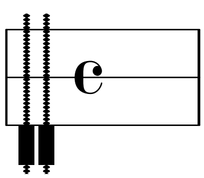
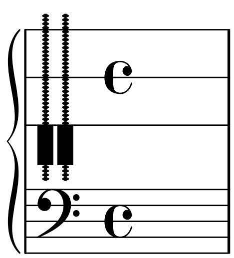
Chapter 3

Clefs

3.1 Piano Strings Clef





3.1.1 Description

In tandem with the customized staff lines, this clef indicates the position(s) of the piano strings to be manipulated by hand inside the piano. The clef shows two piano strings with grooves (intended for low piano strings), as well as their dampers, indicated by the filled rectangles.

3.1.2 Grammar

```
\stopStaff
\pianoStringMutingChartClefStaff
\startStaff
...
\stopStaff
\pianoStaffRevert
\startStaff
```

3.1.3 Code

```
\version "2.26.0"

pianoStrings = #(ly:make-stencil (list
                                  'embedded-ps
                                  "gsave

/groove {
0.1 setlinewidth
0 0 0.5 -0.125 0 -0.25 rcurveto
closepath
fill
} def

/reversegroove {
0.1 setlinewidth
0 0 -0.5 -0.125 0 -0.25 rcurveto
closepath
fill

} def

/stringshape {
currentpoint translate
newpath
0.15 setlinewidth
0.5 5 moveto
0 -10.5 rlineto
stroke


5 -0.25 -5.25 { /i exch def
newpath
0.5 i moveto
groove } for

5 -0.25 -5.25 { /i exch def
newpath
0.5 i moveto
reversegroove
 } for

newpath
0 3 translate
0 -5 moveto
0        -2.5 rlineto
1 0 rlineto
```

```
0 2.5 rlineto
closepath
fill
} def

stringshape
0 0 moveto
1.25 -3 rmoveto
stringshape



grestore
")
                                 (cons 2 0)
                                 (cons 7.5 -7.5))

clefSpace = #(ly:make-stencil "" (cons -0 0 ) (cons 0 3) )

pianoStringMutingChartClefDesign =
#(ly:stencil-add
  (ly:stencil-translate pianoStrings (cons 0 0))
  (ly:stencil-translate clefSpace (cons 2 0))
  (ly:stencil-translate clefSpace (cons 0 -5)
                        ;adjust the numbers for second 'space' to
                        ;vertically adjust the alignment
                        )
  )




pianoStringMutingChartClefStaff = {
 \override Staff.Clef.stencil = \pianoStringMutingChartClefDesign
 \override Staff.StaffSymbol.line-positions = #'(6 0  -6)
 % \override Staff.Clef.show-horizontal-skylines = ##t
 % \override Staff.Clef.show-vertical-skylines = ##t
 \override Staff.NoteHead.no-ledgers = ##t
}

pianoStaffRevert = {
 \revert Staff.Clef.stencil
 \revert Staff.StaffSymbol.line-positions
 % \override Staff.Clef.show-horizontal-skylines = ##t
 % \override Staff.Clef.show-vertical-skylines = ##t
 \revert Staff.NoteHead.no-ledgers
}
```

```


\new Staff {

 \stopStaff
 \pianoStringMutingChartClefStaff
 \startStaff
 s1

}



\new PianoStaff <<
 \new Staff {
  \stopStaff
  \pianoStringMutingChartClefStaff
  \startStaff
  s1
 }
 \new Staff {
  \clef "F" s1
 }
>>

```

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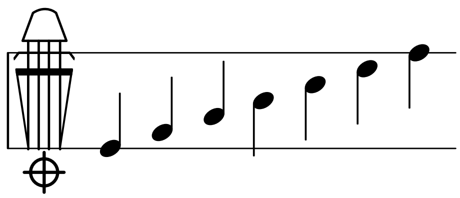
3.2 String Position Clef



3.2.1 Description

String position clef to indicate bowing position. See Discussion for the associated command, `\normalClef`.

3.2.2 Grammar

`\stringPositionClef`

3.2.3 Code

```
\version "2.26.0"

\pointAndClickOff
stringPositionClefDesign =
#(ly:make-stencil (list 'embedded-ps
                        "gsave
currentpoint translate
/fingboardpath
{
newpath

-0.55 7.5 moveto
0 -3 rlineto
1 -6.5 rlineto
-1 -1 rlineto
0 -3 rlineto
4.1 0 rlineto
0 3 rlineto
-1 1 rlineto
1 6.5 rlineto
0 3 rlineto
closepath

} def

fingboardpath clip
newpath
0.15 setlinewidth
```

```
0.5 4.75 moveto
0 -6.8 rlineto
-0.75 5 rlineto
3.5 0 rlineto
-0.75 -5 rlineto
0. 6.8 rlineto
stroke
0.35 setlinewidth
-0.4 2.75 moveto
3.75 0 rlineto
stroke

%inner two line
newpath
0.15 setlinewidth
1.16 4.75 moveto
0. -6.8 rlineto
1.8 4.75 moveto
0. -6.8 rlineto
stroke

%bridge
newpath
-0.4 3.6 moveto
0.3 0.4 rlineto
3.2 0 rlineto
0.3 -0.4 rlineto
stroke

%tailpiece
0.15 4.75 moveto
1 setlinecap
1 setlinejoin
2.75 0 rlineto
-0.65 1.75 rlineto
-0 -0  -0.6 0.55 -1.45 0  rcurveto
closepath
stroke

%mutesign
newpath
0.2 setlinewidth
1 setlinecap
1.5 -2.25 moveto
0 -2.5 rlineto
0.25 -3.5 moveto
2.5 0 rlineto
stroke
```

```
newpath
1.5 -3.5 0.85 0 360 arc
stroke
grestore")
                  (cons 0 3)
                  (cons 0 1))

stringPositionClefSize =
#(lambda (grob)
  (let* ((sPCS (ly:grob-property grob 'font-size 0.0))
         (mult (magstep sPCS)))
   (ly:stencil-scale
    stringPositionClef
    mult mult)))

stringPositionClef = {
 \override Staff.Clef.stencil = \stringPositionClefDesign
}

normalClef = {
 \revert Staff.Clef.stencil
}

{
 \override Staff.StaffSymbol.line-positions = #'(6  -6)
 \override Staff.LedgerLineSpanner.stencil = ##f
 \override Staff.TimeSignature.stencil = ##f
 \override Staff.BarLine.stencil = ##f
 \stringPositionClef c'4^\markup {
  \translate #'(-3 . 2)
  \musicglyph "space"
 }
 _\markup {
  \translate #'(-3 . -3)
  \musicglyph "space"
 }
 e' g' b' d'' f'' a''
}
```

3.2.4 Discussion

1. With the current design, `c'` would place a note at the lower end of the fingerboard. `a''` would place a note on the same line as the bridge.
2. The current design comes with the mute sign. If the mute sign is not needed, remove the following portion of the code above:

```
%mutesign
newpath
```

```
0.2 setlinewidth
1 setlinecap
1.5 -2.25 moveto
0 -2.5 rlineto
0.25 -3.5 moveto
2.5 0 rlineto
stroke
newpath
1.5 -3.5 0.85 0 360 arc
stroke
```

3. Once `\stringPositionClef` is used, in order to revert back to the normal clef, `\normalClef` must be used.

4. See Prescriptive Notation for String Instruments for a possible use of this clef.

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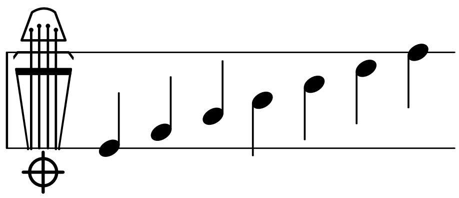
3.3 String Position Clef (revised)



3.3.1 Description

This is a revised version of the String Position Clef, where the fine-tuner pins are graphically represented, as well as the four strings are arranged tighter than the previous version.

3.3.2 Grammar

`\stringPositionClef`

3.3.3 Code

```
\version "2.26.0"

% revision june 16 2026

stringPositionClefDesignRev =
#(ly:make-stencil (list 'embedded-ps
                        "gsave
currentpoint translate
/fingboardpath
{

newpath
%0 1 .7 0 setcmykcolor
-0.5 7.5 moveto
0 -3 rlineto
1 -6.5 rlineto
-1 -1 rlineto
0 -3 rlineto
4.05 0 rlineto
0 3 rlineto
-1 1 rlineto
1 6.5 rlineto
0 3 rlineto
closepath

} def

% fingboardpath
```

```
 fingboardpath clip

newpath
0.15 setlinewidth
0.75 5.25 moveto
0 -7.3 rlineto
-0.2 0 rmoveto
-0.75 5 rlineto
3.45 0 rlineto
-0.75 -5 rlineto
-0.2 0 rmoveto
0. 7.3 rlineto
stroke
0.35 setlinewidth
-0.4 2.75 moveto
3.75 0 rlineto
stroke

%inner two lines
newpath
0.15 setlinewidth
1.25 5.5 moveto
0. -7.5 rlineto
1.8 5.5 moveto
0. -7.5 rlineto
stroke


%finetuner pins
0.75 5.4 0.14 0 360 arc
fill
1.25 5.65 0.14 0 360 arc
fill
1.8 5.65 0.14 0 360 arc
fill
2.3 5.4 0.14 0 360 arc
fill



%bridge
newpath
-0.4 3.6 moveto
0.3 0.4 rlineto
3.2 0 rlineto
0.3 -0.4 rlineto
stroke

```

```
%tailpiece
0.15 4.75 moveto
1 setlinecap
1 setlinejoin
2.75 0 rlineto
-0.65 1.75 rlineto
-0 -0  -0.6 0.55 -1.45 0  rcurveto
closepath
stroke

%mute sign, delete if not needed
newpath
0.2 setlinewidth
1 setlinecap
1.5 -2.25 moveto
0 -2.5 rlineto
0.25 -3.5 moveto
2.5 0 rlineto
stroke
newpath
1.5 -3.5 0.85 0 360 arc
stroke

grestore")
                  (cons 0 3)
                  (cons 0 1))



stringPositionClefDesignRev = {
 \override Staff.Clef.stencil =
 \stringPositionClefDesignRev
}

normalClef = {
 \revert Staff.Clef.stencil
}

{
 \override Staff.StaffSymbol.line-positions = #'(6  -6)
 \override Staff.LedgerLineSpanner.stencil = ##f
 \override Staff.TimeSignature.stencil = ##f
 \override Staff.BarLine.stencil = ##f
 \stringPositionClefDesignRev c'4^\markup {
  \translate #'(-3 . 2)
  \musicglyph "space"
 }
 _\markup {
```

```
    \translate #'(-3 . -3)
    \musicglyph "space"
  }
  e' g' b' d'' f'' a''
}
```

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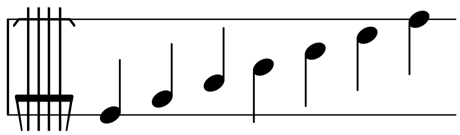
3.4 String Position Clef 2



3.4.1 Description

String position clef to indicate bowing position, but this one provides more space between bridge and the edge of the fingerboard, allowing the visual-timbre correspondence between *sul ponticello* and *sul tasto*.

3.4.2 Grammar

`\stringPositionClef_two`

3.4.3 Code

```
\version "2.26.0"

\pointAndClickOff
stringPositionClefDesign_two =
#(ly:make-stencil (list 'embedded-ps
                        "gsave
currentpoint translate
/fingboardpath
{

newpath
-0.45 4.75 moveto
0 -5 rlineto
0.5 -2.75 rlineto
2.9 0 rlineto
0.5 2.75 rlineto
0 5 rlineto
closepath
} def

fingboardpath clip
newpath
0.15 setlinewidth
0.5 8 moveto
0 -13.8 rlineto
-0.75 5 rlineto
3.5 0 rlineto
```

```
-0.75 -5 rlineto
0 11 rlineto
stroke
0.35 setlinewidth
-0.4 -1 moveto
3.75 0 rlineto
stroke

%inner two line
newpath
0.15 setlinewidth
1.16 4.75 moveto
0. -7.75 rlineto
1.8 4.75 moveto
0. -7.75 rlineto
stroke

%bridge
newpath
-0.4 3.6 moveto
0.3 0.4 rlineto
3.2 0 rlineto
0.3 -0.4 rlineto
stroke

grestore")
                  (cons 0 3)
                  (cons 0 1))

stringPositionClef_two = {
 \override Staff.Clef.stencil =
 \stringPositionClefDesign_two
}

normalClef = {
 \revert Staff.Clef.stencil
}

{
 \override Staff.StaffSymbol.line-positions = #'(6  -6)
 \override Staff.LedgerLineSpanner.stencil = ##f
 \override Staff.TimeSignature.stencil = ##f
 \override Staff.BarLine.stencil = ##f
 \stringPositionClef_two c'4^\markup {
  \translate #'(-3 . 2)
  \musicglyph "space"
 }
 _\markup {
```

```
    \translate #'(-3 . -3)
    \musicglyph "space"
  }
  e' g' b' d'' f'' a''
}
```

3.4.4 Discussion

1. With the current design, `c'` would place a note at the lower end of the fingerboard. `a''` would place a note on the same line as the bridge.
2. Once `\stringPositionClef_two` is used, in order to revert back to the normal clef, `\normalClef` must be used.

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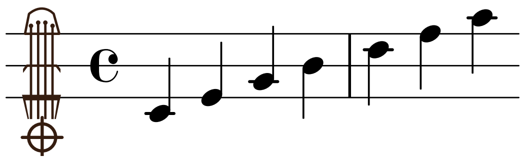
3.5 String Position Clef 3a



3.5.1 Description

In contrast to the two types of String Position Clefs introduced earlier, this clef helps facilitate the showing of the positions between the edge of the fingerboard and bridge, as well as between the bridge and the edge of the tailpiece.

3.5.2 Grammar

\stringPositionClef_three

3.5.3 Code

```
\version "2.26.0"

stringPositionClef_three_Design =
#(ly:make-stencil (list 'embedded-ps
                        "gsave
currentpoint translate
/fingboardpath
{
newpath
0 1 .7 0 setcmykcolor
0.3 4.75 moveto
0 -4.5 rlineto
-0.2 -0.5 rlineto
0.5 -2.15 rlineto
-1 0 rlineto
0 -3 rlineto
3.75 0 rlineto
0 3 rlineto
-1 0 rlineto
0.45 2.15 rlineto
-0.15 0.5 rlineto
0 4.5 rlineto
closepath
%stroke
.1 .4 .5 .9 setcmykcolor
} def

%fingboardpath
```

```
fingboardpath clip
newpath
0.15 setlinewidth
0.8 3.5 moveto
0 -5.85 rlineto
-0.2 0 rmoveto
-0.25 1.3 rlineto
2.2 0 rlineto
-0.2 -1.3 rlineto
-0.2 0 rmoveto
0. 5.85 rlineto
stroke
0.35 setlinewidth
0.15 -1 moveto
3.2 0 rlineto
stroke

%inner two line
newpath
0.15 setlinewidth
1.25 3.6 moveto
0. -5.95 rlineto
1.7 3.6 moveto
0. -5.95 rlineto
stroke

0.8 3.5 0.14 0 360 arc
fill
2.1525 3.5 0.14 0 360 arc
fill
1.25 3.7 0.14 0 360 arc
fill
1.7 3.7 0.14 0 360 arc
fill



%bridge
newpath
0.25 0.6 moveto
0.3 0.4 rlineto
1.85 0 rlineto
0.3 -0.4 rlineto
stroke

%tailpiece
0.425 3 moveto
```

```
1 setlinecap
1 setlinejoin
2.15 0 rlineto
-0.35 1.25 rlineto
-0 -0  -0.65 0.75 -1.55 0  rcurveto
closepath
stroke

%%% mute sign; commentify if not needed %%%
newpath
0.2 setlinewidth
1 setlinecap
1.5 -2.25 moveto
0 -2.5 rlineto
0.25 -3.5 moveto
2.5 0 rlineto
stroke
newpath
1.5 -3.5 0.85 0 360 arc
stroke
%%% end of mute sign for commenting/uncommenting %%%

grestore

")
                  (cons 0 3)
                  (cons 0 1))

stringPositionClefSize =
#(lambda (grob)
  (let* ((bCS (ly:grob-property grob 'font-size 0.0))
         (mult (magstep bCS)))
   (ly:stencil-scale
    stringPositionClef
    mult mult)))

stringPositionClef_three = {
 \override Staff.Clef.stencil =
 \stringPositionClef_three_Design
}

{
 \override Staff.StaffSymbol.line-positions = #'(4 0  -4)
 \stringPositionClef_three
 c'4^\markup {
  \translate #'(-3 . 2)
  \musicglyph "space"
 }
```

```
  _\markup {
   \translate #'(-3 . -3)
   \musicglyph "space"
  }
  e' g' b' d'' f'' a''
}


```

3.5.4 Discussion

1. With the current design, `e'` places a note at the lower end of the fingerboard. `b'` places a note at the bridge line, and `f''` places a note on the line indicating the edge of the tailpiece.
2. The current design comes with the mute sign. If the mute sign is not needed, remove the following portion of the code above:

```
%%% mute sign; commentify if not needed %%%
newpath
0.2 setlinewidth
1 setlinecap
1.5 -2.25 moveto
0 -2.5 rlineto
0.25 -3.5 moveto
2.5 0 rlineto
stroke
newpath
1.5 -3.5 0.85 0 360 arc
stroke
%%% end of mute sign for commenting/uncommenting %%%
```

3. If you do not wish the ledger lines to appear within the staff line, consider using:

   `\override Staff.NoteHead.no-ledgers = ##t`.

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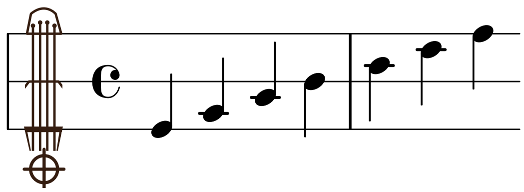
3.6 String Position Clef 3b (Longer Span)



3.6.1 Description

The design of this clef is similar to String Position Clef 3a; however, here the distance among the edge of the fingerboard, bridge, and the edge of the tailpiece is set wider.

3.6.2 Grammar

\stringPositionClef_three

3.6.3 Code

```
\version "2.26.0"

stringPositionClef_three_longer_Design =
#(ly:make-stencil (list 'embedded-ps
                        "gsave
currentpoint translate
/fingboardpath
{
newpath
0 1 .7 0 setcmykcolor
0.3 5.75 moveto
0 -6.5 rlineto
-0.2 -0.5 rlineto
0.5 -2.15 rlineto
-0.5 0 rlineto
0 -2.5 rlineto
2.75 0 rlineto
0 2.5 rlineto
-0.5 0 rlineto
0.45 2.15 rlineto
-0.15 0.5 rlineto
0 6.5 rlineto
closepath
%stroke
.1 .4 .5 .9 setcmykcolor
} def

%fingboardpath
```

```

fingboardpath clip
newpath
0.15 setlinewidth
0.8 4.5 moveto
0 -7.85 rlineto
-0.2 0 rmoveto
-0.25 1.3 rlineto
2.2 0 rlineto
-0.2 -1.3 rlineto
-0.2 0 rmoveto
0. 7.85 rlineto
stroke
0.35 setlinewidth
0.15 -2 moveto
3.2 0 rlineto
stroke

%inner two line
newpath
0.15 setlinewidth
1.25 4.6 moveto
0. -7.95 rlineto
1.7 4.6 moveto
0. -7.95 rlineto
stroke

0.8 4.5 0.14 0 360 arc
fill
2.1525 4.5 0.14 0 360 arc
fill
1.25 4.7 0.14 0 360 arc
fill
1.7 4.7 0.14 0 360 arc
fill



%bridge
newpath
0.25 0.6 moveto
0.3 0.4 rlineto
1.85 0 rlineto
0.3 -0.4 rlineto
stroke

%tailpiece
0.425 4 moveto
```

```
1 setlinecap
1 setlinejoin
2.15 0 rlineto
-0.35 1.25 rlineto
-0 -0  -0.65 0.75 -1.55 0  rcurveto
closepath
stroke
%{
%%% mute sign; commentify if not needed %%%
newpath
0.2 setlinewidth
1 setlinecap
1.5 -3.25 moveto
0 -2.5 rlineto
0.25 -4.5 moveto
2.5 0 rlineto
stroke
newpath
1.5 -4.5 0.85 0 360 arc
stroke
%%% end of mute sign for commenting/uncommenting %%%
%}
grestore

")
                  (cons 0 3)
                  (cons 0 1))

stringPositionClefSize =
#(lambda (grob)
  (let* ((bCS (ly:grob-property grob 'font-size 0.0))
         (mult (magstep bCS)))
   (ly:stencil-scale
    stringPositionClef
    mult mult)))

stringPositionClef_three_longer = {
 \override Staff.Clef.stencil =
 \stringPositionClef_three_longer_Design
}

{
 \override Staff.StaffSymbol.line-positions = #'(6 0  -6)
 \stringPositionClef_three_longer
 c'4^\markup {
  \translate #'(-3 . 2)
  \musicglyph "space"
 }
```

```
  _\markup {
   \translate #'(-3 . -3)
   \musicglyph "space"
  }
  e' g' b' d'' f'' a''
}
```

3.6.4 Discussion

1. With the current design, `c'` places a note at the lower end of the fingerboard. `b'` places a note at the bridge line, and `a''` places a note on the line indicating the edge of the tailpiece.

2. The current design comes with the mute sign. If the mute sign is not needed, remove the following portion of the code above:

```
%%% mute sign; commentify if not needed %%%
newpath
0.2 setlinewidth
1 setlinecap
1.5 -3.25 moveto
0 -2.5 rlineto
0.25 -4.5 moveto
2.5 0 rlineto
stroke
newpath
1.5 -4.5 0.85 0 360 arc
stroke
%%% end of mute sign for commenting/uncommenting %%%
```

3. If you do not wish the ledger lines to appear within the staff line, consider using:

   `\override Staff.NoteHead.no-ledgers = ##t`.

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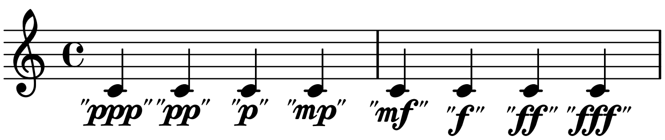
Chapter 4

Dynamics

4.1 Dynamics in Quotation Marks



4.1.1 Description

Updated: November 12, 2025: hairpins will stop closer to the dynamics in quotation marks.

Dynamics in quotation marks, also known as *effort dynamics*, indicate those with which certain techniques must be carried on, understanding that the perceived dynamics will be quieter than what are indicated. Examples abound in scores by Helmut Lachenmann and others for such techniques as air sound, bowing directly on the bridge, etc..

4.1.2 Grammar

```
NOTE \qppp
NOTE \qpp
NOTE \qp
NOTE \qmp
NOTE \qmf
NOTE \qf
NOTE \qff
NOTE \qfff
```

4.1.3 Code

```
\version "2.26.0"

\pointAndClickOff
qmp = #(make-dynamic-script
        (markup
```

```
         #:pad-x -1
         ( #:combine
           #:combine
           #:translate '(-0.85 . -0.1)
           #:normal-text (#:italic #:fontsize 0.75 "\"")
           #:dynamic "mp"
           #:translate '(3.25 . -0.1)
           #:normal-text (#:italic #:fontsize 0.75 "\""))))
qp = #(make-dynamic-script
       (markup
        #:pad-x -1
        (
         #:combine
         #:combine
         #:translate '(-0.95 . -0.1)
         #:normal-text (#:italic #:fontsize 0.75 "\"")
         #:dynamic "p"
         #:translate '(1.35 . -0.1)
         #:normal-text (#:italic #:fontsize 0.75 "\""))))
qpp = #(make-dynamic-script
        (markup
         #:pad-x -1
         (  #:combine
            #:combine
            #:translate '(-0.95 . -0.1)
            #:normal-text (#:italic #:fontsize 0.75 "\"")
            #:dynamic "pp"
            #:translate '(2.75 . -0.1)
            #:normal-text (#:italic #:fontsize 0.75 "\""))))
qppp = #(make-dynamic-script
         (markup
          #:pad-x -1
          (   #:combine
              #:combine
              #:translate '(-0.95 . -0.1)
              #:normal-text (#:italic #:fontsize 0.75 "\"")
              #:dynamic "ppp"
              #:translate '(4.25 . -0.1)
              #:normal-text (#:italic #:fontsize 0.75 "\""))))
qmf = #(make-dynamic-script
        (markup
         #:pad-x -1
         (   #:combine
             #:combine
             #:translate '(-0.85 . 0)
             #:normal-text (#:italic #:fontsize 0.75 "\"")
             #:dynamic "mf"
             #:translate '(3.25 . 0)
```

```
             #:normal-text (#:italic #:fontsize 0.75 "\""))))
qf = #(make-dynamic-script
       (markup
        #:pad-x -1
        ( #:combine
          #:combine
          #:translate '(-0.75 . 0)
          #:normal-text (#:italic #:fontsize 0.75 "\"")
          #:dynamic "f"
          #:translate '(1.65 . 0)
          #:normal-text (#:italic #:fontsize 0.75 "\""))))
qff = #(make-dynamic-script
        (markup
         #:pad-x -1
         ( #:combine
           #:combine
           #:translate '(-0.75 . 0)
           #:normal-text (#:italic #:fontsize 0.75 "\"")
           #:dynamic "ff"
           #:translate '(2.75 . 0)
           #:normal-text (#:italic #:fontsize 0.75 "\""))))
qfff = #(make-dynamic-script
         (markup
          #:pad-x -1
          (  #:combine
             #:combine
             #:translate '(-0.75 . 0)
             #:normal-text (#:italic #:fontsize 0.75 "\"")
             #:dynamic "fff"
             #:translate '(3.85 . 0)
             #:normal-text (#:italic #:fontsize 0.75 "\""))))

{

 c'4\qppp
 c'4\qpp
 c'4\qp
 c'4\qmp

 c'4\qmf
 c'4\qf
 c'4\qff
 c'4\qfff

}

\layout {
 \context {
```

```
    \Score    proportionalNotationDuration = #1/9
  }
}
```

4.1.4 Discussion

In scores by Lachenmann, in concordance with German quotation marks (*Anführungszeichen*), the opening quotation mark points left, and placed on the bottom line, and the closing quotation mark points right and sits at the top of the last character. It would be possible to achieve this by adjusting the parameters in the Scheme code.[1]

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1. See: https://lilypond.org/doc/v2.24/Documentation/extending/markup-construction-in-scheme

4.2 Enhanced Constante Hairpins I (Under The Staff)

4.2.1 Description

NB: Slight revision (December 23, 2025) has been carried out to make sure the hairpin terminates at the end of the staff line when it breaks out to the next system.

A constante hairpin, a horizontal line with a short hook at the end, indicates that a dynamic remains unchanged throughout the notes to which it is applied. There are many contemporary scores that utilize this type of hairpins, e.g. works by Brian Ferneyhough. In LilyPond, it is possible to specify constante hairpins by writing an override function:

```
\override Hairpin.stencil = #constante-hairpin
```

However, similar to the issues raised in the Flared Hairpins, LilyPond's default constante hairpins could use some enhancements, particularly in regards to accommodating three separate components when they break into new systems: 1. beginning; 2. intermediate; and; 3. ending. Furthermore, when the constante hairpins break into a new system, it would sometimes be helpful to have the dynamic in parentheses to remind what the dynamic had been in the previous system, which could be taking place before the page turn.

4.2.2 Grammar

```
NOTE \constante \DYNAMIC NOTE(S) \DYNAMIC OR \!
```

This `\constante` accepts three optional arguments:

1. Vertical offset of the constante hairpin, indicated by a direction `#UP`, `#CENTER`, or `#DOWN`. Without specifying this argument, the default is `#CENTER`. You may use `#UP` when it is followed by a decrescendo hairpin, while you may use `#DOWN` when it is followed by a crescendo hairpin.
2. Whether or not to use cautionary dynamics in subsequent systems when the hairpin breaks into new systems, indicated by `##f` or `##t`. The default is `##f`.
3. Extra vertical adjustment is possible with the third argument by specifying a number. The default is `#0`.

4.2.3 Code

```
\version "2.26.0"


qf = #(make-dynamic-script
       (markup
        #:pad-x -1
        ( #:combine
          #:combine
          #:translate '(-0.75 . 0)
          #:normal-text (#:italic #:fontsize 0.75 "\"")
          #:dynamic "f"
          #:translate '(1.65 . 0)
          #:normal-text (#:italic #:fontsize 0.75 "\""))))

qmp = #(make-dynamic-script
        (markup
         #:pad-x -1
         ( #:combine
           #:combine
           #:translate '(-0.85 . -0.1)
           #:normal-text (#:italic #:fontsize 0.75 "\"")
           #:dynamic "mp"
           #:translate '(3.25 . -0.1)
           #:normal-text (#:italic #:fontsize 0.75 "\""))))


constante =
#(define-music-function (vOff cautionary moreOff dyn)
  ( (ly:dir? CENTER)  (boolean? #f) (number? 0) ly:event?   )
  (define-public
   ((constante-hairpin-revised coords mirrored?) grob)
   "This scheme code was originally the scheme code 'elbowed-hairpin
taken from output-lib.scm, and was revised to suit the need."
   (define (scale-coords coords-list x y)
    (map
```

```
    (lambda (coord) (cons (* x (car coord)) (* y (cdr coord))))
    coords-list))

  (define (hairpin::print-part points decresc? me)
   (let ((stil (make-connected-line points me)))
    (ly:stencil-scale stil 1 1)))

  ;; outer let to trigger suicide
  (let ((sten (ly:hairpin::print grob)))
   (if (grob::is-live? grob)
       (let*
        (
         ;I've left decresc? function but removed all the if clause
         ;associated with it, to simplify the code a little bit
         (verticalOffset vOff)
         (useCautionary? cautionary)
         (moreOffset moreOff)
         (hairpinHeight (ly:grob-property grob 'height))
         (decresc? (eqv? (ly:grob-property grob 'grow-direction) LEFT))
         (toBarline? (ly:grob-property grob 'to-barline))
         ;    (circled-tip? (ly:grob-property grob 'circled-tip))
         (hairpinTypeA?
          (and (eqv? (end-broken-spanner? grob )              #f)
               (eqv? (first-broken-spanner? grob )            #f)
               (eqv? (middle-broken-spanner? grob )           #f)
               (eqv? (not-first-broken-spanner? grob )        #f)
               (eqv? (not-last-broken-spanner? grob )         #f)
               (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
               (eqv? (unbroken-or-last-broken-spanner? grob )  #t)
               (eqv? (unbroken-spanner? grob )                #t)
               )) ;Hairpins that end within one line without
         ;reaching to the end of the system
         (hairpinTypeB?
          (or (and (eqv? (end-broken-spanner? grob )              #t)
                   (eqv? (first-broken-spanner? grob )            #t)
                   (eqv? (middle-broken-spanner? grob )           #f)
                   (eqv? (not-first-broken-spanner? grob )        #f)
                   (eqv? (not-last-broken-spanner? grob )         #f)
                   (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
                   (eqv? (unbroken-or-last-broken-spanner? grob )  #t)
                   (eqv? (unbroken-spanner? grob )                #f)
                   )
              (and (eqv? (end-broken-spanner? grob )              #f)
                   (eqv? (first-broken-spanner? grob )            #t)
                   (eqv? (middle-broken-spanner? grob )           #f)
                   (eqv? (not-first-broken-spanner? grob )        #f)
                   (eqv? (not-last-broken-spanner? grob )         #t)
                   (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
```

```
                  (eqv? (unbroken-or-last-broken-spanner? grob )   #f)
                  (eqv? (unbroken-spanner? grob )                 #f)
                  )
             )) ;Hairpins that breaks into next line
        (hairpinTypeC?
         (and (eqv? (end-broken-spanner? grob )               #f)
              (eqv? (first-broken-spanner? grob )             #f)
              (eqv? (middle-broken-spanner? grob )            #t)
              (eqv? (not-first-broken-spanner? grob )         #t)
              (eqv? (not-last-broken-spanner? grob )          #t)
              (eqv? (unbroken-or-first-broken-spanner? grob ) #f)
              (eqv? (unbroken-or-last-broken-spanner? grob )   #f)
              (eqv? (unbroken-spanner? grob )                 #f)
              )) ;Midway-through Hairpins
        (hairpinTypeD?
         (and (eqv? (end-broken-spanner? grob )               #t)
              (eqv? (first-broken-spanner? grob )             #f)
              (eqv? (middle-broken-spanner? grob )            #f)
              (eqv? (not-first-broken-spanner? grob )         #t)
              (eqv? (not-last-broken-spanner? grob )          #f)
              (eqv? (unbroken-or-first-broken-spanner? grob ) #f)
              (eqv? (unbroken-or-last-broken-spanner? grob )   #t)
              (eqv? (unbroken-spanner? grob )                 #f)
              )) ;multi-system Hairpin that terminates
        (xex
         (cons (car (ly:stencil-extent sten X))
               (- (cdr (ly:stencil-extent sten X)) 1.25))
         )
        (lenx (interval-length xex))
        (yex (ly:stencil-extent sten Y))
        (leny (interval-length yex))
        (xtrans (+ (car xex)  0))
        (ytrans (car yex))
        (uplist (scale-coords coords lenx (/ leny 2)))

        (crescStencil

         (if toBarline?
             ;I added here to make it default to stretch
             ;the hairpin immediately to the dynamics

             (ly:stencil-translate
              (hairpin::print-part
               (scale-coords coords
                             (+ lenx 2.35) (/ leny 2))
               decresc? grob)

              '(-0.5 . 0))
```

```
             ;if not to barline
             (ly:stencil-translate
              (hairpin::print-part
               (scale-coords coords
                             (+ lenx 2.35)  (/ leny 2))
               decresc? grob)

              '(-0.5 . 0))

             )
         )
        (crescStencilBrokenBeginning

         (if toBarline?
             ;I added here to make it default to stretch
             ;the hairpin immediately to the dynamics

             (hairpin::print-part (scale-coords
                                   '((0 . 0) (1 . 0))
                                   (+ lenx 1) (/ leny 2))
                                  decresc? grob)
             ;if not to barline
             (ly:stencil-translate
              (hairpin::print-part
               (scale-coords '((0 . 0) (1 . 0))
                             (+ lenx 1.5)  (/ leny 2))
               decresc? grob)

              '(-0.5 . 0))

             )
         )
        (crescStencilBrokenContinuation
         (ly:stencil-combine-at-edge
          ;if the cautionary accidentals are turned on
          (if cautionary
              (if (list? (ly:music-property dyn 'text))
                  ;this means, when the custom-made dynamics
                  ;(e.g. \qmp) are used
                  (grob-interpret-markup
                   grob
                   (markup
                    #:hspace 2
                    #:whiteout
                    #:center-align
                    (
                     #:concat
                     (#:normal-text
```

```
                          (#:translate '(0 . -0.25)
                                       #:italic (#:fontsize -0.5 "( "
                                                            ))
                          #:pad-x
                          0.35
                          #:dynamic (ly:music-property dyn 'text)
                          #:normal-text
                          (#:translate '(0.5 . -0.25)
                                       #:italic (#:fontsize -0.5 ")")))
                         )
                        )
                       )
                      ;when in-house dynamics (e.g. \p) are used
                      (grob-interpret-markup
                       grob
                       (markup
                        #:hspace 2
                        #:center-align
                        #:whiteout
                        #:concat
                        (#:normal-text
                         (#:translate '(0 . -0.25)
                                      #:italic (#:fontsize 0.5 "("
                                                           ))
                         (#:translate '(0 . -0.25)
                                      #:italic (#:fontsize -10 " "
                                                           ))
                         #:pad-x
                         0
                         #:dynamic (ly:music-property dyn 'text)
                         #:normal-text
                         (#:translate '(0 . -0.25)
                                      #:italic (#:fontsize 0.5 ")")))
                        )
                       ))
                  empty-stencil)
              1
              1
              (hairpin::print-part (scale-coords
                                    '((0.01 . 0.15) (1 . 0.15))
                                    (+ lenx 1)  (/ leny 2))
                                   decresc? grob)
              -1.5
              )
             )
            (crescStencilBrokenEnding
             ;if the flare end is applied
             ;on the dynamic at the beginning
```

```
;of the system, meaning lenx is super low

(cond
 ((< lenx 1)
  (ly:stencil-combine-at-edge
   ;if the cautionary accidentals are turned on
   (if cautionary
       (if (list? (ly:music-property dyn 'text))
           ;this means, when the custom-made dynamics
           ;(e.g. \qmp) are used
           (grob-interpret-markup
            grob
            (markup
             #:hspace -1
             #:whiteout
             #:center-align
             #:concat
             (#:normal-text
              (#:translate '(0 . -0.25)
                           #:italic (#:fontsize -0.5 "( "
                                                     ))
              #:pad-x
              0.25
              #:dynamic (ly:music-property dyn 'text)
              #:normal-text
              (#:translate '(0.5 . -0.25)
                           #:italic (#:fontsize -0.5 ")")))
             )
            )
           ;when in-house dynamics (e.g. \p) are used
           (grob-interpret-markup
            grob
            (markup

             #:hspace -1
             #:whiteout
             #:center-align
             (
              #:concat
              (#:normal-text
               (#:translate '(0 . -0.25)
                            #:italic (#:fontsize 0.5 "("
                                                      ))
               (#:translate '(0 . -0.25)
                            #:italic (#:fontsize -10 " "
                                                      ))
               #:pad-x 0
               #:dynamic (ly:music-property dyn 'text)
```

```
                          #:normal-text
                          (#:translate '(0 . -0.25)
                                       #:italic (#:fontsize 0.5 ")")))
                         ))
                       ))
                  empty-stencil
                  )
              1
              1
              (hairpin::print-part
               (scale-coords  coords
                              (+ lenx 1.85) (/ leny 2) )
               decresc? grob)
              ;vertical alignment (to be refined possibly...?)
              -1.75
              )
             )
            ((>= lenx 1)
             ;if the normal ending flare crescendo
             ;is applied...
             (ly:stencil-combine-at-edge
              ;if the cautionary accidentals are turned on
              (if cautionary
                  (if (list? (ly:music-property dyn 'text))
                      ;this means, when the custom-made dynamics
                      ;(e.g. \qmp) are used
                      (grob-interpret-markup
                       grob
                       (markup
                        #:hspace 2
                        #:whiteout
                        #:center-align
                        #:concat
                        (#:normal-text
                         (#:translate '(0 . -0.25)
                                      #:italic (#:fontsize -0.5 "( "
                                                                     ))
                         #:pad-x 0.25
                         #:dynamic (ly:music-property dyn 'text)
                         #:normal-text
                         (#:translate '(0.5 . -0.25)
                                      #:italic (#:fontsize -0.5 ")")))
                        )
                       )
                      ;if the in-house dynmamics are used

                      (grob-interpret-markup
                       grob
```

```
                          (markup
                           #:hspace 2
                           #:whiteout
                           #:center-align
                           #:concat
                           (#:normal-text
                            (#:translate '(0 . -0.25)
                                         #:italic (#:fontsize 0.5 "("
                                                                 ))
                            (#:translate '(0 . -0.25)
                                         #:italic (#:fontsize -10 " "
                                                                 ))
                            #:pad-x
                            0
                            #:dynamic (ly:music-property dyn 'text)
                            #:normal-text
                            (#:translate '(0 . -0.25)
                                         #:italic (#:fontsize 0.5 ")")))
                           )
                          ))
                     empty-stencil
                     )
                 1
                 1
                 (hairpin::print-part
                  (scale-coords  coords
                                (+ lenx 1.85) (/ leny 2) )
                  decresc? grob)
                 ;vertical alignment (to be refined possibly...?)
                 -1.75
                 )
                )
               )
              )


             (cresc
              (cond (hairpinTypeA? crescStencil)
                    (hairpinTypeB? crescStencilBrokenBeginning)
                    (hairpinTypeC? crescStencilBrokenContinuation)
                    (hairpinTypeD? crescStencilBrokenEnding)
                    )
              )

             (stil
              (ly:stencil-aligned-to
               (ly:stencil-translate
                cresc
```

```
                (cons xtrans ytrans))
               Y CENTER))
             (stil-y-extent (ly:stencil-extent stil Y)))
            ; FOR DEBUG ONLY
            ; (display (list? (ly:music-property dyn 'text)))
            ; (newline)
            ; (display dyn)
            ; (newline)
            ; (newline)
            ; (display "Is this type A Hairpin? " )
            ; (display hairpinTypeA?)
            ; (newline)
            ; (display "Is this type B Hairpin? " )
            ; (display hairpinTypeB?)
            ; (newline)
            ; (display "Is this type C Hairpin? " )
            ; (display hairpinTypeC?)
            ; (newline)
            ; (display "Is this type D Hairpin? " )
            ; (display hairpinTypeD?)
            ;
            ; (newline)
            ; (display "Is it decrescendo? " )
            ; (display decresc?)
            ; (newline)
            ; (display "end-broken-spanner???  " )
            ; (display (end-broken-spanner? grob ))
            ; (newline)
            ; (display "(first-broken-spanner?) Is spanner broken
            ;         and the first of its broken siblings? " )
            ; (display (first-broken-spanner? grob ) )
            ; (newline)
            ; (display "middle-broken-spanner??? " )
            ; (display (middle-broken-spanner? grob ) )
            ; (newline)
            ;
            ; (display "Is spanner broken and not the first of
            ;         its broken siblings? " )
            ; (display (not-first-broken-spanner? grob ) )
            ; (newline)
            ;
            ; (display "Is spanner broken and not the last of
            ;         its broken siblings? " )
            ; (display (not-last-broken-spanner? grob ) )
            ; (newline)
            ;
            ; (display "unbroken-or-first-broken-spanner??? " )
            ; (display (unbroken-or-first-broken-spanner? grob ) )
```

```
          ; (newline)
          ;
          ; (display "unbroken-or-first-broken-spanner??? " )
          ; (display (unbroken-or-last-broken-spanner? grob ) )
          ; (newline)
          ; (display "unbroken spanner?? " )
          ; (display (unbroken-spanner? grob ) )
          ; (newline)
          ; (newline)
          ; (display "distance of xex " )
          ; (display lenx )
          ; (newline)
          ; (newline)
          (ly:make-stencil (ly:stencil-expr stil) xex stil-y-extent))
         ;; return empty, if no Hairpin.stencil present.
         '())))

   #{
    #dyn
    \tweak Hairpin.stencil
    #(lambda (grob)
      ;depending on the hairpin opening width,
      ;you can adjust the value here vvv
      (constante-hairpin-revised '((0 . 0) (1.0 . 0) (1.0 . 0.9)) #f)
      )
    % This calculates the tweak value for
    % the value of the constante V-offset
    \tweak Y-offset #(lambda (grob)
                  (+ (* vOff (ly:grob-property grob 'height))
                     (* (ly:grob-property grob 'height) -0.5 vOff )
                     moreOff
                     )
                  )

    \<
   #}
   )
 \layout {
  indent = #0
  line-width = #100
  ragged-last = ##f
 }
 {

  \override Hairpin.height = #0.5
  \override Hairpin.to-barline = ##f
  % \override Hairpin.stencil = #constante-hairpin
  \override Hairpin.after-line-breaking = ##t
```

```
  c'2  \constante #UP \mf  c'2 c'2 \>  c'2
  c'2 \constante #DOWN ##t #0.3 \p c'2
  \break  c'4 4 4 4 c'4 4 4 4 \break
  c''2 \< c'2 \! \constante #DOWN ##f \qf \break
  c'2 c'2\< c'2 c'2 c'2
  \revert Hairpin.to-barline
  \afterGrace 31/32
  { c'2 \constante #CENTER ##f \ff}  s32\!  \break
  c'2\f c'2\< c'2 c'2 c'2
  \revert Hairpin.to-barline
  \after 1*25/64 \!
  { c'2 \constante #CENTER ##f \ff }
  \bar "|."
}

```

4.2.4 Discussion

For the purpose of using it in my own musical typesetting, this constante hairpin snippet has been designed to work with a narrower hairpin height, e.g.

`\override Hairpin.height = #0.5` (the default is `0.666`.)

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4.3 Enhanced Constante Hairpins II (Over The Staff)

4.3.1 Description

NB: Revision (August 19, 2026) has been carried out in modifying where the default parenthesized dynamics will be placed. Before, the parenthesized dynamics with whiteout would be placed on top of the constante line after the line break, but at a position that exposes the beginning part of the constante line. I fixed it so that the dynamics would hide the beginning part of the constante line.
- Slight revision (December 23, 2025) has been carried out to make sure the hairpin terminates at the end of the staff line when it breaks out to the next system.

The motivation for creating an enhancement to the default constante hairpin provided by LilyPond was that it would not allow the hook to be inverted when placed over the staff.

See Enhanced Constante Hairpins I (Under The Staff) for more information.

4.3.2 Grammar

```
NOTE \constante \DYNAMIC NOTE(S) \DYNAMIC OR \!
```

This `\constante` accepts three optional arguments:

1. Vertical offset of the constante hairpin, indicated by a direction `#UP`, `#CENTER`, or `#DOWN`. Without specifying this argument, the default is `#CENTER`. You may use `#DOWN` when it is followed by a decrescendo hairpin, while you may use `#UP` when it is followed by a crescendo hairpin.[2]

2. Note that the directions are flipped from Enhanced Constante Hairpins I!

2. Whether or not to use cautionary dynamics in subsequent systems when the hairpin breaks into new systems, indicated by `##f` or `##t`. The default is `##f`.
3. Extra vertical adjustment is possible with the third argument by specifying a number. The default is `#0`.

Furthermore, when using the dynamic ***p***, ***pp***, etc., it is possible that the initial horizontal line may be too high. In this case, use `\tweak Y-offset #0` to be placed between the numerical value for the extra vertical adjustment and the dynamics.

4.3.3 Code

```
\version "2.26.0"


qf = #(make-dynamic-script
       (markup
        #:pad-x -1
        ( #:combine
          #:combine
          #:translate '(-0.75 . 0)
          #:normal-text (#:italic #:fontsize 0.75 "\"")
          #:dynamic "f"
          #:translate '(1.65 . 0)
          #:normal-text (#:italic #:fontsize 0.75 "\""))))

qmp = #(make-dynamic-script
        (markup
         #:pad-x -1
         ( #:combine
           #:combine
           #:translate '(-0.85 . -0.1)
           #:normal-text (#:italic #:fontsize 0.75 "\"")
           #:dynamic "mp"
           #:translate '(3.25 . -0.1)
           #:normal-text (#:italic #:fontsize 0.75 "\""))))


constanteUP =
#(define-music-function ( vOff cautionary moreOff dyn)
  ( (ly:dir? CENTER)  (boolean? #t) (number? 0) ly:event?   )
  (define-public
   ((constante-hairpin-revised coords mirrored?) grob)
   "This scheme code was originally the scheme code 'elbowed-hairpin
taken from output-lib.scm, and was revised to suit the need."
   (define (scale-coords coords-list x y)
    (map
     (lambda (coord) (cons (* x (car coord)) (* y (cdr coord))))
     coords-list))
```

```

(define (hairpin::print-part points decresc? me)
 (let ((stil (make-connected-line points me)))
  (ly:stencil-scale stil 1 1)))

;; outer let to trigger suicide
(let ((sten (ly:hairpin::print grob)))
 (if (grob::is-live? grob)
     (let*
      (
       ;I've left decresc? function but removed all the if clause
       ;associated with it, to simplify the code a little bit
       (verticalOffset vOff)
       (moreOffset moreOff)
       (hairpinHeight (ly:grob-property grob 'height))
       (decresc? (eqv? (ly:grob-property grob 'grow-direction) LEFT))
       (toBarline? (ly:grob-property grob 'to-barline))
       ;    (circled-tip? (ly:grob-property grob 'circled-tip))
       (hairpinTypeA?
        (and (eqv? (end-broken-spanner? grob )              #f)
             (eqv? (first-broken-spanner? grob )            #f)
             (eqv? (middle-broken-spanner? grob )           #f)
             (eqv? (not-first-broken-spanner? grob )        #f)
             (eqv? (not-last-broken-spanner? grob )         #f)
             (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
             (eqv? (unbroken-or-last-broken-spanner? grob )  #t)
             (eqv? (unbroken-spanner? grob )                #t)
             )) ;Hairpins that end within one line without
       ;reaching to the end of the system
       (hairpinTypeB?
        (or (and (eqv? (end-broken-spanner? grob )              #t)
                 (eqv? (first-broken-spanner? grob )            #t)
                 (eqv? (middle-broken-spanner? grob )           #f)
                 (eqv? (not-first-broken-spanner? grob )        #f)
                 (eqv? (not-last-broken-spanner? grob )         #f)
                 (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
                 (eqv? (unbroken-or-last-broken-spanner? grob )  #t)
                 (eqv? (unbroken-spanner? grob )                #f)
                 )
            (and (eqv? (end-broken-spanner? grob )              #f)
                 (eqv? (first-broken-spanner? grob )            #t)
                 (eqv? (middle-broken-spanner? grob )           #f)
                 (eqv? (not-first-broken-spanner? grob )        #f)
                 (eqv? (not-last-broken-spanner? grob )         #t)
                 (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
                 (eqv? (unbroken-or-last-broken-spanner? grob )   #f)
                 (eqv? (unbroken-spanner? grob )                #f)
                 )
```

```
               )) ;Hairpins that breaks into next line
          (hairpinTypeC?
           (and (eqv? (end-broken-spanner? grob )              #f)
                (eqv? (first-broken-spanner? grob )            #f)
                (eqv? (middle-broken-spanner? grob )           #t)
                (eqv? (not-first-broken-spanner? grob )        #t)
                (eqv? (not-last-broken-spanner? grob )         #t)
                (eqv? (unbroken-or-first-broken-spanner? grob ) #f)
                (eqv? (unbroken-or-last-broken-spanner? grob )   #f)
                (eqv? (unbroken-spanner? grob )                #f)
                )) ;Midway-through Hairpins
          (hairpinTypeD?
           (and (eqv? (end-broken-spanner? grob )              #t)
                (eqv? (first-broken-spanner? grob )            #f)
                (eqv? (middle-broken-spanner? grob )           #f)
                (eqv? (not-first-broken-spanner? grob )        #t)
                (eqv? (not-last-broken-spanner? grob )         #f)
                (eqv? (unbroken-or-first-broken-spanner? grob ) #f)
                (eqv? (unbroken-or-last-broken-spanner? grob )   #t)
                (eqv? (unbroken-spanner? grob )                #f)
                )) ;multi-system Hairpin that terminates
          (xex
           (cons (car (ly:stencil-extent sten X))
                 (- (cdr (ly:stencil-extent sten X)) 1.25))
           )
          (lenx (interval-length xex))
          (yex (ly:stencil-extent sten Y))
          (leny (interval-length yex))
          (xtrans (+ (car xex)  0))
          (ytrans (car yex))
          (uplist (scale-coords coords lenx (/ leny 2)))

          (crescStencil

           (if toBarline?
               ;I added here to make it default to stretch
               ;the hairpin immediately to the dynamics

               (ly:stencil-translate
                (hairpin::print-part
                 (scale-coords coords
                               (+ lenx 2.35) (/ leny 2))
                 decresc? grob)

                '(-0.5 . 0))
               ;if not to barline
               (ly:stencil-translate
                (hairpin::print-part
```

```
                   (scale-coords coords
                                 (+ lenx 2.35)  (/ leny 2))
                   decresc? grob)

                  '(-0.5 . 0))

                 )
             )
            (crescStencilBrokenBeginning

             (if toBarline?
                 ;I added here to make it default to stretch
                 ;the hairpin immediately to the dynamics

                 (hairpin::print-part (scale-coords
                                       '((0 . 0) (1 . 0))
                                       (+ lenx 1) (/ leny 2))
                                      decresc? grob)
                 ;if not to barline
                 (ly:stencil-translate
                  (hairpin::print-part
                   (scale-coords '((0 . 0) (1 . 0))
                                 (+ lenx 1.5) (/ leny 2))
                   decresc? grob)

                  '(-0.5 . 0))

                 )
             )
            (crescStencilBrokenContinuation
             (ly:stencil-combine-at-edge
              ;if the cautionary accidentals are turned on
              (if cautionary
                  (if (list? (ly:music-property dyn 'text))
                      ;this means, when the custom-made dynamics
                      ;(e.g. \qmp) are used
                      (grob-interpret-markup
                       grob
                       (markup
                        #:hspace 2
                        #:whiteout
                        #:center-align
                        (
                         #:concat
                         (#:normal-text
                          (#:translate '(0 . -0.25)
                                       #:italic (#:fontsize -0.5 "( "
                                                                     ))
```

```
              #:pad-x
              0.35
              #:dynamic (ly:music-property dyn 'text)
              #:normal-text
              (#:translate '(0.5 . -0.25)
                           #:italic (#:fontsize -0.5 ")")))
             )
            )
           )
          ;when in-house dynamics (e.g. \p) are used
          (grob-interpret-markup
           grob
           (markup
            #:hspace 2
            #:center-align
            #:whiteout
            #:concat
            (#:normal-text
             (#:translate '(0 . -0.25)
                          #:italic (#:fontsize 0.5 "("
                                               ))
             (#:translate '(0 . -0.25)
                          #:italic (#:fontsize -10 " "
                                               ))
             #:pad-x
             0
             #:dynamic (ly:music-property dyn 'text)
             #:normal-text
             (#:translate '(0 . -0.25)
                          #:italic (#:fontsize 0.5 ")")))
            )
           ))
      empty-stencil)
  1
  1
  (hairpin::print-part (scale-coords
                        '((0.01 . 0.15) (1 . 0.15))
                        (+ lenx 1)  (/ leny 2))
                       decresc? grob)
  -1.5
  )
 )
(crescStencilBrokenEnding
 ;if the flare end is applied
 ;on the dynamic at the beginning
 ;of the system, meaning lenx is super low

 (cond
```

```
((< lenx 1)
 (ly:stencil-combine-at-edge
  ;if the cautionary accidentals are turned on
  (if cautionary
      (if (list? (ly:music-property dyn 'text))
          ;this means, when the custom-made dynamics
          ;(e.g. \qmp) are used
          (grob-interpret-markup
           grob
           (markup
            #:hspace 0
            #:whiteout
            #:center-align
            #:concat
            (#:normal-text
             (#:translate '(0 . -0.25)
                          #:italic (#:fontsize -0.5 "( "
                                               ))
             #:pad-x
             0.25
             #:dynamic (ly:music-property dyn 'text)
             #:normal-text
             (#:translate '(0.5 . -0.25)
                          #:italic (#:fontsize -0.5 ")")))
            )
           )
          ;when in-house dynamics (e.g. \p) are used
          (grob-interpret-markup
           grob
           (markup
            #:hspace 0

            #:whiteout
            #:center-align
            (
             #:concat
             (#:normal-text
              (#:translate '(0 . -0.25)
                           #:italic (#:fontsize 0.5 "("
                                                ))
              (#:translate '(0 . -0.25)
                           #:italic (#:fontsize -10 " "
                                                ))
              #:pad-x
              0
              #:dynamic (ly:music-property dyn 'text)
              #:normal-text
              (#:translate '(0 . -0.25)
```

```
                           #:italic (#:fontsize 0.5 ")")))
             ))
           ))
      empty-stencil
      )
  1
  1
  (hairpin::print-part
   (scale-coords  coords
                  (+ lenx 1.85) (/ leny 2) )
   decresc? grob)
  ;vertical alignment (to be refined possibly...?)
  -1.75
  )
 )
((>= lenx 1)
 ;if the normal ending flare crescendo
 ;is applied...
 (ly:stencil-combine-at-edge
  ;if the cautionary accidentals are turned on
  (if cautionary
      (if (list? (ly:music-property dyn 'text))
          ;this means, when the custom-made dynamics
          ;(e.g. \qmp) are used
          (grob-interpret-markup
           grob
           (markup
            #:hspace 1.5
            #:whiteout
            #:center-align
            #:concat
            (#:normal-text
             (#:translate '(0 . -0.25)
                          #:italic (#:fontsize -0.5 "( "
                                                     ))
             #:pad-x 0.25
             #:dynamic (ly:music-property dyn 'text)
             #:normal-text
             (#:translate '(0.5 . -0.25)
                          #:italic (#:fontsize -0.5 ")")))
            )
           )
          ;if the in-house dynmamics are used

          (grob-interpret-markup
           grob
           (markup
            #:hspace 1.5
```

```
                      #:whiteout
                      #:center-align
                      #:concat
                      (#:normal-text
                       (#:translate '(0 . -0.25)
                                    #:italic (#:fontsize 0.5 "("
                                                         ))
                       (#:translate '(0 . -0.25)
                                    #:italic (#:fontsize -10 " "
                                                         ))
                       #:pad-x
                       0
                       #:dynamic (ly:music-property dyn 'text)
                       #:normal-text
                       (#:translate '(0 . -0.25)
                                    #:italic (#:fontsize 0.5 ")")))
                      )
                     ))
                empty-stencil
                )
            1
            1
            (hairpin::print-part
             (scale-coords  coords
                            (+ lenx 1.85) (/ leny 2) )
             decresc? grob)
            ;vertical alignment (to be refined possibly...?)
            -1.75
            )
           )
          )
         )


        (cresc
         (cond (hairpinTypeA? crescStencil)
               (hairpinTypeB? crescStencilBrokenBeginning)
               (hairpinTypeC? crescStencilBrokenContinuation)
               (hairpinTypeD? crescStencilBrokenEnding)
               )
         )

        (stil
         (ly:stencil-aligned-to
          (ly:stencil-translate
           cresc
           (cons xtrans ytrans))
          Y CENTER))
```

```
             (stil-y-extent (ly:stencil-extent stil Y)))
            ; FOR DEBUG ONLY
            ; (display (list? (ly:music-property dyn 'text)))
            ; (newline)
            ; (display dyn)
            ; (newline)
            ; (newline)
            ; (display "Is this type A Hairpin? " )
            ; (display hairpinTypeA?)
            ; (newline)
            ; (display "Is this type B Hairpin? " )
            ; (display hairpinTypeB?)
            ; (newline)
            ; (display "Is this type C Hairpin? " )
            ; (display hairpinTypeC?)
            ; (newline)
            ; (display "Is this type D Hairpin? " )
            ; (display hairpinTypeD?)
            ;
            ; (newline)
            ; (display "Is it decrescendo? " )
            ; (display decresc?)
            ; (newline)
            ; (display "end-broken-spanner???  " )
            ; (display (end-broken-spanner? grob ))
            ; (newline)
            ; (display "(first-broken-spanner?) Is spanner broken
            ;         and the first of its broken siblings? " )
            ; (display (first-broken-spanner? grob ) )
            ; (newline)
            ; (display "middle-broken-spanner??? " )
            ; (display (middle-broken-spanner? grob ) )
            ; (newline)
            ;
            ; (display "Is spanner broken and not the first of
            ;         its broken siblings? " )
            ; (display (not-first-broken-spanner? grob ) )
            ; (newline)
            ;
            ; (display "Is spanner broken and not the last of
            ;         its broken siblings? " )
            ; (display (not-last-broken-spanner? grob ) )
            ; (newline)
            ;
            ; (display "unbroken-or-first-broken-spanner??? " )
            ; (display (unbroken-or-first-broken-spanner? grob ) )
            ; (newline)
            ;
```

```
         ; (display "unbroken-or-first-broken-spanner??? " )
         ; (display (unbroken-or-last-broken-spanner? grob ) )
         ; (newline)
         ; (display "unbroken spanner?? " )
         ; (display (unbroken-spanner? grob ) )
         ; (newline)
         ; (newline)
         ; (display "distance of xex " )
         ; (display lenx )
         ; (newline)
         ; (newline)
         (ly:make-stencil (ly:stencil-expr stil) xex stil-y-extent))
        ;; return empty, if no Hairpin.stencil present.
        '())))

  #{
   #dyn
   \tweak Hairpin.stencil
   #(lambda (grob)
     ;depending on the hairpin opening width,
     ;you can adjust the value here vvv
     (constante-hairpin-revised '((0 . 0) (1.0 . 0) (1.0 . -0.9)) #f)

     )
   % This calculates the tweak value for
   % the value of the constante V-offset
   \tweak Y-offset #(lambda (grob)
                     (+ (* vOff (ly:grob-property grob 'height))
                        (* (ly:grob-property grob 'height) -0.5 vOff )
                        moreOff
                        )
                     )

   \<
  #}
  )
\layout {
 indent = #0
 line-width = #100
 ragged-last = ##f
}
{
 \dynamicUp
 \override Hairpin.height = #0.5
 \override Hairpin.to-barline = ##t
 % \override Hairpin.stencil = #constante-hairpin
 \override Hairpin.after-line-breaking = ##t
 c'2  \constanteUP #DOWN \mf  c'2 c'2 \>  c'2
```

```
  c'2 \constanteUP #UP ##t #0.3 \tweak Y-offset #0 \p c'2
  \break  c'4 4 4 4 c'4 4 4 4 \break
  c''2 \< c'2 \! \constanteUP #UP ##f \qf \break
  c'2 c'2\< c'2 c'2 c'2
  \revert Hairpin.to-barline
  \afterGrace 31/32
  { c'2 \constanteUP #CENTER ##f \ff}  s32\!  \break
  c'2\f c'2\< c'2 c'2 c'2
  \revert Hairpin.to-barline
  \after 1*25/64 \!
  { c'2 \constanteUP #CENTER ##f \ff }
  \bar "|."
}

```

4.3.4 Discussion

For the purpose of using it in my own musical typesetting, this constante hairpin snippet has been designed to work with a narrower hairpin height, e.g.

`\override Hairpin.height = #0.5` (the default is `0.666`.)

Table of Contents

4.4 Enhanced Flared Hairpins I (Original Version)

4.4.1 Description

NB: Significantly updated on December 23 and 24, 2025, with new commands being added on January 15 2026.

(See **Enhanced Flared Hairpins II** for the flared hairpins with curved flares.)

In LilyPond, it is possible to specify flared hairpins by writing an override command:

```
\override Hairpin.stencil = #flared-hairpin
```

There are several aspects to LilyPond's default flared hairpin that could be seen as limitations and would be best overcome:

1. Because the flare of the hairpin is drawn using the scheme code `'elbowed-hairpin` in the file `output-lib.scm` that uses a list of points from `0` (beginning) to `1` (end), the longer the hairpin is, the longer its flare becomes. While there may be a case where the longer flare is preferred, in the case of a long hairpin, it becomes more difficult to immediately discern a flared hairpin.

2. While LilyPond's ordinary hairpins do have three separate components when they break into new systems: 1. beginning; 2. intermediate; and; 3. ending, the default flared hairpins draw

identical "flared" hairpins when the lines break. This could invite confusions as it may tempt players to do flared crescendo/diminuendos at the end of each system.

3. When the default flared hairpins are activated, `\override Hairpin.circled-tips` (i.e. to apply circled tips at the onset of the crescendo or at the end of decrescendo) ceases to work.

In developing this enhanced flared hairpin, I have attempted to accommodate different case scenarios, e.g. having `\override Hairpin.to-barline`, `\override Hairpin.after-line-breaking` set to either `##t` or `##f`. It is subject to further refinements. I have also coded the circled-tip dynamic sign as well as tweak functions to be used when the circled-tip dynamic signs are used in tandem with the hairpins.

4.4.2 Grammar

```
\override Hairpin.to-barline = ##f OR ##t
\override Hairpin.after-line-breaking = ##t OR ##f

NOTE \DYNAMIC \flaredCresc \< NOTE(S) \DYNAMIC OR \!

NOTE \DYNAMIC \flaredDecresc \> NOTE(S) \DYNAMIC OR \!
```

If you wish to use circled tips, instead of using `\override Hairpin.circled-tip = ##t`, write:

```
NOTE \o \flaredCresc \< NOTE(S) \DYNAMIC OR \! \flaredDecresc \> NOTE(S) \o
```

Update: January 15, 2026 If you are using the flared crescendo/diminuendo to a note that also has `\after`, use:

```
\flaredCrescOverride NOTE \DYNAMIC \<
\flaredDecrescOverride NOTE \DYNAMIC \>
```

Considering that flared hairpins are often used in tandem with ordinary hairpins, I have embedded `\tweak Hairpin.stencil = #flared-hairpin-new` in `\flaredCresc` and `\flaredDecresc`. Therefore, it is not necessary to write an `\override` command to change the stencil used.

Update: December 23-24, 2025 `\flaredCresc` and `\flaredDecresc` accepts a number as an optional argument. This functionality has been added to allow the adjustment of flaring end of the hairpin, as well as the horizontal placement of the small piece of the arrival hairpin when `\override to-barline` is set `##f`. While I have striven to set the default setting in such a way that these micro-adjustments are not necessary, it seemed helpful to add this functionality when further refinement is needed. Think of this as an `X-offset` for the arrival point. See **Code**.

Update: January 15, 2026 I have discovered that `\flaredCresc` and `\flaredDecresc` do not work when it is applied to a note that also has `\after` preceding it, e.g. when the hairpin is terminated within the given note. In order to make such a case work, I have created additional commands `\flaredCrescOverride` and `\flaredDecrescOverride`, which use `\once \override` commands instead of `\tweak`. These commands also accept an optional argument that behaves like `X-offset`.

4.4.3 Code

```
\version "2.26.0"

```

```

% similar to the original version

#(define-public ((elbowed-hairpin-revised coords mirrored?) grob)
  "This scheme code was originally the scheme code 'elbowed-hairpin
taken from output-lib.scm, and was revised to suit the need.
"
  (define (scale-coords coords-list x y)
   (map
    (lambda (coord) (cons (* x (car coord)) (* y (cdr coord))))
    coords-list))

  (define (hairpin::print-part points decresc? me)
   (let ((stil (make-connected-line points me)))
    (if decresc? (ly:stencil-scale stil -1 1)
        (ly:stencil-scale stil 1 1))))

  ;; outer let to trigger suicide
  (let ((sten (ly:hairpin::print grob)))
   (if (grob::is-live? grob)
       (let*
        ((decresc? (eqv? (ly:grob-property grob 'grow-direction) LEFT))
         (hairpinHeight (ly:grob-property grob 'height))
         (toBarline? (ly:grob-property grob 'to-barline))
         (orig (ly:grob-original grob))
         (siblings (if (ly:grob? orig)
                       (ly:spanner-broken-into orig)
                       '()))
         (hairpinTypeA?
          (and (eqv? (end-broken-spanner? grob )               #f)
               (eqv? (first-broken-spanner? grob )             #f)
               (eqv? (middle-broken-spanner? grob )            #f)
               (eqv? (not-first-broken-spanner? grob )         #f)
               (eqv? (not-last-broken-spanner? grob )          #f)
               (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
               (eqv? (unbroken-or-last-broken-spanner? grob )  #t)
               (eqv? (unbroken-spanner? grob )                 #t)
               )) ;Hairpins that end within one line without reaching
         ;to the end of the system
         (hairpinTypeB?
          (or (and (eqv? (end-broken-spanner? grob )               #t)
                   (eqv? (first-broken-spanner? grob )             #t)
                   (eqv? (middle-broken-spanner? grob )            #f)
                   (eqv? (not-first-broken-spanner? grob )         #f)
                   (eqv? (not-last-broken-spanner? grob )          #f)
                   (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
                   (eqv? (unbroken-or-last-broken-spanner? grob )  #t)
                   (eqv? (unbroken-spanner? grob )                 #f)
```

```
                    )
               (and (eqv? (end-broken-spanner? grob )               #f)
                    (eqv? (first-broken-spanner? grob )             #t)
                    (eqv? (middle-broken-spanner? grob )            #f)
                    (eqv? (not-first-broken-spanner? grob )         #f)
                    (eqv? (not-last-broken-spanner? grob )          #t)
                    (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
                    (eqv? (unbroken-or-last-broken-spanner? grob )   #f)
                    (eqv? (unbroken-spanner? grob )                 #f)
                    )
               )) ;Hairpins that breaks into next line
          (hairpinTypeC?
           (and (eqv? (end-broken-spanner? grob )               #f)
                (eqv? (first-broken-spanner? grob )             #f)
                (eqv? (middle-broken-spanner? grob )            #t)
                (eqv? (not-first-broken-spanner? grob )         #t)
                (eqv? (not-last-broken-spanner? grob )          #t)
                (eqv? (unbroken-or-first-broken-spanner? grob ) #f)
                (eqv? (unbroken-or-last-broken-spanner? grob )   #f)
                (eqv? (unbroken-spanner? grob )                 #f)
                )) ;Midway-through Hairpins
          (hairpinTypeD?
           (and (eqv? (end-broken-spanner? grob )               #t)
                (eqv? (first-broken-spanner? grob )             #f)
                (eqv? (middle-broken-spanner? grob )            #f)
                (eqv? (not-first-broken-spanner? grob )         #t)
                (eqv? (not-last-broken-spanner? grob )          #f)
                (eqv? (unbroken-or-first-broken-spanner? grob ) #f)
                (eqv? (unbroken-or-last-broken-spanner? grob )   #t)
                (eqv? (unbroken-spanner? grob )                 #f)
                )) ;multi-system Hairpin that terminates
          (xex (if decresc?
                   (cons (+ (car (ly:stencil-extent sten X)) 0.75)
                         (cdr (ly:stencil-extent sten X)))
                   (cons (car (ly:stencil-extent sten X))
                         (- (cdr (ly:stencil-extent sten X)) 1.25))
                   ))
          (lenx (interval-length xex))
          (yex (ly:stencil-extent sten Y))
          (leny (interval-length yex))
          (xtrans (+ (car xex) (if decresc? lenx 0)))
          (ytrans (car yex))
          (uplist (scale-coords coords lenx (/ leny 2)))
          (downlist (scale-coords coords lenx (/ leny -2)))
          ; crescStencil is used when the flared crescendo takes place
          ; and finishes within one line.
          (crescStencil
           (if toBarline?
```

```
;I added here to make it default to stretch
;the hairpin immediately to the dynamics
(ly:stencil-combine-at-edge
 (ly:stencil-combine-at-edge
  (hairpin::print-part
   (scale-coords
    coords
    (+ lenx 0.95) (/ leny 2))  decresc? grob)
  1
  -1
  (if mirrored?
      (hairpin::print-part
       (scale-coords
        coords
        (+ lenx 0.95) (/ leny -2))  decresc? grob)
      empty-stencil)
  -0.1 ;vertical padding value here
  )
 0
 1
 (make-path-stencil
  (list
   'moveto 0 (* 0.2 (/ hairpinHeight 0.666) )
   'rlineto 0.25 0.3
   'moveto 0 (* -0.2 (/ hairpinHeight 0.666) )
   'rlineto 0.25 -0.3
   )
  0.1
  2
  2
  #f
  )
 -0.075
 )
;if not to barline
(ly:stencil-combine-at-edge
 (ly:stencil-combine-at-edge
  (hairpin::print-part
   (scale-coords
    coords
    (+ lenx 0.95) (/ leny 2))  decresc? grob)
  1
  -1
  (if mirrored?
      (hairpin::print-part
       (scale-coords
        coords
        (+ lenx 0.95) (/ leny -2))  decresc? grob)
```

```
           empty-stencil)
       -0.1 ;vertical padding value here
       )
      0
      1
      (make-path-stencil

       (list
        'moveto 0 (* 0.2 (/ hairpinHeight 0.666) )
        'rlineto 0.25 0.3
        'moveto 0 (* -0.2 (/ hairpinHeight 0.666) )
        'rlineto 0.25 -0.3
        )
       0.1
       2
       2
       #f
       )
      -0.075
      ))
 )
; when crescendo begins but breaks off to
; the subsequent system(s)
(crescStencilBrokenBeginning
 ; if the hairpin is set only to barline...
 (if toBarline?
     ; ... AND if the hairpin finishes at the beginning
     ; of the next line
     (if  (= (length siblings) 1)
          (ly:stencil-combine-at-edge
           (ly:stencil-combine-at-edge
            (hairpin::print-part
             (scale-coords
              coords
              (+ lenx 0.95) (/ leny 2))  decresc? grob)
            1
            -1
            (if mirrored?
                (hairpin::print-part
                 (scale-coords
                  coords
                  (+ lenx 0.95) (/ leny -2))  decresc? grob)
                empty-stencil)
            -0.1 ;vertical padding value here
            )
           0
           1
           (make-path-stencil
```

```
           (list
            'moveto 0 (* 0.2 (/ hairpinHeight 0.666) )
            'rlineto 0.25 0.3
            'moveto 0 (* -0.2 (/ hairpinHeight 0.666) )
            'rlineto 0.25 -0.3
            )
           0.1
           2
           2
           #f
           )
          -0.075
          )
         (ly:stencil-combine-at-edge
          (hairpin::print-part
           (scale-coords
            coords
            (+ lenx 1.6) (/ leny 1.5)) decresc? grob)
          1
          -1
          (if mirrored?
              (hairpin::print-part
               (scale-coords
                coords

                (+ lenx 1.6) (/ leny -1.5))  decresc? grob)
              empty-stencil)
          -0.1 ;vertical padding value here
          )
         )
    (ly:stencil-combine-at-edge
     (hairpin::print-part
      (scale-coords
       coords
       (+ lenx 1) (/ leny 1.5)) decresc? grob)
     1
     -1
     (if mirrored?
         (hairpin::print-part
          (scale-coords
           coords

           (+ lenx 1) (/ leny -1.5))  decresc? grob)
         empty-stencil)
     -0.1 ;vertical padding value here
     )
    )
)
```

```
(crescStencilBrokenContinuation
 (ly:stencil-combine-at-edge
  (hairpin::print-part
   (scale-coords
    coords
    (+ lenx 1) (/ leny 2)) decresc? grob)
  1
  -1
  (if mirrored?
      (hairpin::print-part
       (scale-coords
        coords
        (+ lenx 1) (/ leny -2))  decresc? grob)
      empty-stencil)
  0.1 ;vertical padding value here
  )
 )
(crescStencilBrokenEnding
 ;if the flare end is applied
 ;on the dynamic at the beginning
 ;of the system, meaning lenx is super low
 (cond ((< lenx 0.2)
        (make-path-stencil
         (list 'moveto 0.45 (* 0.2 (/ hairpinHeight 0.666) )
               'rlineto -0.65 -0.1
               'moveto 0.45 (* 0.2 (/ hairpinHeight 0.666) )
               'rlineto 0.25 0.3
               'moveto 0.45 (* -0.2 (/ hairpinHeight 0.666) )
               'rlineto -0.65 0.1
               'moveto 0.45 (* -0.2 (/ hairpinHeight 0.666) )
               'rlineto 0.25 -0.3
               )
         0.1
         2
         2
         #f
         ))
       ((and (>= lenx 0.2)(< lenx 2))
        ;if the flare end is applied
        ;on the note at the beginning of the system
        ;without the dynamics (lenx is about 1 to 2)
        (ly:stencil-combine-at-edge
         (ly:stencil-combine-at-edge
          (hairpin::print-part
           (scale-coords
            (list
             (cons (- (car (car coords)) 0)
                   (cdr (car coords )))
```

```
                      (cons (- (car (cadr coords)) 0)
                            (cdr (cadr coords))))
                     (+ lenx 0.25) (/ leny 2.75)) decresc? grob)
                   1
                   -1
                   (if mirrored?
                       (hairpin::print-part
                        (scale-coords
                         (list
                          (cons (- (car (car coords)) 0)
                                (cdr (car coords )))
                          (cons (- (car (cadr coords)) 0)
                                (cdr (cadr coords))))
                         (+ lenx 0.25) (/ leny -2.75))  decresc? grob)
                       empty-stencil)
                   0.1 ;vertical padding value here
                   )
                  0
                  1
                  (make-path-stencil
                   (list   'moveto 0 (* 0.15 (/ hairpinHeight 0.666) )
                           'rlineto 0.25 0.3
                           'moveto 0
                           (- (* -0.15 (/ hairpinHeight 0.666) ) 0.1)
                           'rlineto 0.25 -0.3
                           )
                   0.1
                   2
                   2
                   #f
                   )
                  -0.075
                  ))
                ((>= lenx 2)
                 ;if the normal ending flare crescendo
                 ;is applied...
                 (ly:stencil-combine-at-edge
                  (ly:stencil-combine-at-edge
                   (hairpin::print-part
                    (scale-coords
                     coords
                     (+ lenx 0.25) (/ leny 3)) decresc? grob)
                   1
                   -1
                   (if mirrored?
                       (hairpin::print-part
                        (scale-coords
                         coords
```

```
                          (+ lenx 0.25) (/ leny -3))  decresc? grob)
                        empty-stencil)
                    0.1 ;vertical padding value here
                    )
                   0
                   1
                   (make-path-stencil
                    (list   'moveto 0 (* 0.133 (/ hairpinHeight 0.666) )
                            'rlineto 0.25 0.3
                            'moveto 0
                            (- (* -0.133 (/ hairpinHeight 0.666) ) 0.1)
                            'rlineto 0.25 -0.3
                            )
                    0.1
                    2
                    2
                    #f
                    )
                   -0.075
                   ))
                 )
           )
          (decrescStencil
           (if toBarline?
               ;I added here to make it default to stretch
               ;the hairpin immediately to the dynamics
               (ly:stencil-combine-at-edge
                (ly:stencil-combine-at-edge
                 (hairpin::print-part
                  (scale-coords
                   coords
                   (+ lenx 0.25) (/ leny 2)) decresc? grob)
                 1
                 -1
                 (if mirrored?
                     (hairpin::print-part
                      (scale-coords
                       coords
                       (+ lenx 0.25) (/ leny -2))  decresc? grob)
                     empty-stencil)
                 -0.1 ;vertical padding value here
                 )
                0
                -1
                (make-path-stencil
                 (list   'moveto 0 (* 0.19 (/ hairpinHeight 0.666) )
                         'rlineto -0.25 0.3
                         'moveto 0 (* -0.19 (/ hairpinHeight 0.666) )
```

```
                         'rlineto -0.25 -0.3
                         )
                 0.1
                 2
                 2
                 #f
                 )
                -0.075
                )
               ;if not to barline
               (ly:stencil-combine-at-edge
                (ly:stencil-combine-at-edge
                 (hairpin::print-part
                  (scale-coords
                   coords
                   (+ lenx 0.25) (/ leny 2))  decresc? grob)
                 1
                 -1
                 (if mirrored?
                     (hairpin::print-part
                      (scale-coords
                       coords
                       (+ lenx 0.25) (/ leny -2))  decresc? grob)
                     empty-stencil)
                 -0.1 ;vertical padding value here
                 )
                0
                -1
                (make-path-stencil
                 (list
                  'moveto 0 (* 0.2 (/ hairpinHeight 0.666) )
                  'rlineto -0.25 0.3
                  'moveto 0 (* -0.2 (/ hairpinHeight 0.666) )
                  'rlineto -0.25 -0.3
                  )
                 0.1
                 2
                 2
                 #f
                 )
                -0.075
                ))
           )

          (decrescStencilBrokenBeginning
           (ly:stencil-combine-at-edge
            (ly:stencil-combine-at-edge
             (hairpin::print-part
```

```
              (scale-coords
               coords
               (+ lenx 0.2) (/ leny 2.5))
              decresc? grob)
             1
             -1
             (if mirrored?
                 (hairpin::print-part
                  (scale-coords
                   coords
                   (+ lenx 0.2) (/ leny -2.5))  decresc? grob)
                 empty-stencil)
             0.1 ;vertical padding value here
             )
            0
            -1
            (make-path-stencil
             (list   'moveto 0 (* 0.15 (/ hairpinHeight 0.666) )
                     'rlineto -0.25 0.3
                     'moveto 0
                     (- (* -0.15 (/ hairpinHeight 0.666) ) 0.1)
                     'rlineto -0.25 -0.3
                     )
             0.1
             2
             2
             #f
             )
            -0.075
            )
           )
          (decrescStencilBrokenEnding
           (ly:stencil-combine-at-edge
            (hairpin::print-part uplist decresc? grob)
            1
            -1
            (if mirrored?
                (hairpin::print-part downlist decresc? grob)
                empty-stencil)
            -0.1 ;vertical padding value here
            )
           )
          (cresc
           (cond (hairpinTypeA? crescStencil)
                 (hairpinTypeB? crescStencilBrokenBeginning)
                 (hairpinTypeC? crescStencilBrokenContinuation)
                 (hairpinTypeD? crescStencilBrokenEnding)
                 )
```

```
        )
       (decres
        (cond (hairpinTypeA? decrescStencil)
              (hairpinTypeB? decrescStencilBrokenBeginning)
              (hairpinTypeC? crescStencilBrokenContinuation)
              ;crescStencilBrokenContinuation can be used for decresc
              ;as well, because it gets inverted.
              (hairpinTypeD? decrescStencilBrokenEnding)
              )
        )
       (stil
        (ly:stencil-aligned-to
         (ly:stencil-translate
          (if decresc?
              decres
              cresc
              )
          (cons xtrans ytrans))
         Y CENTER))
       (stil-y-extent (ly:stencil-extent stil Y)))
      ; for debugging purposes
      ;         (display "Is this type A Hairpin? " )
      ;         (display hairpinTypeA?)
      ;         (newline)
      ;         (display "Is this type B Hairpin? " )
      ;         (display hairpinTypeB?)
      ;         (newline)
      ;         (display "Is this type C Hairpin? " )
      ;         (display hairpinTypeC?)
      ;         (newline)
      ;         (display "Is this type D Hairpin? " )
      ;         (display hairpinTypeD?)
      ;
      ;         (newline)
      ;         (display "Is it decrescendo? " )
      ;         (display decresc?)
      ;         (newline)
      ;         (display "end-broken-spanner???  " )
      ;         (display (end-broken-spanner? grob ))
      ;         (newline)
      ;         (display "(first-broken-spanner?) Is spanner broken
      ;and the first of its broken siblings? " )
      ;         (display (first-broken-spanner? grob ) )
      ;         (newline)
      ;         (display "middle-broken-spanner??? " )
      ;         (display (middle-broken-spanner? grob ) )
      ;         (newline)
      ;
```

```
        ;         (display "Is spanner broken and not the first of
        ;its broken siblings? " )
        ;         (display (not-first-broken-spanner? grob ) )
        ;         (newline)
        ;
        ;         (display "Is spanner broken and not the last of
        ;its broken siblings? " )
        ;         (display (not-last-broken-spanner? grob ) )
        ;         (newline)
        ;
        ;         (display "unbroken-or-first-broken-spanner??? " )
        ;         (display (unbroken-or-first-broken-spanner? grob ) )
        ;         (newline)
        ;
        ;         (display "unbroken-or-first-broken-spanner??? " )
        ;         (display (unbroken-or-last-broken-spanner? grob ) )
        ;         (newline)
        ;         (display "unbroken spanner?? " )
        ;         (display (unbroken-spanner? grob ) )
        ;         (newline)
        ;         (newline)
        ;         (display "distance of xex " )
        ;         (display lenx )
        ;         (newline)
        ;         (newline)
        (ly:make-stencil (ly:stencil-expr stil) xex stil-y-extent))
       ;; return empty, if no Hairpin.stencil present.
       '())))

#(define-public flared-hairpin-new
  (elbowed-hairpin-revised '((0 . 0) (1 . 0.5))
                           #t))

#(define-public o
  (make-music 'AbsoluteDynamicEvent
              'text
              (markup
               #:pad-x -inf.0
               #:concat (
                         #:pad-to-box '(0 . 0.1) '(-0.75 . 1.75)
                         #:translate '(0 . 0.6)
                         #:musicglyph "scripts.flageolet"
                         ))
              ))

flaredCresc =
#(define-music-function (num dyn) ((number? 0) ly:event? )
  (let* ((hOffset num))
```

```
#{
 \tweak Hairpin.stencil #flared-hairpin-new
 \tweak Hairpin.shorten-pair
 #(lambda (grob)
   (let* (
          ;; have we been split?
          (orig (ly:grob-original grob))

          ;; if yes, get the split pieces (our siblings)
          (siblings (if (ly:grob? orig)
                        (ly:spanner-broken-into orig)
                        '())))

    (cond
     ((= (length siblings) 0)
      (cons -0.6 hOffset)
      )
     ((and (>= (length siblings) 1)
           (eq? (car siblings) grob)
           )
      (cons -0.6 0))
     )



    )
   )
 \tweak Hairpin.after-line-breaking
 #(lambda (grob)
   (let* (
          ;; have we been split?
          (orig (ly:grob-original grob))

          ;; if yes, get the split pieces (our siblings)
          (siblings (if (ly:grob? orig)
                        (ly:spanner-broken-into orig)
                        '()))
          (toBarline? (ly:grob-property grob 'to-barline))
          )

    (cond  ((< (length siblings) 2)
            (ly:grob-set-property!
             grob 'shorten-pair
             (cons -0.6 (* hOffset -1)))

            )

           ((and (>= (length siblings) 2)
```

```
                    (eq? (car (last-pair siblings)) grob)
                    toBarline? )

               (ly:grob-set-property!
                grob 'shorten-pair
                (cons 0 (- -0.6 hOffset)))


               )
              ((and (>= (length siblings) 2)
                    (eq? (car (last-pair siblings)) grob)
                    (not toBarline?) )
               (ly:grob-set-property!
                grob 'shorten-pair
                (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))

               )
              )

       )
      ) #dyn  #}))

flaredCrescOverride =
#(define-music-function (num mus) ((number? 0) ly:music? )
  (let* ((hOffset num))
   #{
    \once \override Hairpin.stencil = #flared-hairpin-new
    \once \override Hairpin.shorten-pair =
    #(lambda (grob)
      (let* (
             ;; have we been split?
             (orig (ly:grob-original grob))

             ;; if yes, get the split pieces (our siblings)
             (siblings (if (ly:grob? orig)
                           (ly:spanner-broken-into orig)
                           '())))

       (cond
        ((= (length siblings) 0)
         (cons -0.6 hOffset)
         )
        ((and (>= (length siblings) 1)
              (eq? (car siblings) grob)
              )
         (cons -0.6 0))
        )

```

```


          )
         )
       \once \override  Hairpin.after-line-breaking =
       #(lambda (grob)
         (let* (
                ;; have we been split?
                (orig (ly:grob-original grob))

                ;; if yes, get the split pieces (our siblings)
                (siblings (if (ly:grob? orig)
                               (ly:spanner-broken-into orig)
                               '()))
                (toBarline? (ly:grob-property grob 'to-barline))
                )

          (cond  ((< (length siblings) 2)
                  (ly:grob-set-property!
                   grob 'shorten-pair
                   (cons -0.6 (* hOffset -1)))

                  )

                 ((and (>= (length siblings) 2)
                       (eq? (car (last-pair siblings)) grob)
                       toBarline? )

                  (ly:grob-set-property!
                   grob 'shorten-pair
                   (cons 0 (- -0.6 hOffset)))


                  )
                 ((and (>= (length siblings) 2)
                       (eq? (car (last-pair siblings)) grob)
                       (not toBarline?) )
                  (ly:grob-set-property!
                   grob 'shorten-pair
                   (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))

                  )
                 )

          )
         ) #mus  #}))

flaredDecresc =
```

```
#(define-music-function (num dyn) ((number? 0) ly:event? )
  (let* ((hOffset num))
   #{
    \tweak Hairpin.stencil #flared-hairpin-new
    \tweak Hairpin.shorten-pair
    #(lambda (grob)
      (let* (
             ;; have we been split?
             (orig (ly:grob-original grob))

             ;; if yes, get the split pieces (our siblings)
             (siblings (if (ly:grob? orig)
                           (ly:spanner-broken-into orig)
                           '()))
             (toBarline? (ly:grob-property grob 'to-barline))

             )

       (cond
        ((and (zero? (length siblings)) (not toBarline?))
         (cons hOffset -0.6)
         )
        ((and (zero? (length siblings)) toBarline?)
         (cons hOffset -0.6)
         )
        ((and (>= (length siblings) 1)
              (eq? (car siblings) grob)
              )
         (cons -0.6 0))
        )
       )
      )
    \tweak Hairpin.after-line-breaking
    #(lambda (grob)
      (let* (
             ;; have we been split?
             (orig (ly:grob-original grob))

             ;; if yes, get the split pieces (our siblings)
             (siblings (if (ly:grob? orig)
                           (ly:spanner-broken-into orig)
                           '())))

       (cond
        ((and (>= (length siblings) 1)
              (eq? (car (last-pair siblings)) grob))
         (ly:grob-set-property!
          grob 'shorten-pair
```

```
          (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))
         )

        ((zero? (length siblings)  )
         #f
         )

        ((and (>= (length siblings) 2)
              (eq? (car (last-pair siblings)) grob)
              toBarline? )
         (ly:grob-set-property!
          grob 'shorten-pair
          (cons 0 (- -0.6 hOffset)))
         )

        ((and (>= (length siblings) 2)
              (eq? (car siblings) grob)
              )
         (ly:grob-set-property!
          grob 'shorten-pair
          (cons   0 0))
         )

        ((and (>= (length siblings) 2)
              (eq? (car (last-pair siblings)) grob)
              (not toBarline?) )
         (ly:grob-set-property!
          grob 'shorten-pair
          (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))
         )
        )


       )
      ) #dyn #}))

flaredDecrescOverride =
#(define-music-function (num mus) ((number? 0) ly:music? )
  (let* ((hOffset num))
   #{
    \once \override Hairpin.stencil = #flared-hairpin-new
    \once \override  Hairpin.shorten-pair =
    #(lambda (grob)
      (let* (
             ;; have we been split?
             (orig (ly:grob-original grob))

             ;; if yes, get the split pieces (our siblings)
```

```
              (siblings (if (ly:grob? orig)
                            (ly:spanner-broken-into orig)
                            '()))
              (toBarline? (ly:grob-property grob 'to-barline))

              )

        (cond
         ((and (zero? (length siblings)) (not toBarline?))
          (cons hOffset -0.6)
          )
         ((and (zero? (length siblings)) toBarline?)
          (cons hOffset -0.6)
          )
         ((and (>= (length siblings) 1)
               (eq? (car siblings) grob)
               )
          (cons -0.6 0))
         )
        )
       )
     \once \override  Hairpin.after-line-breaking =
     #(lambda (grob)
       (let* (
              ;; have we been split?
              (orig (ly:grob-original grob))

              ;; if yes, get the split pieces (our siblings)
              (siblings (if (ly:grob? orig)
                            (ly:spanner-broken-into orig)
                            '())))

        (cond
         ((and (>= (length siblings) 1)
               (eq? (car (last-pair siblings)) grob))
          (ly:grob-set-property!
           grob 'shorten-pair
           (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))
          )

         ((zero? (length siblings)  )
          #f
          )

         ((and (>= (length siblings) 2)
               (eq? (car (last-pair siblings)) grob)
               toBarline? )
          (ly:grob-set-property!
```

```
          grob 'shorten-pair
          (cons 0 (- -0.6 hOffset)))
         )

        ((and (>= (length siblings) 2)
              (eq? (car siblings) grob)
              )
         (ly:grob-set-property!
          grob 'shorten-pair
          (cons   0 0))
         )

        ((and (>= (length siblings) 2)
              (eq? (car (last-pair siblings)) grob)
              (not toBarline?) )
         (ly:grob-set-property!
          grob 'shorten-pair
          (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))
         )
        )


      )
     ) #mus #}))

\layout {
 indent = #0
 line-width = #100
 ragged-last = ##f
}



{
 \override Hairpin.height = #0.5
 % \override Hairpin.stencil = #flared-hairpin-new
 \override Hairpin.to-barline = ##f
 \override Hairpin.after-line-breaking = ##f

 c' \fff \flaredCresc \< c' c' c' \o \flaredDecresc #-0.25 \>
 c' c'\o\flaredCresc #-0.6  \< c' c' \break
 \repeat unfold 3  {c' c' c' c'} \break
 c'\ff \flaredDecresc \> c' c' c'
 c' \p \flaredCresc \< c' c' \flaredDecresc \> c'
 c' \p \flaredCresc \< c'  c'\! \flaredDecresc \> c' \break
 \repeat unfold 3  {c' c' c' c'} \break
 c' \f  \flaredDecresc \> c' c' c'\o \flaredCresc #-1 \<
 c' c' c' c' \break c'\! s4 s4 s4 \bar"."
```

```
  \once \override Hairpin.extra-offset = #'(1.5 . 0)
  \flaredCrescOverride #1.5 \after 2. \f c'1 \<
  \flaredDecrescOverride #1 \after 2. \p c'1  \> \bar"."
}
```

4.4.4 Discussion

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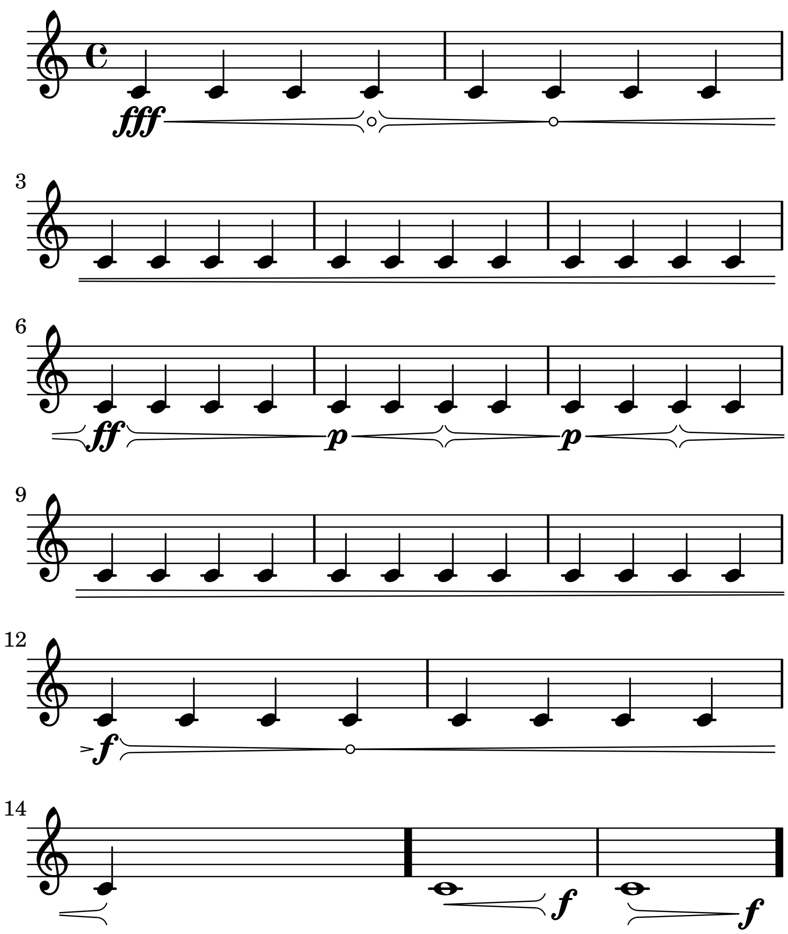
4.5 Enhanced Flared Hairpins II (Curvy Version)



4.5.1 Description

NB: Significantly updated on December 23, 2025 and cosmetic alteration undertaken on December 24, 2025, as well as new commands being added on January 15 2026.

See **Enhanced Flared Hairpins I** for the flared hairpins similar to **LilyPond's default flared hairpins.**

This version implements hairpins with flares that are curved. Initially Santiago Beis inspired me to take advantage of Phi Curve in making the curvy flares as natural as possible, though I have departed from that approach as I found using `rcurve` useful in `make-path-stencil`, which I then combined with the crescendo lines using `ly:stencil-combine-at-edge`.

4.5.2 Grammar

```
\override Hairpin.to-barline = ##f OR ##t
\override Hairpin.after-line-breaking = ##t OR ##f

NOTE \DYNAMIC \flaredCrescCurvy \< NOTE(S) \DYNAMIC OR \!
```

`NOTE \DYNAMIC \flaredDecrescCurvy \> NOTE(S) \DYNAMIC OR \!`

If you wish to use circled tips, instead of using `\override Hairpin.circled-tip = ##t`, write:

`NOTE \o \flaredCrescCurvy \< NOTE(S) \DYNAMIC OR \! \flaredDecrescCurvy \> NOTE(S) \o`

Update: January 15, 2026 If you are using the flared crescendo/diminuendo to a note that also has `\after`, use:

```
\flaredCrescCurvyOverride NOTE \DYNAMIC \<
\flaredDecrescCurvyOverride NOTE \DYNAMIC \>
```

Considering that flared hairpins are often used in tandem with ordinary hairpins, I have embedded `\tweak Hairpin.stencil = #flared-hairpin-curvy` in `\flaredCrescCurvy` and `\flaredDecrescCurvy`. Therefore, it is not necessary to write an `\override` command to change the stencil used.

Update: December 23-24, 2025 `\flaredCresc` and `\flaredDecresc` accepts a number as an optional argument. This functionality has been added to allow the adjustment of flaring end of the hairpin, as well as the horizontal placement of the small piece of the arrival hairpin when `\override to-barline` is set `##f`. While I have striven to set the default setting in such a way that these micro-adjustments are not necessary, it seemed helpful to add this functionality when further refinement is needed. Think of this as an `X-offset` for the arrival point. See **Code**.

Update: January 15, 2026 I have discovered that `\flaredCrescCurvy` and `\flaredDecrescCurvy` do not work when it is applied to a note that also has `\after` preceding it, e.g. when the hairpin is terminated within the given note. In order to make such a case work, I have created additional commands `\flaredCrescCurvyOverride` and `\flaredDecrescCurvyOverride`, which use `\once \override` commands instead of `\tweak`. These commands also accept an optional argument that behaves like `X-offset`.

4.5.3 Code

```
\version "2.26.0"

% curvy version

#(define-public ((curvy-elbowed-hairpin coords mirrored?) grob)
  "This scheme code was originally the scheme code 'elbowed-hairpin
taken from output-lib.scm, and was revised to suit the need.
"
  (define (scale-coords coords-list x y)
   (map
    (lambda (coord) (cons (* x (car coord)) (* y (cdr coord))))
    coords-list))

  (define (hairpin::print-part points decresc? me)
   (let ((stil (make-connected-line points me)))
    (if decresc? (ly:stencil-scale stil -1 1)
        (ly:stencil-scale stil 1 1))))

  ;; outer let to trigger suicide
```

```
(let ((sten (ly:hairpin::print grob)))
 (if (grob::is-live? grob)
     (let*
      ((decresc? (eqv?
                  (ly:grob-property grob 'grow-direction) LEFT))
       (hairpinHeight (ly:grob-property grob 'height))
       (toBarline? (ly:grob-property grob 'to-barline))
       (orig (ly:grob-original grob))
       (siblings (if (ly:grob? orig)
                     (ly:spanner-broken-into orig)
                     '()))
       (hairpinTypeA?
        (and (eqv? (end-broken-spanner? grob )               #f)
             (eqv? (first-broken-spanner? grob )             #f)
             (eqv? (middle-broken-spanner? grob )            #f)
             (eqv? (not-first-broken-spanner? grob )         #f)
             (eqv? (not-last-broken-spanner? grob )          #f)
             (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
             (eqv? (unbroken-or-last-broken-spanner? grob )  #t)
             (eqv? (unbroken-spanner? grob )                 #t)
             )) ;Hairpins that end within one line without
       ;reaching to the end of the system
       (hairpinTypeB?
        (or (and (eqv? (end-broken-spanner? grob )               #t)
                 (eqv? (first-broken-spanner? grob )             #t)
                 (eqv? (middle-broken-spanner? grob )            #f)
                 (eqv? (not-first-broken-spanner? grob )         #f)
                 (eqv? (not-last-broken-spanner? grob )          #f)
                 (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
                 (eqv? (unbroken-or-last-broken-spanner? grob )  #t)
                 (eqv? (unbroken-spanner? grob )                 #f)
                 )
            (and (eqv? (end-broken-spanner? grob )               #f)
                 (eqv? (first-broken-spanner? grob )             #t)
                 (eqv? (middle-broken-spanner? grob )            #f)
                 (eqv? (not-first-broken-spanner? grob )         #f)
                 (eqv? (not-last-broken-spanner? grob )          #t)
                 (eqv? (unbroken-or-first-broken-spanner? grob ) #t)
                 (eqv? (unbroken-or-last-broken-spanner? grob )  #f)
                 (eqv? (unbroken-spanner? grob )                 #f)
                 )
            )) ;Hairpins that breaks into next line
       (hairpinTypeC?
        (and (eqv? (end-broken-spanner? grob )               #f)
             (eqv? (first-broken-spanner? grob )             #f)
             (eqv? (middle-broken-spanner? grob )            #t)
             (eqv? (not-first-broken-spanner? grob )         #t)
             (eqv? (not-last-broken-spanner? grob )          #t)
```

```
                (eqv? (unbroken-or-first-broken-spanner? grob ) #f)
                (eqv? (unbroken-or-last-broken-spanner? grob )  #f)
                (eqv? (unbroken-spanner? grob )                 #f)
                )) ;Midway-through Hairpins
          (hairpinTypeD?
           (and (eqv? (end-broken-spanner? grob )               #t)
                (eqv? (first-broken-spanner? grob )             #f)
                (eqv? (middle-broken-spanner? grob )            #f)
                (eqv? (not-first-broken-spanner? grob )         #t)
                (eqv? (not-last-broken-spanner? grob )          #f)
                (eqv? (unbroken-or-first-broken-spanner? grob ) #f)
                (eqv? (unbroken-or-last-broken-spanner? grob )  #t)
                (eqv? (unbroken-spanner? grob )                 #f)
                )) ;multi-system Hairpin that terminates
          (xex (if decresc?
                   (cons (+ (car (ly:stencil-extent sten X)) 0.75)
                         (cdr (ly:stencil-extent sten X)))
                   (cons (car (ly:stencil-extent sten X))
                         (- (cdr (ly:stencil-extent sten X)) 1.75))
                   ))
          (lenx (interval-length xex))
          (yex (ly:stencil-extent sten Y))
          (leny (interval-length yex))
          (xtrans (+ (car xex) (if decresc? lenx 0)))
          (ytrans (car yex))
          (uplist (scale-coords coords lenx (/ leny 2)))
          (downlist (scale-coords coords lenx (/ leny -2)))
          ; crescStencil is used when the flared crescendo takes place
          ; and finishes within one line.
          (crescStencil
           (if toBarline?
               ;I added here to make it default to stretch
               ;the hairpin immediately to the dynamics
               (ly:stencil-combine-at-edge
                (ly:stencil-combine-at-edge
                 (hairpin::print-part
                  (scale-coords
                   coords
                   (+ lenx 0.95) (/ leny 2))  decresc? grob)
                 1
                 -1
                 (if mirrored?
                     (hairpin::print-part
                      (scale-coords
                       coords
                       (+ lenx 0.95) (/ leny -2))  decresc? grob)
                     empty-stencil)
                 -0.1 ;vertical padding value here
```

```
  )
 0
 1
 (make-path-stencil
  (list
   'moveto 0
   (+ (* 0.1665 (/ hairpinHeight 0.666) ) 0.015)
   'rcurveto 0 0 0.3 -0.0 0.4 0.3
   'moveto 0
   (- (* -0.1665 (/ hairpinHeight 0.666) ) 0.015)
   'rcurveto 0 0 0.3 0.0 0.4 -0.3
   )
  0.1
  2
  2
  #f
  )
 -0.075
 )
;if not to barline
(ly:stencil-combine-at-edge
 (ly:stencil-combine-at-edge
  (hairpin::print-part
   (scale-coords
    coords
    (+ lenx 0.95) (/ leny 2))  decresc? grob)
  1
  -1
  (if mirrored?
      (hairpin::print-part
       (scale-coords
        coords
        (+ lenx 0.95) (/ leny -2))  decresc? grob)
      empty-stencil)
  -0.1 ;vertical padding value here
  )
 0
 1
 (make-path-stencil

  (list
   'moveto 0
   (+ (* 0.1665 (/ hairpinHeight 0.666) ) 0.015)
   'rcurveto 0 0 0.3 -0.0 0.4 0.3
   'moveto 0
   (- (* -0.1665 (/ hairpinHeight 0.666) ) 0.015)
   'rcurveto 0 0 0.3 0.0 0.4 -0.3
   )
```

```
                0.1
                2
                2
                #f
                )
               -0.075
               ))
          )
         ; when crescendo begins but breaks off to
         ; the subsequent system(s)
         (crescStencilBrokenBeginning
          ; if the hairpin is set only to barline...
          (if toBarline?
              ; ... AND if the hairpin finishes at the beginning
              ; of the next line
              ( if  (= (length siblings) 1)
                    (ly:stencil-combine-at-edge
                     (ly:stencil-combine-at-edge
                      (hairpin::print-part
                       (scale-coords
                        coords
                        (+ lenx 0.95) (/ leny 2)) decresc? grob)
                      1
                      -1
                      (if mirrored?
                          (hairpin::print-part
                           (scale-coords
                            coords
                            (+ lenx 0.95) (/ leny -2))  decresc? grob)
                          empty-stencil)
                      -0.1 ;vertical padding value here
                      )
                     0
                     1
                     (make-path-stencil
                      (list
                       'moveto 0
                       (+ (* 0.1665 (/ hairpinHeight 0.666) ) 0.015)
                       'rcurveto 0 0 0.3 -0.0 0.4 0.3
                       'moveto 0
                       (- (* -0.1665 (/ hairpinHeight 0.666) ) 0.015)
                       'rcurveto 0 0 0.3 0.0 0.4 -0.3
                       )
                      0.1
                      2
                      2
                      #f
                      )
```

```
                        -0.075
                        )
                       (ly:stencil-combine-at-edge
                        (hairpin::print-part
                         (scale-coords
                          coords
                          (+ lenx 1) (/ leny 1.5)) decresc? grob)
                        1
                        -1
                        (if mirrored?
                            (hairpin::print-part
                             (scale-coords
                              coords

                              (+ lenx 1) (/ leny -1.5))  decresc? grob)
                            empty-stencil)
                        -0.1 ;vertical padding value here
                        )
                       )
                 (ly:stencil-combine-at-edge
                  (hairpin::print-part
                   (scale-coords
                    coords
                    (+ lenx 1) (/ leny 1.5)) decresc? grob)
                  1
                  -1
                  (if mirrored?
                      (hairpin::print-part
                       (scale-coords
                        coords

                        (+ lenx 1) (/ leny -1.5))  decresc? grob)
                      empty-stencil)
                  -0.1 ;vertical padding value here
                  )
                 )
             )
            (crescStencilBrokenContinuation
             (ly:stencil-combine-at-edge
              (hairpin::print-part
               (scale-coords
                coords
                (+ lenx 1) (/ leny 2)) decresc? grob)
              1
              -1
              (if mirrored?
                  (hairpin::print-part
                   (scale-coords
```

```
                    coords
                    (+ lenx 1) (/ leny -2))  decresc? grob)
                  empty-stencil)
              0.1 ;vertical padding value here
              )
             )
            (crescStencilBrokenEnding
             ;if the flare end is applied
             ;on the dynamic at the beginning
             ;of the system, meaning lenx is super low
             (cond ((< lenx 0.2)
                    (make-path-stencil
                     (list 'moveto 0.45 (* 0.2 (/ hairpinHeight 0.666) )
                           'lineto -0.5 0.1
                           'moveto 0.45 (* 0.2 (/ hairpinHeight 0.666) )
                           'rcurveto 0 0 0.3 0 0.4 0.3
                           'moveto 0.45 (* -0.2 (/ hairpinHeight 0.666) )
                           'lineto -0.5 -0.1
                           'moveto 0.45 (* -0.2 (/ hairpinHeight 0.666) )
                           'rcurveto 0 0 0.3 0 0.4 -0.3
                           )
                     0.1
                     2
                     2
                     #f
                     ))
                   ((and (>= lenx 0.2)(< lenx 2))
                    ;if the flare end is applied
                    ;on the note at the beginning of the system
                    ;without the dynamics (lenx is about 1 to 2)
                    (ly:stencil-combine-at-edge
                     (ly:stencil-combine-at-edge
                      (hairpin::print-part
                       (scale-coords
                        (list
                         (cons (- (car (car coords)) 0)
                               (cdr (car coords )))
                         (cons (- (car (cadr coords)) 0)
                               (cdr (cadr coords))))
                        (+ lenx 0.25) (/ leny 2.75)) decresc? grob)
                      1
                      -1
                      (if mirrored?
                          (hairpin::print-part
                           (scale-coords
                            (list
                             (cons (- (car (car coords)) 0)
                                   (cdr (car coords )))
```

```
                              (cons (- (car (cadr coords)) 0)
                                    (cdr (cadr coords))))
                             (+ lenx 0.25) (/ leny -2.75)) decresc? grob)
                           empty-stencil)
                       0.1 ;vertical padding value here
                       )
                      0
                      1
                      (make-path-stencil
                       (list 'moveto 0 (* 0.15 (/ hairpinHeight 0.666) )
                             'rcurveto 0 0 0.3 -0.0 0.4 0.3
                             'moveto 0
                             (- (* -0.15 (/ hairpinHeight 0.666) ) 0.1)
                             'rcurveto 0 0 0.3 0.0 0.4 -0.3
                             )
                       0.1
                       2
                       2
                       #f
                       )
                      -0.075
                      ))
                    ((>= lenx 2)
                     ;if the normal ending flare crescendo
                     ;is applied...

                     (ly:stencil-combine-at-edge
                      (ly:stencil-combine-at-edge
                       (hairpin::print-part
                        (scale-coords
                         coords
                         (+ lenx 0.75) (/ leny 3)) decresc? grob)
                       1
                       -1
                       (if mirrored?
                           (hairpin::print-part
                            (scale-coords
                             coords
                             (+ lenx 0.75) (/ leny -3))  decresc? grob)
                           empty-stencil)
                       0.1 ;vertical padding value here
                       )
                      0
                      1
                      (make-path-stencil
                       (list   'moveto 0
                               (+ (* 0.111 (/ hairpinHeight 0.666) ) 0.01 )
                               'rcurveto 0 0 0.3 -0.0 0.4 0.3
```

```
                            'moveto 0
                            (- (* -0.111 (/ hairpinHeight 0.666) ) 0.11)
                            'rcurveto 0 0 0.3 0.0 0.4 -0.3
                            )
                    0.1
                    2
                    2
                    #f
                    )
                   -0.075
                   ))
                 )
           )
          (decrescStencil
           (if toBarline?
               ;I added here to make it default to stretch
               ;the hairpin immediately to the dynamics
               (ly:stencil-combine-at-edge
                (ly:stencil-combine-at-edge
                 (hairpin::print-part
                  (scale-coords
                   coords
                   (+ lenx 0.25) (/ leny 2)) decresc? grob)
                 1
                 -1
                 (if mirrored?
                     (hairpin::print-part
                      (scale-coords
                       coords
                       (+ lenx 0.25) (/ leny -2)) decresc? grob)
                     empty-stencil)
                 -0.1 ;vertical padding value here
                 )
                0
                -1
                (make-path-stencil
                 (list 'moveto 0
                       (+ (* 0.1665 (/ hairpinHeight 0.666) ) 0.015)
                       'rcurveto 0 0 -0.3 -0.0 -0.4 0.3
                       'moveto 0
                       (- (* -0.1665 (/ hairpinHeight 0.666) ) 0.015)
                       'rcurveto 0 0 -0.3 0.0 -0.4 -0.3
                       )
                 0.1
                 2
                 2
                 #f
                 )
```

```
                  -0.075
                  )
                 ;if not to barline
                 (ly:stencil-combine-at-edge
                  (ly:stencil-combine-at-edge
                   (hairpin::print-part
                    (scale-coords
                     coords
                     (+ lenx 0.25) (/ leny 2))  decresc? grob)
                   1
                   -1
                   (if mirrored?
                       (hairpin::print-part
                        (scale-coords
                         coords
                         (+ lenx 0.25) (/ leny -2))  decresc? grob)
                       empty-stencil)
                   -0.1 ;vertical padding value here
                   )
                  0
                  -1
                  (make-path-stencil
                   (list
                    'moveto 0
                    (+ (* 0.1665 (/ hairpinHeight 0.666) ) 0.015)
                    'rcurveto 0 0 -0.3 -0.0 -0.4 0.3
                    'moveto 0
                    (- (* -0.1665 (/ hairpinHeight 0.666) ) 0.015)
                    'rcurveto 0 0 -0.3 0.0 -0.4 -0.3
                    )
                   0.1
                   2
                   2
                   #f
                   )
                  -0.075
                  ))
             )

            (decrescStencilBrokenBeginning
             (ly:stencil-combine-at-edge
              (ly:stencil-combine-at-edge
               (hairpin::print-part
                (scale-coords
                 coords
                 (+ lenx 0.2) (/ leny 2.5))
                decresc? grob)
               1
```

```
           -1
           (if mirrored?
               (hairpin::print-part
                (scale-coords
                 coords
                 (+ lenx 0.2) (/ leny -2.5))  decresc? grob)
               empty-stencil)
           0.1 ;vertical padding value here
           )
          0
          -1
          (make-path-stencil
           (list 'moveto 0
                 (+ (* 0.135 (/ hairpinHeight 0.666) ) 0.01 )
                 'rcurveto 0 0 -0.3 -0.0 -0.4 0.3
                 'moveto 0
                 (- (* -0.135 (/ hairpinHeight 0.666) ) 0.11)
                 'rcurveto 0 0 -0.3 0.0 -0.4 -0.3
                 )
           0.1
           2
           2
           #f
           )
          -0.075
          )
         )
        (decrescStencilBrokenEnding
         (ly:stencil-combine-at-edge
          (hairpin::print-part uplist decresc? grob)
          1
          -1
          (if mirrored?
              (hairpin::print-part downlist decresc? grob)
              empty-stencil)
          -0.1 ;vertical padding value here
          )
         )
        (cresc
         (cond (hairpinTypeA? crescStencil)
               (hairpinTypeB? crescStencilBrokenBeginning)
               (hairpinTypeC? crescStencilBrokenContinuation)
               (hairpinTypeD? crescStencilBrokenEnding)
               )
         )
        (decres
         (cond (hairpinTypeA? decrescStencil)
               (hairpinTypeB? decrescStencilBrokenBeginning)
```

```
        (hairpinTypeC? crescStencilBrokenContinuation)
        ;crescStencilBrokenContinuation can be used for
        ;decresc as well, because it gets inverted.
        (hairpinTypeD? decrescStencilBrokenEnding)
        )
  )
 (stil
  (ly:stencil-aligned-to
   (ly:stencil-translate
    (if decresc?
        decres
        cresc
        )
    (cons xtrans ytrans))
   Y CENTER))
 (stil-y-extent (ly:stencil-extent stil Y)))
; for debugging purposes
;         (display "Is this type A Hairpin? " )
;         (display hairpinTypeA?)
;         (newline)
;         (display "Is this type B Hairpin? " )
;         (display hairpinTypeB?)
;         (newline)
;         (display "Is this type C Hairpin? " )
;         (display hairpinTypeC?)
;         (newline)
;         (display "Is this type D Hairpin? " )
;         (display hairpinTypeD?)
;
;         (newline)
;         (display "Is it decrescendo? " )
;         (display decresc?)
;         (newline)
;         (display "end-broken-spanner???  " )
;         (display (end-broken-spanner? grob ))
;         (newline)
;         (display "(first-broken-spanner?) Is spanner broken
;and the first of its broken siblings? " )
;         (display (first-broken-spanner? grob ) )
;         (newline)
;         (display "middle-broken-spanner??? " )
;         (display (middle-broken-spanner? grob ) )
;         (newline)
;
;         (display "Is spanner broken and not the first of
;its broken siblings? " )
;         (display (not-first-broken-spanner? grob ) )
;         (newline)
```

```
        ;
        ;         (display "Is spanner broken and not the last of
        ;its broken siblings? " )
        ;         (display (not-last-broken-spanner? grob ) )
        ;         (newline)
        ;
        ;         (display "unbroken-or-first-broken-spanner??? " )
        ;         (display (unbroken-or-first-broken-spanner? grob ) )
        ;         (newline)
        ;
        ;         (display "unbroken-or-first-broken-spanner??? " )
        ;         (display (unbroken-or-last-broken-spanner? grob ) )
        ;         (newline)
        ;         (display "unbroken spanner?? " )
        ;         (display (unbroken-spanner? grob ) )
        ;         (newline)
        ;         (newline)
        ;         (display "distance of xex " )
        ;         (display lenx )
        ;         (newline)
        ;         (newline)
        (ly:make-stencil (ly:stencil-expr stil) xex stil-y-extent))
       ;; return empty, if no Hairpin.stencil present.
       '())))

#(define-public flared-hairpin-curvy
  (curvy-elbowed-hairpin '((0 . 0) (1 . 0.5))
                         #t))

#(define-public o
  (make-music 'AbsoluteDynamicEvent
              'text
              (markup
               #:pad-x -inf.0
               #:concat (
                         #:pad-to-box '(0 . 0.1) '(-0.75 . 1.75)
                         #:translate '(0 . 0.6)
                         #:musicglyph "scripts.flageolet"
                         ))
              ))
flaredCrescCurvy =
#(define-music-function (num dyn) ((number? 0) ly:event? )
  (let* ((hOffset num))
   #{
    \tweak Hairpin.stencil #flared-hairpin-curvy
    \tweak Hairpin.shorten-pair
    #(lambda (grob)
      (let* (
```

```
              ;; have we been split?
              (orig (ly:grob-original grob))

              ;; if yes, get the split pieces (our siblings)
              (siblings (if (ly:grob? orig)
                             (ly:spanner-broken-into orig)
                             '())))

        (cond
         ((= (length siblings) 0)
          (cons -0.6 hOffset)
          )
         ((and (>= (length siblings) 1)
               (eq? (car siblings) grob)
               )
          (cons -0.6 0))
         )



        )
       )
     \tweak Hairpin.after-line-breaking
     #(lambda (grob)
       (let* (
              ;; have we been split?
              (orig (ly:grob-original grob))

              ;; if yes, get the split pieces (our siblings)
              (siblings (if (ly:grob? orig)
                             (ly:spanner-broken-into orig)
                             '()))
              (toBarline? (ly:grob-property grob 'to-barline))
              )

        (cond  ((< (length siblings) 2)
                (ly:grob-set-property!
                 grob 'shorten-pair
                 (cons -0.6 (* hOffset -1)))

                )

               ((and (>= (length siblings) 2)
                     (eq? (car (last-pair siblings)) grob)
                     toBarline? )

                (ly:grob-set-property!
                 grob 'shorten-pair
```

```
                 (cons 0 (- -0.6 hOffset)))


               )
              ((and (>= (length siblings) 2)
                    (eq? (car (last-pair siblings)) grob)
                    (not toBarline?) )
               (ly:grob-set-property!
                grob 'shorten-pair
                 (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))

               )
              )

       )
      ) #dyn  #}))

flaredDecrescCurvy =
#(define-music-function (num dyn) ((number? 0) ly:event? )
  (let* ((hOffset num))
   #{
    \tweak Hairpin.stencil #flared-hairpin-curvy
    \tweak Hairpin.shorten-pair
    #(lambda (grob)
      (let* (
             ;; have we been split?
             (orig (ly:grob-original grob))

             ;; if yes, get the split pieces (our siblings)
             (siblings (if (ly:grob? orig)
                            (ly:spanner-broken-into orig)
                            '()))
             (toBarline? (ly:grob-property grob 'to-barline))

             )

       (cond
        ((and (zero? (length siblings)) (not toBarline?))
         (cons hOffset -0.6)
         )
        ((and (zero? (length siblings)) toBarline?)
         (cons hOffset -0.6)
         )
        ((and (>= (length siblings) 1)
              (eq? (car siblings) grob)
              )
         (cons -0.6 0))
        )
```

```
       )
      )
    \tweak Hairpin.after-line-breaking
    #(lambda (grob)
      (let* (
             ;; have we been split?
             (orig (ly:grob-original grob))

             ;; if yes, get the split pieces (our siblings)
             (siblings (if (ly:grob? orig)
                           (ly:spanner-broken-into orig)
                           '())))

       (cond
        ((and (>= (length siblings) 1)
              (eq? (car (last-pair siblings)) grob))
         (ly:grob-set-property!
          grob 'shorten-pair
          (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))
         )

        ((zero? (length siblings)  )
         #f
         )

        ((and (>= (length siblings) 2)
              (eq? (car (last-pair siblings)) grob)
              toBarline? )
         (ly:grob-set-property!
          grob 'shorten-pair
          (cons 0 (- -0.6 hOffset)))
         )

        ((and (>= (length siblings) 2)
              (eq? (car siblings) grob)
              )
         (ly:grob-set-property!
          grob 'shorten-pair
          (cons   0 0))
         )

        ((and (>= (length siblings) 2)
              (eq? (car (last-pair siblings)) grob)
              (not toBarline?) )
         (ly:grob-set-property!
          grob 'shorten-pair
          (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))
         )
```

```
        )


       )
      ) #dyn #}))

\layout {
 indent = #0
 line-width = #100
 ragged-last = ##t
}


flaredCrescCurvyOverride =
#(define-music-function (num mus) ((number? 0) ly:music? )
  (let* ((hOffset num))
   #{
    \once \override Hairpin.stencil = #flared-hairpin-curvy
    \once \override Hairpin.shorten-pair =
    #(lambda (grob)
      (let* (
             ;; have we been split?
             (orig (ly:grob-original grob))

             ;; if yes, get the split pieces (our siblings)
             (siblings (if (ly:grob? orig)
                           (ly:spanner-broken-into orig)
                           '())))
       (cond
        ((= (length siblings) 0)
         (cons -0.6 hOffset)
         )
        ((and (>= (length siblings) 1)
              (eq? (car siblings) grob)
              )
         (cons -0.6 0))
        )
       )
      )
    \once \override Hairpin.after-line-breaking =
    #(lambda (grob)
      (let* (
             ;; have we been split?
             (orig (ly:grob-original grob))

             ;; if yes, get the split pieces (our siblings)
             (siblings (if (ly:grob? orig)
                           (ly:spanner-broken-into orig)
```

```
                           '()))
             (toBarline? (ly:grob-property grob 'to-barline))
             )

       (cond  ((< (length siblings) 2)
               (ly:grob-set-property!
                grob 'shorten-pair
                (cons -0.6 (* hOffset -1)))

               )

              ((and (>= (length siblings) 2)
                    (eq? (car (last-pair siblings)) grob)
                    toBarline? )

               (ly:grob-set-property!
                grob 'shorten-pair
                (cons 0 (- -0.6 hOffset)))


               )
              ((and (>= (length siblings) 2)
                    (eq? (car (last-pair siblings)) grob)
                    (not toBarline?) )
               (ly:grob-set-property!
                grob 'shorten-pair
                (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))

               )
              )

       )
      ) #mus  #}))



flaredDecrescCurvyOverride =
#(define-music-function (num mus) ((number? 0) ly:music? )
  (let* ((hOffset num))
   #{
    \once \override Hairpin.stencil = #flared-hairpin-curvy
    \once \override Hairpin.shorten-pair =
    #(lambda (grob)
      (let* (
             ;; have we been split?
             (orig (ly:grob-original grob))

             ;; if yes, get the split pieces (our siblings)
```

```
              (siblings (if (ly:grob? orig)
                            (ly:spanner-broken-into orig)
                            '()))
              (toBarline? (ly:grob-property grob 'to-barline))

              )

        (cond
         ((and (zero? (length siblings)) (not toBarline?))
          (cons hOffset -0.6)
          )
         ((and (zero? (length siblings)) toBarline?)
          (cons hOffset -0.6)
          )
         ((and (>= (length siblings) 1)
               (eq? (car siblings) grob)
              )
          (cons -0.6 0))
         )
        )
      )
    \once \override Hairpin.after-line-breaking =
    #(lambda (grob)
      (let* (
              ;; have we been split?
              (orig (ly:grob-original grob))

              ;; if yes, get the split pieces (our siblings)
              (siblings (if (ly:grob? orig)
                            (ly:spanner-broken-into orig)
                            '())))

        (cond
         ((and (>= (length siblings) 1)
               (eq? (car (last-pair siblings)) grob))
          (ly:grob-set-property!
           grob 'shorten-pair
           (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))
          )

         ((zero? (length siblings)  )
          #f
          )

         ((and (>= (length siblings) 2)
               (eq? (car (last-pair siblings)) grob)
              toBarline? )
          (ly:grob-set-property!
```

```
          grob 'shorten-pair
          (cons 0 (- -0.6 hOffset)))
         )

        ((and (>= (length siblings) 2)
              (eq? (car siblings) grob)
              )
         (ly:grob-set-property!
          grob 'shorten-pair
          (cons   0 0))
         )

        ((and (>= (length siblings) 2)
              (eq? (car (last-pair siblings)) grob)
              (not toBarline?) )
         (ly:grob-set-property!
          grob 'shorten-pair
          (cons  (+ -0.6 hOffset) (- -0.6 hOffset)))
         )
        )


       )
      ) #mus #}))
\layout {
 ragged-last = ##f
}
{
 \override Hairpin.height = #0.5
 % \override Hairpin.stencil = #flared-hairpin-curvy
 \override Hairpin.to-barline = ##f
 \override Hairpin.after-line-breaking = ##f

 c' \fff \flaredCrescCurvy #0.25 \< c' c' c' \o \flaredDecrescCurvy #0 \>
 c' c'\o\flaredCrescCurvy #-0.6  \< c' c' \break
 \repeat unfold 3  {c' c' c' c'} \break
 c'\ff \flaredDecrescCurvy \> c' c' c'
 c' \p \flaredCrescCurvy \< c' c' \flaredDecrescCurvy \> c'
 c' \p \flaredCrescCurvy \< c'  c'\! \flaredDecrescCurvy \> c' \break
 \repeat unfold 3  {c' c' c' c'} \break
 c' \f  \flaredDecrescCurvy \> c' c' c'\o \flaredCrescCurvy #-1 \<
 c' c' c' c' \break c'\! s4 s4 s4  \bar "."
 \once \override Hairpin.extra-offset = #'(1.5 . 0)
 \flaredCrescCurvyOverride #1.5 \after 2. \f c'1 \<
 \flaredDecrescCurvyOverride #1 \after 2. \f c'1 \> \bar"."
}
```

4.5.4 Discussion

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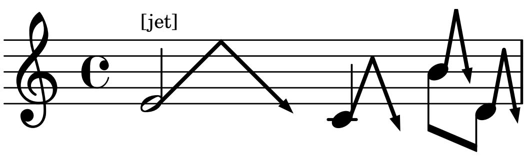
Chapter 5

Noteheads

5.1 Jet Whistle (for flute) (deprecated)



5.1.1 Description

This snippet has been deprecated. The snippet itself works for `2.26.0`; however, readers are advised to use Version 2 of the implementation (initially released Aug. 10, 2025), which will continue to be updated.

Implementation of the jet whistle, as described in *The Techniques of Flute Playing* by Carin Levine and Christina Mitropoulos-Bott.[1]

5.1.2 Grammar

`\jet NOTE #X-length`

5.1.3 Code

```
jet = #(define-music-function (pitchthing width) (ly:music? number?)
          (define p1 (ly:music-property pitchthing 'pitch))
          (define steps (+ -6 (ly:pitch-steps p1)))
          (define radToDeg (* 180 (/ 1 3.141592653589793)))
          #{ #pitchthing ^\markup {
            \postscript
            #(string-append "gsave newpath 0.2 setlinewidth 1.15 "
                             (number->string
```

1. Carin Levine and Christina Mitropoulos-Bott, *The techniques of flute playing = Die Spieltechnik der Flöte* (Kassel ; New York: Bärenreiter, 2003), 18.

```
                                   (+ -2.5 (* 0.5 steps))) " moveto "
                           (number->string
                                   (* 0.5 width)) " 4 rlineto "
                           (number->string
                                   (* 0.5 width)) " -4 rlineto
                  stroke
                  newpath
                  0.1 setlinewidth "
                           (number->string (+ 1.15 width)) " "
                           (number->string (+ -2.55 (* 0.5 steps)))
                           " moveto "
                           (number->string
                            (* radToDeg (atan (/ (* width 0.5) 4))))
                           " rotate
                  0 -1 rlineto
                  -0.35 1 rlineto
                  0.7 0 rlineto
                  -0.35 -1 rlineto
                  closepath
                  fill
                  grestore
                  ")
         } #})

\score {
  {
    \jet e'2^\markup {\fontsize #-5 {[jet]}} #8
    \jet c'4 #3
    \stemDown  \jet b'8 #1.5
    \jet d'8 #1.5
  }

  \layout {
    \context {
      \Score proportionalNotationDuration = #(ly:make-moment 1/10)
      \override SpacingSpanner.uniform-stretching = ##t
    }
  }
}
```

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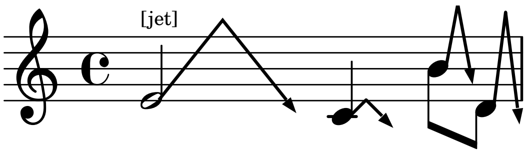
5.2 Jet Whistle (for flute) (Version 2)



5.2.1 Description

Implementation of the jet whistle, as described in *The Techniques of Flute Playing* by Carin Levine and Christina Mitropoulos-Bott.[2]

This is an improved version of the original version of the jet whistle implementation. This version adds the functionality to adjust the height, as well as the width of the jet whistle sign.

5.2.2 Grammar

`\jet NOTE #Y-length #X-length`

5.2.3 Code

```

\version "2.26.0"

jet =  #(define-music-function (pitchthing height width) (ly:music? number? number?)
         (define p1 (ly:music-property pitchthing 'pitch))
         (define steps (+ -6 (ly:pitch-steps p1)))
         (define radToDeg (* 180 (/ 1 3.141592653589793)))


         #{ #pitchthing ^\markup {
          \postscript #(string-append
                        "gsave
newpath
0.2 setlinewidth
1.15 " (number->string (+ -2.5 (* 0.5 steps))) " moveto "

                        (number->string (* 0.5 width)) " "
                        (number->string height) " rlineto "
                        (number->string (* 0.5 width)) " "
                        (number->string (* height -1))
                        " rlineto
stroke
newpath
0.1 setlinewidth
 "
                        (number->string (+ 1.15 width))
```

2. Levine and Mitropoulos-Bott, *The techniques of flute playing = Die Spieltechnik der Flöte*, 18.

```
                        " " (number->string (+ -2.55 (* 0.5 steps)))
                        " moveto "
                        (number->string
                         (* radToDeg (atan (/ (* width 0.5) height))))
                        " rotate
0 -1 rlineto
-0.35 1 rlineto
0.7 0 rlineto
-0.35 -1 rlineto
closepath
fill
grestore
")
         } #})


\score {
 {
  \jet e'2^\markup {\fontsize #-5 {[jet]}} #5 #8
  \jet c'4 #1 #2
  \stemDown
  \jet b'8  #4 #1.5
  \jet d'8  #6 #1.5
 }

 \layout {
  \context {
   \Score proportionalNotationDuration = #1/10
   \override SpacingSpanner.uniform-stretching = ##t
  }
 }
}

```

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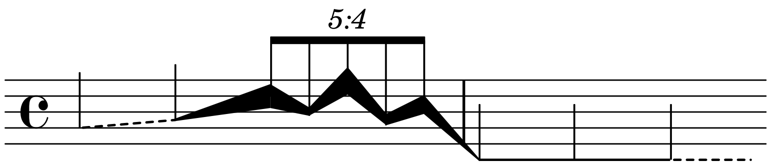
5.3 Line as a Notehead



5.3.1 Description

These functions replace an ordinary notehead with a dashed or a continuous line. For the continuous line, it is possible to adjust the beginning and ending thicknesses.

5.3.2 Grammar

`\dashedLineNotehead NOTE1 PITCH #x-dist`
`\modularLineNotehead NOTE1 PITCH #beginningThick #endingThick #x-dist`

NB

1. `NOTE1` specifies with which note the line starts. If necessary, the duration must be set, as well.
2. `PITCH` specifies with which pitch the line ends. Enter only the pitch; this information is used to determine the angle of the line, and it has no effect in displaying the rhythm.
3. `x-dist` specifies how long the line is.
4. `beginningThick` (for `modularLineNotehead` only) specifies how thick the beginning part of the line should be. `#15` gives a thin line, similar to the `\dashedLineNotehead` line. `#100` is as thick as a space between two neighboring lines of a staff.
5. `endingThick` (for `modularLineNotehead` only) specifies how thick the ending part of the line should be. `#15` gives a thin line, similar to the `\dashedLineNotehead` line. `#100` is as thick as a space between two neighboring lines of a staff.

5.3.3 Code

```
\version "2.26.0"

noteheadless = \once \override Voice.NoteHead.stencil = #point-stencil

dashedLineNotehead =
#(define-music-function
  (beginning end x-distance) (ly:music? ly:music? number?)
  (let*
   (
    (p1 (ly:music-property beginning 'pitch))
    (p2 (ly:music-property end 'pitch))
    (steps
     (-
      (+ (* 7 (ly:pitch-octave p2)) (ly:pitch-notename p2))
      (+ (* 7 (ly:pitch-octave p1)) (ly:pitch-notename p1))
```

```
      )
     )
    )
   #{
    {
     \once \override Voice.NoteHead.stencil = #ly:text-interface::print
     \once \override Voice.NoteHead.stem-attachment = #'(0 . 0)
     \once \override Staff.LedgerLineSpanner.stencil = ##f
     \once \override Voice.NoteHead.text = \markup         {
      % \translate #(cons 0 0)
      \postscript
      #(string-append
        "newpath 1 setlinecap
              0.15 setlinewidth
              0 0 moveto
              [.4 .4 .4 .4] 3 setdash "
        (number->string x-distance)  " " (number->string (* steps 0.5))
        " rlineto stroke"
        )
     }
     #beginning
     \revert Voice.NoteHead.stencil
     \revert Staff.LedgerLineSpanner.stencil
    }
   #})
  )

modularLineNotehead =
#(define-music-function
  (beginning end beginningThickness endingThickness x-distance)
  (ly:music? ly:music? number? number? number?)
  (let*
   (
    (p1 (ly:music-property beginning 'pitch))
    (p2 (ly:music-property end 'pitch))
    (steps
     (-
      (+ (* 7 (ly:pitch-octave p2)) (ly:pitch-notename p2))
      (+ (* 7 (ly:pitch-octave p1)) (ly:pitch-notename p1))
      )
     )
    )
   #{
    {

     \once \override Voice.NoteHead.stencil = #ly:text-interface::print
     \once \override Voice.NoteHead.stem-attachment = #'(0 . 0)
     \once \override Voice.LedgerLineSpanner.transparent = ##t
```

```
      \once \override Voice.NoteHead.text = \markup         {
       \postscript
       #(string-append
         "newpath 1 setlinecap 0.1 setlinewidth -0.05 0 moveto 0 "
         (number->string (* beginningThickness 0.005)) " rlineto "
         (number->string x-distance) " "
         (number->string (+ (- (* endingThickness 0.005)
                               (* beginningThickness 0.005) )
                            (* steps 0.5)))
         " rlineto 0 "
         (number->string  (* endingThickness -0.01))  " rlineto "
         (number->string (* -1 x-distance))  " "
         (number->string (- (* endingThickness 0.005)
                            (* beginningThickness 0.005)
                            (* steps 0.5)))
         " rlineto
               closepath
               fill"
         )
      }
      #beginning
      \revert Voice.NoteHead.stencil
      \revert Staff.LedgerLineSpanner.stencil
     }
    #})
   )


\score {
 {
  \omit Staff.Clef
  \dashedLineNotehead g'4 a' #6
  \modularLineNotehead a' d'' #15 #150 #6
  \override TupletNumber.text = #tuplet-number::calc-fraction-text

  \stemUp  \tuplet 5/4 {
   \modularLineNotehead d''8 b' #150 #50 #2.5
   \modularLineNotehead b' f'' #50 #175 #2.5
   \modularLineNotehead f'' a' #175 #70 #2.5
   \modularLineNotehead a' c'' #70 #120 #2.5
   \modularLineNotehead c'' c' #120 #15 #3.5
  }
  |
  \modularLineNotehead c'4 c' #15 #15 #12
  \noteheadless c'
  \dashedLineNotehead c' c' #5
 }
```

```
  \layout {
   \context {
    \Score proportionalNotationDuration = #1/10
    \override SpacingSpanner.uniform-stretching = ##t
   }
  }
}

```

5.3.4 Discussion

See Prescriptive Notation for String Instruments for a possible use of this notehead.

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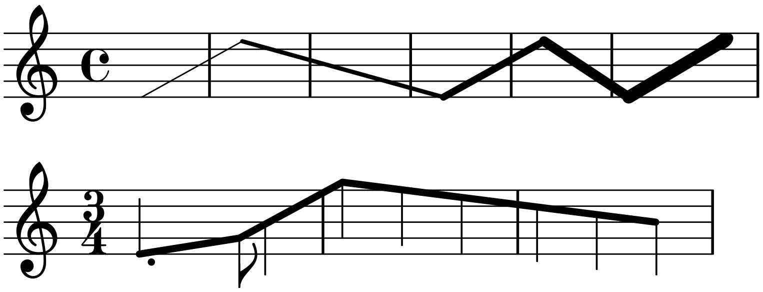
5.4 Line as a Notehead 2



5.4.1 Description

These functions replace ordinary noteheads with a dashed or a continuous line. However, unlike the **First Version**, these functions use `\glissando` as the basis for drawing the line.

5.4.2 Grammar

```
\lineNotehead #THICKNESS NOTE
\lineNoteheadOn #THICKNESS STARTING_NOTE NOTES...
\lineNoteheadOff ARRIVING_NOTE
\lineNoteheadWithRhythm #THICKNESS NOTE
\lineNoteheadWithRhythmOn #THICKNESS STARTING_NOTE NOTES...
\lineNoteheadWithRhythmOff ARRIVING_NOTE
```

NB

1. `\lineNotehead` only shows the line on the staff.
2. `\lineNoteheadWithRhythm` retains the rhythmic information.
3. `\lineNotehead` and `\lineNoteheadWithRhythm` applies the line from one note to another, without the line spanning multiple notes.
4. If the line must span over more than a note, use `\lineNoteheadOn` or `\lineNoteheadWithRhythmOn`.
5. In order to exit the line-as-a-notehead mode, use `\lineNoteheadOff` for both `\lineNotehead` and `\lineNoteheadWithRhythm`. In case the notehead must be disguised at the arrival, you may reduce the font size of the Notehead very drastically. See the Code for an example of this.
6. When using `\lineNoteheadWithRhythm` and `\lineNoteheadWithRhythmOn`, cautions must be paid to the placements of the augmentation dots and the intermediate stems. In the Code, I use:
   `\once \override Voice.Dots.extra-offset = #'(0 . -1)`
   And place this *before* the `\lineNoteheadWithRhythmOn`.

5.4.3 Code

```
\version "2.26.0"

```

```
% revised on January 25 2025

lineNotehead =
#(define-music-function (thickness note) (number? ly:music? )
  #{
   \once \override NoteHead.stencil = #ly:text-interface::print
   \once \override NoteHead.text = \markup{  \char ##x200A }
   \once \override Dots.stencil = ##f
   \once \override Glissando.breakable = ##t
   \once \override Glissando.after-line-breaking = ##t
   \once \override Glissando.thickness = #thickness
   \once \override Glissando.bound-details =
   #'(
      (left (padding . 0))
      (right (padding . 0))
      )
   #note
   \glissando

  #})

lineNoteheadOn =
#(define-music-function (thickness note) (number? ly:music?)
  #{
   \override Stem.stencil = ##f
   \override Flag.stencil = ##f
   \override TupletBracket.stencil = ##f
   \override TupletNumber.stencil = ##f
   \override Beam.stencil = ##f
   \override NoteHead.stencil = #ly:text-interface::print
   \override NoteHead.text = \markup{  \char ##x200A }
   \override Dots.stencil = ##f
   \override Glissando.breakable = ##t
   \override Glissando.after-line-breaking = ##t
   \override Glissando.thickness = #thickness
   \override Glissando.bound-details =
   #'(
      (left (padding . 0))
      (right (padding . 0))
      )
   #note
   \glissando
   \override NoteColumn.glissando-skip = ##t
  #})


lineNoteheadWithRhythm =
#(define-music-function (thickness note) (number? ly:music?)
```

```
  #{
   \once \override NoteHead.stencil = #ly:text-interface::print
   \once \override NoteHead.text = \markup{  \char ##x200A }
   \once \override Glissando.breakable = ##t
   \once \override Glissando.after-line-breaking = ##t
   \once \override Glissando.thickness = #thickness
   \once \override Glissando.bound-details =
   #'(
      (left (padding . 0))
      (right (padding . 0))
      )
   #note
   \glissando

  #})

lineNoteheadWithRhythmOn =
#(define-music-function (thickness note) (number? ly:music?)
  #{
   \override NoteHead.stencil = #ly:text-interface::print
   \override NoteHead.text = \markup{  \char ##x200A }
   \override Glissando.breakable = ##t
   \override Glissando.after-line-breaking = ##t
   \override Glissando.thickness = #thickness
   \override Glissando.bound-details =
   #'(
      (left (padding . 0))
      (right (padding . 0))
      )
   #note
   \glissando
   \override NoteColumn.glissando-skip = ##t
  #})


lineNoteheadOff =
{
 \revert Stem.stencil
 \revert Flag.stencil
 \revert Beam.stencil
 \revert NoteHead.stencil
 \revert Dots.stencil
 \revert Glissando.breakable
 \revert Glissando.after-line-breaking
 \revert Glissando.thickness
 \revert Glissando.bound-details
 \revert NoteColumn.glissando-skip
 \revert TupletBracket.stencil
```

```
 \revert TupletNumber.stencil
 \revert Beam.stencil
}


{

 \lineNotehead #1  e'1
 \lineNoteheadOn #3
 e''1
 b'1
 \lineNoteheadOff
 \lineNotehead #5
 e'1
 \lineNotehead #7
 e''1
 \lineNoteheadOn #9 e'4
 e''4.   e'8
 \lineNoteheadOff
 \omit Stem
 e''4

}

\score {
 {
  \time 3/4
  \once \override Voice.Dots.extra-offset = #'(0 . -1)
  \lineNoteheadWithRhythm #5  e'4.
  \stemDown
  \lineNoteheadWithRhythmOn #5

  g'8
  b'4
  \lineNoteheadOff
  \lineNoteheadWithRhythmOn #5
  g''4
  f''4
  e''4
  d''4
  c''4
  \lineNoteheadOff
  \once \override NoteHead.font-size = #-30
  b'4
 }
 \layout {
  \context{
   \Score   proportionalNotationDuration = #1/8
```

```
    }
  }
}
```

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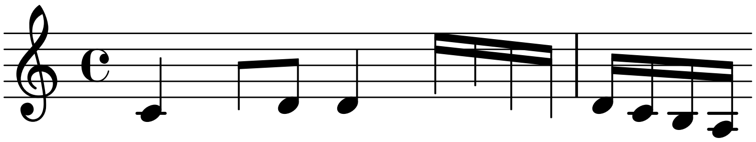
5.5 Noteheadless



5.5.1 Description

This snippet is hardly my own idea, as I largely quoted this technique from one of the snippets available on LilyPond Wiki, formerly LSR.[3] However, I list it here because:

1. it took a while for me to find the workaround for maintaining the musical spacing as a result of omitting noteheads. It is worth noting that because merely disabling `NoteHead.stencil` will render the spacing to be squished, the approach of specifying `##t` for `NoteHead.transparent` (which itself will *not* eliminate the ledger lines) then `##t` for `NoteHead.no-ledgers` is effective in maintaining the general spacing.

See Discussion.

5.5.2 Grammar

`\noteheadless NOTE`
`\noteheadlessOn NOTE`
`\noteheadlessOff`

NB

1. `\noteheadless` affects only one note immediately following.
2. For a group of notes, use `\noteheadlessOn` to toggle on the function. `\noteheadlessOff` will toggle off the function.

5.5.3 Code

```
\version "2.26.0"

%% Inspired by:
%% http://lsr.di.unimi.it/LSR/Item?id=796


noteheadless = {
 \once \override Voice.NoteHead.transparent = ##t
 \once \override Voice.NoteHead.no-ledgers = ##t
}

noteheadlessOn = {
 \override Voice.NoteHead.transparent = ##t
 \override Voice.NoteHead.no-ledgers = ##t
}
```

3. See: https://wiki.lilypond.community/wiki/Creating_harp_glissandi

```
noteheadlessOff = {
 \revert Voice.NoteHead.transparent
 \revert Voice.NoteHead.no-ledgers
}


{
 c'4 \noteheadless c'8 d' d'4
 \noteheadlessOn e'16 f' c' b |
 \noteheadlessOff d' c' b a
}

```

5.5.4 Discussion

As discussed in the Learning Manual, overriding `NoteHead.stencil` to `##f` will cause an error when the processing requires the dimensions of the NoteHead. I now recommend:

```
\once \override NoteHead.stencil = #point-stencil
```

There is also another special stencil, namely `#empty-stencil`, but this appears to cause the processing to assume that there is no pitch assigned to the particular note. In the aforementioned snippet found on LilyPond Wiki, it causes the glissando line to only reach and emerge out of the middle of the five-line staff.

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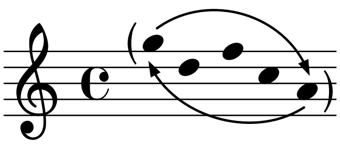
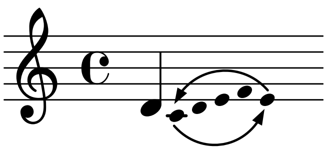
5.6 Parenthesized Group of Noteheads with Arrowed Arcs





5.6.1 Description

Implementation of parentheses enclosing a group of noteheads, accompanied by arrowed arcs. This function takes advantage of manipulating the `\parenthesize` function, and can be applied to both regular and grace notes.

There are two options that can be set:

1. The arrowed arcs can be adjusted à la `extra-offset` (i.e., inputting the values using a pair of values)
2. You can choose the parentheses not to be printed, so that only arrowed arcs are displayed.

A group of small noteheads, accompanied by arrowed arcs, suggests that such a group of notes are to be repeatedly played as fast as possible. This notation can be found in pieces such as *Sequenza II* for harp by Luciano Berio.

In addition to the examples seen above, see **Repeat Note Patterns As Fast As Possible** in Chapter 12.

5.6.2 Grammar

```
\parenthesisBeginWithArrow WIDTH HEIGHT ARCANGLE OFFSET PARENTHESIS? NOTE
\parenthesisEndWithArrow WIDTH HEIGHT ARCANGLE OFFSET PARENTHESIS? NOTE
```

NB

- `WIDTH` specifies the width of the arrowed arc.
- `HEIGHT` specifies the height of the arrowed arc, i.e. how tall the arc will be in the middle of the line.
- `ARCANGLE` specifies how much the arrowed arc should be rotated. A positive value will rotate the arc counterclockwise, while a negative value will rotate it clockwise.
- **OPTIONAL:** `OFFSET` can be used to offset the arrowed arc. If you opt to specify this argument, enter a value in a pair of numbers, e.g. `#'(3 . -2)`.
- **OPTIONAL:** `PARENTHESIS?` can be specified if you opt to make the parenthesis disappear. `##f` will cause only the arrowed arc to be shown.

- Importantly, when there are other objects (markups and spanners shown alongside the arrowed arcs) being in place alongside this object, it may cause the other objects to appear farther from the staff line. Setting `outside-staff-property` to `##f` for these other objects, either via `\tweak` or `\override` should help avoid the problem, but if the problem persists, moving them using `extra-offset` is always an option.

5.6.3 Code

```
\version "2.26.0"

parenthesisBeginWithArrow =
#(define-music-function
  (width height arcAngle offset parenthesis? pitchthing)
  (number? number? number? (pair? (cons 0 0)) (boolean? #t) ly:music?)
  (define p1 (ly:music-property pitchthing 'pitch))
  (define steps (+ -6 (ly:pitch-steps p1)))
  (define radToDeg (* 180 (/ 1 3.141592653589793)))
  #{
   \once \override Parentheses.font-size = #2
   \once \override Parentheses.stencils =
   #(lambda (grob)
     (let ((par-list
            (parentheses-interface::calc-parenthesis-stencils grob)))
      (if parenthesis?
          (list (car par-list) point-stencil)
          (list point-stencil point-stencil)
          )
      )
     )
   \parenthesize
   #pitchthing
   -\tweak ignore-collision ##t
   -\tweak TextScript.outside-staff-priority ##f
   -\tweak TextScript.parent-alignment-X #1
   -\tweak TextScript.extra-offset
   #(cons  (+ -0.5 (car offset)) (+ -1.25 (* steps 0.5) (cdr offset)))
   -\tweak TextScript.rotation #`(,arcAngle -1 -1)
   ^\markup {
    \rotate #3.5
    \concat {
     \path #0.15
     #`((curveto ,(* width 0.25) ,height ,(* width 0.75)
                 ,height ,width 0))
     \hspace #-0.1 \vcenter \postscript
     #(string-append
       "
/arrowhead {
gsave
```

```
"  (number->string
       (+ (* radToDeg (atan (/ (* height -1)  (* width 0.25)))) -90))
       " rotate
0 0.5 rlineto
-0.35 -1 rlineto
0.7 0 rlineto
-0.35 1 rlineto
 closepath
fill
grestore } def

0 0 moveto
arrowhead
")
    }
  } #})



parenthesisEndWithArrow =
#(define-music-function
  (width height arcAngle offset parenthesis? pitchthing)
  (number? number? number? (pair? (cons 0 0)) (boolean? #t) ly:music?)
  (define p1 (ly:music-property pitchthing 'pitch))
  (define steps (+ -6 (ly:pitch-steps p1)))
  (define radToDeg (* 180 (/ 1 3.141592653589793)))
  #{
   \once  \override Parentheses.font-size = #2
   \once \override Parentheses.stencils =
   #(lambda (grob)
     (let ((par-list
            (parentheses-interface::calc-parenthesis-stencils grob)))
      (if parenthesis?
          (list  point-stencil (cadr par-list))
          (list point-stencil point-stencil)
          )
      )
     )
   \parenthesize
   #pitchthing
   -\tweak ignore-collision ##t
   -\tweak TextScript.outside-staff-priority ##f
   -\tweak TextScript.parent-alignment-X #-1
   -\tweak TextScript.extra-offset
   #(cons  (+ 0.75 (car offset)) (+ 1.75 (* steps 0.5) (cdr offset)))
   -\tweak TextScript.rotation #`(,arcAngle -1 0)
   _\markup \right-align {
    \rotate #3.5
```

```
    \concat  {

     \postscript
     #(string-append
       "
/arrowhead {
gsave
"  (number->string
       (+ (* radToDeg (atan (/ (* height -1)  (* width 0.25)))) 270))
       " rotate
" (if (< height 0) (number->string 0.1) (number->string -0.1)) "
 -0.3 rmoveto
-0.35 1 rlineto
0.7 0 rlineto
-0.35 -1 rlineto
closepath
fill
grestore } def

0 0 moveto
arrowhead
")
     \center-align
     \path #0.15
     #`((curveto ,(* width 0.25) ,(* height -1) ,(* width 0.75)
                 ,(* height -1) ,width 0))

    }
  } #})


{
 \omit Stem
 \omit Beam
 \omit Flag

 \once \override Hairpin.outside-staff-priority = ##f
 \parenthesisBeginWithArrow #9.75 #3 #-18.5 #'(0 . -2.75)
 g''16  d''  f'' c''
 \parenthesisEndWithArrow #10 #2.75 #-18.5 #'(0 . 2.75) a'

}


{
 \afterGrace 1/16 d'4 {
  \omit Stem
  \omit Beam
```

```
    \parenthesisBeginWithArrow #5.5 #-2 #3 #'(0 . -3) ##f c'16
    d' e' f'
    \parenthesisEndWithArrow #5.75 #-2 #2  #'(-0.2 . 2.5) ##f e'
  }
}
```

5.6.4 Discussion

See **Repeat Note Patterns As Fast As Possible** in Chapter 12 for an applied use of this function.

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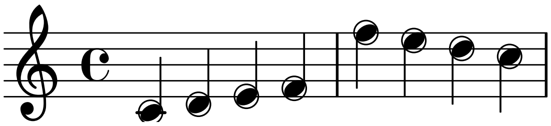
5.7 Slap Tongue, Type A



5.7.1 Description

In my music, I use encircled noteheads to denote slap tongues. Type A, encircled filled notehead, is used for a slap tongue with a regular note immediately following.

5.7.2 Grammar

`\slapA NOTE`

NB It only affects one note, owing to the `\once \override` functions within the code.

5.7.3 Code

```
\version "2.26.0"

slapA = #(define-music-function (note)   (ly:music?)
          #{ \once \override Voice.NoteHead.stencil =
             #ly:text-interface::print
             \once \override Voice.NoteHead.text =
             \markup {
              \concat {
               \musicglyph "noteheads.s2"
               \postscript "newpath
-0.675 0.025 0.75 0 360 arc
closepath
stroke"
              }
             }
             $note #})


{
 \slapA c'4 \slapA d' \slapA e' \slapA f'
 \slapA f'' \slapA e'' \slapA d'' \slapA c''
}
```

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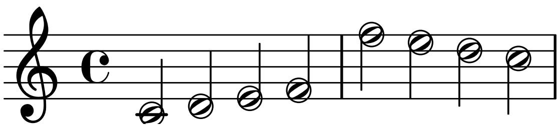
5.8 Slap Tongue, Type B



5.8.1 Description

In my music, I use encircled noteheads to denote slap tongues. Type B, encircled hollow notehead, is used for a slap tongue with an air sound immediately following.

5.8.2 Grammar

`\SlapB NOTE`

NB It only affects one note, owing to the `\once \override` functions within the code.

5.8.3 Code

```
\version "2.26.0"

slapB = #(define-music-function (note)   (ly:music?)
          #{ \once \override Voice.NoteHead.stencil =
             #ly:text-interface::print
             \once \override Voice.NoteHead.text =
             \markup {
              \concat {
               \musicglyph "noteheads.s1"
               \postscript "newpath
-0.675 0.025 0.75 0 360 arc
closepath
stroke"
              }
             }
             $note #})

{
 \slapB c'4 \slapB d' \slapB e' \slapB f'
 \slapB f'' \slapB e'' \slapB d'' \slapB c''
}

```

5.8.4 Discussion

As the musical example shows, when the Type B Slap Tongue notehead is applied to a quarter note, it could invite confusion in terms of rhythm. As a slap tongue itself is a short sound, I only use the slap tongue noteheads on eighth notes or shorter note durations.

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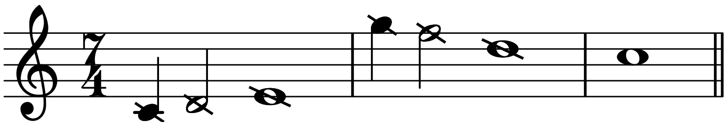
5.9 Slashed Notehead



5.9.1 Description

Noteheads with backslashes applied.[4] I use this notehead to indicate, for example, notes on the piano whose strings are prepared, thus producing pitch/sound different from what is expected normally.

5.9.2 Grammar

```
\slashNote NOTE
\slashNoteOn NOTE
\slashNoteOff
```

NB `\slashNote` only affects one note, owing to the `\once \override` functions within the code. For a group of notes to have slashes applied, use `\slashNoteOn`. `\slashNoteOff` cancels the application.

5.9.3 Code

```
\version "2.26.0"

% Inspired by the code provided by Jean Abou Samra
% https://lists.gnu.org/archive/html/lilypond-user/2022-11/msg00333.html

slashNote =
\once \override Voice.NoteHead.stencil =
#(grob-transformer
  'stencil
  (lambda (grob original)
   (let* ((added-markup
           #{
            \markup \general-align #Y #CENTER
            #(case (ly:grob-property grob 'duration-log)
              ((0) #{ \markup \concat {
               \musicglyph "noteheads.s0"
               \postscript
               "gsave
                    0.17 setlinewidth
                    -2.3 0.6 moveto
                    0.3 -0.6 lineto
                    stroke
                    grestore"
```

4. The code provided by Jean Abou Samra in the following discussion thread on lilypond-user was very helpful in creating this code: https://lists.gnu.org/archive/html/lilypond-user/2022-11/msg00333.html

```
                   } #})

              ((1) #{ \markup \concat {
               \musicglyph "noteheads.s1"
               \postscript
               "gsave
                    0.17 setlinewidth
                    -1.5 0.6 moveto
                    0.3 -0.6 lineto
                    stroke
                    grestore"
                   } #})

              ((2) #{ \markup \concat {
               \musicglyph "noteheads.s2"
               \postscript
               "gsave
                    0.17 setlinewidth
                    -1.5 0.6 moveto
                    0.3 -0.6 lineto
                    stroke
                    grestore"
                   } #}))
          #})
         (added-stencil (grob-interpret-markup grob added-markup)))
    (if (ly:stencil? original)
        (ly:stencil-add original added-stencil)
        added-stencil))))



slashNoteOn =
\override Voice.NoteHead.stencil =
#(grob-transformer
  'stencil
  (lambda (grob original)
   (let* ((added-markup
          #{
           \markup \general-align #Y #CENTER
           #(case (ly:grob-property grob 'duration-log)
             ((0) #{ \markup \concat {
               \musicglyph "noteheads.s0"
               \postscript
               "gsave
                    0.17 setlinewidth
                    -2.3 0.6 moveto
                    0.3 -0.6 lineto
                    stroke
```

```
                        grestore"
                       } #})
                  ((1) #{ \markup \concat {
                   \musicglyph "noteheads.s1"
                   \postscript
                   "gsave
                        0.17 setlinewidth
                        -1.5 0.6 moveto
                        0.3 -0.6 lineto
                        stroke
                        grestore"
                       } #})
                  ((2) #{ \markup \concat {
                   \musicglyph "noteheads.s2"
                   \postscript
                   "gsave
                        0.17 setlinewidth
                        -1.5 0.6 moveto
                        0.3 -0.6 lineto
                        stroke
                        grestore"
                       } #}))
              #})
             (added-stencil (grob-interpret-markup grob added-markup)))
      (if (ly:stencil? original)
          (ly:stencil-add original added-stencil)
          added-stencil))))


slashNoteOff = \revert Voice.NoteHead.stencil


{
 \time 7/4
 \slashNote c'4
 \slashNote d'2
 \slashNote e'1
 \slashNoteOn g''4 f''2 d''1
 \slashNoteOff c''1 \bar "||"
}

```

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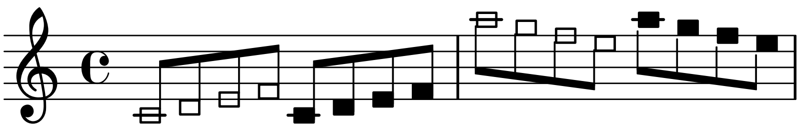
5.10 Square Notehead



5.10.1 Description

Filled and hollow square noteheads.

5.10.2 Grammar

```
\squareHollowNotehead NOTE
\squareHollowNoteheadOn NOTES
\squareHollowNoteheadOff
\squareFilledNotehead NOTE
\squareFilledNoteheadOn NOTES
\squareFilledNoteheadOff

\slashNoteOn NOTE
\slashNoteOff
```

NB `\squareHollowNotehead` and `\squareFilledNotehead` only affect one note, owing to the `\once \override` functions within the code. For a group of notes, use `\squareHollowNoteheadOn` or `\squareFilledNoteheadOn`. `\squareHollowNoteheadOff` and `\squareFilledNoteheadOff` cancel the application.

5.10.3 Code

```
\version "2.26.0"

% See also: https://lsr.di.unimi.it/LSR/Item?id=516
% UPDATED June 13 2025

squareHollowNoteheadDesign =
#(ly:make-stencil '(path 0.15 (moveto  0.0 0.425
                                       rlineto 1.2 0
                                       rlineto 0 -0.875
                                       rlineto -1.2 0
                                       closepath)
                          )
                  (cons -0.15 1.275)
                  (cons -1 1))

squareHollowNotehead =
#(define-music-function (note) (ly:music?)
  #{\once \override Voice.NoteHead.stencil =
    \squareHollowNoteheadDesign $note #})
```

```

squareHollowNoteheadOn =
#(define-music-function (note) (ly:music?)
  #{\override Voice.NoteHead.stencil =
    \squareHollowNoteheadDesign $note #})

squareHollowNoteheadOff = \revert Voice.NoteHead.stencil

squareFilledNoteheadDesign =
#(ly:make-stencil '(path 0.15  (moveto  0.0 0.425
                                        rlineto 1.2 0
                                        rlineto 0 -0.875
                                        rlineto -1.2 0
                                        closepath)

                         round
                         round
                         #t)
                  (cons -0.15 1.275)
                  (cons -1 0))

squareFilledNotehead =
#(define-music-function (note) (ly:music?)
  #{\once \override Voice.NoteHead.stencil =
    \squareFilledNoteheadDesign $note #})
squareFilledNoteheadOn =
#(define-music-function (note) (ly:music?)
  #{\override Voice.NoteHead.stencil =
    \squareFilledNoteheadDesign $note #})

squareFilledNoteheadOff = \revert Voice.NoteHead.stencil

{
 \squareHollowNotehead c'8
 \squareHollowNoteheadOn d' e' f'
 \squareHollowNoteheadOff
 \squareFilledNotehead c'8
 \squareFilledNoteheadOn d' e' f'
 \squareFilledNoteheadOff
 \squareHollowNotehead a''8
 \squareHollowNoteheadOn g'' f'' e''
 \squareHollowNoteheadOff
 \squareFilledNotehead a''8
 \squareFilledNoteheadOn g'' f'' e''
 \squareFilledNoteheadOff
}
```

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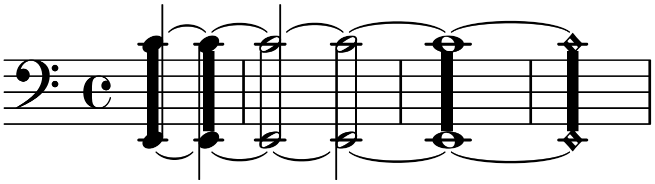
5.11 Tone Cluster



5.11.1 Description

Inspired by the tone cluster notation of Henry Cowell and others. See **Discussion**.

5.11.2 Grammar

```
\toneClusterBar NOTE1 NOTE2 yOffset yLengthAdjust
\toneClusterBarHollow NOTE1 NOTE2 yOffset yLengthAdjust
\toneClusterBarWhole NOTE1 NOTE2 yOffset yLengthAdjust
```

NB

1. The order of pitch boundaries as shown by `NOTE1` and `NOTE2` does not matter; `NOTE1` can be upper or lower pitch boundary, and vice versa for `NOTE2`. See **Code**.
2. `yOffset` indicates where the upper part of the cluster sign begins. When set to `#0`, it starts right at the top line of the ordinary 5-line staff. Each positive/negative integer will bring the beginning point up/down by a space of two neighboring lines of the staff.
3. `yLengthAdjust` indicates any value by which the cluster bar may be extended or reduced. When set to `#0`, the cluster bar will be as long as the distance between the lower boundary of the upper notehead and upper boundary of the lower notehead. Each positive/negative integer will add/reduce the length of the bar by a space of two neighboring lines of the staff.

   For this reason, when the tone cluster sign is applied to a quarter-note dyad, you may wish to set the upper part of the cluster bar right in the middle of the notehead. In the snippet shown, the first cluster's `yOffset` is set to `#1`. `yLengthAdjust` is also set to `#1`, meaning that the cluster bar will go down to the center of the lower notehead. The second cluster intentionally shows what happens when the bar only touches the two boundaries of the noteheads.
4. `\toneClusterBarHollow` shows the notation (quite à la Cowell) specifically for hollowed noteheads. Some people may prefer this notation, instead.
5. `\toneClusterBarWhole` is specifically for the tone cluster notation as applied to a whole-note dyad, owing to width being wider than the quarter or half noteheads.
6. These functions may be used in tandem with other noteheads, as well as ties. See **Code**.

5.11.3 Code

```
\version "2.26.0"

toneClusterBar =
#(define-music-function (note1 note2 yOffset yLengthAdjust)
  (ly:music? ly:music? number? number?)
```

```
  (let* (
         (note1p (ly:music-property note1 'pitch))
         (note2p (ly:music-property note2 'pitch))
         (note1pnumber (+ (* 7 (ly:pitch-octave note1p))
                          (ly:pitch-notename note1p)))
         (note2pnumber (+ (* 7 (ly:pitch-octave note2p))
                          (ly:pitch-notename note2p)))
         (pitchDistance (abs (- note1pnumber note2pnumber)))
         )
   #{
    < #note1
    #note2 > ^\markup {
     \postscript
     #(string-append
       "gsave
            newpath
            0.3 " (number->string (- yOffset 0.5)) " moveto
            0.7 0 rlineto
            0 " (number->string (- (* -0.5 pitchDistance)
                               (- yLengthAdjust 1))) " rlineto
            -0.7 0 rlineto
            closepath
            fill
            grestore")
    }
   #}
   )
  )


toneClusterBarHollow =
#(define-music-function (note1 note2 yOffset yLengthAdjust)
  (ly:music? ly:music? number? number?)
  (let* (
         (note1p (ly:music-property note1 'pitch))
         (note2p (ly:music-property note2 'pitch))
         (note1pnumber (+ (* 7 (ly:pitch-octave note1p))
                          (ly:pitch-notename note1p)))
         (note2pnumber (+ (* 7 (ly:pitch-octave note2p))
                          (ly:pitch-notename note2p)))
         (pitchDistance (abs (- note1pnumber note2pnumber)))
         )
   #{
    < #note1
    #note2 > ^\markup {
     \postscript
     #(string-append
       "gsave
```

```
            newpath
            0.1 " (number->string (- yOffset 0.5)) " moveto
            0 " (number->string (- (* -0.5 pitchDistance)
                                  (+ 0.5 yLengthAdjust))) " rlineto
            0.125 setlinewidth
            1.3 "(number->string (+ 0.75 (- yOffset 0.5))) " moveto
            0 " (number->string (- (* -0.5 pitchDistance)
       (+ 0.75 yLengthAdjust))) " rlineto
            stroke
            grestore")
    }
   #}
   )
  )


toneClusterBarWhole =
#(define-music-function (note1 note2 yOffset yLengthAdjust)
  (ly:music? ly:music? number? number?)
  (let* (
         (note1p (ly:music-property note1 'pitch))
         (note2p (ly:music-property note2 'pitch))
         (note1pnumber (+ (* 7 (ly:pitch-octave note1p))
                             (ly:pitch-notename note1p)))
         (note2pnumber (+ (* 7 (ly:pitch-octave note2p))
                             (ly:pitch-notename note2p)))
         (pitchDistance (abs (- note1pnumber note2pnumber)))
         )
   #{
    < #note1
    #note2 > ^\markup {
     \postscript
     #(string-append
       "gsave
            newpath
            0.125 setlinewidth
            0.55 " (number->string (- yOffset 0.5)) " moveto
            0 " (number->string (- (* -0.5 pitchDistance)
                                  (- yLengthAdjust 1))) " rlineto
            0.75 0 rlineto
            0 " (number->string (abs (- (* -0.5 pitchDistance)
            (- yLengthAdjust 1)))) " rlineto
            closepath fill
            grestore")
    }
   #}
   )
  )
```

```
{
 \time 4/4
 \partial 2
 \clef "F"
 \stemUp \toneClusterBar c'4~ e,~ #1 #1
 \stemDown \toneClusterBar e,~ c'4~ #0.5 #0
 \stemUp \toneClusterBarHollow c'2~ e,~ #0.5 #-0.5
 \stemDown \toneClusterBarHollow c'2~ e,~ #0.5 #-0.5
 \toneClusterBarWhole c'1~ e,~ #0.5 #0
 \toneClusterBar c'1~\harmonic e,~\harmonic #0.5 #0
}
```

5.11.4 Discussion

There have been some discussions on `lilypond-user` mailing list in the past that readers may consult for further ideas on implementing different types of tone cluster notation:

- https://lists.gnu.org/archive/html/lilypond-user/2008-10/msg00484.html (This one in particular lists other notational conventions established by other composers)
- https://lists.gnu.org/archive/html/lilypond-user/2020-12/msg00130.html

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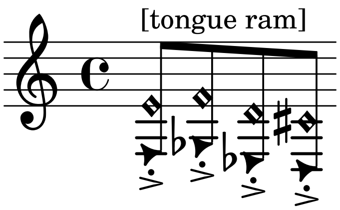
5.12 Tongue Ram (for flute)



5.12.1 Description

Implementation of the tongue ram notation, as described in *The Techniques of Flute Playing* by Carin Levine and Christina Mitropoulos-Bott.[5] See also: FluteXpansions.

Please note that the scheme function I had in previous versions of this document is no longer functional and deprecated. See **Discussion** for more details.

5.12.2 Grammar

```
<HIGH_PITCH \harmonic
\tweak NoteHead.stencil #ly:text-interface::print
\tweak NoteHead.text
\markup \musicglyph "noteheads.s2triangle" LOW_PITCH>
```

5.12.3 Code

```
\version "2.26.0"


{
 <e' \harmonic
 \tweak NoteHead.stencil #ly:text-interface::print
 \tweak NoteHead.text
 \markup \musicglyph "noteheads.s2triangle" f>8-.-> ^\markup { \tiny "[tongue ram]" }
 <f' \harmonic
 \tweak NoteHead.stencil #ly:text-interface::print
 \tweak NoteHead.text
 \markup \musicglyph "noteheads.s2triangle" ges>8-.->
 <d' \harmonic
 \tweak NoteHead.stencil #ly:text-interface::print
 \tweak NoteHead.text
 \markup \musicglyph "noteheads.s2triangle" ees>8-.->
 <cis' \harmonic
 \tweak NoteHead.stencil #ly:text-interface::print
 \tweak NoteHead.text
 \markup \musicglyph "noteheads.s2triangle" d>8-.->
}

```

5. Levine and Mitropoulos-Bott, *The techniques of flute playing = Die Spieltechnik der Flöte*, 28.

```
% The following is deprecated and is no longer functional. I hope to
% revisit this and improve upon it.
% tgrWithIndication =
% #(define-music-function (note1) (ly:music?)
%   (let*         (
%           (p1 #{ #(ly:music-deep-copy note1) \harmonic #})
%           (p2 #{ \transpose c df, #(ly:music-property note1 'pitch)#})
%           (d1  (ly:music-property note1 'duration))
%           )
%    #{ < $p1
%       \single \override NoteHead.stencil = #ly:text-interface::print
%       \single \override NoteHead.text =
%       \markup \musicglyph "noteheads.s2triangle"
%       \single \override Stem.stencil =
%       $p2 > $d1 ^\markup {\override #'(font-size . -2) {[T.R.]} }
%    #}
%    ))
%
% tgr =
% #(define-music-function (note1) (ly:music?)
%   (let*         (
%           (p1 #{ #(ly:music-deep-copy note1) \harmonic #})
%           (p2 #{ \transpose c df, #(ly:music-property note1 'pitch)#})
%           (d1  (ly:music-property note1 'duration))
%           )
%    #{ < $p1
%       \single \override NoteHead.stencil = #ly:text-interface::print
%       \single \override NoteHead.text =
%       \markup \musicglyph "noteheads.s2triangle"
%       \single \override Stem.stencil =
%       $p2 > $d1  #}
%    ))
%
% {
%  \language "english" \tgrWithIndication d'4-.->
%  \tgr cs'4-.-> \tgr ef'4-.->
% }
%
```

5.12.4 Discussion

I leave the deprecated code here, in hopes that I will revisit and improve upon it. For the time being, however, I have used the following in my recent pieces calling for this technique, and it worked well. I hope to eventually automate it.

```
<e' \harmonic
   \tweak NoteHead.stencil #ly:text-interface::print
   \tweak NoteHead.text
   \markup \musicglyph "noteheads.s2triangle" f>8
```

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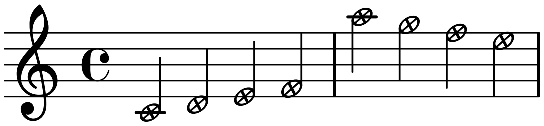
5.13 X In A Hollow Notehead



5.13.1 Description

While LilyPond Notation Reference provides an example of an X-in-a-circle notehead, its shape differs from the regular notehead.[6] This implementation simulates a hollow notehead with which the X notehead is combined.

5.13.2 Grammar

```
\cirX NOTE
```

5.13.3 Code

```
\version "2.26.0"

% Stem attachment function inspired by:
% https://lsr.di.unimi.it/LSR/Snippet?id=518

cirX = #(define-music-function (note) (ly:music?)
          #{
            \temporary \override NoteHead.stencil =
            #ly:text-interface::print
            \temporary \override NoteHead.text =
            \markup
            \translate #'(0.6 . 0)
            \pad-x #-0.22
            \rotate #35
            \scale #'(1 . 0.65)
            \combine \combine \combine \combine
            \override #'(thickness . 2)
            \draw-line #'(0.05 . 0.6)
            \override #'(thickness . 2)
            \draw-line #'(-0.05 . -0.6)
            \override #'(thickness . 2)
            \draw-line #'(0.6 . 0.1 )
            \override #'(thickness . 2)
            \draw-line #'(-0.6 . -0.1 )
            \draw-circle #0.65 #0.175 ##f

            \temporary \override NoteHead.stem-attachment =
            #(lambda (grob)
               (let* ((stem (ly:grob-object grob 'stem))
```

6. https://lilypond.org/doc/v2.24/Documentation/notation/modifying-stencils

```
                      (dir (ly:grob-property stem 'direction UP))
                      (is-up (eqv? dir UP)))
                 (cons dir (if is-up 0.2 -0.2))))
            #note
            \revert NoteHead.stencil
            \revert NoteHead.text
            \revert NoteHead.stem-attachment
          #})


{
  \cirX c'4 \cirX d' \cirX e' \cirX f'
  \cirX a''4 \cirX g'' \cirX f'' \cirX  e''
}

```

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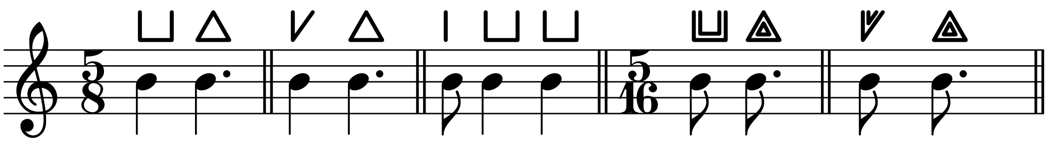
Chapter 6

Markups

6.1 Conducting Patterns



6.1.1 Description

Conducting patterns. While there are several examples of conducting patterns available on LSR,[1] the conducting shapes in my implementation are not affected by the horizontal length of given durations.

6.1.2 Grammar

```
NOTE \condOne
NOTE \condTwoA
NOTE \condTwoB
NOTE \condThree
NOTE \condDoubleTwoA
NOTE \condDoubleTwoB
NOTE \condDoubleThree
```

6.1.3 Code

```
\version "2.26.0"

condOnePattern =
#'((moveto 0.25 1.75)
   (rlineto 0 -1.75))
```

1. See: https://lsr.di.unimi.it/LSR/Item?id=523 and https://lsr.di.unimi.it/LSR/Item?id=259

```

condTwoPatternA =
#'((moveto 0.25 1.75)
   (rlineto 0 -1.75)
   (rlineto 2 0)
   (rlineto 0 1.75))

condDoubleTwoPatternA =
#'((moveto 0.25 1.75)
   (rlineto 0 -1.75)
   (rlineto 2 0)
   (rlineto 0 1.75)
   (moveto 0.65 1.75)
   (rlineto 0 -1.35)
   (rlineto 1.2 0)
   (rlineto 0 1.35))

condTwoPatternB =
#'((moveto 0.25 1.75)
   (rlineto 0 -1.75)
   (rlineto 1.25 1.75))

condDoubleTwoPatternB =
#'((moveto 0.25 1.75)
   (rlineto 0 -1.75)
   (rlineto 1.25 1.75)
   (moveto 0.6 1.75)
   (rlineto 0 -0.7)
   (rlineto 0.5 0.7))

condThreePattern =
#'((moveto 1.15 1.75)
   (rlineto -1 -1.75)
   (rlineto 2 0)
   (closepath))

condDoubleThreePattern =
#'((moveto 1.15 1.75)
   (rlineto -1 -1.75)
   (rlineto 2 0)
   (closepath)
   (moveto 1.15 1.05)
   (rlineto -0.385 -0.7)
   (rlineto 0.75 0)
   (closepath))


condOne = ^\markup {
```

```
 \override #'(line-join-style . round)
 \path #0.25 #condOnePattern
}

condTwoA = ^\markup {
 \override #'(line-join-style . round)
 \path #0.25 #condTwoPatternA
}
condTwoB = ^\markup {
 \override #'(line-join-style . round)
 \path #0.25 #condTwoPatternB
}
condDoubleTwoA = ^\markup {
 \override #'(line-join-style . round)
 \path #0.25 #condDoubleTwoPatternA
}

condDoubleTwoB = ^\markup {
 \override #'(line-join-style . round)
 \path #0.25 #condDoubleTwoPatternB
}

condThree = ^\markup {
 \override #'(line-join-style . round)
 \path #0.25 #condThreePattern
}

condDoubleThree = ^\markup {
 \override #'(line-join-style . round)
 \path #0.25 #condDoubleThreePattern
}

%% Source inspired by
%% and adapted from: http://lsr.di.unimi.it/LSR/Item?id=629
spacerVoice = \new Voice {
 \override MultiMeasureRest.transparent = ##t
 \override MultiMeasureRest.minimum-length = #14
 R16*5
}


\score {
 {
  \time 5/8
  b'4 \condTwoA b'4. \condThree \bar "||"
  b'4 \condTwoB b'4. \condThree \bar "||"
  b'8 \condOne b'4 \condTwoA b'4 \condTwoA \bar "||"
  \time 5/16
```

```
    << {b'8 \condDoubleTwoA b'8. \condDoubleThree} \spacerVoice >> \bar "||"
    << {b'8 \condDoubleTwoB b'8. \condDoubleThree} \spacerVoice >> \bar "||"
  }

}

```

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6.2 Mute Sign

6.2.1 Description

Implementation of the mute sign, used to indicate that vibrating strings must be dampened at a specified moment. Its provenance can be traced back to Carlos Salzedo's *Modern Study of the Harp*.[2]

6.2.2 Grammar

`NOTE/REST^\mutesign`

6.2.3 Code

```
\version "2.26.0"

mutesign = \markup {
 \translate #'(0.5 . 0)
 \postscript

 "newpath
0.2 setlinewidth
1 setlinecap
0 0 moveto
0 2.5 rlineto
-1.25 1.25 moveto
2.5 0 rlineto
stroke
newpath
0 1.25 0.85 0 360 arc
stroke"

}

{
 c'2. r4^\mutesign^\markup {
  \translate #'(-3 . 2)
  \musicglyph "space"
 }

}
```

2. Carlos Salzedo, *L'étude moderne de la harpe... Modern study of the harp* (New York - Boston, G. Schirmer, 1921), 19.

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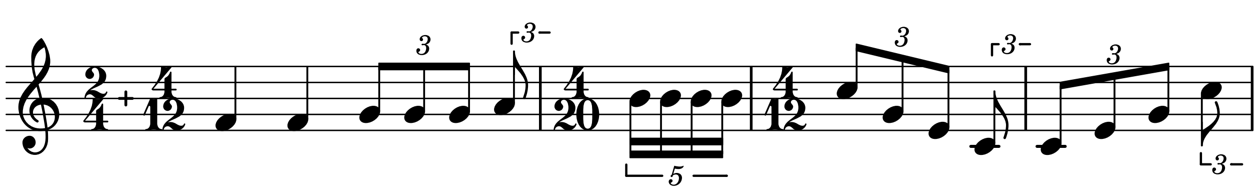
Chapter 7

Rhythm

7.1 Incomplete Tuplet Bracket for Irrational Time Signatures



7.1.1 Description

This entry implements the irrational time signatures[1] as seen on LSR.[2]. Concerning the irrational time signatures, in her *Behind Bars: the Definitive Guide to Music Notation*, Elaine Gould suggests the use of denominator as any division of the semibreve/whole note..[3] However, in these pages there has not been a mention of the use of tuplet brackets while the non-conventional time signature is in place. There are examples, such as *Asyla* for large orchestra by Thomas Adès,,[4] where tuplet brackets are placed atop "incomplete" tuplets.

While it is still prudent to spend a paragraph explaining the nature of the irrational time signatures in the preface, my preference has also been to utilize incomplete tuplet brackets, in order to allow the reading of the rhythm consistent and smooth from bars with ordinary time signatures. It is also helpful to have the brackets shown in cases of compound time signatures that use irrational time signatures in part (see the first measure of the example).

7.1.2 Grammar

```
\incompleteTupletBracket \tuplet ...
\incompleteSmallTupletBracket \tuplet ...
```

NB

1. For incomplete tuplets with two or more notes, use `\incompleteTupletBracket`.

---

1. See Chapter **Time Signatures** for discussion on the variants of the irrational/fractional time signatures.
2. https://lsr.di.unimi.it/LSR/Snippet?id=552
3. Elaine Gould, *Behind bars : the definitive guide to music notation* (London: Faber Music, 2011), 180–181, Book.
4. Thomas Adès, *Asyla : for large orchestra* (Faber Music, 1997).

2. For incomplete tuplets with one note, use `\incompleteSmallTupletBracket`. This was created specifically to ensure that the brackets appear properly in tight space that one-note tuplet customarily gives.

7.1.3 Code

```
\version "2.26.0"

%% "suppressWarning" function comes from:
%% http://lsr.di.unimi.it/LSR/Item?id=552

% Warnings may be suppressed using 'ly:expect-warning'
% Or use the here defined 'suppressWarning'-function, working since 2.20.

suppressWarning =
#(define-void-function (amount message)(number? string?)
   (for-each
    (lambda (warning)
      (ly:expect-warning message))
    (iota amount 1 1)))

\suppressWarning 3 "strange time signature found"

incompleteTupletBracket =  {
  \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
  \once \override Voice.TupletBracket.bracket-visibility = ##t

}
incompleteSmallTupletBracket =  {
  \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
  \once \override Voice.TupletBracket.bracket-visibility = ##t
  \once \override Voice.TupletNumber.X-offset =
  #(lambda (grob)
     (if (= UP (ly:grob-property grob 'direction))
         2.2
         1.2))

  \once \override Voice.TupletBracket.shorten-pair =
  #(lambda (grob)
     (if (= UP (ly:grob-property grob 'direction))
         '(-0.7 . 0.15)
         '(-0.3 . 0.8)))
  \once \override Voice.TupletBracket.X-positions =
  #(lambda (grob)
     (if (= UP (ly:grob-property grob 'direction))
         '(1.8 . 3)
         '(0.3 . 2.7)))
}
```

```
{

  \timeAbbrev #'((2 4) (4 12))
  f'4 f'
  \tuplet 3/2 {g'8[ g' g']}
  \incompleteSmallTupletBracket
  \tuplet 3/2 {a'8 }|
   \set Score.proportionalNotationDuration = #1/16
  \time 4/20
  \incompleteTupletBracket
  \tuplet 5/4 {b'16[ b' b' b']} |

  \time 4/12
  \tuplet 3/2 {c''8[ g' e']}
  \incompleteSmallTupletBracket
  \tuplet 3/2 {c'8} |
  \tuplet 3/2 {c'8[ e' g']}
  \incompleteSmallTupletBracket
  \tuplet 3/2 {c''8} |
}
```

7.1.4 Discussion

In the preceding code, I have opted to notate the tuplets within the bars with irrational time signatures in an ordinary manner, using `\tuplet`. This is to ensure that the incomplete tuplet bracket appears. Compare this with the quoted LSR No. 552, which has a different way of reducing the note duration in order to fit them into the bar with irrational time signature. Observe the way duration is multiplied by fractions, e.g. Line 6.

```
1   {
2    \time 4/4
3    \tempo 4 = 60
4    fis4 fis fis fis
5    \time 2/6
6    g4*2/3 g |
7    g4*2/3 g |
8    \time 4/5
9    as4*4/5 as as as8*4/5 g |
10   \tuplet 3/2 { as4*4/5 as as } as4*4/5 as8*4/5 g |
11   \time 3/7
12   fis4*4/7 fis fis |
13   fis4*4/7 fis fis |
14  }
```

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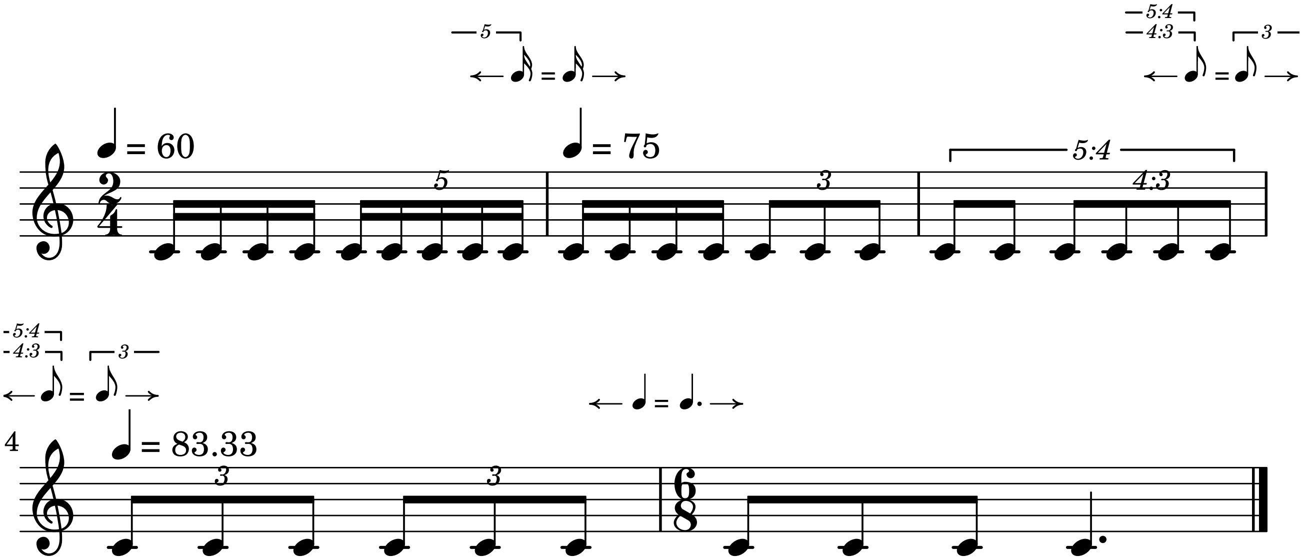
7.2 Metric Modulation Equation (Regular Flag)



7.2.1 Description

This entry implements a metric modulation formula that indicates a note value of one measure being equal to another note value of the subsequent measure. While this notation has existed for a very long time, it was in the twentieth century where composers such as Elliott Carter used them extensively in their works. This code follows the convention commonly seen in Carter's scores, where the formula is accompanied by the right and left arrows to state explicitly that the note value on the left refers to that of the preceding measure, while the one on the right refers to that of the subsequent measure.[5]

Another version with notes with straight flags can be found **here**.

7.2.2 Grammar

```
\MModEquation
  LEFT_NOTE_DURATION #'((LEFT_TUPLET_1)(LEFT_TUPLET_2))
  RIGHT_NOTE_DURATION #'((RIGHT_TUPLET_1)(RIGHT_TUPLET_2))
  #V_OFFSET #H_OFFSET
\MModEquationBegin
  LEFT_NOTE_DURATION #'((LEFT_TUPLET_1)(LEFT_TUPLET_2))
  RIGHT_NOTE_DURATION #'((RIGHT_TUPLET_1)(RIGHT_TUPLET_2))
  #V_OFFSET
```

NB

5. While it may be beyond the scope of this Cookbook, stating the orientation of the note value clarifies any ambiguities, as the convention in the classical period had the two note values flipped, where the note on the left indicated the note value of the forthcoming measure, and the one on the right, that of the preceding measure. See Arthur Weisberg, *Performing twentieth-century music : a handbook for conductors and instrumentalists* (New Haven: Yale University Press, 1996), 52.

1. `\MModEquation` utilizes LilyPond's `\textEndMark` functions. It is therefore meant to be used at the end of a measure where the metric modulation is about to take place. Then you may use `\tempo` function on LilyPond to indicate the new tempo in the subsequent measure.

2. `\MModEquationBegin`, on the other hand, uses LilyPond's `\textMark` functions, which places texts at the beginning of a measure. It is meant to be used as part of the editing process. Because `\MModEquation` places the formula at the end of a measure, when that measure is at the end of a system, requiring another formula at the beginning of the subsequent measure (i.e. at the beginning of a new system) may be plausible. Using `\MModEquationBegin` may be the solution.

3. For `LEFT_NOTE_DURATION` and `RIGHT_NOTE_DURATION`, use a numerical value *without* `#`. You may add dot(s) to indicate augmentation dot(s).

4. For the variables 2 and 4, you may put up to two instances of tuplet values, in the form of list(s) within a list. At the minimum, you will need to supply an empty list, i.e. `#'()`.

   For example, if it is a simple triplet you wish to implement, you would write:

   `#'( ( 3 ) )`

   If you wish to indicate tuplet in ratio, e.g. quintuplet in the span of 4 of the note values, you would create a list with two items, i.e.:

   `#'( ( 5 4 ) )`

   You may set another tuplet instance *above* the initial tuplet, so the following list:

   `#'( ( 4 3 ) ( 5 4 ) )`

   ...would produce the tuplet indication at the end of measure 3 in the example shown above.

5. For `#V_OFFSET` and `#H_OFFSET` (only for `\MModEquation`), enter numbers preceded by `#`. These numbers adjust the placement of the formula. When in doubt, start with `#0`.

7.2.3 Code

```
\version "2.26.0"

MModEquation =
#(define-music-function
  (notevalue1 ratio1 notevalue2 ratio2 verticaloffset horizontaloffset)
  (ly:duration? list? ly:duration? list? number? number?)
  (let* (
         (noteone notevalue1)
         (notetwo notevalue2)
         (ratioone ratio1)
         (ratiotwo ratio2)
         )
   #{
    \tweak X-offset #(- horizontaloffset 0.35)
    \tweak Y-offset #verticaloffset
```

```
      \textEndMark  \markup \right-align {
       \hspace #-0.5
       \raise #0
       \fontsize #-4.5
       \override #'(self-alignment . LEFT)
       {
        {
         \note { $noteone } #(if (= (ly:duration-log noteone) 4) 1 1 )
        }
        {
         \hspace #(if (< (ly:duration-log noteone) 3) -4.15
                      (+ (* (ly:duration-dot-count noteone) -0.5) -4.5))
         \concat {
          \combine
          \musicglyph "arrowheads.open.0M1"
          \draw-line #'(2 . 0) \fontsize #-5 \musicglyph "space"
         }
        }
        {
         \hspace #(cond ((<= (ly:duration-log noteone) 2) 0.75)
                        ((> (ly:duration-log noteone) 2) 0.25) )
         \raise #(cond
                  ((= (ly:duration-log noteone) 0) -2)
                  ((and (> (ly:duration-log noteone) 0)
                        (<= (ly:duration-log noteone) 4) ) -.5)
                  ((> (ly:duration-log noteone) 4)
                   (+ (* (- (ly:duration-log noteone) 5) 0.6) 0))
                  )
         #(cond
           ((= (length ratioone) 1)
            (markup
             #:line
             (#:hspace
              -4.5
              #:raise
              2.75
              (#:center-column
               (#:concat
                (#:translate
                 (cons 3.5 0.5)
                 (#:override
                  (cons 'thickness 1.5)
                  (#:translate
                   (cons -3.5 0)
                   (#:draw-line (cons -4.25 0))))
                 #:hspace
                 -0.1
                 #:override
```

```
                (cons 'thickness 1.5)
                (#:translate
                 (cons 0 0.5)
                 (#:draw-line (cons 0 -0.5))))
               #:vspace
               -0.45
               #:whiteout
               (#:halign
                1.75
                (#:concat
                 (#:fontsize
                  -1
                  (#:italic (#:fontsize -5 (#:musicglyph "space")))
                  #:fontsize
                  -1
                  #:italic
                  (if (= (length (car ratioone)) 2)
                      (begin (string-append
                              (number->string (car (car ratioone)))
                              ":"
                              (number->string (cadr (car ratioone))))  )
                      (number->string (car (car ratioone)))
                      )

                  (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))))

          ((= (length ratioone) 2)
           (markup
            #:line
            (#:hspace
             -4.5
             #:raise
             2.75
             (#:center-column
              (#:concat
               (#:translate
                (cons 3.5 0.5)
                (#:override
                 (cons 'thickness 1.5)
                 (#:translate
                  (cons -3.5 0)
                  (#:draw-line (cons -4.25 0))))
                #:hspace
                -0.1
                #:override
                (cons 'thickness 1.5)
                (#:translate
                 (cons 0 0.5)
```

```
     (#:draw-line (cons 0 -0.5))))
   #:vspace
   -0.45
   #:whiteout
   (#:halign
    1.75
    (#:concat
     (#:fontsize
      -1
      (#:italic (#:fontsize -5 (#:musicglyph "space")))
      #:fontsize
      -1
      #:italic
      (if (= (length (car ratioone)) 2)
          (begin (string-append
                  (number->string (car (car ratioone)))
                  ":"
                  (number->string (cadr (car ratioone))))  )
          (number->string (car (car ratioone)))
          )
      (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))
#:line
(#:hspace
 -4.8
 #:raise
 4
 (#:center-column
  (#:concat
   (#:translate
    (cons 3.5 0.5)
    (#:override
     (cons 'thickness 1.5)
     (#:translate
      (cons -3.5 0)
      (#:draw-line (cons -4.25 0))))
    #:hspace
    -0.1
    #:override
    (cons 'thickness 1.5)
    (#:translate
     (cons 0 0.5)
     (#:draw-line (cons 0 -0.5))))
   #:vspace
   -0.45
   #:whiteout
   (#:halign
    1.75
    (#:concat
```

```
            (#:fontsize
             -1
             (#:italic (#:fontsize -5 (#:musicglyph "space")))
             #:fontsize
             -1
             #:italic
             (if (= (length (cadr ratioone)) 2)
                 (begin (string-append
                         (number->string (car (cadr ratioone)))
                         ":"
                         (number->string (cadr (cadr ratioone))))  )
                 (number->string (car (cadr ratioone)))
                 )
             (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))
       ))
     )
  }
 }
}
\tweak self-alignment-X #CENTER
\tweak Y-offset #(- verticaloffset 0.4)
\tweak X-offset #(- horizontaloffset 0.35)
\textEndMark \markup {  \fontsize #-3 "="}

\tweak Y-offset #verticaloffset
\tweak self-alignment-X #RIGHT
\tweak X-offset #(- horizontaloffset 0.35)
\tweak self-alignment-Y -1
\textEndMark \markup \left-align {
 \hspace #1.9
 \raise #0
 \fontsize #-4.5
 \concat {
  {
   \hspace #(cond ((<= (ly:duration-log notetwo) 2) 4)
                  ((> (ly:duration-log notetwo) 2) 3.75) )
   \raise #(cond
            ((= (ly:duration-log notetwo) 0) -2)
            ((and (> (ly:duration-log notetwo) 0)
                  (<= (ly:duration-log notetwo) 4) ) -.5)
            ((> (ly:duration-log notetwo) 4)
             (+ (* (- (ly:duration-log notetwo) 5) 0.6) 0))
            )
   #(cond
     ((= (length ratiotwo) 1)
      (markup
       #:line
       (#:hspace
```

```
               -4.5
               #:raise
               3.25
               (#:center-column
                (#:concat
                 (#:translate
                  (cons 3.5 0.5)
                  (#:override
                   (cons 'thickness 1.5)
                   (#:translate
                    (cons 0 -0.5 )
                    (#:draw-line (cons 0 -0.5)))
                   )
                  #:hspace
                  -0.1
                  #:override
                  (cons 'thickness 1.5)
                  (#:translate
                   (cons -3.5 0)
                   (#:draw-line (cons -4.25 0))))
                 #:vspace
                 -0.35
                 #:whiteout
                 (#:halign
                  1.75
                  (#:concat
                   (#:fontsize
                    -1
                    (#:italic (#:fontsize -5 (#:musicglyph "space")))
                    #:fontsize
                    -1
                    #:italic
                    (if (= (length (car ratiotwo)) 2)
                        (begin (string-append
                                (number->string (car (car ratiotwo)))
                                ":"
                                (number->string (cadr (car ratiotwo))))  )
                        (number->string (car (car ratiotwo)))
                        )
                    (#:italic
                     (#:fontsize -5
                                 (#:musicglyph "space"))))))))))
            ((= (length ratiotwo) 2)
             (markup
              #:line
              (#:hspace
               -4.5
               #:raise
```

```
  3.25
  (#:center-column
   (#:concat
    (#:translate
     (cons 3.5 0.5)
     (#:override
      (cons 'thickness 1.5)

      (#:translate
       (cons 0 -0.5 )
       (#:draw-line (cons 0 -0.5)))
      )
     #:hspace
     -0.1
     #:override
     (cons 'thickness 1.5)
     (#:translate
      (cons -3.5 0)
      (#:draw-line (cons -4.25 0))))
    #:vspace
    -0.35
    #:whiteout
    (#:halign
     1.75
     (#:concat
      (#:fontsize
       -1
       (#:italic (#:fontsize -5 (#:musicglyph "space")))
       #:fontsize
       -1
       #:italic
       (if (= (length (car ratiotwo)) 2)
           (begin (string-append
                   (number->string (car (car ratiotwo)))
                   ":"
                   (number->string (cadr (car ratiotwo))))  )
           (number->string (car (car ratiotwo)))
           )
       (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))
 #:line
 (#:hspace
  -4.8
  #:raise
  4.5
  (#:center-column
   (#:concat
    (#:translate
     (cons 3.5 0.5)
```

```
                 (#:override
                  (cons 'thickness 1.5)
                  (#:translate
                   (cons 0 -0.5 )
                   (#:draw-line (cons 0 -0.5)))
                  )
                 #:hspace
                 -0.1
                 #:override
                 (cons 'thickness 1.5)
                 (#:translate
                  (cons -3.5 0)
                  (#:draw-line (cons -4.25 0)))
                 )
                #:vspace
                -0.35
                #:whiteout
                (#:halign
                 1.75
                 (#:concat
                  (#:fontsize
                   -1
                   (#:italic (#:fontsize -5 (#:musicglyph "space")))
                   #:fontsize
                   -1
                   #:italic
                   (if (= (length (cadr ratiotwo)) 2)
                       (begin (string-append
                               (number->string (car (cadr ratiotwo)))
                               ":"
                               (number->string (cadr (cadr ratiotwo)))))
                       (number->string (car (cadr ratiotwo)))
                       )

                   (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))
             ))
           )
        }
        {
         \hspace #-4.25
         \note { $notetwo } #(if (= (ly:duration-log notetwo) 4) 1 1 )
        }
        {
         \hspace #0.5
         \combine
         \draw-line #'(-2 . 0)
         \musicglyph "arrowheads.open.01"
        }
```

```
      }
     }
    #})
   )



MModEquationBegin =
#(define-music-function
   (notevalue1 ratio1 notevalue2 ratio2 verticaloffset )
   (ly:duration? list? ly:duration? list? number? )
   (let* (
          (noteone notevalue1)
          (notetwo notevalue2)
          (ratioone ratio1)
          (ratiotwo ratio2)
          )
    #{

     \tweak Y-offset #verticaloffset
     \textMark  \markup \left-align {

      \raise #0
      \fontsize #-4.5
      \override #'(self-alignment . LEFT)
      {
       {
        \hspace #(if (< (ly:duration-log noteone) 3) -4.15
                     (+ (* (ly:duration-dot-count noteone) -0.5) -4.5))
        \concat {
         \combine
         \musicglyph "arrowheads.open.0M1"
         \draw-line #'(2 . 0) \fontsize #-5 \musicglyph "space"
        }
       }

       {
        \hspace #-0.5
        \note { $noteone } #(if (= (ly:duration-log noteone) 4) 1 1 )
       }


       {
        \hspace #(cond
                  ((<= (ly:duration-dot-count noteone) 1) -0.5)
                  ((> (ly:duration-dot-count noteone) 1)
                   (+ (* (ly:duration-dot-count noteone) -0.5) -0.25)))
        \right-align
```

```
        \raise #(cond
                 ((= (ly:duration-log noteone) 0) -2)
                 ((and (> (ly:duration-log noteone) 0)
                       (<= (ly:duration-log noteone) 4) ) -.5)
                 ((> (ly:duration-log noteone) 4)
                  (+ (* (- (ly:duration-log noteone) 5) 0.6) 0))
                 )

        #(cond
          ((= (length ratioone) 1)
           (markup
            #:line
            (#:hspace
             -4.5
             #:raise
             2.75
             (#:center-column
              (#:concat
               (#:translate
                (cons 3.5 0.5)
                (#:override
                 (cons 'thickness 1.5)
                 (#:translate
                  (cons -3.5 0)
                  (#:draw-line (cons -4.25 0))))
                #:hspace
                -0.1
                #:override
                (cons 'thickness 1.5)
                (#:translate
                 (cons 0 0.5)
                 (#:draw-line (cons 0 -0.5))))
               #:vspace
               -0.45
               #:whiteout
               (#:halign
                1.75
                (#:concat
                 (#:fontsize
                  -1
                  (#:italic (#:fontsize -5 (#:musicglyph "space")))
                  #:fontsize
                  -1
                  #:italic
                  (if (= (length (car ratioone)) 2)
                      (begin (string-append
                              (number->string (car (car ratioone)))
                              ":"
```

```
                                (number->string (cadr (car ratioone))))  )
                        (number->string (car (car ratioone)))
                        )

                    (#:italic
                     (#:fontsize -5
                                (#:musicglyph "space")))))))))))

            ((= (length ratioone) 2)
             (markup
              #:line
              (#:hspace
               -4.5
               #:raise
               2.75
               (#:center-column
                (#:concat
                 (#:translate
                  (cons 3.5 0.5)
                  (#:override
                   (cons 'thickness 1.5)
                   (#:translate
                    (cons -3.5 0)
                    (#:draw-line (cons -4.25 0))))
                  #:hspace
                  -0.1
                  #:override
                  (cons 'thickness 1.5)
                  (#:translate
                   (cons 0 0.5)
                   (#:draw-line (cons 0 -0.5))))
                 #:vspace
                 -0.45
                 #:whiteout
                 (#:halign
                  1.75
                  (#:concat
                   (#:fontsize
                    -1
                    (#:italic (#:fontsize -5 (#:musicglyph "space")))
                    #:fontsize
                    -1
                    #:italic
                    (if (= (length (car ratioone)) 2)
                        (begin (string-append
                                (number->string (car (car ratioone)))
                                ":"
                                (number->string (cadr (car ratioone)))))
```

```
                        (number->string (car (car ratioone)))
                        )
                    (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))

              #:line
              (#:hspace
               -4.8
               #:raise
               4
               (#:center-column
                (#:concat
                 (#:translate
                  (cons 3.5 0.5)
                  (#:override
                   (cons 'thickness 1.5)
                   (#:translate
                    (cons -3.5 0)
                    (#:draw-line (cons -4.25 0))))
                  #:hspace
                  -0.1
                  #:override
                  (cons 'thickness 1.5)
                  (#:translate
                   (cons 0 0.5)
                   (#:draw-line (cons 0 -0.5))))
                 #:vspace
                 -0.45
                 #:whiteout
                 (#:halign
                  1.75
                  (#:concat
                   (#:fontsize
                    -1
                    (#:italic (#:fontsize -5 (#:musicglyph "space")))
                    #:fontsize
                    -1
                    #:italic
                    (if (= (length (cadr ratioone)) 2)
                        (begin (string-append
                                (number->string (car (cadr ratioone)))
                                ":"
                                (number->string (cadr (cadr ratioone))))  )
                        (number->string (car (cadr ratioone)))
                        )

                    (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))
              ))
            )
```

```

        }
        {
         \hspace #(+ (* (ly:duration-dot-count noteone) 0.35) -0.2)
         \fontsize #2 \lower #0.5  "="
        }
        \concat {

         {
          \hspace #(cond ((<= (ly:duration-log notetwo) 2) 4.15 )
                         ((> (ly:duration-log notetwo) 2) 4.15) )
          \raise #(cond
                   ((= (ly:duration-log notetwo) 0) -2)
                   ((and (> (ly:duration-log notetwo) 0)
                         (<= (ly:duration-log notetwo) 4) ) -.5)
                   ((> (ly:duration-log notetwo) 4)
                    (+ (* (- (ly:duration-log notetwo) 5) 0.6) 0))
                   )

          #(cond
            ((= (length ratiotwo) 1)
             (markup
              #:line
              (#:hspace
               -4.5
               #:raise
               3.25
               (#:center-column
                (#:concat
                 (#:translate
                  (cons 3.5 0.5)
                  (#:override
                   (cons 'thickness 1.5)

                   (#:translate
                    (cons 0 -0.5 )
                    (#:draw-line (cons 0 -0.5)))
                   )

                  #:hspace
                  -0.1

                  #:override
                  (cons 'thickness 1.5)
                  (#:translate
                   (cons -3.5 0)
                   (#:draw-line (cons -4.25 0))))

```

```
                  #:vspace
                  -0.35
                  #:whiteout
                  (#:halign
                   1.75
                   (#:concat
                    (#:fontsize
                     -1
                     (#:italic (#:fontsize -5 (#:musicglyph "space")))
                     #:fontsize
                     -1
                     #:italic
                     (if (= (length (car ratiotwo)) 2)
                         (begin (string-append
                                 (number->string (car (car ratiotwo)))
                                 ":"
                                 (number->string (cadr (car ratiotwo)))))
                         (number->string (car (car ratiotwo)))
                         )
                     (#:italic
                      (#:fontsize -5
                                  (#:musicglyph "space")))))))))))

             ((= (length ratiotwo) 2)
              (markup
               #:line
               (#:hspace
                -4.5
                #:raise
                3.25
                (#:center-column
                 (#:concat
                  (#:translate
                   (cons 3.5 0.5)
                   (#:override
                    (cons 'thickness 1.5)
                    (#:translate
                     (cons 0 -0.5 )
                     (#:draw-line (cons 0 -0.5)))
                    )
                   #:hspace
                   -0.1
                   #:override
                   (cons 'thickness 1.5)
                   (#:translate
                    (cons -3.5 0)
                    (#:draw-line (cons -4.25 0))))
```

```
   #:vspace
   -0.35
   #:whiteout
   (#:halign
    1.75
    (#:concat
     (#:fontsize
      -1
      (#:italic (#:fontsize -5 (#:musicglyph "space")))
      #:fontsize
      -1
      #:italic
      (if (= (length (car ratiotwo)) 2)
          (begin (string-append
                  (number->string (car (car ratiotwo)))
                  ":"
                  (number->string (cadr (car ratiotwo)))))
          (number->string (car (car ratiotwo)))
          )
      (#:italic
       (#:fontsize -5
                   (#:musicglyph "space")))))))))

#:line
(#:hspace
 -4.8
 #:raise
 4.5
 (#:center-column
  (#:concat
   (#:translate
    (cons 3.5 0.5)
    (#:override
     (cons 'thickness 1.5)
     (#:translate
      (cons 0 -0.5 )
      (#:draw-line (cons 0 -0.5)))
     )
    #:hspace
    -0.1
    #:override
    (cons 'thickness 1.5)
    (#:translate
     (cons -3.5 0)
     (#:draw-line (cons -4.25 0)))
    )
   #:vspace
   -0.35
```

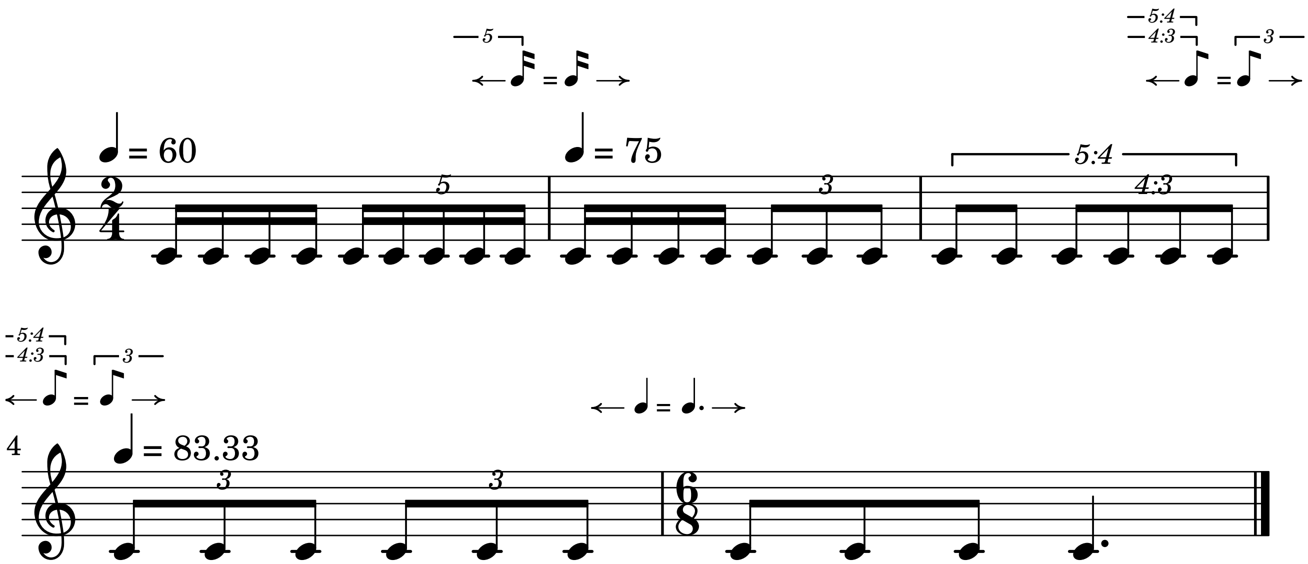
```
               #:whiteout
               (#:halign
                1.75
                (#:concat
                 (#:fontsize
                  -1
                  (#:italic (#:fontsize -5 (#:musicglyph "space")))
                  #:fontsize
                  -1
                  #:italic
                  (if (= (length (cadr ratiotwo)) 2)
                      (begin (string-append
                              (number->string (car (cadr ratiotwo)))
                              ":"
                              (number->string (cadr (cadr ratiotwo)))))
                      (number->string (car (cadr ratiotwo)))
                      )
                  (#:italic
                   (#:fontsize -5
                               (#:musicglyph "space"))))))))
            ))
          )
       }
       {
        \hspace #-4
        \note { $notetwo }
        #(if (= (ly:duration-log notetwo) 4) 1 1 )
       }
       {
        \hspace #0.5
        \combine
        \draw-line #'(-2 . 0)
        \musicglyph "arrowheads.open.01"
       }
      }
     }
    }
   #})
  )

{
 \time 2/4
 \tempo 4 = 60
 c'16
 ^\markup {\translate #'(0 . 10) " "}
 [ c'16 c'16 c'16 ]
 \tuplet 5/4 {c'16 [ c'16 c'16 c'16 c'16 ] }
 \MModEquation 16 #'((5)) 16 #'() #8 #0
```

```
  \tempo 4 = 75
  c'16 [ c'16 c'16 c'16 ]
  \tuplet 3/2 {c'8 [ c'8 c'8 ] }
  \override TupletNumber.text = #tuplet-number::calc-fraction-text
  \tuplet 5/4 {c'8 [ c'8 ] \tuplet 4/3 { c'8 [ c'8 c'8 c'8  ] }}
  \MModEquation 8 #'((4 3)(5 4)) 8 #'((3)) 8 #-3
  \revert TupletNumber.text
  \break
  \MModEquationBegin 8 #'((4 3)(5 4)) 8 #'((3)) #6.5
  \tuplet 3/2 {
   c'8
   ^\markup {\translate #'(0 . 10) " "}
   ^\markup
   {\fontsize #-1 \raise #0.25 \note {4} #1 "= 83.33"}
   [ c'8 c'8 ]
  }
  \tuplet 3/2 {
   c'8 c'8 c'8
   \MModEquation 4 #'() 4. #'() #6 #0
  }
  \time 6/8
  c'8 c'8 c'8 c'4.
  \bar "|."
 }
```

Table of Contents

7.3 Metric Modulation Equation (Straight Flag)



7.3.1 Description

This is an alternative implementation of the metric modulation formula, introduced in the **previous entry**, except this time it uses notes with the modern straight flag.

7.3.2 Grammar

```
\MModEquationSTR
  LEFT_NOTE_DURATION #'((LEFT_TUPLET_1)(LEFT_TUPLET_2))
  RIGHT_NOTE_DURATION #'((RIGHT_TUPLET_1)(RIGHT_TUPLET_2))
  #V_OFFSET #H_OFFSET
\MModEquationBeginSTR
  LEFT_NOTE_DURATION #'((LEFT_TUPLET_1)(LEFT_TUPLET_2))
  RIGHT_NOTE_DURATION #'((RIGHT_TUPLET_1)(RIGHT_TUPLET_2))
  #V_OFFSET
```

NB

1. `\MModEquationSTR` utilizes LilyPond's `\textEndMark` functions. It is therefore meant to be used at the end of a measure where the metric modulation is about to take place. Then you may use `\tempo` function on LilyPond to indicate the new tempo in the subsequent measure.

2. `\MModEquationBeginSTR`, on the other hand, uses LilyPond's `\textMark` functions, which places texts at the beginning of a measure. It is meant to be used as part of the editing process. Because `\MModEquationSTR` places the formula at the end of a measure, when that measure is at the end of a system, requiring another formula at the beginning of the subsequent measure (i.e. at the beginning of a new system) may be plausible. Using `\MModEquationBeginSTR` may be the solution.

3. For `LEFT_NOTE_DURATION` and `RIGHT_NOTE_DURATION`, use a numerical value *without* `#`. You may add dot(s) to indicate augmentation dot(s).

4. For the variables 2 and 4, you may put up to two instances of tuplet values, in the form of list(s) within a list. At the minimum, you will need to supply an empty list, i.e. `#'()`.

   For example, if it is a simple triplet you wish to implement, you would write:

   `#'( ( 3 ) )`

   If you wish to indicate tuplet in ratio, e.g. quintuplet in the span of 4 of the note values, you would create a list with two items, i.e.:

   `#'( ( 5 4 ) )`

   You may set another tuplet instance *above* the initial tuplet, so the following list:

   `#'( ( 4 3 ) ( 5 4 ) )`

   ...would produce the tuplet indication at the end of measure 3 in the example shown above.

5. For `#V_OFFSET` and `#H_OFFSET` (only for `\MModEquationSTR`), enter numbers preceded by `#`. These numbers adjust the placement of the formula. When in doubt, start with `#0`.

7.3.3 Code

```
\version "2.26.0"

MModEquationSTR =
#(define-music-function
  (notevalue1 ratio1 notevalue2 ratio2 verticaloffset horizontaloffset)
  (ly:duration? list? ly:duration? list? number? number?)
  (let* (
         (noteone notevalue1)
         (notetwo notevalue2)
         (ratioone ratio1)
         (ratiotwo ratio2)
         )
   #{
    \tweak X-offset #(- horizontaloffset 0.35)
    \tweak Y-offset #verticaloffset
    \textEndMark  \markup \right-align {
     \hspace #-0.5
     \raise #0
     \fontsize #-4.5
     \override #'(self-alignment . LEFT)
     {
      {
       \override #'(flag-style . modern-straight-flag)
       \note { $noteone } #(if (= (ly:duration-log noteone) 4) 1 1 )
      }
      {
       \hspace #(if (< (ly:duration-log noteone) 3) -4.15
```

```
                      (+ (* (ly:duration-dot-count noteone) -0.5) -4.5))
         \concat {
          \combine
          \musicglyph "arrowheads.open.0M1"
          \draw-line #'(2 . 0) \fontsize #-5 \musicglyph "space"
         }
        }
        {
         \hspace #(cond ((<= (ly:duration-log noteone) 2) 0.75)
                        ((> (ly:duration-log noteone) 2) 0.25) )
         \raise #(cond
                  ((= (ly:duration-log noteone) 0) -2)
                  ((and (> (ly:duration-log noteone) 0)
                        (<= (ly:duration-log noteone) 4) ) -.5)
                  ((> (ly:duration-log noteone) 4)
                   (+ (* (- (ly:duration-log noteone) 5) 0.6) 0))
                  )
         #(cond
           ((= (length ratioone) 1)
            (markup
             #:line
             (#:hspace
              -4.5
              #:raise
              2.75
              (#:center-column
               (#:concat
                (#:translate
                 (cons 3.5 0.5)
                 (#:override
                  (cons 'thickness 1.5)
                  (#:translate
                   (cons -3.5 0)
                   (#:draw-line (cons -4.25 0))))
                 #:hspace
                 -0.1
                 #:override
                 (cons 'thickness 1.5)
                 (#:translate
                  (cons 0 0.5)
                  (#:draw-line (cons 0 -0.5))))
                #:vspace
                -0.45
                #:whiteout
                (#:halign
                 1.75
                 (#:concat
                  (#:fontsize
```

```
                -1
                (#:italic (#:fontsize -5 (#:musicglyph "space")))
                #:fontsize
                -1
                #:italic
                (if (= (length (car ratioone)) 2)
                    (begin (string-append
                            (number->string (car (car ratioone)))
                            ":"
                            (number->string (cadr (car ratioone))))  )
                    (number->string (car (car ratioone)))
                    )

                (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))))

        ((= (length ratioone) 2)
         (markup
          #:line
          (#:hspace
           -4.5
           #:raise
           2.75
           (#:center-column
            (#:concat
             (#:translate
              (cons 3.5 0.5)
              (#:override
               (cons 'thickness 1.5)
               (#:translate
                (cons -3.5 0)
                (#:draw-line (cons -4.25 0))))
              #:hspace
              -0.1
              #:override
              (cons 'thickness 1.5)
              (#:translate
               (cons 0 0.5)
               (#:draw-line (cons 0 -0.5))))
             #:vspace
             -0.45
             #:whiteout
             (#:halign
              1.75
              (#:concat
               (#:fontsize
                -1
                (#:italic (#:fontsize -5 (#:musicglyph "space")))
                #:fontsize
```

```
      -1
      #:italic
      (if (= (length (car ratioone)) 2)
          (begin (string-append
                  (number->string (car (car ratioone)))
                  ":"
                  (number->string (cadr (car ratioone))))  )
          (number->string (car (car ratioone)))
          )
      (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))
#:line
(#:hspace
 -4.8
 #:raise
 4
 (#:center-column
  (#:concat
   (#:translate
    (cons 3.5 0.5)
    (#:override
     (cons 'thickness 1.5)
     (#:translate
      (cons -3.5 0)
      (#:draw-line (cons -4.25 0))))
    #:hspace
    -0.1
    #:override
    (cons 'thickness 1.5)
    (#:translate
     (cons 0 0.5)
     (#:draw-line (cons 0 -0.5))))
   #:vspace
   -0.45
   #:whiteout
   (#:halign
    1.75
    (#:concat
     (#:fontsize
      -1
      (#:italic (#:fontsize -5 (#:musicglyph "space")))
      #:fontsize
      -1
      #:italic
      (if (= (length (cadr ratioone)) 2)
          (begin (string-append
                  (number->string (car (cadr ratioone)))
                  ":"
                  (number->string (cadr (cadr ratioone))))  )
```

```
                 (number->string (car (cadr ratioone)))
                 )
             (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))
       ))
     )
  }
 }
}
\tweak self-alignment-X #CENTER
\tweak Y-offset #(- verticaloffset 0.4)
\tweak X-offset #(- horizontaloffset 0.35)
\textEndMark \markup {  \fontsize #-3 "="}

\tweak Y-offset #verticaloffset
\tweak self-alignment-X #RIGHT
\tweak X-offset #(- horizontaloffset 0.35)
\tweak self-alignment-Y -1
\textEndMark \markup \left-align {
 \hspace #1.9
 \raise #0
 \fontsize #-4.5
 \concat {
  {
   \hspace #(cond ((<= (ly:duration-log notetwo) 2) 4)
                  ((> (ly:duration-log notetwo) 2) 3.75) )
   \raise #(cond
            ((= (ly:duration-log notetwo) 0) -2)
            ((and (> (ly:duration-log notetwo) 0)
                  (<= (ly:duration-log notetwo) 4) ) -.5)
            ((> (ly:duration-log notetwo) 4)
             (+ (* (- (ly:duration-log notetwo) 5) 0.6) 0))
            )
   #(cond
     ((= (length ratiotwo) 1)
      (markup
       #:line
       (#:hspace
        -4.5
        #:raise
        3.25
        (#:center-column
         (#:concat
          (#:translate
           (cons 3.5 0.5)
           (#:override
            (cons 'thickness 1.5)
            (#:translate
             (cons 0 -0.5 )
```

```
                (#:draw-line (cons 0 -0.5)))
               )
              #:hspace
              -0.1
              #:override
              (cons 'thickness 1.5)
              (#:translate
               (cons -3.5 0)
               (#:draw-line (cons -4.25 0))))
             #:vspace
             -0.35
             #:whiteout
             (#:halign
              1.75
              (#:concat
               (#:fontsize
                -1
                (#:italic (#:fontsize -5 (#:musicglyph "space")))
                #:fontsize
                -1
                #:italic
                (if (= (length (car ratiotwo)) 2)
                    (begin (string-append
                            (number->string (car (car ratiotwo)))
                            ":"
                            (number->string (cadr (car ratiotwo))))  )
                    (number->string (car (car ratiotwo)))
                    )
                (#:italic
                 (#:fontsize -5
                             (#:musicglyph "space")))))))))))
        ((= (length ratiotwo) 2)
         (markup
          #:line
          (#:hspace
           -4.5
           #:raise
           3.25
           (#:center-column
            (#:concat
             (#:translate
              (cons 3.5 0.5)
              (#:override
               (cons 'thickness 1.5)

               (#:translate
                (cons 0 -0.5 )
                (#:draw-line (cons 0 -0.5)))
```

```
             )
            #:hspace
            -0.1
            #:override
            (cons 'thickness 1.5)
            (#:translate
             (cons -3.5 0)
             (#:draw-line (cons -4.25 0))))
           #:vspace
           -0.35
           #:whiteout
           (#:halign
            1.75
            (#:concat
             (#:fontsize
              -1
              (#:italic (#:fontsize -5 (#:musicglyph "space")))
              #:fontsize
              -1
              #:italic
              (if (= (length (car ratiotwo)) 2)
                  (begin (string-append
                          (number->string (car (car ratiotwo)))
                          ":"
                          (number->string (cadr (car ratiotwo))))  )
                  (number->string (car (car ratiotwo)))
                  )
              (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))
        #:line
        (#:hspace
         -4.8
         #:raise
         4.5
         (#:center-column
          (#:concat
           (#:translate
            (cons 3.5 0.5)
            (#:override
             (cons 'thickness 1.5)
             (#:translate
              (cons 0 -0.5 )
              (#:draw-line (cons 0 -0.5)))
             )
            #:hspace
            -0.1
            #:override
            (cons 'thickness 1.5)
            (#:translate
```

```
                (cons -3.5 0)
                (#:draw-line (cons -4.25 0)))
               )
              #:vspace
              -0.35
              #:whiteout
              (#:halign
               1.75
               (#:concat
                (#:fontsize
                 -1
                 (#:italic (#:fontsize -5 (#:musicglyph "space")))
                 #:fontsize
                 -1
                 #:italic
                 (if (= (length (cadr ratiotwo)) 2)
                     (begin (string-append
                             (number->string (car (cadr ratiotwo)))
                             ":"
                             (number->string (cadr (cadr ratiotwo)))))
                     (number->string (car (cadr ratiotwo)))
                     )

                 (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))
           ))
         )
      }
      {
       \hspace #-4.25
       \override #'(flag-style . modern-straight-flag)
       \note { $notetwo } #(if (= (ly:duration-log notetwo) 4) 1 1 )
      }
      {
       \hspace #0.5
       \combine
       \draw-line #'(-2 . 0)
       \musicglyph "arrowheads.open.01"
      }
     }
    }
   #})
  )


MModEquationBeginSTR =
#(define-music-function
  (notevalue1 ratio1 notevalue2 ratio2 verticaloffset )
  (ly:duration? list? ly:duration? list? number? )
```

```
   (let* (
          (noteone notevalue1)
          (notetwo notevalue2)
          (ratioone ratio1)
          (ratiotwo ratio2)
          )
    #{

     \tweak Y-offset #verticaloffset
     \textMark  \markup \left-align {

      \raise #0
      \fontsize #-4.5
      \override #'(self-alignment . LEFT)
      {
       {
        \hspace #(if (< (ly:duration-log noteone) 3) -4.15
                     (+ (* (ly:duration-dot-count noteone) -0.5) -4.5))
        \concat {
         \combine
         \musicglyph "arrowheads.open.0M1"
         \draw-line #'(2 . 0) \fontsize #-5 \musicglyph "space"
        }
       }

       {
        \hspace #-0.5
        \override #'(flag-style . modern-straight-flag)
        \note { $noteone } #(if (= (ly:duration-log noteone) 4) 1 1 )
       }


       {
        \hspace #(cond
                  ((<= (ly:duration-dot-count noteone) 1) -0.5)
                  ((> (ly:duration-dot-count noteone) 1)
                   (+ (* (ly:duration-dot-count noteone) -0.5) -0.25)))
        \right-align
        \raise #(cond
                 ((= (ly:duration-log noteone) 0) -2)
                 ((and (> (ly:duration-log noteone) 0)
                       (<= (ly:duration-log noteone) 4) ) -.5)
                 ((> (ly:duration-log noteone) 4)
                  (+ (* (- (ly:duration-log noteone) 5) 0.6) 0))
                 )

        #(cond
          ((= (length ratioone) 1)
```

```
 (markup
  #:line
  (#:hspace
   -4.5
   #:raise
   2.75
   (#:center-column
    (#:concat
     (#:translate
      (cons 3.5 0.5)
      (#:override
       (cons 'thickness 1.5)
       (#:translate
        (cons -3.5 0)
        (#:draw-line (cons -4.25 0))))
      #:hspace
      -0.1
      #:override
      (cons 'thickness 1.5)
      (#:translate
       (cons 0 0.5)
       (#:draw-line (cons 0 -0.5))))
     #:vspace
     -0.45
     #:whiteout
     (#:halign
      1.75
      (#:concat
       (#:fontsize
        -1
        (#:italic (#:fontsize -5 (#:musicglyph "space")))
        #:fontsize
        -1
        #:italic
        (if (= (length (car ratioone)) 2)
            (begin (string-append
                    (number->string (car (car ratioone)))
                    ":"
                    (number->string (cadr (car ratioone))))  )
            (number->string (car (car ratioone)))
            )

        (#:italic
         (#:fontsize -5
                     (#:musicglyph "space")))))))))))

((= (length ratioone) 2)
 (markup
```

```
            #:line
            (#:hspace
             -4.5
             #:raise
             2.75
             (#:center-column
              (#:concat
               (#:translate
                (cons 3.5 0.5)
                (#:override
                 (cons 'thickness 1.5)
                 (#:translate
                  (cons -3.5 0)
                  (#:draw-line (cons -4.25 0))))
                #:hspace
                -0.1
                #:override
                (cons 'thickness 1.5)
                (#:translate
                 (cons 0 0.5)
                 (#:draw-line (cons 0 -0.5))))
               #:vspace
               -0.45
               #:whiteout
               (#:halign
                1.75
                (#:concat
                 (#:fontsize
                  -1
                  (#:italic (#:fontsize -5 (#:musicglyph "space")))
                  #:fontsize
                  -1
                  #:italic
                  (if (= (length (car ratioone)) 2)
                      (begin (string-append
                              (number->string (car (car ratioone)))
                              ":"
                              (number->string (cadr (car ratioone)))))
                      (number->string (car (car ratioone)))
                      )
                  (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))

            #:line
            (#:hspace
             -4.8
             #:raise
             4
             (#:center-column
```

```
               (#:concat
                (#:translate
                 (cons 3.5 0.5)
                 (#:override
                  (cons 'thickness 1.5)
                  (#:translate
                   (cons -3.5 0)
                   (#:draw-line (cons -4.25 0))))
                 #:hspace
                 -0.1
                 #:override
                 (cons 'thickness 1.5)
                 (#:translate
                  (cons 0 0.5)
                  (#:draw-line (cons 0 -0.5))))
                #:vspace
                -0.45
                #:whiteout
                (#:halign
                 1.75
                 (#:concat
                  (#:fontsize
                   -1
                   (#:italic (#:fontsize -5 (#:musicglyph "space")))
                   #:fontsize
                   -1
                   #:italic
                   (if (= (length (cadr ratioone)) 2)
                       (begin (string-append
                               (number->string (car (cadr ratioone)))
                               ":"
                               (number->string (cadr (cadr ratioone))))  )
                       (number->string (car (cadr ratioone)))
                       )

                   (#:italic (#:fontsize -5 (#:musicglyph "space"))))))))
             ))
           )

        }
        {
         \hspace #(+ (* (ly:duration-dot-count noteone) 0.35) -0.2)
         \fontsize #2 \lower #0.5  "="
        }
        \concat {

         {
          \hspace #(cond ((<= (ly:duration-log notetwo) 2) 4.15 )
```

```
                         ((> (ly:duration-log notetwo) 2) 4.15) )
          \raise #(cond
                   ((= (ly:duration-log notetwo) 0) -2)
                   ((and (> (ly:duration-log notetwo) 0)
                         (<= (ly:duration-log notetwo) 4) ) -.5)
                   ((> (ly:duration-log notetwo) 4)
                    (+ (* (- (ly:duration-log notetwo) 5) 0.6) 0))
                   )

          #(cond
            ((= (length ratiotwo) 1)
             (markup
              #:line
              (#:hspace
               -4.5
               #:raise
               3.25
               (#:center-column
                (#:concat
                 (#:translate
                  (cons 3.5 0.5)
                  (#:override
                   (cons 'thickness 1.5)

                   (#:translate
                    (cons 0 -0.5 )
                    (#:draw-line (cons 0 -0.5)))
                   )

                  #:hspace
                  -0.1

                  #:override
                  (cons 'thickness 1.5)
                  (#:translate
                   (cons -3.5 0)
                   (#:draw-line (cons -4.25 0))))


                 #:vspace
                 -0.35
                 #:whiteout
                 (#:halign
                  1.75
                  (#:concat
                   (#:fontsize
                    -1
                    (#:italic (#:fontsize -5 (#:musicglyph "space")))
```

```
        #:fontsize
        -1
        #:italic
        (if (= (length (car ratiotwo)) 2)
            (begin (string-append
                    (number->string (car (car ratiotwo)))
                    ":"
                    (number->string (cadr (car ratiotwo)))))
            (number->string (car (car ratiotwo)))
            )
        (#:italic
         (#:fontsize -5
                     (#:musicglyph "space")))))))))))

((= (length ratiotwo) 2)
 (markup
  #:line
  (#:hspace
   -4.5
   #:raise
   3.25
   (#:center-column
    (#:concat
     (#:translate
      (cons 3.5 0.5)
      (#:override
       (cons 'thickness 1.5)
       (#:translate
        (cons 0 -0.5 )
        (#:draw-line (cons 0 -0.5)))
       )
      #:hspace
      -0.1
      #:override
      (cons 'thickness 1.5)
      (#:translate
       (cons -3.5 0)
       (#:draw-line (cons -4.25 0))))
     #:vspace
     -0.35
     #:whiteout
     (#:halign
      1.75
      (#:concat
       (#:fontsize
        -1
        (#:italic (#:fontsize -5 (#:musicglyph "space")))
        #:fontsize
```

```
      -1
      #:italic
      (if (= (length (car ratiotwo)) 2)
          (begin (string-append
                  (number->string (car (car ratiotwo)))
                  ":"
                  (number->string (cadr (car ratiotwo)))))
          (number->string (car (car ratiotwo)))
          )
      (#:italic
       (#:fontsize -5
                   (#:musicglyph "space")))))))))

#:line
(#:hspace
 -4.8
 #:raise
 4.5
 (#:center-column
  (#:concat
   (#:translate
    (cons 3.5 0.5)
    (#:override
     (cons 'thickness 1.5)
     (#:translate
      (cons 0 -0.5 )
      (#:draw-line (cons 0 -0.5)))
     )
    #:hspace
    -0.1
    #:override
    (cons 'thickness 1.5)
    (#:translate
     (cons -3.5 0)
     (#:draw-line (cons -4.25 0)))
    )
   #:vspace
   -0.35
   #:whiteout
   (#:halign
    1.75
    (#:concat
     (#:fontsize
      -1
      (#:italic (#:fontsize -5 (#:musicglyph "space")))
      #:fontsize
      -1
      #:italic
```

```
                    (if (= (length (cadr ratiotwo)) 2)
                        (begin (string-append
                                (number->string (car (cadr ratiotwo)))
                                ":"
                                (number->string (cadr (cadr ratiotwo)))))
                        (number->string (car (cadr ratiotwo)))
                        )
                    (#:italic
                     (#:fontsize -5
                                 (#:musicglyph "space")))))))))
              ))
            )
         }
         {
          \hspace #-4
          \override #'(flag-style . modern-straight-flag)
          \note { $notetwo }
          #(if (= (ly:duration-log notetwo) 4) 1 1 )
         }
         {
          \hspace #0.5
          \combine
          \draw-line #'(-2 . 0)
          \musicglyph "arrowheads.open.01"
         }
        }
       }
      }
     #})
    )


 {
  \time 2/4
  \tempo 4 = 60
  c'16
  ^\markup {\translate #'(0 . 10) " "}
  [ c'16 c'16 c'16 ]
  \tuplet 5/4 {c'16 [ c'16 c'16 c'16 c'16 ] }
  \MModEquationSTR 16 #'((5)) 16 #'() #8 #0
  \tempo 4 = 75
  c'16 [ c'16 c'16 c'16 ]
  \tuplet 3/2 {c'8 [ c'8 c'8 ] }
  \override TupletNumber.text = #tuplet-number::calc-fraction-text
  \tuplet 5/4 {c'8 [ c'8 ] \tuplet 4/3 { c'8 [ c'8 c'8 c'8  ] }}
  \MModEquationSTR 8 #'((4 3)(5 4)) 8 #'((3)) 8 #-3
  \revert TupletNumber.text
  \break
```

```
  \MModEquationBeginSTR 8 #'((4 3)(5 4)) 8 #'((3)) #6.5
  \tuplet 3/2 {
   c'8
   ^\markup {\translate #'(0 . 10) " "}
   ^\markup
   {\fontsize #-1 \raise #0.25 \note {4} #1 "= 83.33"}
   [ c'8 c'8 ]
  }
  \tuplet 3/2 {
   c'8 c'8 c'8
   \MModEquationSTR 4 #'() 4. #'() #6 #0
  }
  \time 6/8
  c'8 c'8 c'8 c'4.
  \bar "|."
 }

 \layout {
  \context{
   \Score
   harmonicDots = ##t
   \override Flag.stencil = #modern-straight-flag
   \override MetronomeMark.flag-style = #'modern-straight-flag
   \override StemTremolo.shape = #'beam-like
   \override StemTremolo.slope = #0.4

  }
 }


```

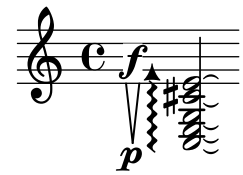
Table of Contents

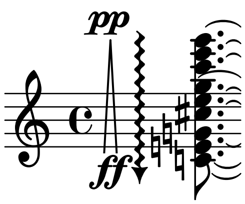
Chapter 8

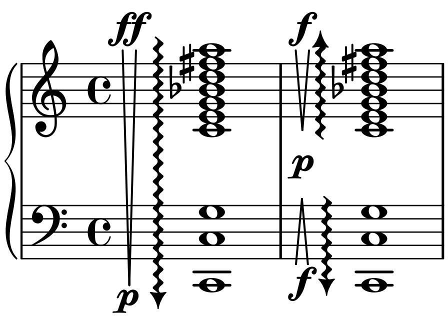
Spanners

This chapter covers snippets that take advantages of spanners (text, line, etc.) in one way or another. Because functions such as `\startTextSpan` and `\stopTextSpan` activate and deactivate these snippets, caution must be paid when using more than one of them at the same time. See Example in **Combinations** to avoid conflicts between or among the spanner snippets.

8.1 Arpeggio with Arrowhead and Vertical Hairpins







8.1.1 Description

Implementation of vertical hairpins associated with the directional arpeggios. LilyPond already provides functionality for the arpeggios with arrowheads.[1] However, I could not find a convenient snippet for the vertical hairpins to be used in tandem with such arpeggios.

In fact, I decided to take the route of drawing vertical hairpin with `\path` function within `\markup`. Should you so desire, you could tweak the value within it to further customize the opening width of the hairpin, for example.

Note that there are two functions, one to be used for single-staff arpeggios, another one for arpeggio that spans over multiple staves. This is because for the latter, the context must be set to `Score` instead of `Staff`.

8.1.2 Grammar

```
 \arpeggioWithVerticalHairpinSingleStaff
  #ARPEGGIO_DIRECTION #CRESCENDO_DIRECTION "DYNAMIC_TOP" "DYNAMIC_BOTTOM" #HAIRPIN_X-
OFFSET
 \arpeggioWithVerticalHairpinMultipleStaves
  #ARPEGGIO_DIRECTION #CRESCENDO_DIRECTION "DYNAMIC_TOP" "DYNAMIC_BOTTOM" #HAIRPIN_X-
OFFSET
```

NB

1. https://lilypond.org/doc/v2.26/Documentation/internals/arpeggio_002dinterface

- `#ARPEGGIO_DIRECTION` sets the direction of the arrowhead. `#UP` places an arrowhead at the top of the arpeggio, while `#DOWN` at the bottom.
- `#CRESCENDO_DIRECTION` defines the direction of the *crescendo as a shape*. To avoid any confusion, it is encouraged to think of this argument independently from the arpeggio direction you previously declared, and to think in a purely visual way (i.e. “How does this crescendo look on the paper, is it looking like `V` or `Λ`?”). For example, `#UP` places a hairpin that looks like `V` while `#DOWN` will place a hairpin that looks like `Λ`.
- `"DYNAMIC_TOP"` and `"DYNAMIC_BOTTOM"` determine which dynamic indications to place at the top and the bottom of the hairpin, respectively. Note that you have to put the dynamics in quotation marks. If you desire no dynamic marking, for either end or both of the hairpin, use `" "`, i.e. surround a space with quotation marks. Declaring `""` will result in warning messages.
- `#HAIRPIN_X-OFFSET` determines the distance between the hairpin and the arpeggio. Normally setting this argument to `#0` should pose no problem.

8.1.3 Code

```
\version "2.26.0"

\language "english"

arpeggioWithVerticalHairpinSingleStaff =
#(define-music-function
  (arpeggioDirection hairpinType dynamicUP dynamicDOWN hairpinOffset )
  (ly:dir? ly:dir? string? string? number? )
  #{
   \once \override Staff.Arpeggio.X-extent =
   #'(-2 . 0.75)
   % NB: Override this value if you wish to adjust
   % the general placement of arpeggio and hairpin
   % as a pair
   \once \override Staff.Arpeggio.stencil =
   #(grob-transformer
     'stencil
     (lambda (grob original)
      (let* ((pos (ly:grob-property grob 'positions))
             (arpeggioArrowStencil
              (if  (eq? arpeggioDirection UP )
                   ;FOR UPWARD ARROW
                   (cons 0.4 (+ (cdr pos) 1.45))
                   ;FOR DOWNWARD ARROW
                   (cons 0.4 (- (car pos) 1.75))
                   )
              )
             (added-markup-one
              (markup
```

```
                 #:overlay
                 (
                  #:translate  arpeggioArrowStencil
                  #:scale  (cons 1.25 1.25)
                  #:arrow-head 1 (if (eq? arpeggioDirection UP ) 1 -1) #t
                  )
                 ))
               (hairpinStencil
                (if (eq? hairpinType UP )
                    ; UNCOMMENT THIS FOR UPWARD CRESCENDO
                    (list (list 'moveto -1  (car pos))
                          (list 'rlineto -0.5   (- (cdr pos) (car pos)))
                          (list 'moveto -1 (car pos))
                          (list 'rlineto 0.5   (- (cdr pos) (car pos)))
                          )
                    ; UNCOMMENT THIS FOR DOWNWARD CRESCENDO
                    (list (list 'moveto -0.5 (car pos))
                          (list 'rlineto -0.5   (- (cdr pos) (car pos)))
                          (list 'moveto -1.5 (car pos))
                          (list 'rlineto 0.5   (- (cdr pos) (car pos)))
                          )
                    ))
               (added-markup-two
                (markup
                 #:overlay
                 (
                  #:general-align X CENTER
                  #:translate-scaled (cons -1.5  (+ (cdr pos) 1))
                  #:whiteout
                  #:dynamic
                  dynamicUP

                  #:general-align X CENTER
                  #:path 0.15 hairpinStencil

                  #:general-align X CENTER
                  #:translate-scaled (cons  -1.5  (- (car pos) 1.5))

                  #:whiteout
                  #:dynamic
                  dynamicDOWN
                  )
                 )
                )
               (added-stencil-one
                (ly:stencil-add original
                                (grob-interpret-markup
                                 grob added-markup-one)))
```

```
            (added-stencil-two
             (grob-interpret-markup grob added-markup-two))
            )

      (if (ly:stencil? original)
          (ly:stencil-combine-at-edge added-stencil-one
                                      X -1
                                      added-stencil-two
                                      hairpinOffset)
          added-stencil
          ))))
  #})

arpeggioWithVerticalHairpinMultipleStaves =
#(define-music-function
  (arpeggioDirection hairpinType dynamicUP dynamicDOWN hairpinOffset )
  (ly:dir? ly:dir? string? string? number? )
  #{
   \once \override Score.Arpeggio.X-extent =
   #'(-2 . 0.75)
   \once \override Score.Arpeggio.stencil =
   #(grob-transformer
     'stencil
     (lambda (grob original)
      (let* ((pos (ly:grob-property grob 'positions))
             (arpeggioArrowStencil
              (if  (eq? arpeggioDirection UP )
                   ;FOR UPWARD ARROW
                   (cons 0.4 (+ (cdr pos) 1.45))
                   ;FOR DOWNWARD ARROW
                   (cons 0.4 (- (car pos) 1.75))
                   )
              )
             (added-markup-one
              (markup
               #:overlay
               (
                #:translate  arpeggioArrowStencil
                #:scale  (cons 1.25 1.25)
                #:arrow-head 1 (if (eq? arpeggioDirection UP ) 1 -1) #t
                )
               ))
             (hairpinStencil
              (if (eq? hairpinType UP )
                  ; UNCOMMENT THIS FOR UPWARD CRESCENDO
                  (list (list 'moveto -1  (car pos))
                        (list 'rlineto -0.5   (- (cdr pos) (car pos)))
                        (list 'moveto -1 (car pos))
```

```
                      (list 'rlineto 0.5   (- (cdr pos) (car pos)))
                      )
                ; UNCOMMENT THIS FOR DOWNWARD CRESCENDO
                (list (list 'moveto -0.5 (car pos))
                      (list 'rlineto -0.5   (- (cdr pos) (car pos)))
                      (list 'moveto -1.5 (car pos))
                      (list 'rlineto 0.5   (- (cdr pos) (car pos)))
                      )
                ))
           (added-markup-two
            (markup
             #:overlay
             (
              #:general-align X CENTER
              #:translate-scaled (cons -1.5  (+ (cdr pos) 1))
              #:whiteout
              #:dynamic
              dynamicUP

              #:general-align X CENTER
              #:path 0.15 hairpinStencil

              #:general-align X CENTER
              #:translate-scaled (cons  -1.5  (- (car pos) 1.5))

              #:whiteout
              #:dynamic
              dynamicDOWN
              )
             )
            )
           (added-stencil-one
            (ly:stencil-add original
                            (grob-interpret-markup
                             grob added-markup-one)))
           (added-stencil-two
            (grob-interpret-markup grob added-markup-two))
           )

     (if (ly:stencil? original)
         (ly:stencil-combine-at-edge added-stencil-one
                                     X -1
                                     added-stencil-two
                                     hairpinOffset)
         added-stencil
         ))))
#})
```

```
{
 \arpeggioWithVerticalHairpinSingleStaff #UP #UP "f" "p" #0
 <c e g cs' e'>2 \arpeggio \laissezVibrer
}
{
 \arpeggioWithVerticalHairpinSingleStaff #DOWN #DOWN "pp" "ff" #0
 \transpose c' c'' { <c! e! g! cs' e' g' c'' e'' g''>8. }
 \arpeggio \laissezVibrer
}

\score {
 \new PianoStaff \with {

 } <<
  \new Staff = "right" \with {  }
  {
   \arpeggioWithVerticalHairpinMultipleStaves #DOWN #UP "ff" "p" #0.25
   \once \set Score.connectArpeggios = ##t
   <c' e' g' bf' d'' fs'' a''>1 \arpeggio |
   \arpeggioWithVerticalHairpinSingleStaff #UP #UP "f" " " #0
   % NB: When you wish to place no dynamic,
   % make sure you include an empty space between quotation marks,
   % otherwise 'warning: Found infinity or nan in output.  Substituting 0.0'
   % message will result.
   <c' e' g' bf' d'' fs'' a''>1 \arpeggio
   \tweak extra-offset #'(-5.25 . -0.75) \p
  }

  \new Staff = "left" \with {
  } {
   \clef bass <c, c g >1 \arpeggio
   \arpeggioWithVerticalHairpinSingleStaff #DOWN #DOWN " " "f" #0.5
   \once \override Staff.Arpeggio.extra-offset = #'(-1.75 . 1)
   % NB: in order to align the hairpin applied to the right hand staff,
   % extra-offset has been applied to offset the arpeggio and hairpin.
   <c, c g >1 \arpeggio
  }
 >>
}
```

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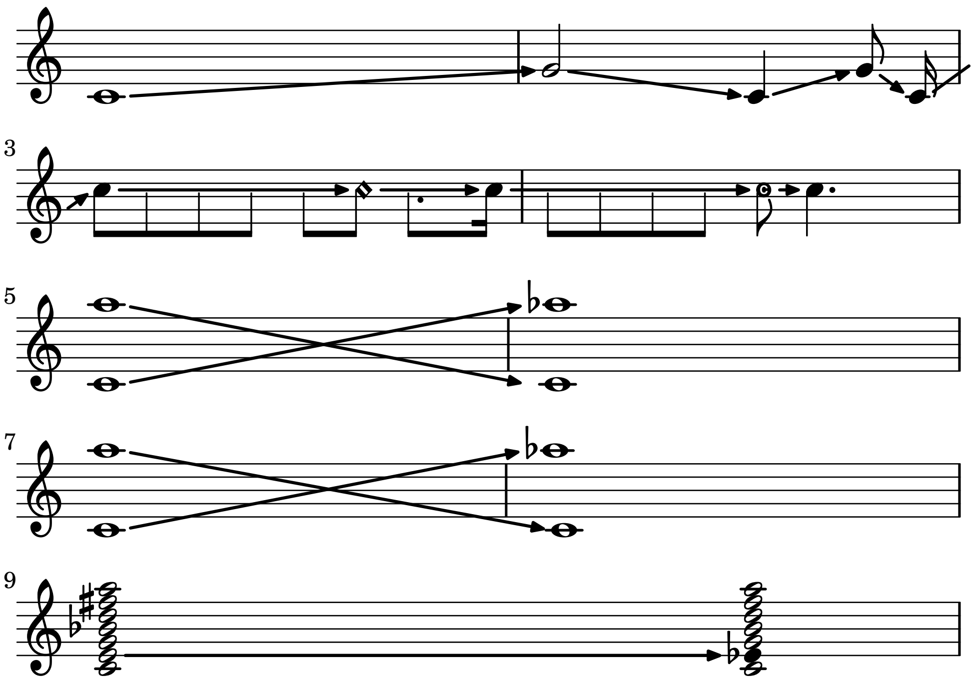
8.2 Arrow Lines



8.2.1 Description

Implementation of arrow lines. It takes advantage of the `\glissando` function. It is possible to have the arrow line span over multiple notes, as `glissando-skip` parameter is set to `##t`. When the arrow line spans over multiple systems, the arrow mark will not appaer at the end of the system.[2] Furthermore, it is possible to use the function on dyads and chords. The placement of the beginning of the arrow is adjusted according to the different types of notehead.

8.2.2 Grammar

```
\arrowLineOn STARTING_NOTE (NOTES...)
\arrowLineOff ARRIVAL_NOTE
```

8.2.3 Code

```

\version "2.26.0"

arrowLineOn =
#(define-music-function (note)(ly:music?)
  (define paddingvalue (if (music-is-of-type? note 'event-chord)
                           (ly:duration-log
                            (ly:music-property
                             (first
                              (ly:music-property note 'elements))
                             'duration))
```

2. See **Discussion** for more details.

```
                            (ly:duration-log
                             (ly:music-property note 'duration))))

   #{

    \override Glissando.breakable = ##t
    \override Glissando.after-line-breaking = ##t
    \override Glissando.thickness = #2.35
    \override Glissando.bound-details.right.arrow = ##t
    \override Glissando.bound-details.right-broken.arrow = ##f
    \override Glissando.bound-details.right-broken.padding = #-1
    \override Glissando.bound-details.left.padding =
    #(cond ((= paddingvalue 0) 0.85)
           ((= paddingvalue 1) 0.65)
           ((>= paddingvalue 2) 0.65))

    \override Glissando.bound-details.right.padding = #0.25
    #note
    \glissando  \override NoteColumn.glissando-skip = ##t
   #})


arrowLineOff =
{
 \revert Glissando.breakable
 \revert Glissando.after-line-breaking
 \revert Glissando.thickness
 \revert Glissando.bound-details.right.arrow
 \revert Glissando.bound-details.right-broken.arrow
 \revert Glissando.bound-details.right-broken.padding
 \revert Glissando.bound-details.left.padding
 \revert Glissando.bound-details.right.padding
 \revert NoteColumn.glissando-skip
}



\score {

 {
  \override Score.TimeSignature.stencil = ##f

  \arrowLineOn
  c'1
  \arrowLineOff

  \arrowLineOn
  g'2
```

```
\arrowLineOff

\arrowLineOn
c'4
\arrowLineOff

\arrowLineOn
g'8 \noBeam
\arrowLineOff
\arrowLineOn
c'16 s16 |

\break
\arrowLineOff
\arrowLineOn
c''8
\override Voice.NoteHead.transparent = ##t
8 8 8 8
\revert Voice.NoteHead.transparent
\arrowLineOff

\arrowLineOn
8 \harmonic
\override Voice.NoteHead.transparent = ##t
\once \override Voice.Dots.extra-offset = #'(-1 . -0.75)

8.
\revert Voice.NoteHead.transparent
\arrowLineOff

\arrowLineOn
16

\override Voice.NoteHead.transparent = ##t
8 8 8 8
\revert Voice.NoteHead.transparent
\arrowLineOff
\easyHeadsOn
\arrowLineOn
8
\arrowLineOff
\easyHeadsOff
4.
\break
\arrowLineOn
<c' a''>1

\arrowLineOff
```

```
    <aes'' c'>1

    <<
     { \arrowLineOn a''1 \arrowLineOff c'1} \\
     {\arrowLineOn c'1 \arrowLineOff aes''1}
    >>

    \break
    \override Voice.Stem.stencil = ##f
    \override Voice.NoteHead.stencil = #ly:text-interface::print
    \override Voice.NoteHead.text =\markup{\musicglyph "noteheads.s1"}
    \set glissandoMap = #'((1 . 1) (1 . 1))
    \arrowLineOn
    <c' e' g' bes' d'' fis'' a''>2
    s4
    \arrowLineOff
    <c'
    \single \override NoteHead.text =
    \markup{\musicglyph "noteheads.s2"}  es'
    g' bes' d'' fis'' a''>4

  }


  \layout {

    indent = #0
    line-width = #125
    ragged-last = ##f

    \context {
     \Score
     proportionalNotationDuration = #1/7
    }
  }
}

```

8.2.4 Discussion

There are a few things to consider when using the arrow lines on dyads and chords:

- By default, all pair of notes will have arrow lines. In order to selectively show the arrow lines, use `\set glissandoMap`. See **1.3.3 Expressive marks as lines** in LilyPond's *Notation Reference* for details.
- Just as the ordinary `\glissando` function, the X coordinate of the terminating point for *all* of the lines between two dyads/chords is determined by the presence of accidentals in the arrival dyads/chords. Thus, if there is an accidental on one or more of the pitches in the

arriving dyad/chord, there may be a space between the tip of arrow and the pitches *without* the accidentals. Should this be avoided, it is best to apply the arrow lines in different layers, so that each of the layers will have a different X-coordinate value of the terminating point of the arrow lines.

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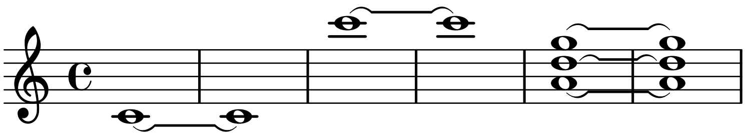
8.3 Boulez Tie



8.3.1 Description

Implementation of ties seen in Movement 5 *Tombeau* of *Pli selon pli* by Pierre Boulez.[3] The shape is roughly a combination of a short, *laissez vibrer* sign on both originating and arriving notes, connected by a horizontal line. In the score, these are manually written in, and the shape is somewhat irregular.

8.3.2 Grammar

```
\override Tie.stencil = \boulezTie
```

8.3.3 Code

```
\version "2.26.0"

boulezTie =
#(lambda (grob)

  (let (
        ;defines four control points, each of which containing
        ;x and y, defined from a to h...
        (a (caar (ly:grob-property grob 'control-points)) )
        (b  (cdar (ly:grob-property grob 'control-points)) )
        (c (car (cadr (ly:grob-property grob 'control-points))))
        (d (cdr (cadr (ly:grob-property grob 'control-points))))
        (e (car (caddr (ly:grob-property grob 'control-points))))
        (f (cdr (caddr (ly:grob-property grob 'control-points))))
        (g (car (cadddr (ly:grob-property grob 'control-points))))
        (h (cdr (cadddr (ly:grob-property grob 'control-points))))
        )
   (make-path-stencil

    (list 'moveto a b
          'curveto a b (* (+ a (+ c 0.5)) 0.5) d (+ c 0.5) b
          'lineto (- e 0.5) b
          'curveto (- e 0.5) b (+ (- e 0.5) (* (- g (- e 0.5)) 0.5)) f g h

          'curveto g h (+ (- e 0.5) (* (- g (- e 0.5)) 0.6))
          (+ f 0.1) (- e 0.5) (+ b 0.15)
```

3. Pierre Boulez, *Pli selon pli: V. Tombeau* (London: Universal Edition, 1971).

```
            'lineto (+ c 0.5) (+ b 0.15)
            'curveto (+ c 0.5) (+ b 0.15) (* (+ a (+ c 0.5)) 0.5) (+ d 0.1) a b
            'closepath
            )

      0.01
      1
      1
      #t
      )
     )
    )

{

 \override Tie.stencil = \boulezTie

 c'1 ~ c'1
 c''' ~ c'''
 <a' d'' g''> ~ <a' d'' g''>
}
```

8.3.4 Discussion

It is possible to adjust the shape of the tie using `\shape` and `\vshape` commands.

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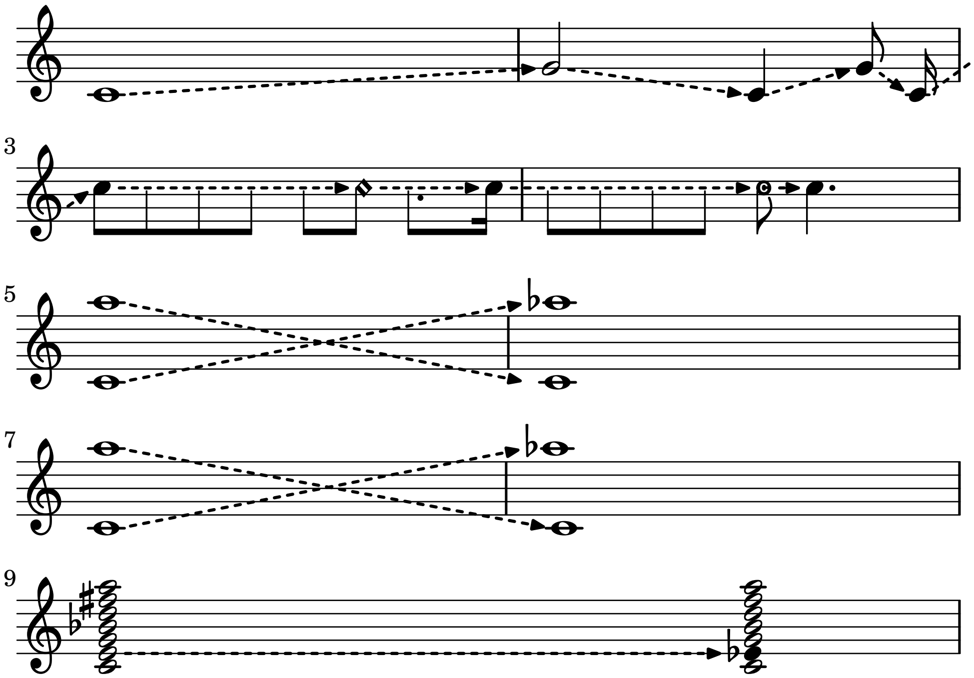
8.4 Dashed Arrow Lines



8.4.1 Description

Implementation of dashed arrow lines. Its design is nearly identical to that of the **Arrow Lines**.

8.4.2 Grammar

```
\dashedArrowLineOn STARTING_NOTE (NOTES...)
\dashedArrowLineOff ARRIVAL_NOTE
```

8.4.3 Code

```
\version "2.26.0"

dashedArrowLineOn =
#(define-music-function (note)(ly:music?)
  (define paddingvalue (if (music-is-of-type? note 'event-chord)
                           (ly:duration-log
                            (ly:music-property
                             (first
                              (ly:music-property note 'elements))
                             'duration))
                           (ly:duration-log
                            (ly:music-property note 'duration))))

  #{

   \override Glissando.breakable = ##t
```

```
   \override Glissando.after-line-breaking = ##t
   \override Glissando.thickness = #2.35
   \override Glissando.style = #'dashed-line
   \override Glissando.bound-details.right.arrow = ##t
   \override Glissando.bound-details.right-broken.arrow = ##f
   \override Glissando.bound-details.right-broken.padding = #-1
   \override Glissando.bound-details.left.padding =
   #(cond ((= paddingvalue 0) 0.85)
          ((= paddingvalue 1) 0.65)
          ((>= paddingvalue 2) 0.65))

   \override Glissando.bound-details.right.padding = #0.25
   #note
   \glissando  \override NoteColumn.glissando-skip = ##t
  #})


dashedArrowLineOff =
{
 \revert Glissando.breakable
 \revert Glissando.after-line-breaking
 \revert Glissando.thickness
 \revert Glissando.style
 \revert Glissando.bound-details.right.arrow
 \revert Glissando.bound-details.right-broken.arrow
 \revert Glissando.bound-details.right-broken.padding
 \revert Glissando.bound-details.left.padding
 \revert Glissando.bound-details.right.padding
 \revert NoteColumn.glissando-skip
}



\score {

 {
  \override Score.TimeSignature.stencil = ##f

  \dashedArrowLineOn
  c'1
  \dashedArrowLineOff

  \dashedArrowLineOn
  g'2
  \dashedArrowLineOff

  \dashedArrowLineOn
  c'4
```

```
\dashedArrowLineOff

\dashedArrowLineOn
g'8 \noBeam
\dashedArrowLineOff
\dashedArrowLineOn
c'16 s16 |

\break
\dashedArrowLineOff
\dashedArrowLineOn
c''8
\override Voice.NoteHead.transparent = ##t
8 8 8 8
\revert Voice.NoteHead.transparent
\dashedArrowLineOff

\dashedArrowLineOn
8 \harmonic
\override Voice.NoteHead.transparent = ##t
\once \override Voice.Dots.extra-offset = #'(-1 . -0.75)

8.
\revert Voice.NoteHead.transparent
\dashedArrowLineOff

\dashedArrowLineOn
16

\override Voice.NoteHead.transparent = ##t
8 8 8 8
\revert Voice.NoteHead.transparent
\dashedArrowLineOff
\easyHeadsOn
\dashedArrowLineOn
8
\dashedArrowLineOff
\easyHeadsOff
4.
\break
\dashedArrowLineOn
<c' a''>1

\dashedArrowLineOff
<aes'' c'>1

<<
 { \dashedArrowLineOn a''1 \dashedArrowLineOff c'1} \\
```

```
    {\dashedArrowLineOn c'1 \dashedArrowLineOff aes''1}
   >>
   \break
   \override Voice.Stem.stencil = ##f
   \override Voice.NoteHead.stencil = #ly:text-interface::print
   \override Voice.NoteHead.text =\markup{\musicglyph "noteheads.s1"}
   \set glissandoMap = #'((1 . 1) (1 . 1))
   \dashedArrowLineOn
   <c' e' g' bes' d'' fis'' a''>2
   s4
   \dashedArrowLineOff
   <c'
   \single \override NoteHead.text =
   \markup{\musicglyph "noteheads.s2"}  es'
   g' bes' d'' fis'' a''>4

  }

  \layout {

   indent = #0
   line-width = #125

   ragged-last = ##f

   \context {
    \Score
    proportionalNotationDuration = #1/7
   }
  }
 }
```

8.4.4 Discussion

See **Discussion** of the Arrow Lines. **Table of Contents**

8.5 Dashed Flat Slurs

8.5.1 Description

See **Flat Slurs** for the ordinary non-dashed version.

Because the **Flat Slurs** is based on a rewritten stencil, further overrides within `Slur`, such as Making slurs with complex dash structure will not work. This particular implementation addresses the issue of achieving dashed lines for the flat slurs. Here, three different stencils are created, defined as variables, then combined using `ly:stencil-add`.

See **Flat Slurs** for the ordinary non-dashed version, as well as **Discussion** for hints on tweaking.

8.5.2 Grammar

`\override Slur.stencil = \flatSlur`

8.5.3 Code

```
\version "2.26.0"

dashedFlatSlur =
#(lambda (grob)

  (let* (
          ;defines four control points, each of which containing
          ;x and y, defined from a to h...
          (a (caar (ly:grob-property grob 'control-points)) )
          (b (cdar (ly:grob-property grob 'control-points)) )
          (c (car (cadr (ly:grob-property grob 'control-points))))
          (d (cdr (cadr (ly:grob-property grob 'control-points))))
          (e (car (caddr (ly:grob-property grob 'control-points))))
```

```
       (f (cdr (caddr (ly:grob-property grob 'control-points))))
       (g (car (cadddr (ly:grob-property grob 'control-points))))
       (h (cdr (cadddr (ly:grob-property grob 'control-points))))
       (firstCurve (make-path-stencil
                    (list 'moveto a b
                          'curveto a b a (- d 0.2) c (- d 0.2)
                          'lineto c (- d 0.4)
                          'curveto c (- d 0.4) (+ a 0.1) (- d 0.3) a b
                          'closepath
                          )
                    0.01
                    1
                    1
                    #t
                    ))
       (secondCurve (make-path-stencil
                     (list 'moveto e (- f 0.2)
                           'curveto e (- f 0.2) g (- f 0.2) g h
                           'curveto g h (+ g 0.1) (- f 0.5) e (- f 0.4)
                           'closepath
                           )
                     0.01
                     1
                     1
                     #t
                     ))

       (dashedLine (grob-interpret-markup
                    grob
                    (markup
                     #:line
                     (#:postscript
                      (string-append
                       "0.2 setlinewidth [0.4 0.4] 0 setdash "
                       (number->string (+ c 0.2) ) " "
                       (number->string (- d 0.3))
                       " moveto "
                       (number->string (- e 0.2)) " "
                       (number->string (- f 0.3))
                       " lineto stroke"))
                     ))
                   )
       )
 (ly:stencil-add firstCurve dashedLine secondCurve

                 )
 )
)
```

```
\layout {
 indent = 0
 line-width = 80
}
{
 \time 2/4
 \override Slur.stencil = \dashedFlatSlur

 \override Slur.details =
 #'(
    (max-slope . 0)
    (max-slope-factor . 100)
    (free-head-distance . 0.2)
    (free-slur-distance . 0)
    (extra-encompass-free-distance . 0.3)
    (extra-encompass-collision-distance . 0.8)
    (head-slur-distance-max-ratio . 3)
    (head-slur-distance-factor . 10)
    )

 \shape #'(
           (0 . 1)
           (0 . 0.5)
           (0 . 0.5)
           (0 . 0.25)
           ) Slur

 e'16 ( d' c' d' e' d' c'8 )

 \shape #'(
           (0 . 0)
           (0 . -0.5)
           (0 . -0.5)
           (0 . -1)
           ) Slur
 g''16 ( f'' e'' f'' g'' f'' e''8 ) \break


 \shape #'(
           (( 0 . 0) (0 . 0) (0 . 0) (0 . 0))
           ((-1.25 . 3.5) (-1.25 . 3.75)(0 . 3.75)(0 . 2.5))
           ) Slur
 c''4 ( b' a' g' ~ | \break

 g'8 )

 \shape #'(
           (0 . 0)
```

```
            (0 . 0.2)
            (0.75 . 2.45)
            (0 . 3)
            ) Slur


  a ( c' e' g' b' ) r4 |


 }
```

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8.6 Flat Slurs

8.6.1 Description

Implementation of a straight-line slur in which only the beginning and ending parts are curved. This type of slurs can be found in scores by such composers as Brian Ferneyhough, Claus-Steffen Mahnkopf, and Donald Martino, where such slurs were manually drawn using a ruler. Even in the age of digital musical engraving, composers such as Stefan Beyer have emulated this slur using the musical engraving software programs.

This flat slur requires careful tweaking in order to make it look optimal on the score. This is discussed further in **Discussion**.

See also **Dashed Flat Slurs**.

8.6.2 Grammar

```
\override Slur.stencil = \flatSlur
```

8.6.3 Code

```
\version "2.26.0"

flatSlur =
#(lambda (grob)

  (let (
        ;defines four control points, each of which containing
        ;x and y, defined from a to h...
        (a (caar (ly:grob-property grob 'control-points)) )
        (b  (cdar (ly:grob-property grob 'control-points)) )
        (c (car (cadr (ly:grob-property grob 'control-points))))
        (d (cdr (cadr (ly:grob-property grob 'control-points))))
        (e (car (caddr (ly:grob-property grob 'control-points))))
```

```
         (f (cdr (caddr (ly:grob-property grob 'control-points))))
         (g (car (cadddr (ly:grob-property grob 'control-points))))
         (h (cdr (cadddr (ly:grob-property grob 'control-points))))
         )
    (make-path-stencil
     (if (= (ly:grob-property grob 'direction) 1)
         ;this is if the slur/tie direction is up...
         (list 'moveto a b
               'curveto a b a (- d 0.2) c (- d 0.2)
               'lineto e (- f 0.2)
               'curveto e (- f 0.2) g (- f 0.2) g h

               'curveto g h (+ g 0.1) (- f 0.5) e (- f 0.4)
               'lineto c (- d 0.4)
               'curveto c (- d 0.4) (+ a 0.1) (- d 0.3) a b
               'closepath
               )
         ;if the direction is down...
         (list 'moveto a b
               'curveto a b a (+ d 0.2) c (+ d 0.2)
               'lineto e (+ f 0.2)

               'curveto e (+ f 0.2) g (+ f 0.2) g h

               'curveto g h (- g 0.1) (+ f 0.5) e (+ f 0.4)
               'lineto c (+ d 0.4)
               'curveto c (+ d 0.4) (+ a 0.3) (+ d 0.3) a b
               'closepath
               ))

     0.01
     1
     1
     #t
     )
    )
   )
\layout {
 indent = 0
 line-width = 80
}
{
 \time 2/4
 \override Slur.stencil = \flatSlur

 \override Slur.details =
 #'(
    (max-slope . 0)
```

```
    (max-slope-factor . 100)
    (free-head-distance . 0.2)
    (free-slur-distance . 0)
    (extra-encompass-free-distance . 0.3)
    (extra-encompass-collision-distance . 0.8)
    (head-slur-distance-max-ratio . 3)
    (head-slur-distance-factor . 10)
    )

 \shape #'(
          (0 . 1)
          (0 . 0.5)
          (0 . 0.5)
          (0 . 0.25)
          ) Slur

 e'16 ( d' c' d' e' d' c'8 )

 \shape #'(
          (0 . 0)
          (0 . -0.5)
          (0 . -0.5)
          (0 . -1)
          ) Slur
 g''16 ( f'' e'' f'' g'' f'' e''8 ) \break



 c''4 ( b' a' g' ~ | \break

 g'8 )

 \shape #'(
          (0 . 0)
          (0 . 0.2)
          (0.75 . 2.5)
          (0 . 3)
          ) Slur


 a ( c' e' g' b' ) r4 |


}
```

8.6.4 Discussion

While this code can be used out of box by overriding `Slur.stencil` to `\flatSlur`, you may find that tweak will be needed. Initial experiments have shown that the following overrides of `Slur.details` help:

```
\override Slur.details =
#'(
   (max-slope . 0)
   (max-slope-factor . 100)
   (free-head-distance . 0.2)
   (free-slur-distance . 0)
   (extra-encompass-free-distance . 0.3)
   (extra-encompass-collision-distance . 0.8)
   (head-slur-distance-max-ratio . 3)
   (head-slur-distance-factor . 10)
   )
```

There may be other effective `Slur.details` configurations.

Beyond this, further refinements can be made using `\shape` and `\vshape`.

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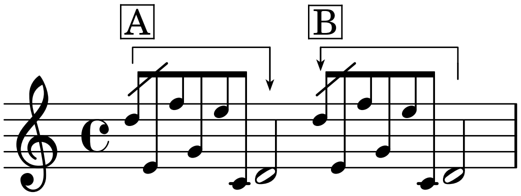
8.7 Grace Note Brackets I



8.7.1 Description

NB: See Grace Note Brackets II for the updated version of this code.) Replication of grace note brackets seen in scores by Pierre Boulez (e.g. *Sur Incises*,[4] *...explosante-fixe...*[5]). Bracket A in the example shows that the grace notes are to be played *before* the beat to which they are applied. Whereas Bracket B shows that the grace notes are to be played *on* the beat to which they are applied.

8.7.2 Grammar

```
\graceNoteBeforeBeatOn NOTE
\graceNoteBeforeBeatOff NOTE
\graceNoteAfterBeatOn NOTE
\graceNoteAfterBeatOff NOTE
```

8.7.3 Code

```
\version "2.26.0"

% This code includes snippet for grace note
% slashes, which has been taken from:
% https://lsr.di.unimi.it/LSR/Item?id=1048


graceNoteBeforeBeatOn =
#(define-music-function (starting_note) (ly:music?)
  #{
   \once \override TextSpanner.style = #'line
   \once \override TextSpanner.bound-details.left.text =
   \markup { \draw-line #'(0 . -1) }
   \once \override TextSpanner.bound-details.right.text =
   \markup {
    \postscript
    "newpath 0 0 moveto
0 -2.5 rlineto
```

4. Pierre Boulez, *Sur incises : pour trois pianos, trois harpes et trois percussions-claviers (1996/1998)* (Universal Edition, 1998).

5. Pierre Boulez, *... explosante-fixe ... transitoire VII : (version 1991/93)* (Universal Edition, 1991).

```
stroke
newpath
-0.275 -2 moveto
0.275 -0.75 rlineto
0.275 0.75 rlineto
-0.275 -0.2 rlineto
closepath
fill"
   }
   \once \override TextSpanner.Y-offset = #5
   \once \override TextSpanner.bound-details.left.padding = #0.5
   \once \override TextSpanner.bound-details.right.padding = #-0.25
   #starting_note
   \startTextSpan
  #})


graceNoteBeforeBeatOff =
#(define-music-function (ending_note) (ly:music?)
  #{
   #ending_note
   \stopTextSpan
  #})


graceNoteAfterBeatOn =
#(define-music-function (starting_note) (ly:music?)
  #{
   \once \override TextSpanner.style = #'line
   \once \override TextSpanner.bound-details.right.text =
   \markup {
    \combine \draw-line #'(0 . -1)
    \postscript "newpath
0 -1 moveto
0 -1 rlineto
stroke"
   }
   \once \override TextSpanner.bound-details.left.text =
   \markup {
    \postscript
    "newpath 0 0 moveto
0 -1 rlineto
stroke
newpath
-0.275 -0.75 moveto
0.275 -0.75 rlineto
0.275 0.75 rlineto
-0.275 -0.2 rlineto
```

```
closepath
fill"
   }
   \once \override TextSpanner.Y-offset = #2
   \once \override TextSpanner.bound-details.left.padding = #0.5
   \once \override TextSpanner.bound-details.right.padding = #-0.25
   #starting_note
   \startTextSpan
  #})


graceNoteAfterBeatOff =
#(define-music-function (ending_note) (ly:music?)
  #{
   #ending_note
   \stopTextSpan
  #})

%%%%%%%%%%%%%%%%%%LSR SNIPPET START%%%%%%%%%%%%%%%%%

#(define (degrees->radians deg)
  (* PI (/ deg 180.0)))

slash =
#(define-music-function (ang stem-fraction protrusion)
  (number? number? number?)
  (remove-grace-property 'Voice 'Stem 'direction)
  #{
   \once \override Stem.stencil =
   #(lambda (grob)
     (let* ((x-parent (ly:grob-parent grob X))
            (is-rest? (ly:grob?
                       (ly:grob-object x-parent 'rest)))
            (beam (ly:grob-object grob 'beam))
            (stil (ly:stem::print grob)))
      (cond
       (is-rest? empty-stencil)
       ((ly:grob? beam)
        (let* ((refp (ly:grob-system grob))
               (stem-y-ext (ly:grob-extent grob grob Y))
               (stem-length
                (- (cdr stem-y-ext) (car stem-y-ext)))
               (beam-X-pos (ly:grob-property beam 'X-positions))
               (beam-Y-pos (ly:grob-property beam 'positions))
               (beam-slope (/ (- (cdr beam-Y-pos) (car beam-Y-pos))
                              (- (cdr beam-X-pos) (car beam-X-pos))))
               (beam-angle (atan beam-slope))
               (dir (ly:grob-property grob 'direction))
```

```
               (line-dy (* stem-length stem-fraction))
               (line-dy-with-protrusions (if (= dir 1)
                                            (+ (* 4 protrusion) beam-angle)
                                            (- (* 4 protrusion) beam-angle)))
               (ang (if (> beam-slope 0)
                        (if (= dir 1)
                            (+ (degrees->radians ang) (* beam-angle 0.7))
                            (degrees->radians ang))
                        (if (= dir 1)
                            (degrees->radians ang)
                            (- (degrees->radians ang) (* beam-angle 0.7)))))
               (line-dx (/ line-dy-with-protrusions (tan ang)))
               (protrusion-dx (/ protrusion (tan ang)))
               (corr (if (= dir 1) (car stem-y-ext) (cdr stem-y-ext))))
          (ly:stencil-add
           stil
           (grob-interpret-markup grob
                                  (markup
                                   #:translate
                                   (cons (- protrusion-dx)
                                         (+ corr
                                            (* dir
                                               (- stem-length
                                                  (+ stem-fraction protrusion)))))
                                   #:override '(thickness . 1.7)
                                   #:draw-line
                                   (cons line-dx
                                         (* dir line-dy-with-protrusions)))))))
        (else stil))))
   #})

startSlashedGraceMusic = {
 \slash 40 1 0.5
 \override Flag.stroke-style = #"grace"
}
stopSlashedGraceMusic = {
 \revert Flag.stroke-style
}

startAcciaccaturaMusic = {
 \slash 40 1 0.5
 s1*0(
 \override Flag.stroke-style = #"grace"
}
stopAcciaccaturaMusic = {
 \revert Flag.stroke-style
 s1*0)
}
```

```
%%%%%%%%%%%%%%%%%%% LSR SNIPPET END %%%%%%%%%%%%%%%%%%

{
 \grace {
  \startSlashedGraceMusic
  \graceNoteBeforeBeatOn d''8^\markup{\box A} e' f'' g' e'' c'
 }
 \graceNoteBeforeBeatOff d'2
 \grace {
  \startSlashedGraceMusic
  \graceNoteAfterBeatOn d''8^\markup{\box B} e' f'' g' e'' c'
 }
 \graceNoteAfterBeatOff d'2
}
```

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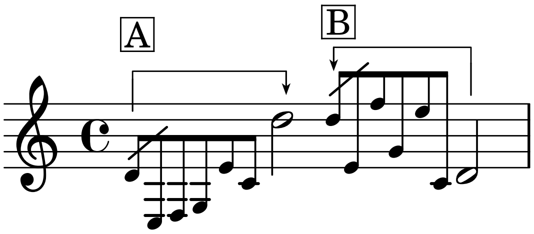
8.8 Grace Note Brackets II



8.8.1 Description

This is an updated version of Grace Note Brackets I. It differs from the original version in that this version takes a list of three parameters in order to finely adjust the shape of the bracket in order to accommodate various shapes of grace notes and the actual note.

8.8.2 Grammar

```
\graceNoteBeforeBeatOn #'(OVERALL LEFT RIGHT) NOTE
\graceNoteBeforeBeatOff #'(OVERALL LEFT RIGHT) NOTE
\graceNoteAfterBeatOn #'(OVERALL LEFT RIGHT) NOTE
\graceNoteAfterBeatOff #'(OVERALL LEFT RIGHT) NOTE
```

NB The list accepts three integers as parameters, i.e.:

1. `OVERALL` is a value of the distance between the top line of the staff and the horizontal line of the grace note bracket. This value cannot be smaller than the skyline value established by the staff line and the notes; when the skyline value is greater than what is specified in this bracket, the skyline value is favored. When in doubt, start with `0`, then increase the amount gradually.
2. `LEFT` and `RIGHT` values (negative value only!) adjust the lengths of the left and right hooks.

8.8.3 Code

```
\version "2.26.0"

% This code includes snippet for grace note
% slashes, which has been taken from:
% https://lsr.di.unimi.it/LSR/Item?id=1048

% Slightly revised, Jan. 19/22 2025 - YO

graceNoteBeforeBeatOn =
#(define-music-function (setting-list starting_note) (list? ly:music? )
  #{
   \once \override TextSpanner.style = #'line
   \once \override TextSpanner.bound-details.left.text =
   \markup {
```

```
    \combine
    \draw-line #(cons 0 -0.5)
    \postscript #(string-append "newpath
0 -0.5 moveto
0 " (number->string (cadr setting-list)) " rlineto
stroke")
   }
   \once \override TextSpanner.bound-details.right.text =
   \markup {
    \postscript
    #(string-append "newpath 0 0 moveto
0 " (number->string (caddr setting-list))  " rlineto
stroke
newpath
-0.275 " (number->string (+ (caddr setting-list) 0.25))  " moveto
0.275 -0.75 rlineto
0.275 0.75 rlineto
-0.275 -0.2 rlineto
closepath
fill")
   }
   \once \override TextSpanner.extra-offset = #(cons 0 (car setting-list))
   \once \override TextSpanner.bound-details.left.padding = #0.5
   \once \override TextSpanner.bound-details.right.padding = #-0.25
   #starting_note
   \startTextSpan
  #})


graceNoteBeforeBeatOff =
#(define-music-function (ending_note) (ly:music?)
  #{
   #ending_note
   \stopTextSpan
  #})


graceNoteAfterBeatOn =
#(define-music-function (setting-list starting_note) (list? ly:music?)
  #{
   \once \override TextSpanner.style = #'line
   \once \override TextSpanner.bound-details.right.text =
   \markup {
    \combine
    \draw-line #(cons 0 -1)
    \postscript #(string-append "newpath
0 -1 moveto
0 " (number->string (caddr setting-list)) " rlineto
```

```
stroke")
   }
   \once \override TextSpanner.bound-details.left.text =
   \markup {
    \postscript
    #(string-append "newpath 0 0 moveto
0 " (number->string (cadr setting-list))  " rlineto
stroke
newpath
-0.275 " (number->string (+ (cadr setting-list) 0.25))  " moveto
0.275 -0.75 rlineto
0.275 0.75 rlineto
-0.275 -0.2 rlineto
closepath
fill")
   }
   \once \override TextSpanner.extra-offset = #(cons 0 (car setting-list))
   \once \override TextSpanner.bound-details.left.padding = #0.5
   \once \override TextSpanner.bound-details.right.padding = #-0.25
   #starting_note
   \startTextSpan
  #})


graceNoteAfterBeatOff =
#(define-music-function (ending_note) (ly:music?)
  #{
   #ending_note
   \stopTextSpan
  #})

%%%%%%%%%%%%%%%%%%%LSR SNIPPET START%%%%%%%%%%%%%%%%%%

#(define (degrees->radians deg)
  (* PI (/ deg 180.0)))

slash =
#(define-music-function (ang stem-fraction protrusion)
  (number? number? number?)
  (remove-grace-property 'Voice 'Stem 'direction)
  #{
   \once \override Stem.stencil =
   #(lambda (grob)
     (let* ((x-parent (ly:grob-parent grob X))
            (is-rest? (ly:grob?
                       (ly:grob-object x-parent 'rest)))
            (beam (ly:grob-object grob 'beam))
            (stil (ly:stem::print grob)))
```

```
      (cond
       (is-rest? empty-stencil)
       ((ly:grob? beam)
        (let* ((refp (ly:grob-system grob))
               (stem-y-ext (ly:grob-extent grob grob Y))
               (stem-length
                (- (cdr stem-y-ext) (car stem-y-ext)))
               (beam-X-pos (ly:grob-property beam 'X-positions))
               (beam-Y-pos (ly:grob-property beam 'positions))
               (beam-slope (/ (- (cdr beam-Y-pos) (car beam-Y-pos))
                              (- (cdr beam-X-pos) (car beam-X-pos))))
               (beam-angle (atan beam-slope))
               (dir (ly:grob-property grob 'direction))
               (line-dy (* stem-length stem-fraction))
               (line-dy-with-protrusions (if (= dir 1)
                                             (+ (* 4 protrusion) beam-angle)
                                             (- (* 4 protrusion) beam-angle)))
               (ang (if (> beam-slope 0)
                        (if (= dir 1)
                            (+ (degrees->radians ang) (* beam-angle 0.7))
                            (degrees->radians ang))
                        (if (= dir 1)
                            (degrees->radians ang)
                            (- (degrees->radians ang) (* beam-angle 0.7)))))
               (line-dx (/ line-dy-with-protrusions (tan ang)))
               (protrusion-dx (/ protrusion (tan ang)))
               (corr (if (= dir 1) (car stem-y-ext) (cdr stem-y-ext))))
         (ly:stencil-add
          stil
          (grob-interpret-markup grob
                                 (markup
                                  #:translate
                                  (cons (- protrusion-dx)
                                        (+ corr
                                           (* dir
                                              (- stem-length
                                                 (+ stem-fraction protrusion)))))
                                  #:override '(thickness . 1.7)
                                  #:draw-line
                                  (cons line-dx
                                        (* dir line-dy-with-protrusions)))))))
       (else stil))))
  #})

startSlashedGraceMusic = {
 \slash 40 1 0.5
 \override Flag.stroke-style = #"grace"
}
```

```
stopSlashedGraceMusic = {
 \revert Flag.stroke-style
}

startAcciaccaturaMusic = {
 \slash 40 1 0.5
 s1*0(
 \override Flag.stroke-style = #"grace"
}
stopAcciaccaturaMusic = {
 \revert Flag.stroke-style
 s1*0)
}
%%%%%%%%%%%%%%%%%%% LSR SNIPPET END %%%%%%%%%%%%%%%%%%

{
 \grace {
  \startSlashedGraceMusic
  \graceNoteBeforeBeatOn #'(1 -2 -1) d'8^\markup{\translate #'(0 . 3) \box A}
  e f g e' c'
 }
 \graceNoteBeforeBeatOff d''2
 \grace {
  \startSlashedGraceMusic
  \graceNoteAfterBeatOn #'(0 -1 -2) d''8^\markup{\box B} e' f'' g' e'' c'
 }
 \graceNoteAfterBeatOff d'2
}

```

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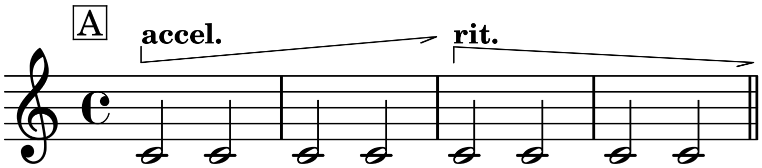
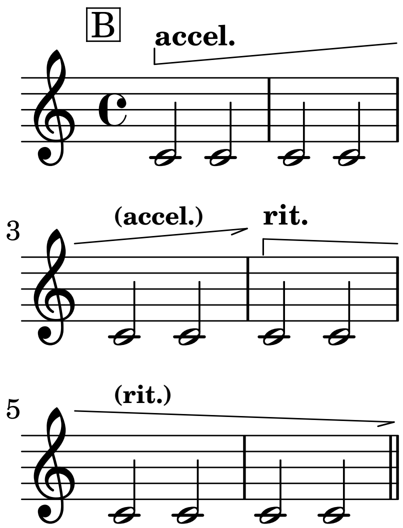
8.9 Tempo Arrows





8.9.1 Description

Replication of accelerando and rallentando arrows chiefly seen in scores by Tōru Takemitsu.[6] The snippets also handle line break.

8.9.2 Grammar

```
\accelArrow #Line_angle ... \stopTextSpan
\rallArrow #Line_angle ... \stopTextSpan
```

NB

1. `#Line_angle` sets how angled the horizontal line should be. `#5` should be more than sufficient for a short line. When it goes over a line break or it extends for a long time, a smaller number may be recommended, such as `#2`.
2. These commands only set the tempo arrows; as such, indications such as `accel.` and `rall.` need to be added separately.

8.9.3 Code

```
\version "2.26.0"

% freely modified from: https://lsr.di.unimi.it/LSR/Item?id=1168
```

6. Examples abound, but see: Tōru Takemitsu, *Fantasma/cantos : for clarinet and orchestra* (Schott ; Schott Japan, 1993) and Tōru Takemitsu, *Les yeux clos II : for piano* (Schott ; Schott Japan, 1990) Other composers from the same publishing company, e.g. Toshio Hosokawa, have also adopted variants of the arrows in their music.

```
% as well as http://lsr.di.unimi.it/LSR/Item?id=1023


accelArrow =
#(define-music-function (line_angle) (number?)

  (define x_value (cos (* (/ 3.14159265358979 180) (- 90 line_angle))))
  (define y_value (sin (* (/ 3.14159265358979 180) (- 90 line_angle))))
  #{
   \tweak direction #up
   \tweak style  #'line
   \tweak thickness  #1
   \tweak to-barline ##t
   \tweak rotation #(list line_angle -1 0 )
   \tweak bound-details.left.stencil #ly:text-interface::print
   \tweak bound-details.left.text \markup \postscript
   #(string-append
     "gsave newpath
0 0 moveto "
     (number->string x_value) " "
     (number->string y_value)
     " rlineto
stroke
grestore")
   \tweak bound-details.left-broken.stencil #ly:text-interface::print
   \tweak bound-details.left-broken.text ##f

   \tweak bound-details.right.stencil #ly:text-interface::print
   \tweak bound-details.right.text \markup \postscript
   "newpath
0 0 moveto
-1 -0.3 rlineto
stroke"
   \tweak bound-details.right-broken.stencil #ly:text-interface::print
   \tweak bound-details.right-broken.text ##f
   \tweak font-shape  #'upright
   \tweak bound-details.left.padding  #0
   \tweak bound-details.right.padding  #0
   \tweak breakable ##t
   \tweak after-line-breaking ##t

   \startTextSpan
  #})

rallArrow =
#(define-music-function (line_angle) (number?)

  (define x_value (cos (* (/ 3.14159265358979 180) (- 90 line_angle))))
```

```
  (define y_value (sin (* (/ 3.14159265358979 180) (- 90 line_angle))))
  #{
   \tweak direction #up
   \tweak style  #'line
   \tweak thickness  #1
   \tweak to-barline ##t
   \tweak rotation #(list (* -1 line_angle) 1 0 )
   \tweak bound-details.left.stencil #ly:text-interface::print
   \tweak bound-details.left.text \markup \postscript
   #(string-append
     "gsave
newpath
0 0 moveto "
     (number->string x_value) " "
     (number->string (* -1 y_value))
     " rlineto
stroke
grestore")
   \tweak bound-details.left-broken.stencil #ly:text-interface::print
   \tweak bound-details.left-broken.text ##f

   \tweak bound-details.right.stencil #ly:text-interface::print
   \tweak bound-details.right.text \markup \postscript
   "newpath
0 0 moveto
-1 -0.3 rlineto
stroke"
   \tweak bound-details.right-broken.stencil #ly:text-interface::print
   \tweak bound-details.right-broken.text ##f
   \tweak font-shape  #'upright
   \tweak bound-details.left.padding  #0
   \tweak bound-details.right.padding  #0
   \tweak breakable ##t
   \tweak after-line-breaking ##t

   \startTextSpan
  #})

\score {
 \layout {
  indent = 0
 }
 {
  c'2^\markup{\translate #'(-4 . 2) \box "A"}
  ^\markup {\translate #'(0 . 1.5) \tiny \bold "accel."} \accelArrow #5   c'2
  c'2 \after 2 \stopTextSpan c'2
  c'2 ^\markup {\translate #'(0 . 1.5) \tiny \bold "rit."}  \rallArrow #3   c'2
  c'2 \after 2 \stopTextSpan c'2 \bar "||"
```

```
 }
}

\score {
 \layout {
  indent = 0
  line-width = 50
 }
 {
  c'2^\markup{\translate #'(-4 . 2) \box "B"}
  ^\markup {\translate #'(0 . 1.5) \tiny \bold "accel."} \accelArrow #5 c'2
  c'2 c'2
  c'2^\markup {\translate #'(0 . 1.5) \teeny \bold "(accel.)"} \after 2 \stopTextSpan c'2
  c'2 ^\markup {\translate #'(0 . 1.5) \tiny \bold "rit."}  \rallArrow #2 c'2 \break
  c'2^\markup {\translate #'(0 . 1.5) \teeny \bold "(rit.)"} c'2
  c'2 \after 2 \stopTextSpan c'2 \bar "||"
 }
}

%^\markup{\translate #'(0 . 2) \bold "accel."}
```

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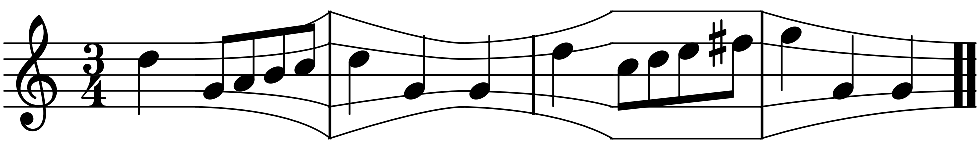
Chapter 9

Staff Lines

9.1 Expanding, Shrinking and Bloated Staff Lines



9.1.1 Description

I made this code as a proof of concept after having read some excellent snippets on LSR.[1]

9.1.2 Grammar

```
\expandingStaff #X-length
\shrinkingStaff #X-length
\bloatedStaff
\normalStaff
```

9.1.3 Code

```
\version "2.26.0"

\language "english"

shrinkingStaff =
#(define-music-function
  (staffDist)
  (number?)

  #{
```

1. See: https://lsr.di.unimi.it/LSR/Item?id=878, https://lsr.di.unimi.it/LSR/Item?id=1005, and https://lsr.di.unimi.it/LSR/Item?id=1007.

```
   \stopStaff
   \once  \override Staff.StaffSymbol.stencil = #ly:text-interface::print
   \once  \override Staff.StaffSymbol.text = \markup {
    \postscript #(string-append
                  "newpath
          0 4 moveto
          0 4 6 2 " (number->string staffDist) " 2 curveto
          0 2 moveto
          0 2 6 1 " (number->string staffDist) " 1 curveto
          0 0 moveto "
                  (number->string staffDist) " 0 lineto
          0 -2 moveto
          0 -2 6 -1 " (number->string staffDist) " -1 curveto
          0 -4 moveto
          0 -4 6 -2 " (number->string staffDist) " -2 curveto
          stroke")


   }
   \override Staff.StaffSymbol.line-positions = #'(-4 -2 0 2 4 )
   \startStaff
  #})

normalStaff = {
 \stopStaff
 \revert Staff.StaffSymbol.line-positions
 \revert Staff.StaffSymbol.stencil
 \startStaff
}

expandingStaff =
#(define-music-function
  (staffDist)
  (number?)

  #{

   \stopStaff
   \once  \override Staff.StaffSymbol.stencil = #ly:text-interface::print
   \once  \override Staff.StaffSymbol.text = \markup {
    \postscript #(string-append
                  "newpath
          0 2 moveto
          0 2 6 2 " (number->string staffDist) " 4 curveto
          0 1 moveto
          0 1 6 1 " (number->string staffDist) " 2 curveto
          0 0 moveto "
                  (number->string staffDist) " 0 lineto
```

```
          0 -1 moveto
          0 -1 6 -1 " (number->string staffDist) " -2 curveto
          0 -2 moveto
          0 -2 6 -2 " (number->string staffDist) " -4 curveto
          stroke ")
   }

   \startStaff
   \override Staff.StaffSymbol.line-positions = #'(-8 -4 0 4 8 )
  #})

bloatedStaff = {
 \stopStaff
 \override Staff.StaffSymbol.line-positions = #'(-8 -4 0 4 8 )
 \override Staff.LedgerLineSpanner.stencil = ##f
 \startStaff
}



% to adjust the length of the individual barlines, see:
% https://lilypond.org/doc/v2.24/Documentation/internals/barline

{

 \override Staff.LedgerLineSpanner.transparent = ##t
 \numericTimeSignature
 \time 3/4
 \once \override Staff.BarLine.bar-extent = #'(-2 . 2)
 d''4 \expandingStaff #8.5

 g'8 a' b' c''
 \once \override Staff.BarLine.bar-extent = #'(-4 . 4)
 \shrinkingStaff #8.5
 d''4 g' \expandingStaff #9.5 g'
 \once \override Staff.BarLine.bar-extent = #'(-2.5 . 2.5)


 e''4 \bloatedStaff c''8 d'' e'' fs''
 \once \override Staff.BarLine.bar-extent = #'(-4 . 4)

 \shrinkingStaff #13.5

 g''4 g' g'
 \bar ".."

}

```

```
\layout {
 \context{
  \Score   proportionalNotationDuration = #1/6
 }
}

```

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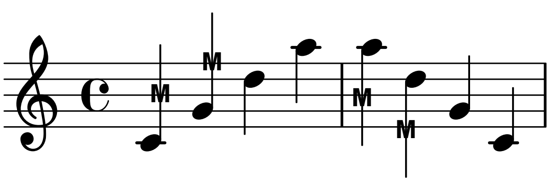
Chapter 10

Stems

10.1 ”M” on Stem



10.1.1 Description

This function attaches ”M” to the stem. I have used this to indicate **M**ultiphonics on woodwind instruments in my pieces. This function lengthens the stem in order to give a balanced look, especially combined with stems/flags.

10.1.2 Grammar

```
\MOnStemOn NOTE ...
\MOnStemOff
```

NB `\MOnStemOn` toggles the feature on, while `\MOnStemOff` toggles it off.

10.1.3 Code

```
\version "2.26.0"

MOnStemOn = {
 \override Stem.length = #12
 \override Stem.details.beamed-lengths = #'(5.5)
 \override Stem.stencil =
 #(lambda (grob)
   (let* ((x-parent (ly:grob-parent grob X))
          (is-rest? (ly:grob? (ly:grob-object x-parent 'rest))))
    (if is-rest?
        empty-stencil
        (ly:stencil-combine-at-edge
```

```
          (ly:stem::print grob)
          Y
          (- (ly:grob-property grob 'direction))
          (grob-interpret-markup grob
                                 (markup
                                  #:center-align
                                  #:teeny #:sans #:bold "M"))
          -3.5))))
}

MOnStemOff = {
 \revert Stem.length
 \revert Stem.details.beamed-lengths
 \revert Stem.stencil
 \revert Flag.stencil
}

{
 \MOnStemOn c'4 g' \MOnStemOff d'' a''
 \MOnStemOn a'' d'' \MOnStemOff g' c'
}
```

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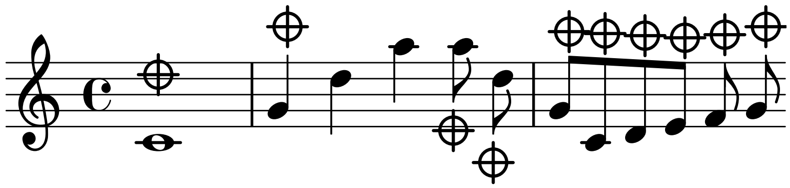
10.2 Mute Sign on Stem



10.2.1 Description

This function attaches a mute sign *above/below* the stem.

10.2.2 Grammar

```
\muteSignOnStemOn NOTE ...
\muteSignOnStemOff
```

NB `\muteSignOnStemOn` toggles the feature on, while `\muteSignOnStemOff` toggles it off.

10.2.3 Code

```

\version "2.26.0"

muteSignOnStemOn = {

 %  \override Stem.length =
 % #(lambda (grob)
 %            (if (= UP (ly:grob-property grob 'direction ))
 %
 %  7.5
 %  7.5))


 %  \override Stem.details.beamed-lengths = #'(5.5)


 \override Stem.stencil =
 #(lambda (grob)
   (let* ((x-parent (ly:grob-parent grob X))
          (is-rest? (ly:grob? (ly:grob-object x-parent 'rest))))
    (if is-rest?
        empty-stencil

        (if (= UP (ly:grob-property grob 'direction))

            (ly:stencil-combine-at-edge
             (ly:stem::print grob)
             Y
```

```
             (+ (ly:grob-property grob 'direction))
             (grob-interpret-markup
              grob
              (markup

               #:postscript
               "newpath
            0.2 setlinewidth
            1 setlinecap
            0 0 moveto
            0 2.5 rlineto
            -1.25 1.25 moveto
            2.5 0 rlineto
            stroke
            newpath
            0 1.25 0.85 0 360 arc
            stroke"
               ))
             0.5)

            (ly:stencil-combine-at-edge
             (ly:stem::print grob)
             Y
             (+ (ly:grob-property grob 'direction))
             (grob-interpret-markup
              grob
              (markup
               #:rotate 180
               #:postscript
               "newpath
            0.2 setlinewidth
            1 setlinecap
            0 0 moveto
            0 2.5 rlineto
            -1.25 1.25 moveto
            2.5 0 rlineto
            stroke
            newpath
            0 1.25 0.85 0 360 arc
            stroke"
               ))
             0.5)
            ))))
}

muteSignOnStemOff = {
 \revert Stem.length
 \revert Stem.details.beamed-lengths
```

```
  \revert Stem.stencil
  \revert Flag.stencil
}

{
  \muteSignOnStemOn c'1^\markup {\translate #'(0 . 3) \musicglyph "space"}

  g'4 \muteSignOnStemOff d'' a''
  \muteSignOnStemOn a''8 \noBeam d'' _\markup {\translate #'(0 . -3) \musicglyph "space" } g'
}

```

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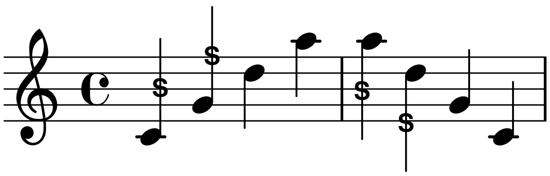
10.3 ”S” on Stem



10.3.1 Description

This function attaches ”S” to the stem. I have used this to indicate **S**plit tone on clarinet/bass clarinet in my pieces. This function lengthens the stem in order to give a balanced look, especially combined with stems/flags.

10.3.2 Grammar

```
\SOnStemOn NOTE ...
\SOnStemOff
```

NB `\SOnStemOn` toggles the feature on, while `\SOnStemOff` toggles it off.

10.3.3 Code

```
\version "2.26.0"

\pointAndClickOff

SOnStemOn = {
 \override Stem.length = #12
 \override Stem.details.beamed-lengths = #'(5.5)
 \override Stem.stencil =
 #(lambda (grob)
   (let* ((x-parent (ly:grob-parent grob X))
          (is-rest? (ly:grob? (ly:grob-object x-parent 'rest))))
    (if is-rest?
        empty-stencil
        (ly:stencil-combine-at-edge
         (ly:stem::print grob)
         Y
         (- (ly:grob-property grob 'direction))
         (grob-interpret-markup grob
                                (markup
                                 #:center-align
                                 #:teeny #:sans #:bold "S"))
         -3.5))))
}

SOnStemOff = {
 \revert Stem.length
 \revert Stem.details.beamed-lengths
```

```
 \revert Stem.stencil
 \revert Flag.stencil
}

{
 \SOnStemOn c'4 g' \SOnStemOff d'' a''
 \SOnStemOn a'' d'' \SOnStemOff g' c'
}
```

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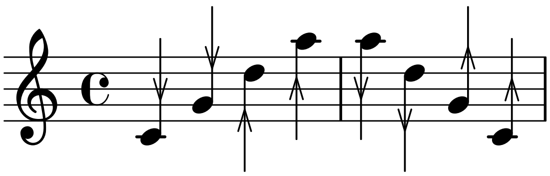
10.4 ”V” on Stem



10.4.1 Description

This function attaches “V” to the stem. I have used this to designate a note with a differentiated timbre from others, for example “brassy tone” for bassoon in my *Gz III* (2019-21) for bass clarinet and bassoon. This function lengthens the stem in order to give a balanced look, especially combined with stems/flags.

I have also added a function where an inverted “V” is attached.

10.4.2 Grammar

```
\VOnStemOn NOTE ...
\VOnStemUpsideDownOn NOTE ...
\VOnStemOff
```

NB `\VOnStemOn` toggles the feature on, while `\VOnStemOff` toggles it off.

10.4.3 Code

```
\version "2.26.0"

VOnStemOn = {
 \override Stem.no-stem-extend = ##f
 \override Stem.length = #12
 \override Stem.details.beamed-lengths = #'(5.5)
 \override Stem.stencil =
 #(lambda (grob)
   (let* ((x-parent (ly:grob-parent grob X))
          (is-rest? (ly:grob? (ly:grob-object x-parent 'rest))))
    (if is-rest?
        empty-stencil
        (ly:stencil-combine-at-edge
         (ly:stem::print grob)
         Y
         (- (ly:grob-property grob 'direction))
         (grob-interpret-markup grob
                                (markup
                                 #:center-align
                                 #:teeny #:sans
                                 #:musicglyph "scripts.uupbow"))
         -3.5))))
}
```

```

VOnStemUpsideDownOn = {
 \override Stem.no-stem-extend = ##f
 \override Stem.length = #12
 \override Stem.details.beamed-lengths = #'(5.5)
 \override Stem.stencil =
 #(lambda (grob)
   (let* ((x-parent (ly:grob-parent grob X))
          (is-rest? (ly:grob? (ly:grob-object x-parent 'rest))))
    (if is-rest?
        empty-stencil
        (ly:stencil-combine-at-edge
         (ly:stem::print grob)
         Y
         (- (ly:grob-property grob 'direction))
         (grob-interpret-markup grob
                                (markup
                                 #:center-align
                                 #:teeny #:sans
                                 #:rotate 180
                                 #:musicglyph "scripts.uupbow"))
         -3.5))))
}

VOnStemOff = {
 \revert Stem.length
 \revert Stem.stencil
 \revert Flag.stencil
}


{
 \VOnStemOn c'4 g' \VOnStemOff
 \VOnStemUpsideDownOn d'' a'' \VOnStemOff
 \VOnStemOn a'' d'' \VOnStemOff
 \VOnStemUpsideDownOn g' c' \VOnStemOff
}

```

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Chapter 11

Time Signatures

New to `2.26.0`: LilyPond's `\time` command accepts fractional time signatures, making the relevant codes discussed in the first nine entries of this chapter obsolete.[1] However, for the time being I will continue to keep the codes current, for they may offer good case studies of how custom Scheme codes interact with overriding stencil functions.

First nine entries of this chapter discuss fractional time signatures (variants of the irrational time signatures) and their compound forms. I have been inspired to create these implementations after chancing upon the email exchanges on `lilypond-user` dated from 2014.[2]

While Gould discourages the use of time signatures with numerators as fractions,[3] there are cases where the use of such time signatures seems justified, particularly when the fractions deal with some form of tuplets. This is a form of time signature notation widely seen in works by Chaya Czernowin, Stefan Beyer, myself, and so many others.

I present the implementation of fractional time signatures in three different styles, A, B, and C. There are implementations for compound meters for each of the styles, in two and three time signatures.

1. https://lilypond.org/doc/v2.26/Documentation/changes/rhythm-improvements and https://lilypond.org/doc/v2.26/Documentation/notation/displaying-rhythms#time-signature

2. See: https://lists.gnu.org/archive/html/lilypond-user/2014-06/msg00209.html. However, in the process of writing this documentation I have come across another email thread on the same mailing list: https://mail.gnu.org/archive/html/lilypond-user/2020-04/msg00423.html

3. Gould, *Behind bars : the definitive guide to music notation*, 180.

11.1 Fractional Time Signatures, Style A

11.1.1 Description

This particular style of fractional time signatures[4] can be seen in scores by Stefan Beyer, for example *Marsch* (2013-14),[5] *Mittel und Zwecke (Boulevard)* (2014),[6] *Bleib hier. Schau zu. Mach kein Geräusch.* (2017),[7] and *Most of My Clients Come Back* (2012-13).[8] In the case of *Mass und Gewicht* (2021), Beyer uses fractions on the denominator of the time signatures.[9]

Because the size the fractions is a half of the ordinary time signatures, it may be difficult to see from afar.[10]

11.1.2 Grammar

```
\fractionalTimeSignatureA
       #'(NUM1 NUM2 NUM3 NUM4) MEASURE_SPAN BEAT_STRUCT
\fractionalTimeSignatureA
       #'(NUM2 NUM3 NUM4) MEASURE_SPAN BEAT_STRUCT
\fractionalTimeSignatureAPlus
       #'(NUM1 NUM2 NUM3 NUM4) MEASURE_SPAN BEAT_STRUCT
\fractionalTimeSignatureAPlus
       #'(NUM2 NUM3 NUM4) MEASURE_SPAN BEAT_STRUCT

\backToNormalTimeSignature
```

4. After having come up with this code, there were other implementations that could be seen on this email thread: https://mail.gnu.org/archive/html/lilypond-user/2020-04/msg00423.html
5. Stefan Beyer, *Marsch* (Manuscript, 2013-14).
6. Stefan Beyer, *Mittel und Zwecke (Boulevard)* (Manuscript, 2014).
7. Stefan Beyer, *Bleib hier. Schau zu. Mach kein Geräusch.* (Manuscript, 2017).
8. Stefan Beyer, *Most of My Clients Come Back* (Manuscript, 2012-13).
9. It would be relatively easy to modify the Scheme code so that the fraction appears next to the denominator of the time signature, instead.
10. It should be noted that in other works such as *Lotte Reiniger's The Sleeping Beauty* (2020-21), Beyer also uses the irrational time signatures as seen in the *Incomplete Tuplet Bracket for Irrational Time Signatures* section of this document.

NB

1. `\fractionalTimeSignatureA` lists time signatures *without* the use of the `+` (plus) sign.
2. `\fractionalTimeSignatureAPlus` lists time signatures with the `+` (plus) sign, when the list with four `NUM`s are given.
3. `NUM1`, `NUM2`, `NUM3`, and `NUM4` can be understood as follows:

$$\frac{1+\frac{2}{3}}{4}$$

   where `NUM1` is optional. The code has `cond` clause, which adjusts the appearance of the time signature according to the length of the list, either having 3 or 4 numbers.
4. `MEASURE_SPAN` denotes how the measure may be written using an *irrational time signature*. In the example snippet, this would be:

$$\frac{3}{4}+\frac{2}{12}=\frac{\mathbf{11}}{\mathbf{12}}$$

5. `BEAT_STRUCT` indicates beat structure, by which the beaming of the measure abides.
6. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

11.1.3 Code

```
% Inspired by:
% https://lists.gnu.org/archive/html/lilypond-user/2014-06/msg00209.html

% Revised Aug 10 2025 to include the function to revert to a
% regular time signature

\version "2.26.0"

suppressWarning =
#(define-void-function (amount message)(number? string?)
  (for-each
   (lambda (warning)
    (ly:expect-warning message))
   (iota amount 1 1)))

\suppressWarning 3 "strange time signature found"

incompleteTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t

}
incompleteSmallTupletBracket =  {
```

```
  \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
  \once \override Voice.TupletBracket.bracket-visibility = ##t
  \once \override Voice.TupletNumber.X-offset =
  #(lambda (grob)
    (if (= UP (ly:grob-property grob 'direction))
        2.2
        1.2))

  \once \override Voice.TupletBracket.shorten-pair =
  #(lambda (grob)
    (if (= UP (ly:grob-property grob 'direction))
        '(-0.7 . 0.15)
        '(-0.3 . 0.8)))
  \once \override Voice.TupletBracket.X-positions =
  #(lambda (grob)
    (if (= UP (ly:grob-property grob 'direction))
        '(1.8 . 3)
        '(0.3 . 2.7)))
}

fractionalTimeSignatureA =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)
  #{
   \time $underlyingMeter
   \set beatStructure = $beatStructure

   \override Staff.TimeSignature.stencil =
   #ly:text-interface::print

   \override Staff.TimeSignature.text =
   #(if (= (length timeSignatureToShow) 4)

        (markup
         #:override
         (cons 'baseline-skip 0)
         (#:center-column
          (#:number
           (#:concat
            (#:simple
             (number->string (car timeSignatureToShow))
             #:halign -1.5
             (#:center-column
              ((#:translate
                (cons 0 1)
                (#:fontsize -6
                            (number->string
```

```
                             (cadr timeSignatureToShow))))
               (#:translate
                (cons 0 0)
                (#:fontsize -6
                            (number->string
                             (caddr timeSignatureToShow)))))))
           #:number
           (number->string (cadddr timeSignatureToShow)))))

        (markup
         #:override
         (cons 'baseline-skip 0)
         (#:center-column
          (#:number
           (#:translate
            (cons 0 1)
            (#:fontsize -6 (number->string
                            (car timeSignatureToShow))))
           #:number
           (#:translate
            (cons 0 0)
            (#:fontsize -6 (number->string
                            (cadr timeSignatureToShow))))
           #:number
           (number->string (caddr timeSignatureToShow)))))

        )
  #})

fractionalTimeSignatureAPlus =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)
  #{
   \time $underlyingMeter
   \set beatStructure = $beatStructure

   \override Staff.TimeSignature.stencil =
   #ly:text-interface::print

   \override Staff.TimeSignature.text =
   #(if (= (length timeSignatureToShow) 4)

        (markup
         #:override
         (cons 'baseline-skip 0)
         (#:center-column
          (#:number
```

```
           (#:concat
            (#:simple
             (number->string (car timeSignatureToShow))

             (#:fontsize -12 (string-append " "))
             (string-append "+")
             (#:fontsize -12 (string-append " "))

             #:center-column
             ((#:translate
               (cons 0 1)
               (#:fontsize -6
                           (number->string
                            (cadr timeSignatureToShow))))
              (#:translate
               (cons 0 0)
               (#:fontsize -6
                           (number->string
                            (caddr timeSignatureToShow)))))))
           #:number
           (number->string (cadddr timeSignatureToShow)))))

        (markup
         #:override
         (cons 'baseline-skip 0)
         (#:center-column
          (#:number
           (#:translate
            (cons 0 1)
            (#:fontsize -6 (number->string
                            (car timeSignatureToShow))))
           #:number
           (#:translate
            (cons 0 0)
            (#:fontsize -6 (number->string
                            (cadr timeSignatureToShow))))
           #:number
           (number->string (caddr timeSignatureToShow)))))

        )
  #})


backToNormalTimeSignature =
{
 \unset beatStructure
 \revert Timing.TimeSignature.stencil
 \revert Timing.TimeSignature.text
```

```
 \revert Staff.TimeSignature.stencil
 \revert Staff.TimeSignature.text
}


\new Staff \with { instrumentName = \markup {\fontsize #4 \box "A"}} {
 \fractionalTimeSignatureA #'(3 2 3 4) 11/12 3,3,3,2
 \tuplet 3/2 { c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c' c' c'}
 \incompleteTupletBracket \tuplet 3/2 {c' c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}
\new Staff \with { instrumentName = \markup {\fontsize #4 \box "B"}} {
 \fractionalTimeSignatureAPlus #'(3 2 3 4) 11/12 3,3,3,2
 \tuplet 3/2 { c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c' c' c'}
 \incompleteTupletBracket \tuplet 3/2 {c' c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}
\new Staff \with { instrumentName = \markup {\fontsize #4 \box "C"}} {
 \fractionalTimeSignatureA #'(11 3 4) 11/12 3,3,3,2
 \tuplet 3/2 { c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c' c' c'}
 \incompleteTupletBracket \tuplet 3/2 {c' c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}

```

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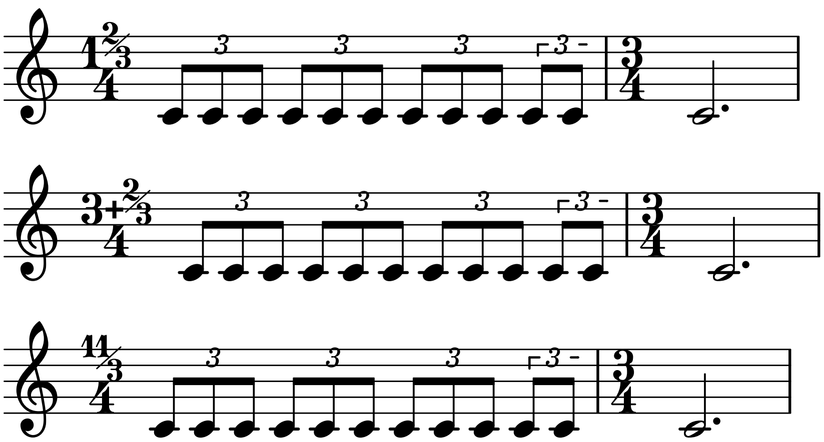
11.2 Fractional Time Signatures, Style B



11.2.1 Description

Style B differs from Style A, as the fraction has a bigger font size. This is similar to the design of fractional time signatures I have used in works such as *Gz II* (2017-22).[11]

11.2.2 Grammar

```
\fractionalTimeSignatureB
       #'(NUM1 NUM2 NUM3 NUM4) MEASURE_SPAN BEAT_STRUCT
\fractionalTimeSignatureB
       #'(NUM2 NUM3 NUM4) MEASURE_SPAN BEAT_STRUCT
\fractionalTimeSignatureBPlus
       #'(NUM1 NUM2 NUM3 NUM4) MEASURE_SPAN BEAT_STRUCT
\fractionalTimeSignatureBPlus
       #'(NUM2 NUM3 NUM4) MEASURE_SPAN BEAT_STRUCT

\backToNormalTimeSignature
```

NB

1. `\fractionalTimeSignatureB` lists time signatures *without* the use of the `+` (plus) sign.
2. `\fractionalTimeSignatureBPlus` lists time signatures with the `+` (plus) sign, when the list with four `NUM`s are given.
3. See *Grammar* of Fractional Time Signatures, Style A for the explanation on the arguments.
4. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

11. Yoshiaki Onishi, *Gz II : for two accordions* (Brühl and Berlin: Edition Gravis, 2024).

11.2.3 Code

```
\version "2.26.0"

% Revised Jan 2 2025 for improving the appearance of fractions
% Revised Aug 10 2025 to include the function to revert to a
% regular time signature

suppressWarning =
#(define-void-function (amount message)(number? string?)
  (for-each
   (lambda (warning)
    (ly:expect-warning message))
   (iota amount 1 1)))

\suppressWarning 3 "strange time signature found"

incompleteTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t

}
incompleteSmallTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t
 \once \override Voice.TupletNumber.X-offset =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       2.2
       1.2))

 \once \override Voice.TupletBracket.shorten-pair =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(-0.7 . 0.15)
       '(-0.3 . 0.8)))
 \once \override Voice.TupletBracket.X-positions =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(1.8 . 3)
       '(0.3 . 2.7)))
}

fractionalTimeSignatureB =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)
  #{
```

```
    \time $underlyingMeter
    \set beatStructure = $beatStructure

    \override Staff.TimeSignature.stencil =
    #ly:text-interface::print
    \override Staff.TimeSignature.text =

    #(if (= (length timeSignatureToShow) 4)

         (markup
          (make-override-markup
           (cons 'baseline-skip 0)
           (make-center-column-markup
            (list
             (make-line-markup
              (list
               (make-number-markup
                (number->string (car timeSignatureToShow)))


               (make-hspace-markup -0.5)
               (make-right-align-markup
                (make-number-markup
                 (make-translate-markup
                  (cons 0 1.5)
                  (make-fontsize-markup
                   -3
                   (number->string (cadr timeSignatureToShow))))))

               (make-hspace-markup -1.5)

               (make-override-markup
                (cons 'alignment 0)
                (make-translate-markup
                 (cons 0 0.8)
                 (make-draw-line-markup (cons 1.5 1.35))))

               (make-hspace-markup -1.5)

               (make-number-markup
                (make-left-align-markup
                 (make-fontsize-markup
                  -3
                  (number->string (caddr timeSignatureToShow)))))))

             (make-number-markup
              (number->string (cadddr timeSignatureToShow)))))))
```

```


        (markup
         (make-override-markup
          (cons 'baseline-skip 0)
          (make-center-column-markup
           (list
            (make-line-markup
             (list
              (make-number-markup
               (make-right-align-markup
                (make-translate-markup
                 (cons 0 1.5)
                 (make-fontsize-markup
                  -3
                  (number->string (car timeSignatureToShow))))))

              (make-hspace-markup -1.5)

              (make-override-markup
               (cons 'alignment 0)
               (make-translate-markup
                (cons 0 0.8)
                (make-draw-line-markup (cons 1.5 1.35))))

              (make-hspace-markup -1.5)

              (make-translate-markup
               (cons 0 0)
               (make-fontsize-markup
                -3
                (make-number-markup
                 (number->string (cadr timeSignatureToShow)))))))

            (make-number-markup
             (number->string (caddr timeSignatureToShow)))))))

        )
  #})

fractionalTimeSignatureBPlus =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)
  #{

   \time $underlyingMeter
   \set beatStructure = $beatStructure
```

```
\override Staff.TimeSignature.stencil =
#ly:text-interface::print
\override Staff.TimeSignature.text =

#(if (= (length timeSignatureToShow) 4)


     (markup
      (make-override-markup
       (cons 'baseline-skip 0)
       (make-center-column-markup
        (list
         (make-line-markup
          (list
           (make-number-markup
            (number->string (car timeSignatureToShow)))
           (make-fontsize-markup
            -12
            (make-simple-markup " "))


           (make-hspace-markup -1.25)
           (make-translate-markup
            (cons 0 0.4)
            (make-bold-markup
             (make-simple-markup "+")))

           (make-hspace-markup -0.25)

           (make-hspace-markup -0.5)
           (make-right-align-markup
            (make-number-markup
             (make-translate-markup
              (cons 0 1.5)
              (make-fontsize-markup
               -3
               (number->string (cadr timeSignatureToShow))))))




           (make-hspace-markup -1.5)

           (make-override-markup
            (cons 'alignment 0)
            (make-translate-markup
             (cons 0 0.8)
```

```
                (make-draw-line-markup (cons 1.5 1.35))))

              (make-hspace-markup -1.5)

              (make-number-markup
               (make-left-align-markup
                (make-fontsize-markup
                 -3
                 (number->string (caddr timeSignatureToShow)))))))

            (make-number-markup
             (number->string (cadddr timeSignatureToShow)))))))


        (markup
         (make-override-markup
          (cons 'baseline-skip 0)
          (make-center-column-markup
           (list
            (make-line-markup
             (list
              (make-number-markup
               (make-right-align-markup
                (make-translate-markup
                 (cons 0 1.6)
                 (make-fontsize-markup
                  -3
                  (number->string (car timeSignatureToShow))))))

              (make-hspace-markup -1.5)

              (make-override-markup
               (cons 'alignment 0)
               (make-translate-markup
                (cons 0 0.8)
                (make-draw-line-markup (cons 1.5 1.35))))

              (make-hspace-markup -1.5)

              (make-translate-markup
               (cons 0 0)
               (make-fontsize-markup
                -3
                (make-number-markup
                 (number->string (cadr timeSignatureToShow)))))))

            (make-number-markup
             (number->string (caddr timeSignatureToShow)))))))
```

```
        )
  #})

backToNormalTimeSignature =
{
 \unset beatStructure
 \revert Timing.TimeSignature.stencil
 \revert Timing.TimeSignature.text
 \revert Staff.TimeSignature.stencil
 \revert Staff.TimeSignature.text
}


{
 \fractionalTimeSignatureB #'(1 2 3 4) 11/12 3,3,3,2
 \tuplet 3/2 {c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c' c' c'}
 \incompleteTupletBracket \tuplet 3/2 {c' c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}

{
 \fractionalTimeSignatureBPlus #'(3 2 3 4) 11/12 3,3,3,2
 \tuplet 3/2 {c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c' c' c'}
 \incompleteTupletBracket \tuplet 3/2 {c' c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}

{
 \fractionalTimeSignatureB #'(11 3 4) 11/12 3,3,3,2
 \tuplet 3/2 {c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c' c' c'}
 \incompleteTupletBracket \tuplet 3/2 {c' c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}

```

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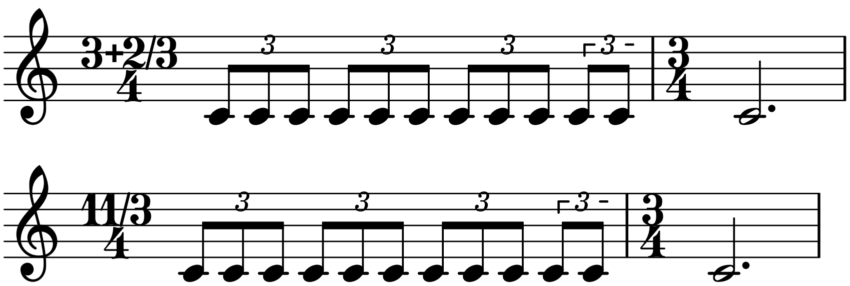
11.3 Fractional Time Signatures, Style C



11.3.1 Description

Style C of the fractional time signatures offers the largest font size for displaying the fractions. This design is commonly seen in scores by Chaya Czernowin, in such works as String Quartet (1995),[12] *Lovesong* (2010),[13] *Streams (Slow Summer Stay I)* (2012),[14] and *At the fringe of our gaze* (2012/13).[15]

11.3.2 Grammar

```
\fractionalTimeSignatureC
       #'(NUM1 NUM2 NUM3 NUM4) MEASURE_SPAN BEAT_STRUCT
\fractionalTimeSignatureC
       #'(NUM2 NUM3 NUM4) MEASURE_SPAN BEAT_STRUCT

\backToNormalTimeSignature
```

NB

1. By default, `\fractionalTimeSignatureC` shows `+` (plus) sign when four `NUM`s are given. As the font size for the ordinary numerator and the fractions is the same, without `+` it becomes very confusing to read the time signature. Thus, contrary to Styles A and B, there is no separate function for the time signature with the `+` sign given.
2. See *Grammar* of Fractional Time Signatures, Style A for the explanation on the arguments.
3. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

11.3.3 Code

```
\version "2.26.0"

% Revised Jan 2 2025 for improving the appearance of fractions
% Revised Aug 10 2025 to include the function to revert to a
```

12. Chaya Czernowin, *String Quartet* (Schott, 1995).
13. Chaya Czernowin, *Lovesong : for mixed ensemble* (Schott, 2010).
14. Chaya Czernowin, *Streams (Slow Summer Stay I) : for 8 players* (Schott, 2012).
15. Chaya Czernowin, *At the fringe of our gaze : for Orchestra and Concertino Group* (Schott, 2012/13).

```
% regular time signature

suppressWarning =
#(define-void-function (amount message)(number? string?)
  (for-each
   (lambda (warning)
    (ly:expect-warning message))
   (iota amount 1 1)))

\suppressWarning 2 "strange time signature found"

incompleteTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t

}
incompleteSmallTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t
 \once \override Voice.TupletNumber.X-offset =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       2.2
       1.2))

 \once \override Voice.TupletBracket.shorten-pair =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(-0.7 . 0.15)
       '(-0.3 . 0.8)))
 \once \override Voice.TupletBracket.X-positions =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(1.8 . 3)
       '(0.3 . 2.7)))
}

fractionalTimeSignatureC =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)
  #{

   \time $underlyingMeter
   \set beatStructure = $beatStructure

   \override Staff.TimeSignature.stencil =
   #ly:text-interface::print
```

```
    \override Staff.TimeSignature.text =

    #(if (= (length timeSignatureToShow) 4)

         (markup
          (make-override-markup
           (cons 'baseline-skip 0)
           (make-center-column-markup
            (list
             (make-line-markup
              (list
               (make-number-markup
                (number->string
                 (car timeSignatureToShow)))
               (make-fontsize-markup
                -12
                (make-simple-markup " "))


               (make-hspace-markup -1.25)
               (make-translate-markup
                (cons 0 0.4)
                (make-bold-markup
                 (make-simple-markup "+")))

               (make-hspace-markup -0.25)

               (make-hspace-markup -0.5)
               (make-right-align-markup
                (make-number-markup
                 (number->string
                  (cadr timeSignatureToShow))))

               (make-hspace-markup -0.6)

               (make-override-markup
                (list (cons 'alignment 0)
                      (cons 'thickness 2))
                (make-draw-line-markup (cons 0.5 2)))

               (make-hspace-markup -0.6)

               (make-number-markup
                (make-left-align-markup
                 (number->string
                  (caddr timeSignatureToShow))))))

             (make-number-markup
```

```
              (number->string
               (cadddr timeSignatureToShow)))))))


         (markup
          (make-override-markup
           (cons 'baseline-skip 0)
           (make-center-column-markup
            (list
             (make-line-markup
              (list

               (make-right-align-markup
                (make-number-markup
                 (number->string
                  (car timeSignatureToShow))))

               (make-hspace-markup -0.6)

               (make-override-markup
                (list (cons 'alignment 0)
                      (cons 'thickness 2))
                (make-draw-line-markup
                 (cons 0.5 2)))

               (make-hspace-markup -0.6)

               (make-number-markup
                (make-left-align-markup
                 (number->string
                  (cadr timeSignatureToShow))))))

             (make-number-markup
              (number->string
               (caddr timeSignatureToShow))))))
          ))
  #})

backToNormalTimeSignature =
{
 \unset beatStructure
 \revert Timing.TimeSignature.stencil
 \revert Timing.TimeSignature.text
 \revert Staff.TimeSignature.stencil
 \revert Staff.TimeSignature.text
}
```

```
{
 \fractionalTimeSignatureC #'(3 2 3 4) 11/12 3,3,3,2
 \tuplet 3/2 { c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c' c' c'}
 \incompleteTupletBracket \tuplet 3/2 {c' c'}

 \backToNormalTimeSignature
 \time 3/4
 c'2.
}


{
 \fractionalTimeSignatureC #'(11 3 4) 11/12 3,3,3,2
 \tuplet 3/2 { c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c' c' c'}
 \incompleteTupletBracket \tuplet 3/2 {c' c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}

```

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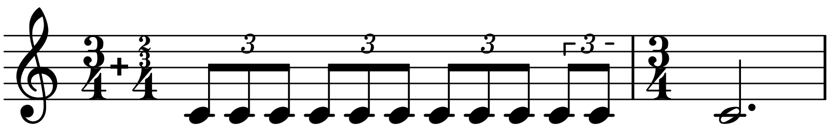
11.4 Compound Meter with Two Fractional Time Signatures, Style A



11.4.1 Description

This is an implementation of a compound meter with two fractional time signatures with Style A.

11.4.2 Grammar

```
\compoundFractionalTimeSignatureATwo
       #'((TIME_SIG1)(TIME_SIG2)) MEASURE_SPAN BEAT_STRUCT

\backToNormalTimeSignature
```

NB

1. Following the convention of `\compoundMeter` to enter the two time signatures, you will create a list of lists. Each `TIME_SIG` accepts:
   - an ordinary time signature (list with two numbers);
   - a time signature with a fraction (list with three numbers), or;
   - a time signature with an ordinary numerator and a fraction.

   See *Grammar* of Fractional Time Signatures, Style A for the explanation on the arguments for the order of arguments to specify time signatures.
2. `MEASURE_SPAN` and `BEAT_STRUCT` follow the same convention as before.
3. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

11.4.3 Code

```
\version "2.26.0"

% Revised Aug 10 2025 to include the function to revert to a
% regular time signature

suppressWarning =
#(define-void-function (amount message)(number? string?)
  (for-each
   (lambda (warning)
    (ly:expect-warning message))
   (iota amount 1 1)))
```

```

\suppressWarning 1 "strange time signature found"

incompleteTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t

}
incompleteSmallTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t
 \once \override Voice.TupletNumber.X-offset =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       2.2
       1.2))

 \once \override Voice.TupletBracket.shorten-pair =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(-0.7 . 0.15)
       '(-0.3 . 0.8)))
 \once \override Voice.TupletBracket.X-positions =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(1.8 . 3)
       '(0.3 . 2.7)))
}

compoundFractionalTimeSignatureATwo =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)
  (define mkup
   (markup
    #:concat
    (
     #:override
     (cons 'baseline-skip 0)
     (cond ((= (length (car timeSignatureToShow)) 2)
            (make-center-column-markup
             (list (make-number-markup
                    (number->string
                     (car (car timeSignatureToShow))))
                   (make-number-markup
                    (number->string
                     (cadr (car timeSignatureToShow)))))))

```

```
          ((= (length (car timeSignatureToShow)) 3)
           (make-override-markup
            (cons 'baseline-skip 0)
            (make-center-column-markup
             (list


              (make-center-column-markup
               (list
                (make-translate-markup
                 (cons 0 1)
                 (make-fontsize-markup
                  -6
                  (make-number-markup
                   (number->string
                    (car (car timeSignatureToShow))))))
                (make-translate-markup
                 (cons 0 0)
                 (make-fontsize-markup
                  -6
                  (make-number-markup
                   (number->string
                    (cadr (car timeSignatureToShow))))))))
              (make-number-markup
               (number->string
                (caddr (car timeSignatureToShow))))))
            ))


          ((= (length (car timeSignatureToShow)) 4)

           (make-override-markup
            (cons 'baseline-skip 0)
            (make-center-column-markup
             (list

              (make-concat-markup
               (list (make-number-markup
                      (number->string
                       (car (car timeSignatureToShow))))
                     (make-halign-markup
                      -1.5
                      (make-center-column-markup
                       (list
                        (make-translate-markup
                         (cons 0 1)
                         (make-fontsize-markup
                          -6
```

```
                          (make-number-markup
                           (number->string
                            (cadr (car timeSignatureToShow))))))
                        (make-translate-markup
                         (cons 0 0)
                         (make-fontsize-markup
                          -6
                          (make-number-markup
                           (number->string
                            (caddr (car timeSignatureToShow))))))))))
              (make-number-markup
               (number->string
                (cadddr (car timeSignatureToShow))))))
            ))
          )


    #:translate
    (cons 0 -0.5)
    (#:fontsize -12 " ")
    #:translate
    (cons 0 -0.5)
    (#:bold "+")
    #:translate
    (cons 0 -0.5)
    (#:fontsize -12 " ")

    #:override
    (cons 'baseline-skip 0)
    (cond ((= (length (cadr timeSignatureToShow)) 2)
           (make-center-column-markup
            (list (make-number-markup
                   (number->string
                    (car (cadr timeSignatureToShow))))
                  (make-number-markup
                   (number->string
                    (cadr (cadr timeSignatureToShow)))))))

          ((= (length (cadr timeSignatureToShow)) 3)

           (make-override-markup
            (cons 'baseline-skip 0)
            (make-center-column-markup
             (list


              (make-center-column-markup
               (list
```

```
              (make-translate-markup
               (cons 0 1)
               (make-fontsize-markup
                -6
                (make-number-markup
                 (number->string
                  (car (cadr timeSignatureToShow))))))
              (make-translate-markup
               (cons 0 0)
               (make-fontsize-markup
                -6
                (make-number-markup
                 (number->string
                  (cadr (cadr timeSignatureToShow)))))))))
            (make-number-markup
             (number->string
              (caddr (cadr timeSignatureToShow))))))
          ))


        ((= (length (cadr timeSignatureToShow)) 4)

         (make-override-markup
          (cons 'baseline-skip 0)
          (make-center-column-markup
           (list

            (make-concat-markup
             (list (make-number-markup
                    (number->string
                     (car (cadr timeSignatureToShow))))
                   (make-halign-markup
                    -1.5
                    (make-center-column-markup
                     (list
                      (make-translate-markup
                       (cons 0 1)
                       (make-fontsize-markup
                        -6
                        (make-number-markup
                         (number->string
                          (cadr (cadr timeSignatureToShow))))))
                      (make-translate-markup
                       (cons 0 0)
                       (make-fontsize-markup
                        -6
                        (make-number-markup
                         (number->string
```

```
                             (caddr (cadr timeSignatureToShow))))))))))
               (make-number-markup
                (number->string
                 (cadddr (cadr timeSignatureToShow))))))
             ))
           )
     )))

  #{
   \time $underlyingMeter
   \set beatStructure = $beatStructure

   \override Timing.TimeSignature.stencil =
   #ly:text-interface::print
   \override Timing.TimeSignature.text =
   #mkup
  #})


backToNormalTimeSignature =
{
 \unset beatStructure
 \revert Timing.TimeSignature.stencil
 \revert Timing.TimeSignature.text
 \revert Staff.TimeSignature.stencil
 \revert Staff.TimeSignature.text
}


{
 \compoundFractionalTimeSignatureATwo #'((3 4)(2 3 4)) 11/12 3,3,3,2
 \tuplet 3/2 { c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c' c' c'}
 \incompleteTupletBracket \tuplet 3/2 {c' c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}

```

11.4.4 Discussion

1. This was a tricky one to make, as I had to resort to building the Scheme code without using the syntactic sugars, i.e. `#:`.[16] If any modification are to be made to this code, it is recommended to carefully examine where the corresponding parenthesis of a starting parenthesis is located. It is also helpful to watch LilyPond Log for any errors, as it seems to give hints for how many

16. See *Known issues and warnings* at: https://lilypond.org/doc/v2.24/Documentation/extending/markup-construction-in-scheme

argument(s) a function is looking for.

2. I am hoping to find ways to simplify the code, as the same bits (with variations in variables that are called upon) of the codes are used to streamline the formatting of the time signature appearances.

3. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

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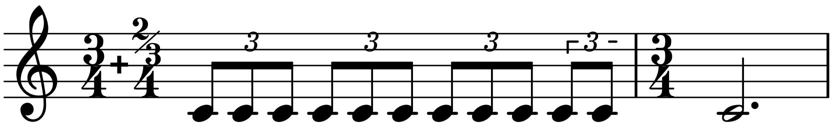
11.5 Compound Meter with Two Fractional Time Signatures, Style B



11.5.1 Description

This is an implementation of a compound meter with two fractional time signatures with Style B.

11.5.2 Grammar

```
\compoundFractionalTimeSignatureBTwo
       #'((TIME_SIG1)(TIME_SIG2)) MEASURE_SPAN BEAT_STRUCT

\backToNormalTimeSignature
```

NB

1. Following the convention of `\compoundMeter` to enter the two time signatures, you will create a list of lists. Each `TIME_SIG` accepts: an ordinary time signature (list with two numbers), a time signature with a fraction (list with three numbers), or a time signature with an ordinary numerator and a fraction. See *Grammar* of Fractional Time Signatures, Style A for the explanation on the arguments for the order of arguments to specify time signatures.

2. `MEASURE_SPAN` and `BEAT_STRUCT` follow the same convention as before.

3. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

11.5.3 Code

```
\version "2.26.0"

% Revised Jan 2 2025 for improving the appearance of fractions
% Revised Aug 10 2025 to include the function to revert to a
% regular time signature

suppressWarning =
#(define-void-function (amount message)(number? string?)
  (for-each
   (lambda (warning)
    (ly:expect-warning message))
   (iota amount 1 1)))

\suppressWarning 1 "strange time signature found"

```

```
incompleteTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t

}
incompleteSmallTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t
 \once \override Voice.TupletNumber.X-offset =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       2.2
       1.2))

 \once \override Voice.TupletBracket.shorten-pair =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(-0.7 . 0.15)
       '(-0.3 . 0.8)))
 \once \override Voice.TupletBracket.X-positions =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(1.8 . 3)
       '(0.3 . 2.7)))
}

compoundFractionalTimeSignatureBTwo =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)
  (define mkup
   (markup
    #:concat
    (

     #:override
     (cons 'baseline-skip 0)
     (cond ((= (length (car timeSignatureToShow)) 2)
            (make-center-column-markup
             (list (make-number-markup
                    (number->string
                     (car (car timeSignatureToShow))))
                   (make-number-markup
                    (number->string
                     (cadr (car timeSignatureToShow)))))))

           ((= (length (car timeSignatureToShow)) 3)
            (make-override-markup
```

```
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list
                  (make-number-markup
                   (make-right-align-markup
                    (make-translate-markup
                     (cons 0 1.6)
                     (make-fontsize-markup
                      -3
                      (number->string
                       (car (car timeSignatureToShow)))))))

                  (make-hspace-markup -1.5)

                  (make-override-markup
                   (cons 'alignment 0)
                   (make-translate-markup
                    (cons 0 0.8)
                    (make-draw-line-markup (cons 1.5 1.35))))

                  (make-hspace-markup -1.5)

                  (make-translate-markup
                   (cons 0 0)
                   (make-fontsize-markup
                    -3
                    (make-number-markup
                     (number->string
                      (cadr (car timeSignatureToShow))))))))

                (make-number-markup
                 (number->string
                  (caddr (car timeSignatureToShow)))))))))


            ((= (length (car timeSignatureToShow)) 4)

             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list
                  (make-number-markup
                   (number->string
                    (car (car timeSignatureToShow))))
```

```
                (make-fontsize-markup
                 -12
                 (make-simple-markup " "))

                (make-hspace-markup -1.25)
                (make-translate-markup
                 (cons 0 0.4)
                 (make-bold-markup
                  (make-simple-markup "+")))

                (make-hspace-markup -0.25)

                (make-hspace-markup -0.5)
                (make-right-align-markup
                 (make-number-markup
                  (make-translate-markup
                   (cons 0 1.5)
                   (make-fontsize-markup
                    -3
                    (number->string
                     (cadr (car timeSignatureToShow)))))))

                (make-hspace-markup -1.5)

                (make-override-markup
                 (cons 'alignment 0)
                 (make-translate-markup
                  (cons 0 0.8)
                  (make-draw-line-markup
                   (cons 1.5 1.35))))

                (make-hspace-markup -1.5)

                (make-number-markup
                 (make-left-align-markup
                  (make-fontsize-markup
                   -3
                   (number->string
                    (caddr (car timeSignatureToShow)))))))

              (make-number-markup
               (number->string
                (cadddr (car timeSignatureToShow))))))))


    #:translate
    (cons 0 -0.5)
    (#:fontsize -12 " ")
```

```
      #:translate
      (cons 0 -0.5)
      (#:bold "+")
      #:translate
      (cons 0 -0.5)
      (#:fontsize -12 " ")

      #:override
      (cons 'baseline-skip 0)

      (cond ((= (length (cadr timeSignatureToShow)) 2)
             (make-center-column-markup
              (list (make-number-markup
                     (number->string
                      (car (cadr timeSignatureToShow))))
                    (make-number-markup
                     (number->string
                      (cadr (cadr timeSignatureToShow)))))))

            ((= (length (cadr timeSignatureToShow)) 3)
             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list
                  (make-number-markup
                   (make-right-align-markup
                    (make-translate-markup
                     (cons 0 1.6)
                     (make-fontsize-markup
                      -3
                      (number->string
                       (car (cadr timeSignatureToShow)))))))

                  (make-hspace-markup -1.5)

                  (make-override-markup
                   (cons 'alignment 0)
                   (make-translate-markup
                    (cons 0 0.8)
                    (make-draw-line-markup (cons 1.5 1.35))))

                  (make-hspace-markup -1.5)

                  (make-translate-markup
                   (cons 0 0)
                   (make-fontsize-markup
```

```
                -3
                (make-number-markup
                 (number->string
                  (cadr (cadr timeSignatureToShow)))))))

            (make-number-markup
             (number->string
              (caddr (cadr timeSignatureToShow)))))))


        ((= (length (cadr timeSignatureToShow)) 4)

         (make-override-markup
          (cons 'baseline-skip 0)
          (make-center-column-markup
           (list
            (make-line-markup
             (list
              (make-number-markup
               (number->string
                (car (cadr timeSignatureToShow))))
              (make-fontsize-markup
               -12
               (make-simple-markup " "))

              (make-hspace-markup -1.25)
              (make-translate-markup
               (cons 0 0.4)
               (make-bold-markup
                (make-simple-markup "+")))

              (make-hspace-markup -0.25)

              (make-hspace-markup -0.5)
              (make-right-align-markup
               (make-number-markup
                (make-translate-markup
                 (cons 0 1.5)
                 (make-fontsize-markup
                  -3
                  (number->string
                   (cadr (cadr timeSignatureToShow)))))))

              (make-hspace-markup -1.5)

              (make-override-markup
               (cons 'alignment 0)
               (make-translate-markup
```

```
                   (cons 0 0.8)
                   (make-draw-line-markup
                    (cons 1.5 1.35))))

                 (make-hspace-markup -1.5)

                 (make-number-markup
                  (make-left-align-markup
                   (make-fontsize-markup
                    -3
                    (number->string
                     (caddr (cadr timeSignatureToShow)))))))

               (make-number-markup
                (number->string
                 (cadddr (cadr timeSignatureToShow))))))))
     )))

  #{
   \time $underlyingMeter
   \set beatStructure = $beatStructure

   \override Timing.TimeSignature.stencil =
   #ly:text-interface::print
   \override Timing.TimeSignature.text =
   #mkup
  #})


backToNormalTimeSignature =
{
 \unset beatStructure
 \revert Timing.TimeSignature.stencil
 \revert Timing.TimeSignature.text
 \revert Staff.TimeSignature.stencil
 \revert Staff.TimeSignature.text
}


{
 \compoundFractionalTimeSignatureBTwo #'((3 4)(2 3 4)) 11/12 3,3,3,2
 \tuplet 3/2 { c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c' c' c'}
 \incompleteTupletBracket \tuplet 3/2 {c' c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}
```

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11.5.4 Discussion

See **Discussion** of the entry *Compound Meter with Two Fractional Time Signatures, Style A.*

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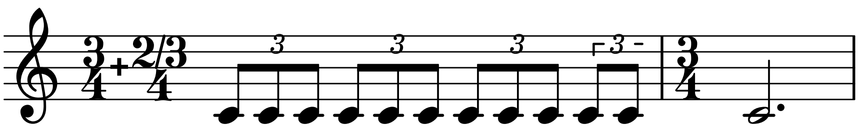
11.6 Compound Meter with Two Fractional Time Signatures, Style C



11.6.1 Description

This is an implementation of a compound meter with two fractional time signatures with Style C.

11.6.2 Grammar

```
\compoundFractionalTimeSignatureCTwo
       #'((TIME_SIG1)(TIME_SIG2)) MEASURE_SPAN BEAT_STRUCT

\backToNormalTimeSignature
```

NB

1. Following the convention of `\compoundMeter` to enter the two time signatures, you will create a list of lists. Each `TIME_SIG` accepts: an ordinary time signature (list with two numbers), a time signature with a fraction (list with three numbers), or a time signature with an ordinary numerator and a fraction. See *Grammar* of Fractional Time Signatures, Style A for the explanation on the arguments for the order of arguments to specify time signatures.
2. `MEASURE_SPAN` and `BEAT_STRUCT` follow the same convention as before.
3. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

11.6.3 Code

```
\version "2.26.0"


% Revised Jan 2 2025 for improving the appearance of fractions
% Revised Aug 10 2025 to include the function to revert to a
% regular time signature

suppressWarning =
#(define-void-function (amount message)(number? string?)
  (for-each
   (lambda (warning)
    (ly:expect-warning message))
   (iota amount 1 1)))

\suppressWarning 1 "strange time signature found"
```

```

incompleteTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t

}
incompleteSmallTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t
 \once \override Voice.TupletNumber.X-offset =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       2.2
       1.2))

 \once \override Voice.TupletBracket.shorten-pair =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(-0.7 . 0.15)
       '(-0.3 . 0.8)))
 \once \override Voice.TupletBracket.X-positions =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(1.8 . 3)
       '(0.3 . 2.7)))
}

compoundFractionalTimeSignatureCTwo =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)
  (define mkup
   (markup
    #:concat
    (

     #:override
     (cons 'baseline-skip 0)
     (cond ((= (length (car timeSignatureToShow)) 2)
            (make-center-column-markup
             (list (make-number-markup
                    (number->string
                     (car (car timeSignatureToShow))))
                   (make-number-markup
                    (number->string
                     (cadr (car timeSignatureToShow)))))))

           ((= (length (car timeSignatureToShow)) 3)
```

```
             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list

                  (make-right-align-markup
                   (make-number-markup
                    (number->string
                     (car (car timeSignatureToShow)))))

                  (make-hspace-markup -0.6)

                  (make-override-markup
                   (list (cons 'alignment 0)
                         (cons 'thickness 2))
                   (make-draw-line-markup
                    (cons 0.5 2)))

                  (make-hspace-markup -0.6)

                  (make-number-markup
                   (make-left-align-markup
                    (number->string
                     (cadr (car timeSignatureToShow)))))))

                (make-number-markup
                 (number->string
                  (caddr (car timeSignatureToShow)))))))


            ((= (length (car timeSignatureToShow)) 4)

             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list
                  (make-number-markup
                   (number->string
                    (car (car timeSignatureToShow))))
                  (make-fontsize-markup
                   -12
                   (make-simple-markup " "))
```

```
                (make-hspace-markup -1.25)
                (make-translate-markup
                 (cons 0 0.4)
                 (make-bold-markup
                  (make-simple-markup "+")))

                (make-hspace-markup -0.25)

                (make-hspace-markup -0.5)
                (make-right-align-markup
                 (make-number-markup
                  (number->string
                   (cadr (car timeSignatureToShow)))))

                (make-hspace-markup -0.6)

                (make-override-markup
                 (list (cons 'alignment 0)
                       (cons 'thickness 2))
                 (make-draw-line-markup (cons 0.5 2)))

                (make-hspace-markup -0.6)

                (make-number-markup
                 (make-left-align-markup
                  (number->string
                   (caddr (car timeSignatureToShow)))))))

              (make-number-markup
               (number->string
                (cadddr (car timeSignatureToShow))))))))


    #:translate
    (cons 0 -0.5)
    (#:fontsize -12 " ")
    #:translate
    (cons 0 -0.5)
    (#:bold "+")
    #:translate
    (cons 0 -0.5)
    (#:fontsize -12 " ")

    #:override
    (cons 'baseline-skip 0)

    (cond ((= (length (cadr timeSignatureToShow)) 2)
           (make-center-column-markup
```

```
              (list (make-number-markup
                     (number->string
                      (car (cadr timeSignatureToShow))))
                    (make-number-markup
                     (number->string
                      (cadr (cadr timeSignatureToShow)))))))

            ((= (length (cadr timeSignatureToShow)) 3)
             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list

                  (make-right-align-markup
                   (make-number-markup
                    (number->string
                     (car (cadr timeSignatureToShow)))))

                  (make-hspace-markup -0.6)

                  (make-override-markup
                   (list (cons 'alignment 0)
                         (cons 'thickness 2))
                   (make-draw-line-markup
                    (cons 0.5 2)))

                  (make-hspace-markup -0.6)

                  (make-number-markup
                   (make-left-align-markup
                    (number->string
                     (cadr (cadr timeSignatureToShow)))))))

                (make-number-markup
                 (number->string
                  (caddr (cadr timeSignatureToShow))))))))


            ((= (length (cadr timeSignatureToShow)) 4)

             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list
```

```
                   (make-number-markup
                    (number->string
                     (car (cadr timeSignatureToShow))))
                   (make-fontsize-markup
                    -12
                    (make-simple-markup " "))


                   (make-hspace-markup -1.25)
                   (make-translate-markup
                    (cons 0 0.4)
                    (make-bold-markup
                     (make-simple-markup "+")))

                   (make-hspace-markup -0.25)

                   (make-hspace-markup -0.5)
                   (make-right-align-markup
                    (make-number-markup
                     (number->string
                      (cadr (cadr timeSignatureToShow)))))

                   (make-hspace-markup -0.6)

                   (make-override-markup
                    (list (cons 'alignment 0)
                          (cons 'thickness 2))
                    (make-draw-line-markup (cons 0.5 2)))

                   (make-hspace-markup -0.6)

                   (make-number-markup
                    (make-left-align-markup
                     (number->string
                      (caddr (cadr timeSignatureToShow))))))

                 (make-number-markup
                  (number->string
                   (cadddr (cadr timeSignatureToShow))))))))
       )))

    #{
     \time $underlyingMeter
     \set beatStructure = $beatStructure
     \override Timing.TimeSignature.stencil =
     #ly:text-interface::print
     \override Timing.TimeSignature.text =
     #mkup
```

```
  #})


backToNormalTimeSignature =
{
 \unset beatStructure
 \revert Timing.TimeSignature.stencil
 \revert Timing.TimeSignature.text
 \revert Staff.TimeSignature.stencil
 \revert Staff.TimeSignature.text
}


{

 \compoundFractionalTimeSignatureCTwo #'((3 4)(2 3 4)) 11/12 3,3,3,2
 \tuplet 3/2 {c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c' c' c'}
 \incompleteTupletBracket \tuplet 3/2 {c' c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}

```

11.6.4 Discussion

See **Discussion** of the entry *Compound Meter with Two Fractional Time Signatures, Style A*.

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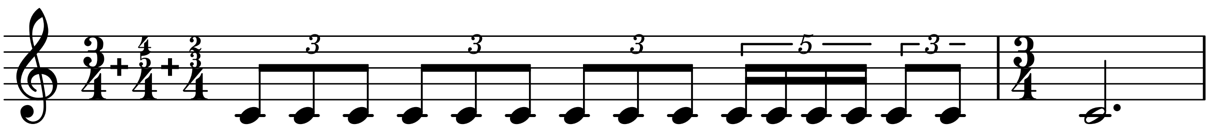
11.7 Compound Meter with Three Fractional Time Signatures, Style A



11.7.1 Description

This is an implementation of a compound meter with three fractional time signatures with Style A.

11.7.2 Grammar

```
\compoundFractionalTimeSignatureAThree
     #'((TIME_SIG1)(TIME_SIG2)(TIME_SIG3)) MEASURE_SPAN BEAT_STRUCT

\backToNormalTimeSignature
```

NB

1. Following the convention of `\compoundMeter` to enter the two time signatures, you will create a list of lists. Each `TIME_SIG` accepts:
   - an ordinary time signature (list with two numbers);
   - a time signature with a fraction (list with three numbers), or;
   - a time signature with an ordinary numerator and a fraction.

   See *Grammar* of Fractional Time Signatures, Style A for the explanation on the arguments for the order of arguments to specify time signatures.
2. In the code of the given snippet, the value for `MEASURE_SPAN` may appear absurd. However, this results from following the same convention as before, i.e. adding the constituent time signatures to give a general irrational time signature for the entire bar. Thus:

$$\frac{3}{4} + \frac{4}{20} + \frac{2}{12} = \mathbf{\frac{67}{60}}$$

3. `BEAT_STRUCT` follows the same convention as before; however, as the given code shows, it may be necessary to still use `[` and `]` to explicitly specify the beaming.
4. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

11.7.3 Code

```
\version "2.26.0"
```

```

% Revised Aug 10 2025 to include the function to revert to a
% regular time signature

suppressWarning =
#(define-void-function (amount message)(number? string?)
  (for-each
   (lambda (warning)
    (ly:expect-warning message))
   (iota amount 1 1)))

\suppressWarning 3 "strange time signature found"

incompleteTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t

}
incompleteSmallTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t
 \once \override Voice.TupletNumber.X-offset =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       2.2
       1.2))

 \once \override Voice.TupletBracket.shorten-pair =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(-0.7 . 0.15)
       '(-0.3 . 0.8)))
 \once \override Voice.TupletBracket.X-positions =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(1.8 . 3)
       '(0.3 . 2.7)))
}

compoundFractionalTimeSignatureAThree =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)
  (define mkup
   (markup
    #:concat
    (
     #:override
```

```
      (cons 'baseline-skip 0)
      (cond ((= (length (car timeSignatureToShow)) 2)
             (make-center-column-markup
              (list (make-number-markup
                     (number->string
                      (car (car timeSignatureToShow))))
                    (make-number-markup
                     (number->string
                      (cadr (car timeSignatureToShow)))))))

            ((= (length (car timeSignatureToShow)) 3)
             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-center-column-markup
                 (list
                  (make-translate-markup
                   (cons 0 1)
                   (make-fontsize-markup
                    -6
                    (make-number-markup
                     (number->string
                      (car (car timeSignatureToShow))))))
                  (make-translate-markup
                   (cons 0 0)
                   (make-fontsize-markup
                    -6
                    (make-number-markup
                     (number->string
                      (cadr (car timeSignatureToShow))))))))
                (make-number-markup
                 (number->string
                  (caddr (car timeSignatureToShow))))))
              ))


            ((= (length (car timeSignatureToShow)) 4)

             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list

                (make-concat-markup
                 (list (make-number-markup
                        (number->string
                         (car (car timeSignatureToShow))))
```

```
                        (make-halign-markup
                         -1.5
                         (make-center-column-markup
                          (list
                           (make-translate-markup
                            (cons 0 1)
                            (make-fontsize-markup
                             -6
                             (make-number-markup
                              (number->string
                               (cadr (car timeSignatureToShow))))))
                           (make-translate-markup
                            (cons 0 0)
                            (make-fontsize-markup
                             -6
                             (make-number-markup
                              (number->string
                               (caddr (car timeSignatureToShow))))))))))
                 (make-number-markup
                  (number->string
                   (cadddr (car timeSignatureToShow))))))
               ))
             )


       #:translate
       (cons 0 -0.5)
       (#:fontsize -12 " ")
       #:translate
       (cons 0 -0.5)
       (#:bold "+")
       #:translate
       (cons 0 -0.5)
       (#:fontsize -12 " ")

       #:override
       (cons 'baseline-skip 0)
       (cond ((= (length (cadr timeSignatureToShow)) 2)
              (make-center-column-markup
               (list (make-number-markup
                      (number->string
                       (car (cadr timeSignatureToShow))))
                     (make-number-markup
                      (number->string
                       (cadr (cadr timeSignatureToShow)))))))

             ((= (length (cadr timeSignatureToShow)) 3)

```

```
              (make-override-markup
               (cons 'baseline-skip 0)
               (make-center-column-markup
                (list


                 (make-center-column-markup
                  (list
                   (make-translate-markup
                    (cons 0 1)
                    (make-fontsize-markup
                     -6
                     (make-number-markup
                      (number->string
                       (car (cadr timeSignatureToShow))))))
                   (make-translate-markup
                    (cons 0 0)
                    (make-fontsize-markup
                     -6
                     (make-number-markup
                      (number->string
                       (cadr (cadr timeSignatureToShow))))))))
                 (make-number-markup
                  (number->string
                   (caddr (cadr timeSignatureToShow))))))
               ))


             ((= (length (cadr timeSignatureToShow)) 4)

              (make-override-markup
               (cons 'baseline-skip 0)
               (make-center-column-markup
                (list

                 (make-concat-markup
                  (list (make-number-markup
                         (number->string
                          (car (cadr timeSignatureToShow))))
                        (make-halign-markup
                         -1.5
                         (make-center-column-markup
                          (list
                           (make-translate-markup
                            (cons 0 1)
                            (make-fontsize-markup
                             -6
                             (make-number-markup
```

```
                             (number->string
                              (cadr (cadr timeSignatureToShow))))))
                          (make-translate-markup
                           (cons 0 0)
                           (make-fontsize-markup
                            -6
                            (make-number-markup
                             (number->string
                              (caddr (cadr timeSignatureToShow))))))))))
                (make-number-markup
                 (number->string
                  (cadddr (cadr timeSignatureToShow))))))
              ))
            )
      #:translate
      (cons 0 -0.5)
      (#:fontsize -12 " ")
      #:translate
      (cons 0 -0.5)
      (#:bold "+")
      #:translate
      (cons 0 -0.5)
      (#:fontsize -12 " ")

      #:override
      (cons 'baseline-skip 0)
      (cond ((= (length (caddr timeSignatureToShow)) 2)
             (make-center-column-markup
              (list (make-number-markup
                     (number->string
                      (car (caddr timeSignatureToShow))))
                    (make-number-markup
                     (number->string
                      (cadr (caddr timeSignatureToShow)))))))

            ((= (length (caddr timeSignatureToShow)) 3)

             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list


                (make-center-column-markup
                 (list
                  (make-translate-markup
                   (cons 0 1)
                   (make-fontsize-markup
```

```
                    -6
                    (make-number-markup
                     (number->string
                      (car (caddr timeSignatureToShow)))))
                  (make-translate-markup
                   (cons 0 0)
                   (make-fontsize-markup
                    -6
                    (make-number-markup
                     (number->string
                      (cadr (caddr timeSignatureToShow)))))))))
                (make-number-markup
                 (number->string
                  (caddr (caddr timeSignatureToShow))))))
              ))


            ((= (length (caddr timeSignatureToShow)) 4)

             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list

                (make-concat-markup
                 (list (make-number-markup
                        (number->string
                         (car (caddr timeSignatureToShow))))
                       (make-halign-markup
                        -1.5
                        (make-center-column-markup
                         (list
                          (make-translate-markup
                           (cons 0 1)
                           (make-fontsize-markup
                            -6
                            (make-number-markup
                             (number->string
                              (cadr (caddr timeSignatureToShow))))))
                          (make-translate-markup
                           (cons 0 0)
                           (make-fontsize-markup
                            -6
                            (make-number-markup
                             (number->string
                              (caddr (caddr timeSignatureToShow))))))))))
                (make-number-markup
                 (number->string
```

```
                 (cadddr (caddr timeSignatureToShow))))))
             ))
           )


     )))

  #{
   \time $underlyingMeter
   \set beatStructure = $beatStructure

   \override Timing.TimeSignature.stencil =
   #ly:text-interface::print
   \override Timing.TimeSignature.text =
   #mkup
  #})

backToNormalTimeSignature =
{
 \unset beatStructure
 \revert Timing.TimeSignature.stencil
 \revert Timing.TimeSignature.text
 \revert Staff.TimeSignature.stencil
 \revert Staff.TimeSignature.text
}


{

 \compoundFractionalTimeSignatureAThree #'((3 4)(4 5 4)(2 3 4)) 67/60 3,3,3,4,2
 \tuplet 3/2 {c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c'[ c' c']}
 \incompleteTupletBracket \tuplet 5/4 {c'16[ c' c' c']}
 \incompleteTupletBracket \tuplet 3/2 {c'8 c'}

 \backToNormalTimeSignature
 \time 3/4
 c'2.
}

```

11.7.4 Discussion

1. As mentioned in the **Grammar** section, it appears that specifying the beaming in the LilyPond code is still necessary. This is probably because of the unusual value of the fraction that needs to be given in the second argument of the function, `MEASURE_SPAN`.

2. As in the case of the other compound meters introduced in this document, I am hoping to

find ways to simplify the code.

3. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

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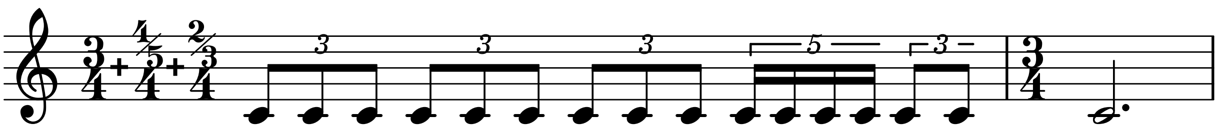
11.8 Compound Meter with Three Fractional Time Signatures, Style B



11.8.1 Description

This is an implementation of a compound meter with three fractional time signatures with Style B.

11.8.2 Grammar

```
\compoundFractionalTimeSignatureBThree
      #'((TIME_SIG1)(TIME_SIG2)(TIME_SIG3)) MEASURE_SPAN BEAT_STRUCT

\backToNormalTimeSignature
```

NB

1. See **Grammar** of the entry *Compound Meter with Three Fractional Time Signatures, Style A.*
2. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

11.8.3 Code

```
\version "2.26.0"



% Revised Jan 2 2025 for improving the appearance of fractions
% Revised Aug 10 2025 to include the function to revert to a
% regular time signature

suppressWarning =
#(define-void-function (amount message)(number? string?)
  (for-each
   (lambda (warning)
    (ly:expect-warning message))
   (iota amount 1 1)))

\suppressWarning 1 "strange time signature found"

incompleteTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
```

```
 \once \override Voice.TupletBracket.bracket-visibility = ##t

}
incompleteSmallTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t
 \once \override Voice.TupletNumber.X-offset =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       2.2
       1.2))

 \once \override Voice.TupletBracket.shorten-pair =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(-0.7 . 0.15)
       '(-0.3 . 0.8)))
 \once \override Voice.TupletBracket.X-positions =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(1.8 . 3)
       '(0.3 . 2.7)))
}

compoundFractionalTimeSignatureBThree =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)
  (define mkup
   (markup
    #:concat

    (

     #:override
     (cons 'baseline-skip 0)
     (cond ((= (length (car timeSignatureToShow)) 2)
            (make-center-column-markup
             (list (make-number-markup
                    (number->string
                     (car (car timeSignatureToShow))))
                   (make-number-markup
                    (number->string
                     (cadr (car timeSignatureToShow)))))))

           ((= (length (car timeSignatureToShow)) 3)
            (make-override-markup
             (cons 'baseline-skip 0)
```

```
              (make-center-column-markup
               (list
                (make-line-markup
                 (list
                  (make-number-markup
                   (make-right-align-markup
                    (make-translate-markup
                     (cons 0 1.6)
                     (make-fontsize-markup
                      -3
                      (number->string
                       (car (car timeSignatureToShow)))))))

                  (make-hspace-markup -1.5)

                  (make-override-markup
                   (cons 'alignment 0)
                   (make-translate-markup
                    (cons 0 0.8)
                    (make-draw-line-markup (cons 1.5 1.35))))

                  (make-hspace-markup -1.5)

                  (make-translate-markup
                   (cons 0 0)
                   (make-fontsize-markup
                    -3
                    (make-number-markup
                     (number->string
                      (cadr (car timeSignatureToShow))))))))

                (make-number-markup
                 (number->string
                  (caddr (car timeSignatureToShow)))))))


            ((= (length (car timeSignatureToShow)) 4)

             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list
                  (make-number-markup
                   (number->string
                    (car (car timeSignatureToShow))))
                  (make-fontsize-markup
```

```
             -12
             (make-simple-markup " "))

            (make-hspace-markup -1.25)
            (make-translate-markup
             (cons 0 0.4)
             (make-bold-markup
              (make-simple-markup "+")))

            (make-hspace-markup -0.25)

            (make-hspace-markup -0.5)
            (make-right-align-markup
             (make-number-markup
              (make-translate-markup
               (cons 0 1.5)
               (make-fontsize-markup
                -3
                (number->string
                 (cadr (car timeSignatureToShow)))))))

            (make-hspace-markup -1.5)

            (make-override-markup
             (cons 'alignment 0)
             (make-translate-markup
              (cons 0 0.8)
              (make-draw-line-markup
               (cons 1.5 1.35))))

            (make-hspace-markup -1.5)

            (make-number-markup
             (make-left-align-markup
              (make-fontsize-markup
               -3
               (number->string
                (caddr (car timeSignatureToShow)))))))

          (make-number-markup
           (number->string
            (cadddr (car timeSignatureToShow))))))))


#:translate
(cons 0 -0.5)
(#:fontsize -12 " ")
#:translate
```

```
      (cons 0 -0.5)
      (#:bold "+")
      #:translate
      (cons 0 -0.5)
      (#:fontsize -12 " ")

      #:override
      (cons 'baseline-skip 0)

      (cond ((= (length (cadr timeSignatureToShow)) 2)
             (make-center-column-markup
              (list (make-number-markup
                     (number->string
                      (car (cadr timeSignatureToShow))))
                    (make-number-markup
                     (number->string
                      (cadr (cadr timeSignatureToShow)))))))

            ((= (length (cadr timeSignatureToShow)) 3)
             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list
                  (make-number-markup
                   (make-right-align-markup
                    (make-translate-markup
                     (cons 0 1.6)
                     (make-fontsize-markup
                      -3
                      (number->string
                       (car (cadr timeSignatureToShow)))))))

                  (make-hspace-markup -1.5)

                  (make-override-markup
                   (cons 'alignment 0)
                   (make-translate-markup
                    (cons 0 0.8)
                    (make-draw-line-markup (cons 1.5 1.35))))

                  (make-hspace-markup -1.5)

                  (make-translate-markup
                   (cons 0 0)
                   (make-fontsize-markup
                    -3
```

```
                (make-number-markup
                 (number->string
                  (cadr (cadr timeSignatureToShow))))))))

            (make-number-markup
             (number->string
              (caddr (cadr timeSignatureToShow))))))))


        ((= (length (cadr timeSignatureToShow)) 4)

         (make-override-markup
          (cons 'baseline-skip 0)
          (make-center-column-markup
           (list
            (make-line-markup
             (list
              (make-number-markup
               (number->string
                (car (cadr timeSignatureToShow))))
              (make-fontsize-markup
               -12
               (make-simple-markup " "))

              (make-hspace-markup -1.25)
              (make-translate-markup
               (cons 0 0.4)
               (make-bold-markup
                (make-simple-markup "+")))

              (make-hspace-markup -0.25)

              (make-hspace-markup -0.5)
              (make-right-align-markup
               (make-number-markup
                (make-translate-markup
                 (cons 0 1.5)
                 (make-fontsize-markup
                  -3
                  (number->string
                   (cadr (cadr timeSignatureToShow)))))))

              (make-hspace-markup -1.5)

              (make-override-markup
               (cons 'alignment 0)
               (make-translate-markup
                (cons 0 0.8)
```

```
              (make-draw-line-markup
               (cons 1.5 1.35))))

            (make-hspace-markup -1.5)

            (make-number-markup
             (make-left-align-markup
              (make-fontsize-markup
               -3
               (number->string
                (caddr (cadr timeSignatureToShow)))))))

          (make-number-markup
           (number->string
            (cadddr (cadr timeSignatureToShow))))))))

#:translate
(cons 0 -0.5)
(#:fontsize -12 " ")
#:translate
(cons 0 -0.5)
(#:bold "+")
#:translate
(cons 0 -0.5)
(#:fontsize -12 " ")

#:override
(cons 'baseline-skip 0)

(cond ((= (length (caddr timeSignatureToShow)) 2)
       (make-center-column-markup
        (list (make-number-markup
               (number->string
                (car (caddr timeSignatureToShow))))
              (make-number-markup
               (number->string
                (cadr (caddr timeSignatureToShow)))))))

      ((= (length (caddr timeSignatureToShow)) 3)
       (make-override-markup
        (cons 'baseline-skip 0)
        (make-center-column-markup
         (list
          (make-line-markup
           (list
            (make-number-markup
             (make-right-align-markup
              (make-translate-markup
```

```
                     (cons 0 1.6)
                     (make-fontsize-markup
                      -3
                      (number->string
                       (car (caddr timeSignatureToShow)))))))

                  (make-hspace-markup -1.5)

                  (make-override-markup
                   (cons 'alignment 0)
                   (make-translate-markup
                    (cons 0 0.8)
                    (make-draw-line-markup (cons 1.5 1.35))))

                  (make-hspace-markup -1.5)

                  (make-translate-markup
                   (cons 0 0)
                   (make-fontsize-markup
                    -3
                    (make-number-markup
                     (number->string
                      (cadr (caddr timeSignatureToShow))))))))

                (make-number-markup
                 (number->string
                  (caddr (caddr timeSignatureToShow))))))))


            ((= (length (caddr timeSignatureToShow)) 4)

             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list
                  (make-number-markup
                   (number->string
                    (car (caddr timeSignatureToShow))))
                  (make-fontsize-markup
                   -12
                   (make-simple-markup " "))

                  (make-hspace-markup -1.25)
                  (make-translate-markup
                   (cons 0 0.4)
                   (make-bold-markup
```

```
                  (make-simple-markup "+")))

                (make-hspace-markup -0.25)

                (make-hspace-markup -0.5)
                (make-right-align-markup
                 (make-number-markup
                  (make-translate-markup
                   (cons 0 1.5)
                   (make-fontsize-markup
                    -3
                    (number->string
                     (cadr (caddr timeSignatureToShow)))))))

                (make-hspace-markup -1.5)

                (make-override-markup
                 (cons 'alignment 0)
                 (make-translate-markup
                  (cons 0 0.8)
                  (make-draw-line-markup
                   (cons 1.5 1.35))))

                (make-hspace-markup -1.5)

                (make-number-markup
                 (make-left-align-markup
                  (make-fontsize-markup
                   -3
                   (number->string
                    (caddr (caddr timeSignatureToShow)))))))

              (make-number-markup
               (number->string
                (cadddr (caddr timeSignatureToShow))))))))
    )

   ))

  #{
   \time $underlyingMeter
   \set beatStructure = $beatStructure

   \override Timing.TimeSignature.stencil =
   #ly:text-interface::print
   \override Timing.TimeSignature.text =
   #mkup
  #})
```

```

backToNormalTimeSignature =
{
 \unset beatStructure
 \revert Timing.TimeSignature.stencil
 \revert Timing.TimeSignature.text
 \revert Staff.TimeSignature.stencil
 \revert Staff.TimeSignature.text
}

{

 \compoundFractionalTimeSignatureBThree #'((3 4)(4 5 4)(2 3 4)) 67/60 3,3,3,4,2
 \tuplet 3/2 {c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c'[ c' c']}
 \incompleteTupletBracket \tuplet 5/4 {c'16[ c' c' c']}
 \incompleteTupletBracket \tuplet 3/2 {c'8 c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.

}
```

11.8.4 Discussion

See **Discussion** of the entry *Compound Meter with Three Fractional Time Signatures, Style A.*

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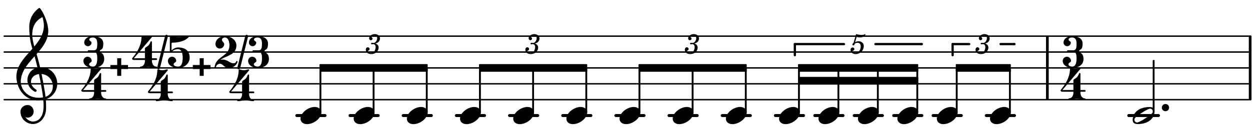
11.9 Compound Meter with Three Fractional Time Signatures, Style C



11.9.1 Description

This is an implementation of a compound meter with three fractional time signatures with Style C.

11.9.2 Grammar

```
\compoundFractionalTimeSignatureCThree
       #'((TIME_SIG1)(TIME_SIG2)(TIME_SIG3)) MEASURE_SPAN BEAT_STRUCT

\backToNormalTimeSignature
```

NB

1. See **Grammar** of the entry *Compound Meter with Three Fractional Time Signatures, Style A.*
2. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

11.9.3 Code

```
\version "2.26.0"



% Revised Jan 2 2025 for improving the appearance of fractions
% Revised Aug 10 2025 to include the function to revert to a
% regular time signature

suppressWarning =
#(define-void-function (amount message)(number? string?)
  (for-each
   (lambda (warning)
    (ly:expect-warning message))
   (iota amount 1 1)))

\suppressWarning 1 "strange time signature found"

incompleteTupletBracket =  {
```

```
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t

}
incompleteSmallTupletBracket =  {
 \once \override Voice.TupletBracket.edge-height = #'(0.7 . 0)
 \once \override Voice.TupletBracket.bracket-visibility = ##t
 \once \override Voice.TupletNumber.X-offset =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       2.2
       1.2))

 \once \override Voice.TupletBracket.shorten-pair =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(-0.7 . 0.15)
       '(-0.3 . 0.8)))
 \once \override Voice.TupletBracket.X-positions =
 #(lambda (grob)
   (if (= UP (ly:grob-property grob 'direction))
       '(1.8 . 3)
       '(0.3 . 2.7)))
}

compoundFractionalTimeSignatureCThree =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)
  (define mkup
   (markup
    #:concat
    (

     #:override
     (cons 'baseline-skip 0)
     (cond ((= (length (car timeSignatureToShow)) 2)
            (make-center-column-markup
             (list (make-number-markup
                    (number->string
                     (car (car timeSignatureToShow))))
                   (make-number-markup
                    (number->string
                     (cadr (car timeSignatureToShow)))))))

           ((= (length (car timeSignatureToShow)) 3)
            (make-override-markup
             (cons 'baseline-skip 0)
```

```
              (make-center-column-markup
               (list
                (make-line-markup
                 (list

                  (make-right-align-markup
                   (make-number-markup
                    (number->string
                     (car (car timeSignatureToShow)))))

                  (make-hspace-markup -0.6)

                  (make-override-markup
                   (list (cons 'alignment 0)
                         (cons 'thickness 2))
                   (make-draw-line-markup
                    (cons 0.5 2)))

                  (make-hspace-markup -0.6)

                  (make-number-markup
                   (make-left-align-markup
                    (number->string
                     (cadr (car timeSignatureToShow)))))))

                (make-number-markup
                 (number->string
                  (caddr (car timeSignatureToShow)))))))


            ((= (length (car timeSignatureToShow)) 4)

             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list
                  (make-number-markup
                   (number->string
                    (car (car timeSignatureToShow))))
                  (make-fontsize-markup
                   -12
                   (make-simple-markup " "))


                  (make-hspace-markup -1.25)
                  (make-translate-markup
```

```
                    (cons 0 0.4)
                    (make-bold-markup
                     (make-simple-markup "+")))

                   (make-hspace-markup -0.25)

                   (make-hspace-markup -0.5)
                   (make-right-align-markup
                    (make-number-markup
                     (number->string
                      (cadr (car timeSignatureToShow)))))

                   (make-hspace-markup -0.6)

                   (make-override-markup
                    (list (cons 'alignment 0)
                          (cons 'thickness 2))
                    (make-draw-line-markup (cons 0.5 2)))

                   (make-hspace-markup -0.6)

                   (make-number-markup
                    (make-left-align-markup
                     (number->string
                      (caddr (car timeSignatureToShow))))))

                 (make-number-markup
                  (number->string
                   (cadddr (car timeSignatureToShow))))))))


      #:translate
      (cons 0 -0.5)
      (#:fontsize -12 " ")
      #:translate
      (cons 0 -0.5)
      (#:bold "+")
      #:translate
      (cons 0 -0.5)
      (#:fontsize -12 " ")

      #:override
      (cons 'baseline-skip 0)

      (cond ((= (length (cadr timeSignatureToShow)) 2)
             (make-center-column-markup
              (list (make-number-markup
                     (number->string
```

```
                      (car (cadr timeSignatureToShow))))
                    (make-number-markup
                     (number->string
                      (cadr (cadr timeSignatureToShow)))))))

            ((= (length (cadr timeSignatureToShow)) 3)
             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list

                  (make-right-align-markup
                   (make-number-markup
                    (number->string
                     (car (cadr timeSignatureToShow)))))

                  (make-hspace-markup -0.6)

                  (make-override-markup
                   (list (cons 'alignment 0)
                         (cons 'thickness 2))
                   (make-draw-line-markup
                    (cons 0.5 2)))

                  (make-hspace-markup -0.6)

                  (make-number-markup
                   (make-left-align-markup
                    (number->string
                     (cadr (cadr timeSignatureToShow)))))))

                (make-number-markup
                 (number->string
                  (caddr (cadr timeSignatureToShow))))))))


            ((= (length (cadr timeSignatureToShow)) 4)

             (make-override-markup
              (cons 'baseline-skip 0)
              (make-center-column-markup
               (list
                (make-line-markup
                 (list
                  (make-number-markup
                   (number->string
```

```
                    (car (cadr timeSignatureToShow))))
                  (make-fontsize-markup
                   -12
                   (make-simple-markup " "))


                  (make-hspace-markup -1.25)
                  (make-translate-markup
                   (cons 0 0.4)
                   (make-bold-markup
                    (make-simple-markup "+")))

                  (make-hspace-markup -0.25)

                  (make-hspace-markup -0.5)
                  (make-right-align-markup
                   (make-number-markup
                    (number->string
                     (cadr (cadr timeSignatureToShow)))))

                  (make-hspace-markup -0.6)

                  (make-override-markup
                   (list (cons 'alignment 0)
                         (cons 'thickness 2))
                   (make-draw-line-markup (cons 0.5 2)))

                  (make-hspace-markup -0.6)

                  (make-number-markup
                   (make-left-align-markup
                    (number->string
                     (caddr (cadr timeSignatureToShow))))))

                (make-number-markup
                 (number->string
                  (cadddr (cadr timeSignatureToShow))))))))

      #:translate
      (cons 0 -0.5)
      (#:fontsize -12 " ")
      #:translate
      (cons 0 -0.5)
      (#:bold "+")
      #:translate
      (cons 0 -0.5)
      (#:fontsize -12 " ")
```

```
    #:override
    (cons 'baseline-skip 0)

    (cond ((= (length (caddr timeSignatureToShow)) 2)
           (make-center-column-markup
            (list (make-number-markup
                   (number->string
                    (car (caddr timeSignatureToShow))))
                  (make-number-markup
                   (number->string
                    (cadr (caddr timeSignatureToShow)))))))

          ((= (length (caddr timeSignatureToShow)) 3)
           (make-override-markup
            (cons 'baseline-skip 0)
            (make-center-column-markup
             (list
              (make-line-markup
               (list

                (make-right-align-markup
                 (make-number-markup
                  (number->string
                   (car (caddr timeSignatureToShow)))))

                (make-hspace-markup -0.6)

                (make-override-markup
                 (list (cons 'alignment 0)
                       (cons 'thickness 2))
                 (make-draw-line-markup
                  (cons 0.5 2)))

                (make-hspace-markup -0.6)

                (make-number-markup
                 (make-left-align-markup
                  (number->string
                   (cadr (caddr timeSignatureToShow))))))

              (make-number-markup
               (number->string
                (caddr (caddr timeSignatureToShow))))))))


          ((= (length (caddr timeSignatureToShow)) 4)

           (make-override-markup
```

```
           (cons 'baseline-skip 0)
           (make-center-column-markup
            (list
             (make-line-markup
              (list
               (make-number-markup
                (number->string
                 (car (caddr timeSignatureToShow))))
               (make-fontsize-markup
                -12
                (make-simple-markup " "))


               (make-hspace-markup -1.25)
               (make-translate-markup
                (cons 0 0.4)
                (make-bold-markup
                 (make-simple-markup "+")))

               (make-hspace-markup -0.25)

               (make-hspace-markup -0.5)
               (make-right-align-markup
                (make-number-markup
                 (number->string
                  (cadr (caddr timeSignatureToShow)))))

               (make-hspace-markup -0.6)

               (make-override-markup
                (list (cons 'alignment 0)
                      (cons 'thickness 2))
                (make-draw-line-markup (cons 0.5 2)))

               (make-hspace-markup -0.6)

               (make-number-markup
                (make-left-align-markup
                 (number->string
                  (caddr (caddr timeSignatureToShow))))))

             (make-number-markup
              (number->string
               (cadddr (caddr timeSignatureToShow))))))))
   )
  ))

#{
```

```
   \time $underlyingMeter
   \set beatStructure = $beatStructure

   \override Timing.TimeSignature.stencil =
   #ly:text-interface::print
   \override Timing.TimeSignature.text =
   #mkup
  #})

backToNormalTimeSignature =
{
 \unset beatStructure
 \revert Timing.TimeSignature.stencil
 \revert Timing.TimeSignature.text
 \revert Staff.TimeSignature.stencil
 \revert Staff.TimeSignature.text
}


{

 \compoundFractionalTimeSignatureCThree #'((3 4)(4 5 4)(2 3 4)) 67/60 3,3,3,4,2
 \tuplet 3/2 {c'8 c' c'} \tuplet 3/2 {c' c' c'}
 \tuplet 3/2 {c'[ c' c']}
 \incompleteTupletBracket \tuplet 5/4 {c'16[ c' c' c']}
 \incompleteTupletBracket \tuplet 3/2 {c'8 c'}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}

```

11.9.4 Discussion

See **Discussion** of the entry *Compound Meter with Three Fractional Time Signatures, Style A.*

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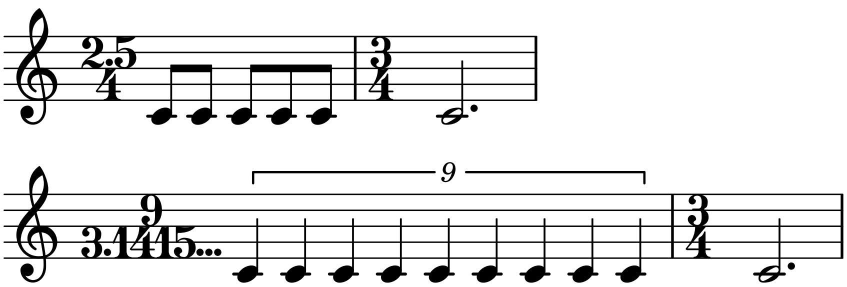
11.10 Time Signature with Decimals



11.10.1 Description

This is an implementation of a time signature with decimals. This function allows the user to use decimals for both numerator and denominator values of the time signature. While there are many examples where the numerator value contains decimals, notable examples for denominators that contain decimals include works by Mark Andre, such as *Un-fini I* (1995) for harp,[17] *Un-fini II* (1994-95) for harpsichord,[18] and *Contrapunctus* (1998/99) for piano..[19] This function also has the option of showing ellipsis, as discussed in **Grammar**.

11.10.2 Grammar

```
\decimalPointTimeSignature
        #'((NUMERATOR)(DENOMINATOR)) MEASURE_SPAN BEAT_STRUCT

\backToNormalTimeSignature
```

NB

1. The first argument takes a list of two lists. For both `NUMERATOR` and `DENOMINATOR`, one or two numbers can be placed.
   (a) If only one number is placed, it is treated as an integer. For example, `#'((3)(4))` would print: $\frac{3}{4}$.
   (b) If two numbers are placed, the first number is the integer portion of the number, and the second number is the decimals. For example, `#'((3 5)(4 232))` would print: $\frac{3.5}{4.232}$.
   (c) Placing a dot `.` at the end of the second number will print the ellipsis `...` at the end. This is useful for notating infinite decimal representations. For example, `#'((3 14159.)(4))` would print: $\frac{3.14159...}{4}$.
2. `MEASURE_SPAN` denotes how the measure may be written without the use of "decimal point" time signature.
3. `BEAT_STRUCT` indicates beat structure, by which the beaming of the measure abides.

17. Mark Andre, *Un-fini I : 1995, für eine Harfenistin/einen Harfenisten (Harfe, Tam-tam, grosse Trommel)*, Neue Musik bei Carus (Stuttgart: Carus, 1997).
18. Mark Andre, *Un-fini II : pour clavecin (1996)* (Paris: Editions Durand, 1998).
19. Mark Andre, *Contrapunctus : pour piano* (Paris: Durand, 2006).

4. When you wish to go back to a regular time signature, use `\backToNormalTimeSignature`, otherwise the identical fractional time signature will keep showing up.

11.10.3 Code

```
\version "2.26.0"



decimalPointTimeSignature =
#(define-music-function
  (timeSignatureToShow underlyingMeter beatStructure)
  (list? fraction? number-list?)

  (define (is-float? x)
   (and (number? x) (inexact? x)))

  #{
   \time $underlyingMeter
   \set beatStructure = $beatStructure
   \override Staff.TimeSignature.stencil =
   #ly:text-interface::print
   \override Staff.TimeSignature.text =
   #(markup
     (make-override-markup
      (cons 'baseline-skip 0)
      (make-center-column-markup
       (list
        (if (= (length (car  timeSignatureToShow)) 1)
            (make-number-markup
             (number->string

              (car (car timeSignatureToShow))))
            (make-line-markup
             (list
              (make-number-markup
               (number->string
                (car (car timeSignatureToShow))))
              (make-hspace-markup -0.5)
              (make-translate-markup
               '(0 . 0.15)
               (make-musicglyph-markup "period"))
              (make-hspace-markup -0.5)
              (if (not (is-float? (cadr (car timeSignatureToShow))))
                  (make-number-markup
                   (number->string
                    (inexact->exact (cadr (car timeSignatureToShow)))
                    ))
```

```
                    (make-line-markup
                     (list (make-number-markup
                            (number->string
                             (inexact->exact
                              (cadr (car timeSignatureToShow)))
                             ))
                           (make-hspace-markup -0.5)
                           (make-translate-markup
                            '(0 . 0.15)
                            (make-musicglyph-markup "period"))
                           (make-hspace-markup -0.5)
                           (make-translate-markup
                            '(0 . 0.15)
                            (make-musicglyph-markup "period"))
                           (make-hspace-markup -0.5)
                           (make-translate-markup
                            '(0 . 0.15)
                            (make-musicglyph-markup "period"))))
                    )
                )
               )
              )
          (if (= (length (cadr  timeSignatureToShow)) 1)
              (make-number-markup
               (number->string
                (car (cadr timeSignatureToShow))))
              (make-line-markup
               (list
                (make-number-markup
                 (number->string
                  (car (cadr timeSignatureToShow))))
                (make-hspace-markup -0.5)
                (make-translate-markup
                 '(0 . 0.15)
                 (make-musicglyph-markup "period"))
                (make-hspace-markup -0.5)
                (if (not (is-float? (cadr (cadr timeSignatureToShow))))
                    (make-number-markup
                     (number->string
                      (inexact->exact (cadr (cadr timeSignatureToShow)))
                      ))
                    (make-line-markup
                     (list (make-number-markup
                            (number->string
                             (inexact->exact
                              (cadr (cadr timeSignatureToShow)))
                             ))
                           (make-hspace-markup -0.5)
```

```
                         (make-translate-markup
                          '(0 . 0.15)
                          (make-musicglyph-markup "period"))
                         (make-hspace-markup -0.5)
                         (make-translate-markup
                          '(0 . 0.15)
                          (make-musicglyph-markup "period"))
                         (make-hspace-markup -0.5)
                         (make-translate-markup
                          '(0 . 0.15)
                          (make-musicglyph-markup "period"))))
                  )
              )
             )
            )
        )
       ))
     (make-hspace-markup -1))
  #})

backToNormalTimeSignature =
{
 \unset beatStructure
 \revert Timing.TimeSignature.stencil
 \revert Timing.TimeSignature.text
 \revert Staff.TimeSignature.stencil
 \revert Staff.TimeSignature.text
}


{
 \decimalPointTimeSignature #'((2 5)(4)) 5/8 2,3
 c'8 c' c' c' c'
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}


{
 \decimalPointTimeSignature #'((9)(3 1415.)) 9/4 3,3,3
 \tuplet 9/9 {c'4 4 4 4 4 4 4 4 4}
 \backToNormalTimeSignature
 \time 3/4
 c'2.
}
```

11.10.4 Discussion

The structure of the code where the user specifies integer and decimal portions of either numerator, denominator, or both, resulted from the fact that the period "." by default appeared too close to the staff line and the denominator, possibly rendering the time signature difficult to read. In the code I made these periods appear via `\translate` feature, where I offset the period upward by the value of 0.15, allowing the period sign to be separated from the staff line and the denominator.

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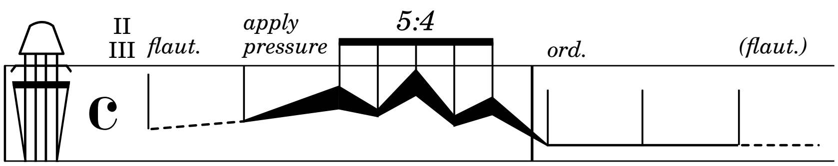
Chapter 12

Combinations

This chapter presents examples that combine several snippets from the previous chapters. **Variables Used** provides a comprehensive list of all the variables required to generate the snippet. Among these, indented variables indicate ”variables of a variable,” i.e., dependent variables necessary for the main variables to function. The **Code** section only lists the score portion of the LilyPond code.

12.1 Prescriptive Notation for String Instruments



12.1.1 Description

An example of a prescriptive notation for a string instrument. Vertical placement of the notehead corresponds to the position at which bowing takes place. Horizontally it shows the change of the bow pressure against the string(s).

12.1.2 Variables Used

```
\stringPositionClef
  \stringPositionClefDesign
\dashedLineNotehead
\modularLineNotehead
\noteheadless
```

12.1.3 Code

```
\version "2.26.0"

strPosClefDesign = #(ly:make-stencil (list 'embedded-ps
```

```
                                            "gsave
currentpoint translate
/fingboardpath
{

newpath

-0.55 7.5 moveto
0 -3 rlineto
1 -6.5 rlineto
-1 -1 rlineto
0 -3 rlineto
4.1 0 rlineto
0 3 rlineto
-1 1 rlineto
1 6.5 rlineto
0 3 rlineto
closepath

} def

fingboardpath clip
newpath
0.15 setlinewidth
0.5 4.75 moveto
0 -6.8 rlineto
-0.75 5 rlineto
3.5 0 rlineto
-0.75 -5 rlineto
0. 6.8 rlineto
stroke
0.35 setlinewidth
-0.4 2.75 moveto
3.75 0 rlineto
stroke

%inner two line
newpath
0.15 setlinewidth
1.16 4.75 moveto
0. -6.8 rlineto
1.8 4.75 moveto
0. -6.8 rlineto
stroke

%bridge
newpath
-0.4 3.6 moveto
```

```
0.3 0.4 rlineto
3.2 0 rlineto
0.3 -0.4 rlineto
stroke

%tailpiece
0.15 4.75 moveto
1 setlinecap
1 setlinejoin
2.75 0 rlineto
-0.65 1.75 rlineto
-0 -0  -0.6 0.55 -1.45 0  rcurveto
closepath
stroke

newpath
0.2 setlinewidth
1 setlinecap
1.5 -2.25 moveto
0 -2.5 rlineto
0.25 -3.5 moveto
2.5 0 rlineto
stroke
newpath
1.5 -3.5 0.85 0 360 arc
stroke
grestore")
                                     (cons 0 3)
                                     (cons 0 1))

strPosClefSize =
#(lambda (grob)
  (let* ((bCS (ly:grob-property grob 'font-size 0.0))
         (mult (magstep bCS)))
   (ly:stencil-scale
    strPosClef
    mult mult)))

strPosClef = {
 \override Staff.Clef.stencil = \strPosClefDesign
}


dashedLineNotehead =
#(define-music-function
  (beginning end x-distance) (ly:music? ly:music? number?)
  (let*
   (
```

```
    (p1 (ly:music-property beginning 'pitch))
    (p2 (ly:music-property end 'pitch))
    (steps
     (-
      (+ (* 7 (ly:pitch-octave p2)) (ly:pitch-notename p2))
      (+ (* 7 (ly:pitch-octave p1)) (ly:pitch-notename p1))
      )
     )
    )
   #{
    {

     \once \override Voice.NoteHead.stencil = #ly:text-interface::print
     \once \override Voice.NoteHead.stem-attachment = #'(0 . 0)
     \once \override Staff.LedgerLineSpanner.stencil = ##f
     \once \override Voice.NoteHead.text = \markup         {
      % \translate #(cons 0 0)
      \postscript
      #(string-append
        "newpath 1 setlinecap
              0.15 setlinewidth
              0 0 moveto
              [.4 .4 .4 .4] 3 setdash "
        (number->string x-distance)  " " (number->string (* steps 0.5))
        " rlineto stroke"
        )
     }
     #beginning
     \revert Voice.NoteHead.stencil
     \revert Staff.LedgerLineSpanner.stencil
    }
   #})
  )

modularLineNotehead =
#(define-music-function
  (beginning end beginningThickness endingThickness x-distance)
  (ly:music? ly:music? number? number? number?)
  (let*
   (
    (p1 (ly:music-property beginning 'pitch))
    (p2 (ly:music-property end 'pitch))
    (steps
     (-
      (+ (* 7 (ly:pitch-octave p2)) (ly:pitch-notename p2))
      (+ (* 7 (ly:pitch-octave p1)) (ly:pitch-notename p1))
      )
     )
```

```
    )
   #{
    {

     \once \override Voice.NoteHead.stencil = #ly:text-interface::print
     \once \override Voice.NoteHead.stem-attachment = #'(0 . 0)
     \once \override Voice.LedgerLineSpanner.transparent = ##t
     \once \override Voice.NoteHead.text = \markup         {
      \postscript
      #(string-append
        "newpath 1 setlinecap 0.1 setlinewidth -0.05 0 moveto 0 "
        (number->string (* beginningThickness 0.005)) " rlineto "
        (number->string x-distance) " "
        (number->string (+ (- (* endingThickness 0.005)
                              (* beginningThickness 0.005) )
                           (* steps 0.5)))
        " rlineto 0 "
        (number->string  (* endingThickness -0.01))  " rlineto "
        (number->string (* -1 x-distance))  " "
        (number->string (- (* endingThickness 0.005)
                           (* beginningThickness 0.005)
                           (* steps 0.5)))
        " rlineto
              closepath
              fill"
        )
     }
     #beginning
     \revert Voice.NoteHead.stencil
     \revert Staff.LedgerLineSpanner.stencil
    }
   #})
  )

noteheadless = \once \override Voice.NoteHead.stencil = ##f

#(define-public ((tuplet-number::append-note-wrapper function note) grob)
  (let ((txt (function grob)))
   (markup txt #:fontsize -5 #:note note UP)))



\score {
 {
  \override Staff.StaffSymbol.line-positions = #'(6  -6)
  \strPosClef \dashedLineNotehead g'4
  ^\markup {\fontsize #-4 \italic flaut.}
  ^\markup \translate #'(-2.5 . -0)
```

```
    \center-column {
     \translate #'(0 . -1.5)
     \fontsize #-4 II \fontsize #-4 III
    }
    a' #6
    \modularLineNotehead a'
    ^\markup \column {
     \translate #'(0 . -1.5)
     \fontsize #-4 \italic apply
     \fontsize #-4 \italic pressure
    }
    d'' #15 #150 #6
    \override TupletNumber.text = #tuplet-number::calc-fraction-text
    \stemUp  \tuplet 5/4 {
     \modularLineNotehead d''8 b' #150 #50 #2.5
     \modularLineNotehead b' f'' #50 #175 #2.5
     \modularLineNotehead f'' a' #175 #70 #2.5
     \modularLineNotehead a' c'' #70 #120 #2.5
     \modularLineNotehead c'' e' #120 #15 #3.5
    }
    |
    \modularLineNotehead e'4
    ^\markup {\fontsize #-4 \italic ord.} e' #15 #15 #12
    \noteheadless e'
    \dashedLineNotehead e'
    ^\markup {\fontsize #-4 \italic (flaut.)} e' #5
   }

   \layout {
    \context {
     \Score proportionalNotationDuration = #1/10
     \override SpacingSpanner.uniform-stretching = ##t
    }
   }
  }

```

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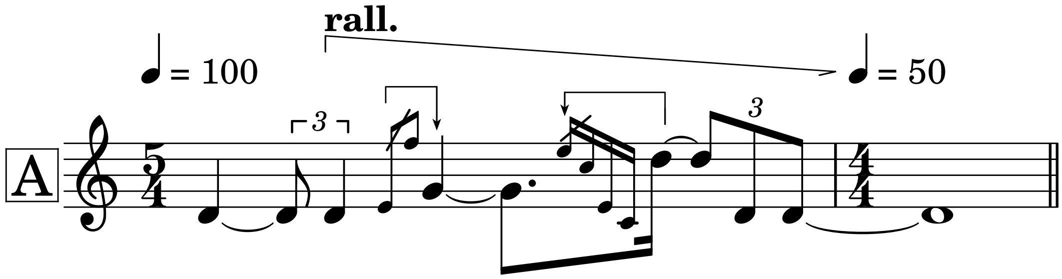
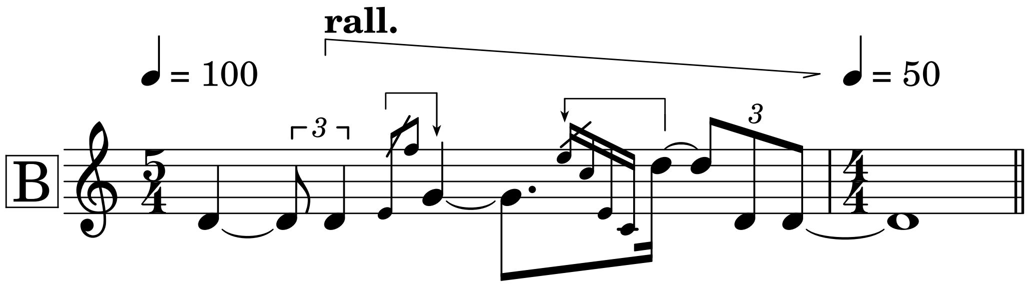
12.2 Multiple Instances Of Spanners At Once





12.2.1 Description

Invoking two or more Text Spanners (that require `\stopTextSpan` for them to finish their processes) all on one single layer could cause the spanners to behave unexpectedly. This entry is an attempt to avoid such unexpected behaviors by invoking a spanner per layer (A), or per staff line (B).

12.2.2 Variables Used

```
\startSlashedGraceMusic
\stopSlashedGraceMusic
\graceNoteBeforeBeatOn
\graceNoteBeforeBeatOff
\graceNoteAfterBeatOn
\graceNoteAfterBeatOff
\rallArrow
```

12.2.3 Code

```
\version "2.26.0"



accelArrow =
#(define-music-function (line_angle) (number?)

  (define x_value (cos (* (/ 3.14159265358979 180) (- 90 line_angle))))
  (define y_value (sin (* (/ 3.14159265358979 180) (- 90 line_angle))))
  #{
```

```
    \tweak direction #up
    \tweak style  #'line
    \tweak thickness  #1
    \tweak to-barline ##t
    \tweak rotation #(list line_angle -1 0 )
    \tweak bound-details.left.stencil #ly:text-interface::print
    \tweak bound-details.left.text \markup \postscript
    #(string-append
      "gsave newpath
0 0 moveto "
      (number->string x_value) " "
      (number->string y_value)
      " rlineto
stroke
grestore")
    \tweak bound-details.left-broken.stencil #ly:text-interface::print
    \tweak bound-details.left-broken.text ##f

    \tweak bound-details.right.stencil #ly:text-interface::print
    \tweak bound-details.right.text \markup \postscript
    "newpath
0 0 moveto
-1 -0.3 rlineto
stroke"
    \tweak bound-details.right-broken.stencil #ly:text-interface::print
    \tweak bound-details.right-broken.text ##f
    \tweak font-shape  #'upright
    \tweak bound-details.left.padding  #0
    \tweak bound-details.right.padding  #0
    \tweak breakable ##t
    \tweak after-line-breaking ##t

    \startTextSpan
  #})

rallArrow =
#(define-music-function (line_angle) (number?)

  (define x_value (cos (* (/ 3.14159265358979 180) (- 90 line_angle))))
  (define y_value (sin (* (/ 3.14159265358979 180) (- 90 line_angle))))
  #{
    \tweak direction #up
    \tweak style  #'line
    \tweak thickness  #1
    \tweak to-barline ##t
    \tweak rotation #(list (* -1 line_angle) 1 0 )

    \tweak bound-details.left.stencil #ly:text-interface::print
```

```
    \tweak bound-details.left.text \markup \postscript
    #(string-append
      "gsave
newpath
0 0 moveto "
      (number->string x_value) " "
      (number->string (* -1 y_value))
      " rlineto
stroke
grestore")
    \tweak bound-details.left-broken.stencil #ly:text-interface::print
    \tweak bound-details.left-broken.text ##f

    \tweak bound-details.right.stencil #ly:text-interface::print
    \tweak bound-details.right.text \markup \postscript
    "newpath
0 0 moveto
-1 -0.3 rlineto
stroke"
    \tweak bound-details.right-broken.stencil #ly:text-interface::print
    \tweak bound-details.right-broken.text ##f
    \tweak font-shape  #'upright
    \tweak bound-details.left.padding  #0
    \tweak bound-details.right.padding  #0
    \tweak breakable ##t
    \tweak after-line-breaking ##t
    \startTextSpan
  #})


graceNoteBeforeBeatOn =
#(define-music-function (starting_note) (ly:music?)
  #{
    \once \override TextSpanner.style = #'line
    \once \override TextSpanner.bound-details.left.text =
    \markup { \draw-line #'(0 . -1) }
    \once \override TextSpanner.bound-details.right.text =
    \markup {
     \postscript
     "newpath 0 0 moveto
0 -2.5 rlineto
stroke
newpath
-0.275 -2 moveto
0.275 -0.75 rlineto
0.275 0.75 rlineto
-0.275 -0.2 rlineto
closepath
```

```
fill"
   }
   \once \override TextSpanner.Y-offset = #5
   \once \override TextSpanner.bound-details.left.padding = #0.5
   \once \override TextSpanner.bound-details.right.padding = #-0.25
   #starting_note
   \startTextSpan
  #})


graceNoteBeforeBeatOff =
#(define-music-function (ending_note) (ly:music?)
  #{
   #ending_note
   \stopTextSpan
  #})


graceNoteAfterBeatOn =
#(define-music-function (starting_note) (ly:music?)
  #{
   \once \override TextSpanner.style = #'line
   \once \override TextSpanner.bound-details.right.text =
   \markup {
    \combine \draw-line #'(0 . -1)
    \postscript "newpath
0 -1 moveto
0 -1 rlineto
stroke"
   }
   \once \override TextSpanner.bound-details.left.text =
   \markup {
    \postscript
    "newpath 0 0 moveto
0 -1 rlineto
stroke
newpath
-0.275 -0.75 moveto
0.275 -0.75 rlineto
0.275 0.75 rlineto
-0.275 -0.2 rlineto
closepath
fill"
   }
   \once \override TextSpanner.Y-offset = #2
   \once \override TextSpanner.bound-details.left.padding = #0.5
   \once \override TextSpanner.bound-details.right.padding = #-0.25

```

```
   #starting_note
   \startTextSpan
  #})


graceNoteAfterBeatOff =
#(define-music-function (ending_note) (ly:music?)
  #{
   #ending_note
   \stopTextSpan
  #})

%%%%%%%%%%%%%%%%%%%LSR SNIPPET START%%%%%%%%%%%%%%%%%

#(define (degrees->radians deg)
  (* PI (/ deg 180.0)))


slash =
#(define-music-function (ang stem-fraction protrusion)
  (number? number? number?)
  (remove-grace-property 'Voice 'Stem 'direction)
  #{
   \once \override Stem.stencil =
   #(lambda (grob)
     (let* ((x-parent (ly:grob-parent grob X))
            (is-rest? (ly:grob?
                       (ly:grob-object x-parent 'rest)))
            (beam (ly:grob-object grob 'beam))
            (stil (ly:stem::print grob)))
      (cond
       (is-rest? empty-stencil)
       ((ly:grob? beam)
        (let* ((refp (ly:grob-system grob))
               (stem-y-ext (ly:grob-extent grob grob Y))
               (stem-length
                (- (cdr stem-y-ext) (car stem-y-ext)))
               (beam-X-pos (ly:grob-property beam 'X-positions))
               (beam-Y-pos (ly:grob-property beam 'positions))
               (beam-slope (/ (- (cdr beam-Y-pos) (car beam-Y-pos))
                              (- (cdr beam-X-pos) (car beam-X-pos))))
               (beam-angle (atan beam-slope))
               (dir (ly:grob-property grob 'direction))
               (line-dy (* stem-length stem-fraction))
               (line-dy-with-protrusions (if (= dir 1)
                                             (+ (* 4 protrusion) beam-angle)
                                             (- (* 4 protrusion) beam-angle)))
               (ang (if (> beam-slope 0)
```

```
                        (if (= dir 1)
                            (+ (degrees->radians ang) (* beam-angle 0.7))
                            (degrees->radians ang))
                        (if (= dir 1)
                            (degrees->radians ang)
                            (- (degrees->radians ang) (* beam-angle 0.7)))))
               (line-dx (/ line-dy-with-protrusions (tan ang)))
               (protrusion-dx (/ protrusion (tan ang)))
               (corr (if (= dir 1) (car stem-y-ext) (cdr stem-y-ext))))
         (ly:stencil-add
          stil
          (grob-interpret-markup grob
                                 (markup
                                  #:translate
                                  (cons (- protrusion-dx)
                                        (+ corr
                                           (* dir
                                              (- stem-length
                                                 (+ stem-fraction protrusion)))))
                                  #:override '(thickness . 1.7)
                                  #:draw-line
                                  (cons line-dx
                                        (* dir line-dy-with-protrusions)))))))
       (else stil))))
  #})

startSlashedGraceMusic = {
 \slash 40 1 0.5
 \override Flag.stroke-style = #"grace"
}
stopSlashedGraceMusic = {
 \revert Flag.stroke-style
}

startAcciaccaturaMusic = {
 \slash 40 1 0.5
 s1*0(
 \override Flag.stroke-style = #"grace"
}
stopAcciaccaturaMusic = {
 \revert Flag.stroke-style
 s1*0)
}
%%%%%%%%%%%%%%%%%%%% LSR SNIPPET END %%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%%%%%%%% A %%%%%%%%%%%%%%%%%%%

\score {
```

```

 \new Staff = "allElementsCombined"
 \with {instrumentName = \markup {\fontsize #4 \box "A"}}  {
  \numericTimeSignature
  \override Score.MetronomeMark.Y-offset = #5.75
  \tempo 4 = 100
  \time 5/4

  <<
   {
    \tieNeutral \stemNeutral
    d'4~
    \tuplet 3/2 {d'8 d'4}
    \stemUp \grace  {
     \startSlashedGraceMusic \graceNoteBeforeBeatOn e'8 f''
     \stopSlashedGraceMusic
    } \graceNoteBeforeBeatOff g'4~
    \stemNeutral g'8.[ \grace {
     \startSlashedGraceMusic \graceNoteAfterBeatOn
     e''16 c'' e' c' \stopSlashedGraceMusic
    }
    \graceNoteAfterBeatOff d''16]~
    \tuplet 3/2 {d''8 d'8 d'8~} |
    \time 4/4
    d'1 \bar"||"
   }
   \\
   {
    s4  \tuplet 3/2 {
     s8 \override Voice.TextSpanner.Y-offset = #6.5
     s4^\markup {\translate #'(0 . 6.5) \bold "rall."}
     \rallArrow #4
    } s2. \tempo 4 = 50 s4*4 \stopTextSpan
   }
  >>
 }
}
%%%%%%%%%%%%%%%%%%% B %%%%%%%%%%%%%%%%%%
\score {
 <<
  \new Staff = "tempoLine" \with {
   \remove Clef_engraver
   \remove Staff_symbol_engraver
   \remove Time_signature_engraver
  }
  {
   \numericTimeSignature
   \override Score.MetronomeMark.Y-offset = #6
```

```
    \tempo 4 = 100
    \time 5/4
    s4  \tuplet 3/2 {
     s8 \override Voice.TextSpanner.Y-offset = #-2.25
     s4^\markup {\translate #'(0 . 0) \bold "rall."}
     \rallArrow #4
    } s2 \after 64*15 \stopTextSpan s8*2 |
    \tempo 4 = 50  s4*4
   }

   \new Staff = "music"
   \with { instrumentName = \markup {\fontsize #4 \box "B"}}
   {
    \tieNeutral \stemNeutral
    d'4~
    \tuplet 3/2 {d'8 d'4}
    \grace {
     \startSlashedGraceMusic \graceNoteBeforeBeatOn e'8 f''
     \stopSlashedGraceMusic
    } \graceNoteBeforeBeatOff g'4~
    g'8.[ \grace {
     \startSlashedGraceMusic \graceNoteAfterBeatOn
     e''16 c'' e' c' \stopSlashedGraceMusic
    }
    \graceNoteAfterBeatOff d''16]~
    \tuplet 3/2 {d''8 d'8 d'8~} |
    \time 4/4
    d'1 \bar"||"
   }
  >>
 }
```

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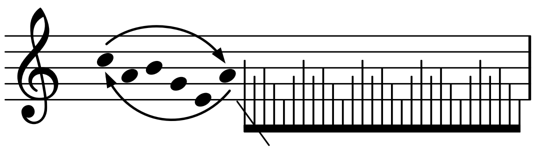
12.3 Repeat Note Patterns As Fast As possible



12.3.1 Description

An implementation of the arrowed arcs, followed by the rapid note patterns without the noteheads. Note that the rapid note patterns are created without the use of grace notes, and the slash is added to imitate the look of the grace notes.

12.3.2 Variables Used

\parenthesisBeginWithArrow
\parenthesisEndWithArrow
\slashBeam

12.3.3 Code

```
\version "2.26.0"

\pointAndClickOff

parenthesisBeginWithArrows =
#(define-music-function
  (width height arcAngle offset parenthesis? pitchthing)
  (number? number? number? (pair? (cons 0 0)) (boolean? #t) ly:music?)
  (define p1 (ly:music-property pitchthing 'pitch))
  (define steps (+ -6 (ly:pitch-steps p1)))
  (define radToDeg (* 180 (/ 1 3.141592653589793)))
  #{
   \once \override Parentheses.font-size = #2
   \once \override Parentheses.stencils =
   #(lambda (grob)
     (let ((par-list
            (parentheses-interface::calc-parenthesis-stencils grob)))
      (if parenthesis?
          (list (car par-list) point-stencil)
          (list point-stencil point-stencil)
          )
      )
     )
   \parenthesize
   #pitchthing
```

```
   -\tweak ignore-collision ##t
   -\tweak TextScript.outside-staff-priority ##f
   -\tweak TextScript.parent-alignment-X #1
   -\tweak TextScript.extra-offset
   #(cons  (+ -0.5 (car offset)) (+ -1.25 (* steps 0.5) (cdr offset)))
   -\tweak TextScript.rotation #`(,arcAngle -1 -1)
   ^\markup {
    \rotate #3.5
    \concat {
     \path #0.15
     #`((curveto ,(* width 0.25) ,height ,(* width 0.75)
                 ,height ,width 0))
     \hspace #-0.1 \vcenter \postscript
     #(string-append
       "
/arrowhead {
gsave
"  (number->string
       (+ (* radToDeg (atan (/ (* height -1)  (* width 0.25)))) -90))
       " rotate
0 0.5 rlineto
-0.35 -1 rlineto
0.7 0 rlineto
-0.35 1 rlineto
 closepath
fill
grestore } def

0 0 moveto
arrowhead
")
    }
  } #})



parenthesisEndWithArrows =
#(define-music-function
  (width height arcAngle offset parenthesis? pitchthing)
  (number? number? number? (pair? (cons 0 0)) (boolean? #t) ly:music?)
  (define p1 (ly:music-property pitchthing 'pitch))
  (define steps (+ -6 (ly:pitch-steps p1)))
  (define radToDeg (* 180 (/ 1 3.141592653589793)))
  #{
   \once  \override Parentheses.font-size = #2
   \once \override Parentheses.stencils =
   #(lambda (grob)
     (let ((par-list
```

```
            (parentheses-interface::calc-parenthesis-stencils grob)))
      (if parenthesis?
          (list  point-stencil (cadr par-list))
          (list point-stencil point-stencil)
          )
      )
     )
   \parenthesize
   #pitchthing
   -\tweak ignore-collision ##t
   -\tweak TextScript.outside-staff-priority ##f
   -\tweak TextScript.parent-alignment-X #-1
   -\tweak TextScript.extra-offset
   #(cons  (+ 0.75 (car offset)) (+ 1.75 (* steps 0.5) (cdr offset)))
   -\tweak TextScript.rotation #`(,arcAngle -1 0)
   _\markup \right-align {
    \rotate #3.5
    \concat  {

     \postscript
     #(string-append
       "
/arrowhead {
gsave
"  (number->string
       (+ (* radToDeg (atan (/ (* height -1)  (* width 0.25)))) 270))
       " rotate
" (if (< height 0) (number->string 0.1) (number->string -0.1)) "
 -0.3 rmoveto
-0.35 1 rlineto
0.7 0 rlineto
-0.35 -1 rlineto
closepath
fill
grestore } def

0 0 moveto
arrowhead
")
     \center-align
     \path #0.15
     #`((curveto ,(* width 0.25) ,(* height -1) ,(* width 0.75)
                 ,(* height -1) ,width 0))

    }
  } #})


```

```
slashBeam = {
 \once \override Stem.stencil =
 #(grob-transformer
   'stencil
   (lambda (grob original)
    (let* ((added-markup
            #{
             #(case (ly:grob-property grob 'direction)
               ( (1) #{ \markup \general-align #X #-0.5
                        {
                         \path #0.1
                         #'((rmoveto 0 0)
                            (rlineto 2 2.75))
                     } #}
                     )
               ( (-1) #{ \markup \general-align #X #-0.5
                         {
                          \path #0.1
                          #'((rmoveto 0 0)
                             (rlineto 2 -2.75))
                      } #}
                      )
               )
            #}
            )
           (added-stencil (grob-interpret-markup grob added-markup)))

     (if (ly:stencil? original)
         (case  (ly:grob-property grob 'direction)
          ((1) (ly:stencil-combine-at-edge
                original 1 1 added-stencil -1.75))
          ((-1) (ly:stencil-combine-at-edge
                 original 1 -1 added-stencil -1.75))
          )
         added-stencil))))
}



{
 \new Staff
 \once \override Staff.TimeSignature.stencil = ##f
 {
  \set Score.proportionalNotationDuration = #1/5

  <<
   {
    \omit Stem
```

```
     \omit Beam
     \omit Flag
     \tiny
     \parenthesisBeginWithArrows #7.25 #2 #-10 #'(0 . -1) ##f c''64
     a' b' g' e'
     \parenthesisEndWithArrows #7.75 #3 #-10 #'(0 . 1)##f a'
    }
    \new Voice {
     \stemDown
     \override NoteHead.stencil = #point-stencil
     % \once \override Beam.beam-thickness = #0.2
     \override Beam.length-fraction = #0.225
     s16  s32 [ \slashBeam  \autoBeamOn
     \repeat unfold 4 {   c''  a' b' g' e' a' }
     c''  a' b' g' e' ]
    }
   >>
  }
 }

```

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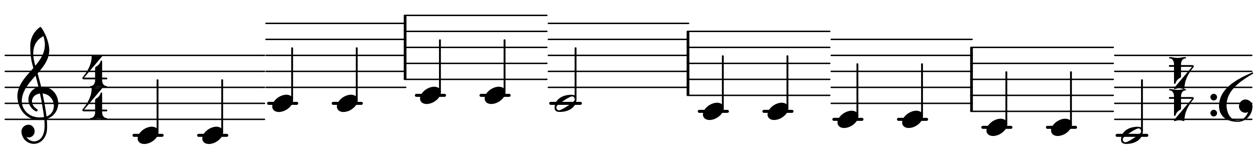
Chapter 13

Miscellanies

This chapter presents snippets that do not really belong to any of the other preceding chapters but I learned tremendously from making. Quite often I have made these snippets as a diversion.

13.1 Shifting Staffs, Rotated Clef and Time Signature



13.1.1 Description

Staff lines that are shifted so that, when the note moves away from the middle C, the staff lines move accordingly. The excerpt ends with a time signature and a clef that are rotated 180 degrees.

13.1.2 Code

```
\version "2.26.0"

staone = {
 \stopStaff
 \override Staff.StaffSymbol.line-positions =
 #'(0 2 4 6 8)
 \startStaff
}
statwo = {
 \stopStaff
 \override Staff.StaffSymbol.line-positions =
 #'(1 3 5 7 9)
 \startStaff
}
stathree = {
 \stopStaff
```

```
 \override Staff.StaffSymbol.line-positions =
 #'(-1 1 3 5 7)
 \startStaff
}
stafour = {
 \stopStaff
 \override Staff.StaffSymbol.line-positions =
 #'(-2 0 2 4 6)
 \startStaff
}
stafive = {
 \stopStaff
 \override Staff.StaffSymbol.line-positions =
 #'(-3 -1 1 3 5)
 \startStaff
}
stanorm = {
 \stopStaff
 \revert Staff.StaffSymbol.line-positions
 \startStaff
}
{
 \numericTimeSignature
 \time 4/4

 c'4 c' \staone g' g' \statwo a' a' \staone g'2
 \stathree f'4 f' \stafour e' e' \stafive d' d' \stanorm
 \override TextScript.outside-staff-priority = ##f
 \once \override TextScript.extra-offset = #'(0 . -4.5)
 c'2 ^\markup \concat {
  {
   \hspace #3 \rotate #180
   {\compound-meter #'(4 4)}
  }
  {
   \translate-scaled #'(1 . 0.5)
   \rotate #180 \musicglyph "clefs.F"
  }
 }
 \bar ""

}

\layout {
 \context{
  \Score   proportionalNotationDuration = #1/7
 }
}
```

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Chapter 14

Exploring Scheme

14.1 Introduction

Scheme, one of the dialects of the Lisp family of programming languages, is used in LilyPond as its extension language. Scheme allows LilyPond users to explore the inner workings of the program, enabling significant customization. The snippets in this document would not exist without taking advantage of it.[1]

However, learning Scheme can be daunting. In his unfinished book on Scheme and LilyPond, Urs Liska refers to its "thorny path."[2] While I have experience with Common Lisp (another Lisp dialect) from my work with OpenMusic, adjusting to Scheme's grammatical nuances still took some time.

This chapter does not aim to be a comprehensive guide to using Scheme in LilyPond.[3] Instead, it offers suggestions for newcomers to familiarize themselves with Scheme.

14.1.1 Step 1a: Focus on the Scheme Language Itself

Scheme is a language distinct from LilyPond, and understanding this distinction is essential. For simpler LilyPond tasks, Scheme may not be necessary. However, when working with internal parameters, Scheme allows deeper customization. It is beneficial to first study Scheme independently, learning its syntax and concepts by writing simple code.

14.1.2 Step 1b: Get Used to Prefix Notation

Scheme, like its Lisp relatives, uses prefix notation (Cambridge Polish Notation). Here are examples:

```
(+ 12 34)
>> This expression results in the value of 46.
```

1. For newcomers: parts of LilyPond code written in Scheme are often enclosed in `#(` and `)`. Numerical values preceded by `#`, and number pairs such as `\#'(1 . -2)`, are also part of the Scheme language.

2. Urs Liska, *Understanding Scheme In LilyPond*, vol. 2024, December 19 (2020), Web Page, https://scheme-book.readthedocs.io/en/latest/.

3. For a deeper dive, refer to the resource by Liska, as well as Jean Abou Samra, *Extending LilyPond*, vol. 2024, December 19 (2021), Web Page, https://extending-lilypond.gitlab.io/en/index.html. LilyPond also provides its own Extending Manual: https://lilypond.org/doc/v2.24/Documentation/extending/index

```
(+ 4 (* 3 9))
>> This expression first resolves the multiplication: (+ 4 27), which is 31.
```

If you are new to this, I recommend starting with Daniel P. Friedman and Matthias Felleisen, *The little Schemer (4th ed.)* (Cambridge, MA, USA: MIT Press, 1996), ISBN: 0262560992. While you might be eager to dive into using Scheme in LilyPond, learning Scheme as a programming language will make the process smoother.[4]

14.1.3 Step 2: Study Lots of Snippets

Once familiar with Scheme, study how it integrates with LilyPond by reviewing snippets from LSR. Start with shorter examples and analyze their structure. Here is an example snippet for adding the *Schleifer* ornament:[5] The corresponding code:[6]

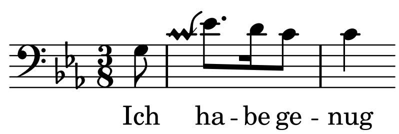

Figure 14.1: LSR No. 1185: *Schleifer* Ornament.

```
% Implementation by Martin Straeten of the Schleifer ornament
% as used by Johann Sebastian Bach, contributed to the user
% mailing list. In this case, it functions like a set of (always?)
% two grace notes, hence using a modified grace note to represent
% it in LilyPond makes sense.
%
% Code styling and user interface by Simon Albrecht 2024.

schleiferMarkup = \markup {
  \large \halign #.2 \raise #0.0
  \combine
  \halign #.8 \musicglyph "scripts.prall"
  \rotate #140 \normalsize \raise #2.4 \musicglyph "flags.u3"
}
schleiferGrace =
#(define-music-function (note) (ly:music?)
   #{
     \grace {
       \once\override NoteHead.stencil = #ly:text-interface::print
       \once\override NoteHead.X-extent = #'(-2 . -0)
       \once\override NoteHead.text = \schleiferMarkup
       \once\omit Stem
       \once\omit Flag
```

4. Liska and Samra's resources serve as excellent refreshers later on.
5. https://lsr.di.unimi.it/LSR/Item?id=1185
6. The mailing list thread referenced in the preamble is available at https://lists.gnu.org/archive/html/lilypond-user/2021-09/msg00352.html

```
24        $note
25      }
26    #})
27
28 \relative {
29   \time 3/8
30   \partial 8
31   \clef bass
32   \key c \minor
33   g8
34   \schleiferGrace c es8. d16 c8
35   c4
36 }
37 \addlyrics {
38   Ich ha -- be ge -- nug
39 }
```

The `\schleiferGrace` variable creates a customized ornament using Scheme's `define-music-function` macro. For a deeper understanding of the macro syntax, refer to the *LilyPond – Internals Reference*.[7]

Taking the variable `\schleiferGrace`, we see that invoking it creates an instance of activating a Scheme function that starts at Line 16. `define-music-function` is a macro that allows you to create a function that operates on LilyPond.

According to *LilyPond – Internals Reference*, the syntax for `define-music-function` is:

```
(define-music-function (arg1 arg2 …)
                       (type1? type2? …)
    function-body)
```

In the code, the argument's name is `note`, and it is tested according to the type specified in `type1?`, which in this case is `ly:music?`. According to the *Internal Reference*, `ly:music?` is a function that checks whether the object—in this case, `note`—is a `Music` object. Thus, it becomes clear that this function will not work unless it is followed by a musical note.

From Line 17 to Line 26, we see that a LilyPond code snippet has been inserted, as `#{` and `#}` signify the boundary of the LilyPond code within the Scheme code. This means that as part of invoking the variable `\schleiferGrace`, it passes through this LilyPond fragment, which is responsible for creating a grace note. Here, the notehead of the grace note is replaced with `\schleiferMarkup`, which is defined in Lines 9 to 14 of the code.[8]

Lines 22 and 23 show that the stem and flag are omitted from the grace note, while Line 24's `$note` signifies that the original argument `note` is called upon.[9] In this way, the *Schleifer* ornament is

7. https://lilypond.org/doc/v2.24/Documentation/internals/scheme-functions

8. The technique of sequential overrides, invoking the Scheme command `#ly:text-interface::print`, sets the `.stencil` of the notehead to use whatever is defined in the `.text` parameter. This technique is frequently used and is very useful in customizing notation. See also: https://lilypond.org/doc/v2.24/Documentation/notation/modifying-stencils.

9. Refer to this page for the difference between `#` and `$`: https://lilypond.org/doc/v2.24/Documentation/extending/lilypond-scheme-syntax

created from a note that follows the variable `\schleiferGrace`. This note is transformed into a grace note with a customized stencil setting, all done within the Scheme code.

14.1.4 Step 3: Hack the Codes

Once you study a code and become familiar with how it operates, experimenting with the code by hacking is a good way to deepen your understanding. Below, I give one example using the preceding *Schleifer* ornament snippet.

The *LilyPond – Internal Reference* reveals that the object `NoteHead` has its own standard settings, as well as support for about a dozen other interfaces.[10] One of them is the `grob-interface`, which makes it possible to change the color of a graphical object, or *Grob*.[11] Further reading in the *LilyPond – Notation Reference* shows that it is possible to override the color of an object.[12] Let us now tweak the *Schleifer* ornament code to allow us to change the ornament's color.

Following the reference, add the following line underneath `\once\override NoteHead.X-extent`:

```
\once\override NoteHead.color = #red
```

Running LilyPond now should produce the following result: Hard-coding a change like this may

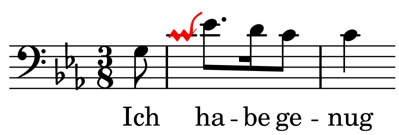

Figure 14.2: LSR No. 1185: *Schleifer* Ornament in red.

be good for testing the waters, but we may want the *Schleifer* ornament in more than just one color. The beauty of extending LilyPond is that we can customize the Scheme code to allow for this flexibility.

Let us move on. We should now let the `define-music-function` know that we are adding an additional argument to specify the color. The first part of the code will look like this:

```
#(define-music-function (note schleiferColor) (ly:music? color?)
```

This adds the argument `schleiferColor`, which only accepts color, as indicated by the corresponding test function `color?`.

Then, implement this argument in the sequence of `\once\override` processes. The line `NoteHead.color` can now be changed to:

```
\once\override NoteHead.color = #schleiferColor
```

Now, the variable `\schleiferGrace` requires one more argument to specify the ornament's color. The entire code should look like this:

```
schleiferMarkup = \markup {
  \large \halign #.2 \raise #0.0
  \combine
  \halign #.8 \musicglyph "scripts.prall"
```

10. https://lilypond.org/doc/v2.24/Documentation/internals/notehead
11. https://lilypond.org/doc/v2.24/Documentation/internals/grob_002dinterface
12. https://lilypond.org/doc/v2.24/Documentation/notation/inside-the-staff#coloring-objects

```
  \rotate #140 \normalsize \raise #2.4 \musicglyph "flags.u3"
}

schleiferGrace =
#(define-music-function (note schleiferColor) (ly:music? color?)
   #{
     \grace {
       \once\override NoteHead.stencil = #ly:text-interface::print
       \once\override NoteHead.X-extent = #'(-2 . 0)
       \once\override NoteHead.color = #schleiferColor
       \once\override NoteHead.text = \schleiferMarkup
       \once\omit Stem
       \once\omit Flag
       $note
     }
   #})
\relative {
  \time 3/8
  \partial 8
  \clef bass
  \key c \minor
  g8
  \schleiferGrace c #green es8. d16 c8
  c4
}
\addlyrics {
  Ich ha -- be ge -- nug
}
```

This produces the following output:

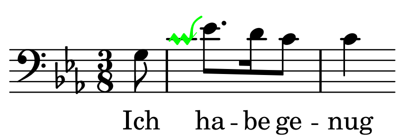

Figure 14.3: LSR No. 1185: *Schleifer* Ornament in green.

Notice that on Line 27, `#green` has been added. You can change this to any of the colors listed under ”Normal Colors” in the *Notation Reference*,[13] such as `#'"lightsalmon"`, `#(x11-color "medium turquoise")`, or even `#'"#5e45ad"`.

As an exercise, try replicating the following excerpt:[14]

Figure 14.4: Can you replicate this?

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13. https://lilypond.org/doc/v2.24/Documentation/notation/list-of-colors
14. See LSR1185e3.ly for the answer.

14.2 Example 1: Creating a Time Signature with Its Compound Meter Form

On January 1, 2025, I came across a post by an anonymous user on Facebook.[15] The post asked if it would be possible to create a time signature that had its beat structure expressed in the form of a compound meter. Something like this:

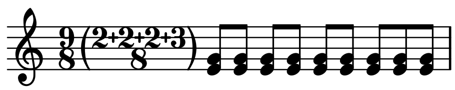

Figure 14.5: What the anonymous user wanted to achieve.

I responded to the post with relevant email threads on `lilypond-user` mailing list. I commented that it would be possible to override `TimeSignature.stencil` with custom-made time signatures. Incidentally, I was making a series of Fractional Time Signatures, which used this method.

The code for the aforementioned example is as follows:

```
\version "2.24.4"

{
  \clef "G"
  \time 9/8
  \set beatStructure = #'(2 2 2 3)
  \once \override Timing.TimeSignature.stencil = #ly:text-interface::print
  \once \override Timing.TimeSignature.text = \markup
  {
    \override #'(baseline-skip . 0)
    \center-column \number {9 8}
    \center-column {\fontsize #6 \musicglyph "accidentals.leftparen"}
    \hspace #-0.75
    \override #'(baseline-skip . 0)
    \center-column \number {{2+2+2+3} 8}
    \hspace #-0.75
    \center-column {\fontsize #6 \musicglyph "accidentals.rightparen"}
    \hspace #-1
  }
  \repeat unfold 9 {<e' g'>8 }
}
```

I realized that, while this might be an acceptable method if such time signatures appeared only once or twice in a piece, it may become problematic if I had to copy and paste this code every time I have such a time signature. Normally this could easily be resolved by making a variable out of `\override` clauses; however, a piece of music may use time signatures of this form in different configurations, just as the following example:

The code:

```
\version "2.24.4"
```

15. https://www.facebook.com/groups/gnulilypond/posts/10162467719483529/

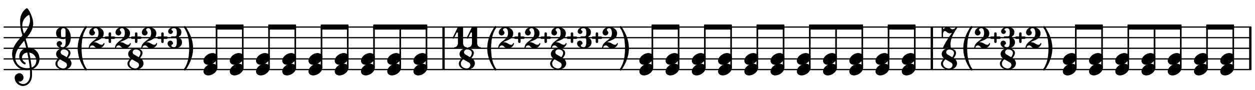

Figure 14.6: More compound meters.

```
{
  \clef "G"
  \time 9/8
  \set beatStructure = #'(2 2 2 3)
  \once \override Timing.TimeSignature.stencil = #ly:text-interface::print
  \once \override Timing.TimeSignature.text = \markup
  {
    \override #'(baseline-skip . 0)
    \center-column \number {9 8}
    \center-column {\fontsize #6 \musicglyph "accidentals.leftparen"}
    \hspace #-0.75
    \override #'(baseline-skip . 0)
    \center-column \number {{2+2+2+3} 8}
    \hspace #-0.75
    \center-column {\fontsize #6 \musicglyph "accidentals.rightparen"}
    \hspace #-1
  }
  \repeat unfold 9 {<e' g'>8}

  \time 11/8
  \set beatStructure = #'(2 2 2 3 2)
  \once \override Timing.TimeSignature.stencil = #ly:text-interface::print
  \once \override Timing.TimeSignature.text = \markup
  {
    \override #'(baseline-skip . 0)
    \center-column \number {11 8}
    \center-column {\fontsize #6 \musicglyph "accidentals.leftparen"}
    \hspace #-0.75
    \override #'(baseline-skip . 0)
    \center-column \number {{2+2+2+3+2} 8}
    \hspace #-0.75
    \center-column {\fontsize #6 \musicglyph "accidentals.rightparen"}
    \hspace #-1
  }
  \repeat unfold 11 {<e' g'>8 }

  \time 7/8
  \set beatStructure = #'(2 3 2)
  \once \override Timing.TimeSignature.stencil = #ly:text-interface::print
  \once \override Timing.TimeSignature.text = \markup
```

```
  {
    \override #'(baseline-skip . 0)
    \center-column \number {7 8}
    \center-column {\fontsize #6 \musicglyph "accidentals.leftparen"}
    \hspace #-0.75
    \override #'(baseline-skip . 0)
    \center-column \number {{2+3+2} 8}
    \hspace #-0.75
    \center-column {\fontsize #6 \musicglyph "accidentals.rightparen"}
    \hspace #-1
  }
  \repeat unfold 7 {<e' g'>8 }

}
```

Writing as long of a code as this (for just three measures!) would be cumbersome, indeed. What could help is to come up with a music function, using the Scheme.

14.2.1 Step 1: Analyze What Could Be Automatized

I quote the code for the first example of this section again. This time, however, I turn the variables that could change each time I create an instance of this kind of time signature, into red:

```
\version "2.24.4"

{
  \clef "G"
  \time 9/8
  \set beatStructure = #'(2 2 2 3)
  \once \override Timing.TimeSignature.stencil = #ly:text-interface::print
  \once \override Timing.TimeSignature.text = \markup
  {
    \override #'(baseline-skip . 0)
    \center-column \number {9 8}
    \center-column {\fontsize #6 \musicglyph "accidentals.leftparen"}
    \hspace #-0.75
    \override #'(baseline-skip . 0)
    \center-column \number {{2+2+2+3}{8}}
    \hspace #-0.75
    \center-column {\fontsize #6 \musicglyph "accidentals.rightparen"}
    \hspace #-1
  }
  \repeat unfold 9 {<e' g'>8 }
}
```

14.2.2 Step 2: Write the Code

It would be good if this function could accept the following as arguments:

- Time signature of the measure as defined normally in the LilyPond function `\time`. For this, I will set `timesig` as the name of the argument, that tests its value with `fraction?`.
- The customized stencil of the time signature. I need to declare how it looks, namely:
  - Overall time signature;
  - Numerator portion of the compound meter, and;
  - Denominator portion of the compound meter.

  It should look similar to how the LilyPond function `\compoundMeter` that accepts a list of lists.For this, I will set `beatstruct` as the name of the argument, that tests its value with `list?`.

I will now build the rest of the function. Notice the way the Scheme code references various locations of a list, using `car`, `cadr`, and so on:

```
\version "2.24.4"

compoundTimeWithBeatStructure =
#(define-music-function (timesig beatstruct) (fraction? list?)
   #{
     \time #timesig
     \set beatStructure = #(cadr beatstruct)
     \once \override Timing.TimeSignature.stencil = #ly:text-interface::print
     \once \override Timing.TimeSignature.text = \markup
     {
       \override #'(baseline-skip . 0)
       \center-column \number
       {
         #(number->string (car (car beatstruct)))
         #(number->string (cadr (car beatstruct)))
       }
       \center-column {\fontsize #6 \musicglyph "accidentals.leftparen"}
       \hspace #-0.75
       \override #'(baseline-skip . 0)
       \center-column \number
       {
         {#(string-join (map number->string (cadr beatstruct)) "+")}
         #(number->string (car (caddr beatstruct)))
       }
       \hspace #-0.75
       \center-column {\fontsize #6 \musicglyph "accidentals.rightparen"}
       \hspace #-1
     }
   #}

   )

{
```

```
  \compoundTimeWithBeatStructure 9/8 #'((9 8)(2 2 2 3)(8))
  \repeat unfold 9 {<e' g'>8}
  \compoundTimeWithBeatStructure 11/8 #'((11 8)(2 2 2 3 2)(8))
  \repeat unfold 11 {<e' g'>8}
  \compoundTimeWithBeatStructure 7/8 #'((7 8)(2 3 2)(8))
  \repeat unfold 7 {<e' g'>8}
}
```

Thus, there is now a function called `\compoundTimeWithBeatStructure`, whose grammar is:

```
\compoundTimeWithBeatStructure
        TIME_SIGNATURE #'((TIME_SIGNATURE)(BEAT_STRUCTURE)(DENOMINATOR))
```

Running the code will result in the identical snippet as the previous figure:

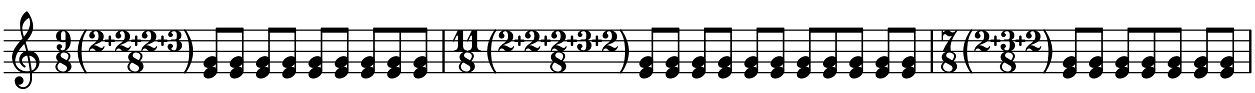

Figure 14.7: The same result as before with a shorter code.

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Appendices

Appendix A: Resources

As I taught LilyPond in a special topic course at the University of Delaware in Fall 2024, I compiled a list of links to useful websites and pages. It is in no way intended as a comprehensive list; instead, I list some essential pages that I have frequently looked up and found very useful. This page is subject to frequent revision.

On LilyPond

- Website: https://lilypond.org/
- Installing: https://lilypond.org/doc/v2.24/Documentation/learning/installing
- Manuals: https://lilypond.org/manuals.html

Text Editor for LilyPond

- Frescobaldi (Editor): https://frescobaldi.org/

Coding LilyPond

- Cheat Sheet: https://lilypond.org/doc/v2.24/Documentation/notation/cheat-sheet
- Snippets: https://lilypond.org/doc/v2.24/Documentation/web/snippets
- LilyPond Snippet Repository: https://lsr.di.unimi.it/

Mailing List

- Mailing list: https://lists.gnu.org/mailman/listinfo/lilypond-user
- Archives 1 https://lists.gnu.org/archive/html/lilypond-user/
- Archives 2 https://www.mail-archive.com/lilypond-user@gnu.org/

Advanced Topic on LilyPond

- LilyPond – Extending v2.24.4: https://lilypond.org/doc/v2.24/Documentation/extending/index#top
- Scheme (in LilyPond): https://scheme-book.readthedocs.io/en/latest/
- Extending LilyPond: https://extending-lilypond.gitlab.io/en/extending/index.html
- Scheme Resources https://www.gnu.org/software/guile/learn/#scheme-resources
- PostScript Manual: https://www.adobe.com/jp/print/postscript/pdfs/PLRM.pdf
- PostScript Tutorial: https://paulbourke.net/dataformats/postscript/

Troubleshooting

- The default text font for LilyPond doesn't seem to work (Mac)
- Frescobaldi freezes upon loading

Miscellaneous Items

- About Emmentaler font: https://lilypond.org/doc/v2.25/Documentation/notation/the-emmentaler-font

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